

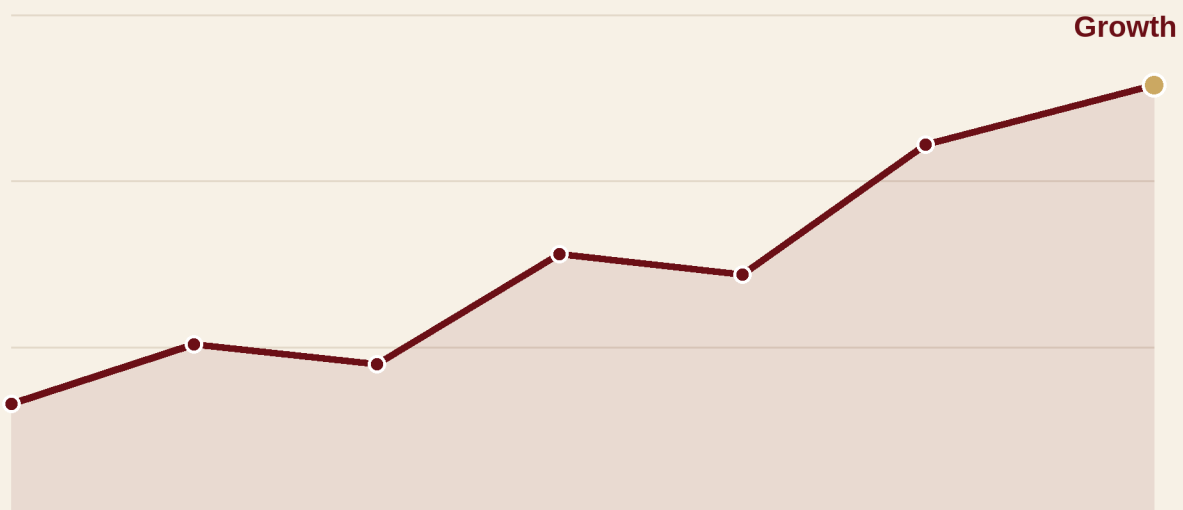
FINANCIAL MANAGEMENT

One Stop Solution

QUESTION BANK | PART 1

PERFORMANCE ANALYSIS

Investment appraisal • Cost of capital • Working capital



Master Index

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How to Approach the ACCA FM Exam

1. Exam format

The FM exam is a 3-hour computer-based exam worth 100 marks, split into three compulsory sections. There is no choice of questions, everything must be attempted.

Section	Structure	Marks
A	15 objective test questions (OTQs), 2 marks each, unrelated to one another	30
B	3 scenario-based question sets; each scenario has 5 linked OTQs worth 2 marks each. A scenario can be drawn from any part of the syllabus and mixes calculation and narrative items	30
C	2 constructed response questions (CRQs), 20 marks each, manually marked, each usually split into several sub-requirements combining calculation and discussion	40

Sections A and B are marked by computer, so only the final answer matters there, no credit is given for workings. Section C is marked by a human, so workings, layout and stated assumptions directly earn marks, even if the final figure is wrong.

2. Overall time management

A simple and reliable rule is 1.8 minutes per mark (180 minutes / 100 marks), which gives a working budget of:

Section	Marks	Target time
A	30	54 minutes
B	30	54 minutes
C	40	72 minutes

Most students do best starting with Section A, then B, then C, since the OTQ sections warm you up and bank marks quickly before the two long Section C questions. If a Section C question looks unfamiliar or difficult, do the more comfortable one first, the order in which the two are answered makes no difference to marking.

The single most common way marks are lost in FM is running out of time on Section C because too long was spent chasing an elusive final answer in Section A or B. If an OTQ is not solved within roughly 2-3 minutes, make a reasoned guess, flag it if the software allows review, and move on. A guessed OTQ has a real chance of being right; an unfinished Section C requirement has none.

3. Approaching Section A and B (objective test questions)

Since only the selected answer is marked, the goal is accuracy under time pressure rather than a full written derivation:

Read the requirement itself before the scenario, so you know exactly what number or statement you are being asked to identify; this avoids wasted reading time and mis-answering a question that had extra, irrelevant data.

Do rough workings on the on-screen scratchpad rather than in your head, OTQs are deliberately designed with plausible "distractor" answers that match a common error (e.g. using the wrong discount factor, or forgetting to tax-adjust interest), so a hasty mental calculation is more likely to land on a distractor than a correct answer.

For "select TWO" or "select THREE" style questions, evaluate every option on its own merits rather than assuming the first plausible-looking ones are correct; these questions are marked as all-or-nothing in most CBE formats, so a single wrong selection costs the whole 2 marks.

For true/false and multi-statement questions, watch for absolute wording ("always", "never", "only"); these are frequently the false statement, since finance theory rarely holds in every case.

In Section B, read the whole scenario once before attempting any of its five items, later items sometimes depend on figures calculated in an earlier item within the same scenario, so working the items out of order can save time.

4. Approaching Section C (constructed response questions)

Section C is where technique has the biggest effect on the final mark, because a human marker is awarding marks for method, not just the final number:

Read the full requirement before starting any calculation, and note the mark allocation for each sub-part; it tells you almost exactly how much time and detail that sub-part deserves. A 4-mark sub-requirement does not need the depth of a 12-mark one.

Show full workings, labelled clearly (e.g. "Cost of equity", "WACC", "NPV"), and lay them out in the order the marker would expect to see them. If the final figure is wrong because of one arithmetic slip, clear, correctly-structured workings still earn most of the method marks; an unlabelled block of numbers with no final answer earns very little.

State assumptions explicitly wherever the question is ambiguous (e.g. "it is assumed that the cash flows arise at the end of each year", "it is assumed that the current level of working capital remains unchanged"), markers are specifically instructed to give credit for reasonable, clearly stated assumptions.

For discussion requirements, write in short, clearly separated points rather than one dense paragraph, and make sure every point is applied to the specific company or scenario in the question rather than left as generic theory, "bookwork" answers that never mention the company's own figures or situation are capped well below full marks.

Attempt every sub-requirement, even partially. A rough, partially correct attempt at a sub-part almost always scores more than leaving it blank, and a blank sub-part guarantees zero on marks that might otherwise be picked up for method alone (e.g. correctly identifying the formula, or correctly setting up a table, even without a completed final figure).

If a figure carried forward from an earlier part turns out to be wrong, keep using it consistently in later parts rather than stopping, FM (like most ACCA papers) awards "own figure" follow-through marks, so one early error does not have to cost marks in every part that depends on it.

5. Topics that recur heavily across all three sections

Certain techniques are tested repeatedly, in different disguises, across Sections A, B and C, so fluency with them pays off disproportionately: discounted cash flow investment appraisal (NPV, IRR, and the effect of tax and inflation); working capital management (receivables/payables/inventory policy and the cash operating cycle); the cost of capital (CAPM, dividend growth model, WACC, and ungearing/re gearing betas); sources of finance and rights issue calculations (TERP, EPS impact); foreign currency and interest rate risk hedging (forward contracts, money market hedges, FRAs); and business valuation (asset-based, P/E, dividend-based and cash-flow-based methods). These map directly onto Chapters 2-6, 7-10, 13-14, 15-18 and 20 of this question bank, and are the areas most worth over-practising rather than treating every chapter as equally likely to appear.

6. In the final weeks before the exam

Practise under timed conditions using full past exam questions (not just isolated OTQs), so that the 1.8-minutes-per-mark pace becomes automatic rather than something to calculate in the exam itself.

Review the FM examiner's reports for recent sittings (linked from ACCA's "Examining team guidance" page for FM), these describe, sitting by sitting, exactly which calculation steps and discussion points candidates most often got wrong, which is far more targeted than generic advice.

Keep a short personal list of formulae and their trigger words (e.g. "irredeemable preference share -> $K_p = \text{dividend}/\text{market price}$ "), and test yourself on recalling them from memory alone, since no formula sheet beyond ACCA's standard exam formulae and tables is provided for anything else.

On the day itself, do a fast first pass through all three sections attempting only what comes immediately, then a second pass to pick up anything skipped; this guarantees that easy marks are never lost purely because time ran out before reaching them.

7. ACCA command verbs

Every FM requirement is built around a specific command verb, and that verb tells you exactly what kind of answer is expected, not just the topic. The table below is ACCA's own verb library. Two verbs in particular catch people out in FM: "Calculate" wants numbers with workings and nothing else, while "Discuss", "Evaluate", "Advise" and "Recommend" want prose built on those numbers, not a repeat of the calculation. Matching your answer style to the verb, rather than writing the same way for every requirement, is one of the simplest ways to stop losing easy marks in Section C.

Verb	What it requires
Define	State the precise meaning of a term or concept concisely, without extra detail.
State	Give a clear, concise fact or point; no analysis or justification is needed.
List	Itemise the elements asked for, without explaining them.
Identify	Pinpoint a specific issue, adjustment, risk or indicator relevant to the scenario.
Describe	Communicate the key features of something by setting out its characteristics.
Summarise	Give a condensed account of the issues, covering the main facts only.
Explain	Make a concept clear and understandable by setting out the logic or reasoning behind it.
Illustrate	Support an explanation with a clear, relevant example, often a small calculation.
Interpret	Convey the meaning of information or data, including why it matters.
Apply	Put a concept, technique or model to work in the context of the scenario given.
Calculate	Work out a numerical answer using the appropriate method, formula or model, showing workings.
Prepare	Produce a document in the format asked for (e.g. a report or briefing note), in a tone suited to the reader.
Analyse	Break the information into its components and examine each in detail to identify the underlying issues.
Compare	Set out how two or more items relate to each other.
Contrast	Set out how two or more items differ from each other.
Assess	Weigh up the importance or value of something, considering both its strengths and weaknesses.
Evaluate	Judge the value or effectiveness of something, covering advantages, disadvantages, risks and opportunities.
Discuss	Examine an issue from more than one side, setting out balanced arguments for and against.
Advise	Give expert guidance that helps the reader reach a well-informed decision, based on the analysis done.
Recommend	Choose the single most appropriate option or action, and state it clearly, following logically from the analysis.

Verb	What it requires
Justify	Give adequate grounds for a decision or conclusion, using the scenario's own facts as supporting evidence.
Conclude	Arrive at a clear, well-reasoned final judgement.

As a rough guide, "Define", "State" and "List" need only a sentence or a short line per point, while "Discuss", "Evaluate" and "Advise" need fully developed points, each explained and applied to the scenario, in line with the marks on offer. If a requirement combines a calculation verb with a discussion verb (for example "Calculate the WACC and advise the board whether the project should proceed"), treat it as two separate tasks with two separate answer styles, not one blended paragraph.

Formulae & Discount Tables

A quick-reference sheet of the core formulae used across this question bank, plus present value and annuity discount tables for offline practice. Grouped by syllabus area, matching the chapter order of the exam kit.

1. Investment appraisal (Chapters 2-6)

Future value: $FV = PV \times (1 + r)^n$

Present value: $PV = FV / (1 + r)^n$

Annuity present value factor (years 1 to n): $[1 - (1 + r)^{-n}] / r$

Perpetuity present value factor: $1 / r$

Growing perpetuity present value factor: $1 / (r - g)$

IRR by linear interpolation: $IRR = a\% + [NPV_a / (NPV_a - NPV_b)] \times (b\% - a\%)$

Return on capital employed (average investment): $ROCE = \text{Average annual accounting profit} / \text{Average investment}$

2. Cost of capital and business valuations (Chapters 15-20)

Dividend growth model (cost of equity): $Ke = [D_0 \times (1 + g)] / P_0 + g$

Historic dividend growth rate: $g = (D_n / D_0)^{1/n} - 1$

Capital Asset Pricing Model: $Ke = R_f + \text{Beta} \times (R_m - R_f)$

Cost of irredeemable debt (after tax): $Kd = [i \times (1 - T)] / MV$

Cost of redeemable debt: IRR of the loan note's post-tax cash flows (interest and redemption)

Weighted average cost of capital: $WACC = [Ke \times Ve / (Ve + Vd)] + [Kd \times (1 - T) \times Vd / (Ve + Vd)]$

Ungearing an equity beta (Hamada): $\text{Beta-asset} = \text{Beta-equity} \times [Ve / (Ve + Vd \times (1 - T))]$

Regearing an asset beta: $\text{Beta-equity} = \text{Beta-asset} \times [(Ve + Vd \times (1 - T)) / Ve]$

Dividend valuation model (share price): $P_0 = D_1 / (Ke - g)$

Price/earnings method (equity value): $\text{Value} = \text{Earnings} \times \text{P/E ratio}$

Dividend yield: $\text{Dividend yield} = D_0 / P_0$

Miller and Modigliani (no tax): $Ke = Keu + (Keu - Kd) \times (Vd / Ve)$

Miller and Modigliani (with tax): $Ke = Keu + (Keu - Kd) \times (1 - T) \times (Vd / Ve)$

3. Working capital management (Chapters 7-10)

Economic order quantity: $EOQ = \sqrt{(2 \times Co \times D) / Ch}$

Miller-Orr spread: $\text{Spread} = 3 \times [(3/4 \times \text{transaction cost} \times \text{variance}) / \text{daily interest rate}]^{1/3}$

Miller-Orr upper limit: $\text{Upper limit} = \text{Lower limit} + \text{Spread}$

Miller-Orr return point: $\text{Return point} = \text{Lower limit} + (\text{Spread} / 3)$

Cash operating (working capital) cycle: Inventory days + Receivables days - Payables days

4. Foreign currency and interest rate risk (Chapters 13-14)

Purchasing power parity (expected future spot rate): $S_1 = S_0 \times [(1 + \text{inflation, country A}) / (1 + \text{inflation, country B})]$

Interest rate parity (forward rate): $F_0 = S_0 \times [(1 + \text{interest rate, country A}) / (1 + \text{interest rate, country B})]$

Money market hedge: Borrow/deposit today in the foreign currency, converting at spot, so the future receipt or payment is matched

Present Value Table

Present value of \$1 = $1 / (1 + r)^n$, shown to 3 decimal places.

n	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239

Rates 1% to 10%, periods 1 to 15.

n	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065

Rates 11% to 20%, periods 1 to 15.

Annuity Table

Present value of an annuity of \$1 per year for n years = $[1 - (1 + r)^{-n}] / r$, shown to 3 decimal places.

n	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606

Rates 1% to 10%, periods 1 to 15.

n	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675

Rates 11% to 20%, periods 1 to 15.

Basic Concept

1. How to Think Through an FM Question

FM questions rarely fail students because the underlying technique is unknown. They fail because the question wasn't broken down properly before the student started calculating. A reliable routine:

Read the requirement before the scenario. The requirement (the actual instruction, usually at the end) tells you exactly what number or statement is wanted. Reading the scenario first without knowing the target often means re-reading it a second time anyway.

Identify the command word. "Calculate" wants a number with workings; "Discuss" or "Evaluate" wants written analysis. Answering a calculation question with prose, or a discussion question with only a number, loses marks regardless of how correct the underlying thinking is. (The full command-verb list is in the How to Approach the Exam guide.)

List out the data actually given, in the order it appears, before touching a formula. Scenarios often bury one crucial figure mid-paragraph (a tax rate, a discount rate, a "payable one year in arrears") that is easy to miss on a first read.

Identify what is missing and has to be derived. Most exam questions do not hand you every input directly, at least one figure (a growth rate, a missing year's cash flow, an implied discount rate) has to be calculated from what is given, using exactly the kind of ratio or rearrangement covered in sections 2-4 below.

Match the scenario to a technique. A question mentioning "payback", "NPV" or "IRR" belongs to investment appraisal (Chapters 2-6); "receivables days" or "EOQ" belongs to working capital (Chapters 7-10); "cost of equity" or "WACC" belongs to cost of capital (Chapter 17); and so on. Recognising the topic quickly is most of the battle.

Show every working step, labelled. Even in objective test questions where only the final answer is marked, doing the workings on paper (or the on-screen scratchpad) in a labelled, logical order is what prevents arithmetic slips, and in Section C it is what earns the marks even when the final figure is wrong.

Sense-check the answer before moving on. A percentage should not be negative unless the question is about a loss or a fall; a payback period should not be longer than the project's life; an NPV that is wildly larger than the initial investment usually signals an error, not a correct answer.

A quick way to spot what a question is really testing: scan it once for numbers with \$ signs (calculation), numbers with % signs (a rate to apply, or a rate being asked for), and any command verb in bold or at the very end (what kind of answer is wanted). Three seconds of scanning like this before reading properly saves far more time than it costs.

Mini scenario: "A machine costing \$40,000 is expected to generate net cash inflows of \$15,000 a year for 4 years. The company's cost of capital is 12%. Calculate the NPV of the investment and advise whether it should proceed."

Requirement: Two tasks in one sentence: Calculate (a number, with workings) and Advise (a short written conclusion).

Data given: Cost \$40,000 (Year 0), inflow \$15,000 a year for 4 years (Years 1-4), cost of capital 12%.

Technique: "NPV" and "cost of capital" both point straight to investment appraisal (Chapters 2-3): discount each cash flow to present value and sum them, then advise based on whether the total is positive or negative.

2. Basic Mathematics the Exam Assumes You Know

BODMAS (order of operations)

Calculations must be done in a fixed order: **Brackets first, then Orders (powers and roots), then Division and Multiplication (left to right), then Addition and Subtraction (left to right)**. Getting this order wrong is one of the most common sources of a wrong answer despite knowing the correct formula.

Example: Calculate $200 + 300 \times (1 + 0.10)^2$

Step 1 (Brackets): $(1 + 0.10) = 1.10$

Step 2 (Orders): $1.10^2 = 1.21$

Step 3 (Multiplication): $300 \times 1.21 = 363$

Step 4 (Addition): $200 + 363 = 563$

Adding $200 + 300$ first (ignoring the brackets and power) would wrongly give $500 \times 1.21 = 605$, a completely different and incorrect answer.

Percentages

A percentage is just a fraction of 100, so $8\% = 8/100 = 0.08$. **To increase a value by a percentage, multiply by $(1 + \text{the rate})$; to decrease it, multiply by $(1 - \text{the rate})$.** To find the percentage change between two values, use $(\text{new value} - \text{old value}) / \text{old value}$.

Example (increase): A share price rises from \$4.00 to \$4.60. What is the percentage increase?

$$(4.60 - 4.00) / 4.00 = 0.60 / 4.00 = 0.15 = 15\%$$

Example (decrease): Operating costs fall from \$12,000 to \$9,600. What is the percentage decrease?

$$(9,600 - 12,000) / 12,000 = -2,400 / 12,000 = -0.20, \text{ i.e. a } 20\% \text{ decrease.}$$

Which figure goes in the denominator? Students often get confused when a question doesn't literally say "old" and "new" (it might say "current" vs "budgeted", "before" vs "after", "this year" vs "last year"). The rule never changes: **the denominator is always the value the change happened FROM**, the starting point, the earlier one in time, the 100% you are comparing everything else against. It does not matter which value the question mentions first in the sentence; what matters is which one existed first in time.

Example: A question says: "the exchange rate is now \$1.65, compared to \$1.50 before the announcement." Even though \$1.65 is mentioned first, \$1.50 is the OLD (earlier) rate, so it goes in the denominator: $(1.65 - 1.50) / 1.50 = 10\%$ increase.

Percentage change is not the same as a percentage point change, this is a very common exam trap. A "percentage point" is a simple, plain difference between two percentages; a "percentage change" (or "percentage increase/decrease") is that difference expressed as a fraction of the original percentage.

Example: A company's profit margin rises from 20% to 25%.

Percentage point change: $25\% - 20\% = 5$ percentage points.

Percentage change: $(25\% - 20\%) / 20\% = 5/20 = 0.25 =$ a 25% increase in the margin.

Always check exactly which one the question is asking for: "by how many percentage points" wants the plain difference (5), while "by what percentage" or "percentage increase" wants the proportional change (25%).

Negative numbers and the brackets convention

In accounting and finance, **a negative number is almost always shown in round brackets rather than with a minus sign**, so \$(1,500) means negative \$1,500, an outflow, a loss, or a reduction, depending on context. This convention is used throughout every chapter of this exam kit and in the real ACCA exam software, so treat (X) as -X automatically when reading any table or cash flow line.

Example: A summary shows Revenue \$80,000 and Costs \$(95,000). What is the net result?

$\$80,000 + \$(95,000) = \$80,000 - \$95,000 = \$(15,000)$, i.e. a net loss of \$15,000.

Rounding, and when to do it

Only round the final answer, never a figure used in a later calculation. Carrying a rounded intermediate figure into the next step (for example rounding a discount factor to 2 decimal places and then using that rounded figure to calculate a present value) is one of the most common causes of an answer being marked wrong despite the method being entirely correct. Keep full precision (or at least 4 decimal places) throughout the workings, and round only at the very last step, to whatever the question specifically asks for ("to the nearest \$", "to one decimal place", "to 3 decimal places", and so on).

Wrong: Round the discount factor early to 0.75, then $0.75 \times \$10,000 = \$7,500$.

Right: Keep the full discount factor 0.7513, then $0.7513 \times \$10,000 = \$7,513$. Only this final figure gets rounded, e.g. to the nearest \$, if the question asks for it.

Reading \$000 and \$m notation

A scenario stating "all figures in \$000" or a table headed "\$'000" means **every number shown should be multiplied by 1,000** to get the actual dollar amount, so a figure of "450" in such a table is actually \$450,000. The same logic applies to "\$m" (multiply by 1,000,000). Losing track of the units, or mixing a \$000 figure with a plain \$ figure in the same calculation, is a frequent and entirely avoidable source of an answer being out by a factor of 1,000.

Example: A table headed "\$'000" shows Revenue 4,500 and Costs 2,800.

Actual figures: Revenue \$4,500,000 and Costs \$2,800,000, so profit = \$1,700,000 (not \$1,700).

3. Financial Statement Terminology and Common Conventions

Before the ratios in the next section make sense, it helps to be fluent in the terminology of the income statement itself, since questions and sources use several different names for exactly the same line, and a few conventions (how a percentage dividend is calculated, how an interest rate should be read) trip students up if they haven't been made explicit once.

Income statement pro-forma

Line item	Also known as	\$
Sales	Revenue, Turnover	xxx
(-) Cost of sales	COGS, Cost of goods sold	(xxx)
Gross profit	Gross income	xxx
(-) Operating expenses	e.g. admin, selling & distribution expense	(xxx)
Operating profit	EBIT, PBIT, Operating income	xxx
(-) Interest expense	Finance costs	(xxx)
Profit before tax	EBT, PBT, Taxable income	xxx
(-) Tax		(xxx)
Profit after tax	EAT, PAT, Net profit, Net income	xxx
(-) Preference dividend		(xxx)
Earnings	Profit attributable to ordinary shareholders	xxx

A margin is not another name for the profit figure itself, it is that profit figure expressed as a percentage of sales:

Gross margin = Gross profit / Sales

Operating margin = Operating profit / Sales

Net margin = Profit after tax / Sales

If a question gives a margin together with the sales figure, multiply the two together to find the missing \$ line, then carry on building the rest of the income statement from there in the usual way, this is a common technique in questions that reconstruct a financial statement from a set of given ratios.

Example: Sales are \$2,000,000 and the gross margin is 35%. What is the cost of sales?

Gross profit = 35% x \$2,000,000 = \$700,000, so cost of sales = \$2,000,000 - \$700,000 = \$1,300,000.

Earnings is not the final step, it is what is used to calculate Earnings Per Share (EPS): EPS = Earnings / Number of ordinary shares in issue. This is one of the most frequently required figures in this

exam kit, feeding directly into the P/E ratio (share price/EPS) and into share price and shareholder wealth calculations throughout Chapters 1, 15, 16, 17 and 18.

Example: Earnings for the year are \$1,600,000 and there are 2,000,000 ordinary shares in issue.
 $EPS = \$1,600,000 / 2,000,000 = \0.80 per share.

Interpretation of dividends

Measure	Based on
Dividend (given as a percentage)	Nominal (face) value of the share
Dividend yield	Market price of the share
Dividend payout ratio	Earnings

A dividend expressed as a percentage (e.g. "a 6% dividend") is a percentage of the share's **nominal (face) value**, not its market price, unless a question specifically states otherwise. Dividend yield, by contrast, is calculated against **market price** (dividend per share / market price per share), and dividend payout ratio is calculated against **earnings** (dividend / earnings, or dividend per share / EPS).

Example: A share has a nominal value of \$1.00 and a market price of \$2.27. A 25% dividend has just been paid.

Dividend per share = $25\% \times \$1.00 = \0.25 . Dividend yield = $\$0.25 / \$2.27 = 11.0\%$ (based on market price, not nominal value).

Interest rate conventions

Unless a question specifically states otherwise, **an interest rate is always assumed to be a per annum (p.a.) rate**, even where it relates to a shorter borrowing, deposit or forward period.

- (i) An interest rate of 10% means 10% p.a.
- (ii) An interest rate on 6 months' borrowing of 10% still means 10% p.a., time-apportioned to $10\% \times 6/12 = 5\%$ for that six-month period.
- (iii) A 6-month LIBOR rate of 10% is likewise 10% p.a., not 10% for the six months.
- (iv) An interest rate of 1% per month, by contrast, is explicitly stated as a monthly (p.m.) rate, and should be used directly rather than treated as an annual rate to be pro-rated.

4. Ratios You're Expected to Recognise on Sight

Chapter 19 covers financial ratio analysis in full, but scenarios throughout earlier chapters (particularly Case Scenario and Constructed Response Questions) **routinely expect you to calculate or interpret a ratio without re-explaining the formula**. This is a quick-reference list of the ones that turn up most often before Chapter 19.

Profitability ratios

Ratio	Formula	What it measures
ROCE	Operating profit / (Total equity + Non-current liabilities)	Overall profitability relative to the capital invested
Gross profit margin	Gross profit / Revenue	Profitability at the trading (before overheads) level
Net profit margin	Profit after tax / Revenue	Overall profitability after all costs, interest and tax

Liquidity and gearing ratios

Ratio	Formula	What it measures
Current ratio	Current assets / Current liabilities	Short-term liquidity, ability to cover near-term obligations
Quick (acid-test) ratio	(Current assets - Inventory) / Current liabilities	Liquidity excluding the slowest-to-convert current asset
Gearing ratio	Debt / (Debt + Equity), by book or market value	Reliance on debt finance relative to equity
Interest cover	Operating profit / Interest expense	How comfortably profit covers the interest bill

Working capital efficiency ratios

Ratio	Formula	What it measures
Receivables days	(Trade receivables / Credit sales) x 365	Average time taken to collect cash from customers
Payables days	(Trade payables / Credit purchases) x 365	Average time taken to pay suppliers
Inventory days	(Inventory / Cost of sales) x 365	Average time inventory is held before sale

Investor ratios

Ratio	Formula	What it measures
EPS	Profit after tax and preference dividends / Number of ordinary shares	Profit attributable to each ordinary share
Dividend cover	EPS / Dividend per share	How many times over profit could

Ratio	Formula	What it measures
		pay the declared dividend
Dividend yield	Dividend per share / Share price	Cash return to a shareholder relative to the share price
Price/earnings (P/E) ratio	Share price / EPS	How many years of current earnings the share price represents

These formulas reappear, sometimes rearranged to solve for a different unknown, throughout Chapters 1, 15, 17, 19 and 20. Section 7 below covers how to rearrange a formula like this when the question gives you the ratio's result and asks for one of the inputs instead.

Example: A company has operating profit \$180,000, total equity \$600,000, non-current liabilities \$400,000, current assets \$150,000 and current liabilities \$100,000.

ROCE: $180,000 / (600,000 + 400,000) = 180,000 / 1,000,000 = 18\%$

Current ratio: $150,000 / 100,000 = 1.5$ (or 1.5:1)

5. Time Value of Money: What is Present Value?

A dollar received today is worth more than a dollar received in a year's time, even with zero inflation and zero risk, simply because today's dollar could be invested now and start earning a return. This is the single idea that Chapters 2 to 6, most of Chapters 13-14, and the business valuation methods in Chapter 20 are all built on.

Future value: what a sum today will grow to

If \$PV is invested today at an annual rate r , compounding for n years, its future value is:

$$FV = PV \times (1 + r)^n$$

Example: \$1,000 invested today at 6% per year for 3 years.

$$FV = 1,000 \times (1.06)^3 = 1,000 \times 1.191 = \$1,191$$

Present value: working backwards from a future sum

Present value is the reverse question: what is a sum receivable in the future actually worth today, given a required rate of return? Rearranging the future value formula:

$$PV = \frac{FV}{(1 + r)^n}$$

Example: What is \$1,191 receivable in 3 years' time worth today, at a discount rate of 6%?

$$PV = 1,191 / (1.06)^3 = 1,191 / 1.191 = \$1,000$$

(This is exactly the reverse of the future value example above, which is exactly the point:

discounting simply undoes compounding.)

Discount factors, and why tables exist

The term $1 / (1 + r)^n$ is called the discount factor for rate r and period n . Rather than recalculating $(1 + r)^n$ by hand for every question, published present value tables (see the Formulae & Tables section of this exam kit) give the discount factor directly, so a present value calculation becomes: $PV = \text{future cash flow} \times \text{discount factor}$. **An annuity factor works the same way, but gives the combined discount factor for a regular cash flow received every year for several years in a row**, so it does not need to be discounted year by year.

Example (annuity): What is the present value of \$500 receivable at the end of each year for 4 years, at a discount rate of 8%?

Annuity factor = $[1 - (1.08)^{-4}] / 0.08 = 3.312$ (or read straight from the Annuity Table in Formulae & Tables)

$PV = 500 \times 3.312 = \$1,656$

Why this matters practically: any time a question gives you a discount rate (also called the cost of capital, required return, or interest rate) and a cash flow at some future point, **the instinct should be to ask "do I need to bring this to today's value (discount it) or project it forward (compound it)?"** Almost every investment appraisal, bond or share valuation, and lease-versus-buy question in this exam kit is built from repeated applications of exactly this one idea.

6. Reading a Scenario Correctly

A large share of avoidable errors come from misreading the scenario rather than misapplying the technique. A few conventions used consistently throughout this exam kit, and in the real exam:

Time 0 (or Year 0) means now, the point at which any initial investment or cost is incurred, and it is never discounted (its discount factor is always 1.000). Year 1 means one year from now, and so on.

"Payable in arrears" means paid at the end of the period it relates to (for example, tax payable one year in arrears on a Year 1 profit is actually paid in Year 2). **"Payable in advance" means paid at the start of the period.**

Unless a question states otherwise, **assume a 365-day year** for ratio calculations (receivables/payables/inventory days) and **ignore taxation entirely if no tax rate is mentioned** anywhere in the question.

A table given in a scenario is there to be used directly. If a table shows five years of sales volumes or costs, the question is almost always expecting those exact figures to feed into the calculation, not a simplified or averaged version of them.

Example (arrears timing): A company's Year 1 taxable profit is \$50,000, tax rate 20%, tax payable one year in arrears.

The $\$50,000 \times 20\% = \$10,000$ tax charge relates to Year 1, but the actual cash outflow (the figure to discount) falls in Year 2, not Year 1.

7. Basic Algebra: Rearranging a Formula

Many questions give you the result of a formula and ask you to find one of the inputs instead of the output, this is just algebra, not a new technique. **The rule is always the same: whatever is done to one side of the equation must be done to the other, until the unknown is on its own.**

Example: The dividend growth model states $P_0 = D_1 / (K_e - g)$. A share is priced at \$2.50, next year's dividend is expected to be \$0.20, and the growth rate is 4%. What is the cost of equity, K_e ?

Step 1: Start from $P_0 = D_1 / (K_e - g)$, and multiply both sides by $(K_e - g)$:

$$P_0 \times (K_e - g) = D_1$$

Step 2: Divide both sides by P_0 :

$$K_e - g = D_1 / P_0$$

Step 3: Add g to both sides to isolate K_e :

$$K_e = D_1 / P_0 + g = 0.20 / 2.50 + 0.04 = 0.08 + 0.04 = 0.12 = 12\%$$

Example (a second rearrangement): A project has an average annual accounting profit of \$24,000 and a target ROCE of 15%. What average investment does this imply?

Step 1: Start from $ROCE = \text{Average annual accounting profit} / \text{Average investment}$:

$$0.15 = 24,000 / \text{Average investment}$$

Step 2: Multiply both sides by Average investment, then divide both sides by 0.15:

$$\text{Average investment} = 24,000 / 0.15 = \$160,000$$

The same rearranging approach applies to the present value formula (solving for the discount rate r , or the number of periods n), to the WACC formula (solving for K_e or K_d if the overall WACC is given instead), and to several of the ratios in Section 4. **If a formula is rearranged correctly, plugging the known values back into the original, unrearranged formula should reproduce the number the question gave you**, which is a reliable way to check the algebra before moving on.

Study Progress Tracker

Tick each box as you complete a pass through the chapter.

Chapter	Studied	Practised	Revised	2nd Revision
Chapter 1: The Financial Management Function	☐	☐	☐	☐
Chapter 2: Basic Investment Appraisal Techniques	☐	☐	☐	☐
Chapter 3: Investment Appraisal: Discounted Cash Flow Techniques	☐	☐	☐	☐
Chapter 4: Investment Appraisal: Further Aspects of Discounted Cash Flows	☐	☐	☐	☐
Chapter 5: Asset Investment Decisions and Capital Rationing	☐	☐	☐	☐
Chapter 6: Investment Appraisal Under Uncertainty	☐	☐	☐	☐
Chapter 7: Working Capital Management	☐	☐	☐	☐
Chapter 8: Working Capital Management: Inventory Control	☐	☐	☐	☐
Chapter 9: Working Capital Management: Accounts Receivable and Payable	☐	☐	☐	☐
Chapter 10: Working Capital Management: Cash and Funding Strategies	☐	☐	☐	☐
Chapter 11: The Economic Environment for Business	☐	☐	☐	☐
Chapter 12: Financial Markets and the Treasury Function	☐	☐	☐	☐
Chapter 13: Foreign Exchange Risk	☐	☐	☐	☐
Chapter 14: Interest Rate Risk	☐	☐	☐	☐
Chapter 15: Sources of Finance	☐	☐	☐	☐
Chapter 16: Dividend Policy	☐	☐	☐	☐
Chapter 17: The Cost of Capital	☐	☐	☐	☐
Chapter 18: Capital Structure	☐	☐	☐	☐
Chapter 19: Financial Ratios	☐	☐	☐	☐
Chapter 20: Business Valuations and Market Efficiency	☐	☐	☐	☐

Chapter 1: The Financial Management Function

Questions

Objective Test Questions

The nature and purpose of financial management

1 In relation to the financial management of a company, which of the following provides the best definition of a firm's primary financial objective?

- A To achieve long-term growth in earnings
- B To maximise the level of annual dividends
- C To maximise the wealth of its ordinary shareholders
- D To maximise the level of annual profits

2 Which THREE of the following are the main types of decision facing the financial manager in a company?

- A Income decision
- B Investment decision
- C Dividend decision
- D Financing decision
- E Appraisal decision
- F Budget decision

3 Which of the following is LEAST likely to fall within financial management?

- A The dividend payment to shareholders is increased
- B Funds are raised to finance an investment project
- C Surplus assets are sold off
- D A report is produced comparing actual results to budget

4 Which of the following tasks would typically be carried out by a member of the financial management team?

- A Evaluating proposed expansion plans
- B Review of overtime spending
- C Depreciation of non-current assets
- D Apportioning overheads to cost units

5 Which of the following would you expect to be the responsibility of financial management?

- A Producing annual accounts
- B Producing monthly management accounts
- C Advising on investment in non-current assets
- D Deciding pay rates for staff

6 Which of the following is LEAST likely to be within the scope of financial management?

- A Determining the amount of dividend to pay to shareholders
- B Raising funds to finance an investment project
- C Selling off surplus assets
- D Appointing non-executive directors to the remuneration committee

7 Which of the following statements describes the main objective of financial management?

- A Efficient acquisition and deployment of financial resources to ensure achievement of objectives
- B Providing information to management for day-to-day functions of control and decision-making
- C Providing information to external users about the historical results of the organisation
- D Maximisation of shareholder wealth

Financial objectives

8 Which of the following is most consistent with maximising shareholder wealth?

- A Profit maximisation
- B Market share growth
- C Minimising the company's cost of capital
- D Maximising earnings per share

9 Which of the following is the best indicator of shareholder wealth?

- A Operating profit
- B Sales revenues
- C Market price of the share
- D Price/earnings ratio

10 Indicate whether the following objectives are financial or non-financial objectives of a company.

Objective	Financial	Non-financial
Maximisation of market share		
Earnings growth		
Sales revenue growth		
Achieving a target level of customer satisfaction		
Achieving a target level of return on capital employed		

11 Which TWO of the following are examples of financial objectives that a company might choose to pursue?

- A Dealing honestly and fairly with customers on all occasions
- B Provision of good working conditions and industrial relations
- C Earning above a particular level of return on capital employed
- D Producing environmentally friendly products
- E Restricting the level of gearing to below a specified target level

12 Indicate whether the following statements are true or false.

Statement	True	False
Financial management is concerned with the long-term raising of finance and the allocation and control of resources		
Management accounting is concerned with providing information for the more day-to-day functions of control and decision-making		
Financial accounting is concerned with providing information about the historical results of past plans and decisions		

13 Which TWO of the following statements are correct?

- A Maximising market share is an example of a financial objective
- B Shareholder wealth maximisation is the primary financial objective for a company listed on a stock exchange
- C Financial objectives should be quantitative so that their achievement can be measured
- D Three E's are used as a performance measure to assess value for money in not-for-profit organisations. The three E's stand for economy, efficiency and environment

14 Increasing which TWO of the following would be associated with the financial objective of shareholder wealth maximisation?

- A Share price
- B Dividend payment
- C Reported profit
- D Earnings per share
- E Weighted average cost of capital

15 Which of the following is NEVER consistent with the objective of maximising shareholder wealth?

- A Following corporate social responsibility
- B Increasing sales levels
- C Satisficing
- D Paying dividends

16 Which of the following is a sign that a company is failing to maximise shareholder wealth?

- A Management has challenging performance bonuses
- B Investment decisions are evaluated on the basis of their net present value
- C The company has not diversified its operations
- D Financing of the company is via equity finance alone

17 Which of the following actions is LEAST likely to increase shareholder wealth?

- A The weighted average cost of capital is decreased by a recent financing decision
- B The financial rewards of directors are linked to increasing earnings per share
- C The board of directors decides to invest in a project with a positive net present value
- D The annual report declares full compliance with the corporate governance code

Stakeholder objectives and conflicts

18 The agency problem is a driving force behind the growing importance attached to sound corporate governance. In this context, who are the agents?

- A Customers
- B Shareholders
- C Managers
- D Auditors

19 Which TWO of the following are examples of internal stakeholders in a firm?

- A Company directors
- B Customers
- C Suppliers
- D Employees
- E Finance providers

20 What is the main purpose of corporate governance?

- A To separate ownership and management control of organisations
- B To maximise shareholder value
- C To facilitate effective management of organisations and to make organisations more visibly accountable to a wider range of stakeholders
- D To ensure that regulatory frameworks are adhered to

21 In which of the following principal/agent relationships is the principal named second?

- A Shareholders and managers
- B Customers and the company
- C Shareholders and debt-holders
- D Employers and employees

22 The directors of Portico Co have recently engaged a firm of consultants to negotiate standard terms of trade for one of its strategic business units. This includes the agreement by Portico Co to pay a 5% penalty on any late invoice settlement.

This policy is an illustration of the company's concern for which major stakeholder?

- A Lenders
- B Suppliers
- C Customers
- D Trade unions

23 Under the terms of the UK Corporate Governance Code, what are the only type of directors permitted to sit on a company's audit committee?

- A Independent non-executive directors
- B Non-executive directors
- C Executive directors
- D Directors with financial experience

24 Which of the following does NOT form part of the objectives of a corporate governance best practice framework?

- A Separation of chairperson and CEO roles
- B Establishment of audit, nomination and remuneration committees
- C Minimisation of risk
- D Employment of non-executive directors

25 Y Co is a listed company considering closing a factory in Country A, resulting in redundancies, and opening a new, more automated factory elsewhere. In a board discussion of the proposal, one director comments: "I think the key issue is improving reported profit; that is the main way we can keep our shareholders happy, which is, after all, our job. Sacking people in Country A will only result in a boost to the share price due to the cost savings, as long as we keep shareholders informed, that is."

Which TWO of the following statements concerning the director's comments are true?

- A Increasing profit is always in shareholders' best interests
- B Directors have a fiduciary duty to put shareholder interests first
- C Sacking staff will have no impact on the share price, other than as a result of the cost savings that result
- D In a semi-strong form market, the share price of Y Co will react when news about the new factory and any redundancies is made public

The role of management and goal congruence

26 Which THREE of the following are characteristics of a carefully designed remuneration package?

- A Linking of rewards to changes in shareholder wealth
- B Matching of managers' time horizons to shareholders' time horizons
- C Possibility of manipulation by managers
- D Encouragement for managers to adopt the same attitudes to risk as shareholders
- E Motivational to primarily achieve short-term goals

27 Indicate which of the following are typical criticisms of executive share option schemes (ESOPs).

Statement	Is a criticism	Is NOT a criticism
When directors exercise their options, they tend to sell the shares almost immediately to cash in on their profits		
If the share price falls when options have been awarded, and the options have no value, they cannot act as an incentive		
Directors may distort reported profits to protect the share price and the value of their share options		

28 Which of the following are NOT usually considered to be agency costs?

- A Costs associated with giving share options to managers
- B Costs of monitoring managers
- C Costs of obtaining a listing on the stock exchange
- D Costs of divergent behaviour by managers

29 Which of the following is LEAST likely to solve the agency problem between shareholders and managers?

- A Giving managers profit-related rewards
- B Using external auditors to gauge company performance
- C Writing restrictive covenants into loan note contracts
- D Monitoring managers' actions

30 Which of the following statements are correct?

(1) Share option schemes always reward good performance by managers

(2) Performance-related pay can encourage dysfunctional behaviour

(3) Value for money as an objective in not-for-profit organisations requires the pursuit of economy, efficiency and effectiveness

- A 1 and 2 only
- B 1 and 3 only
- C 2 and 3 only
- D 1, 2 and 3

(ACCA FM June 2015)

Measuring achievement of corporate objectives

31 Hathaway Co has just paid a dividend of \$0.21 per share and its share price is \$3.50 per share. One year ago its share price was \$3.60 per share.

What is the total shareholder return over the period?

- A 8.9%
- B 8.6%
- C 3.1%
- D 0.9%

32 Identify, by indicating in the table below, whether each of the following actions would increase or decrease shareholder wealth.

Action	Increase	Decrease
The average cost of capital is increased by a recent financing decision		
The company's operating cycle becomes longer		
The board of directors accepts a negative net present value project		

Action	Increase	Decrease
The annual report declares full compliance with the corporate governance code		

33 TKQ Co has just paid a dividend of 21c per share and its share price is \$3.50 per share. One year ago its share price was \$3.10 per share.

What is the total shareholder return over the period?

- A 17.4%
- B 18.2%
- C 18.9%
- D 19.7%

34 Indicate whether the following statements are true or false.

Statement	True	False
Value for money is usually taken to mean economy, efficiency and engagement		
Cum dividend means the buyer of the share is entitled to receive the dividend shortly to be paid		
The dividend payout ratio compares the dividend per share with the market price per share		
The agency problem means that shareholder wealth is not being maximised		

35 Last year ABC Co made profit before tax of \$2,628,000. Tax amounted to \$788,000. ABC Co's share capital was \$2,000,000 (2,000,000 shares of \$1) and \$4,000,000 6% preference shares.

What was the earnings per share (EPS) for the year (to two decimal places)?

\$

36 A company has recently declared a dividend of 12c per share. The share price is \$3.72 cum div and earnings for the most recent year were 60c per share.

What is the P/E ratio (to one decimal place)?

37 The following information relates to a company:

	Year 1	Year 2	Year 3	Year 4
Earnings per share (cents)	30.0	31.8	33.9	35.7
Dividends per share (cents)	13.0	13.2	13.3	15.0
Share price at start of year (\$)	1.95	1.98	2.01	2.25

Which of the following statements is correct?

- A The dividend payout ratio is greater than 40% in every year in the period
- B Mean growth in dividends per share over the period is 4%
- C Total shareholder return for the third year is 26%
- D Mean growth in earnings per share over the period is 6% per year

(ACCA FM June 2015)

38 PT Co has just paid a dividend of 15 cents per share and its share price one year ago was \$3.00 per share. The total shareholder return for the year was 25%.

What is the current share price (to two decimal places)?

\$

(ACCA FM December 2014)

39 H Co's share price is \$3.50 at the end of 20X1 and this includes a capital gain of \$0.75 since the beginning of the period. A dividend of \$0.25 has been paid for 20X1.

What is the shareholder return (to one decimal place)?

%

40 Indicate whether the following statements are true or false.

Statement	True	False
Accounting profit is not the same as economic profit		
Profit takes account of risk		
Accounting profit can be manipulated by managers		

(ACCA FM December 2014)

41 Geeh Co paid an interim dividend of \$0.06 per ordinary share on 31 October 20X6 and declared a final dividend of \$0.08 on 31 December 20X6. The ordinary shares in Geeh Co are trading at a cum-div price of \$1.83.

What is the dividend yield (to one decimal place)?

%

(ACCA FM September 2017)

42 Peach Co's latest results are as follows:

Item	\$
Profit before interest and taxation	2,500,000
Profit before taxation	2,250,000
Profit after tax	1,400,000

Extracts from its statement of financial position are as follows:

Item	\$
Equity	10,000,000
Non-current liabilities	2,500,000

What is Peach Co's return on capital employed (ROCE)?

- A 14%
- B 18%
- C 20%
- D 25%

(ACCA F9 September 2016)

Objective setting in not-for-profit organisations

43 Which of the following actions is consistent with increasing value for money?

- A Using a cheaper source of goods and thereby decreasing the quality of not-for-profit organisation services
- B Searching for ways to diversify the finances of the not-for-profit organisation
- C Decreasing waste in the provision of a service by the not-for-profit organisation
- D Focusing on meeting the financial objectives of the not-for-profit organisation

44 Which TWO of the following are 'efficiency' targets that a not-for-profit organisation might put in place?

- A Reduction of wastage of paper
- B Pay rates for staff of appropriate levels of qualification
- C Staff utilisation
- D Customer satisfaction ratings

45 ARP is a charity providing transport for people visiting hospitals.

Which of the following performance measures would BEST fit with efficiency in a value for money review?

- A Percentage of members who re-use the service
- B Cost per journey to hospital
- C A comparison of actual operating expenses against the budget
- D Number of communities served

46 A state-funded hospital has the objective of providing value for money. It measures the achievement of this objective using the three E's (economy, effectiveness and efficiency).

Which of the following performance indicators measures the hospital's economy?

- A Number of patients admitted per year
- B Total medical staff salary cost
- C Cost per successfully treated patient
- D Average waiting time for patient admission to hospital

47 BeachLitter is a charity that was set up with the objective of cleaning litter from all beaches within the local area.

Which TWO of the following performance measures would best fit with effectiveness in a value for money audit?

- A Average hours worked to clean each beach
- B Number of beaches awarded the government's clean beach certification
- C Average annual expenditure on each successfully cleaned beach
- D Average hours worked by each volunteer per month
- E Amount of litter cleared each year measured in tonnes

48 Which TWO of the following statements about not-for-profit organisations are correct?

- A Not-for-profit organisations often have multiple stakeholders with conflicting objectives
- B The provision of value for money embodies economy, equality and effectiveness
- C Not-for-profit organisations usually have one dominant stakeholder
- D The key financial objective of not-for-profit organisations is to provide value for money

49 Value for money is an important objective for not-for-profit organisations. Which action is LEAST consistent with increasing value for money?

- A Using a cheaper source of goods without decreasing the quality of not-for-profit organisation services
- B Searching for ways to diversify the finances of the not-for-profit organisation
- C Decreasing waste in the provision of a service by the not-for-profit organisation
- D Focusing on meeting the objectives of the not-for-profit organisation

50 A government is trying to assess schools by using a range of financial and non-financial factors. One of the chosen methods is the percentage of students passing five exams or more.

Which of the three E's in the value for money framework is being measured here?

- A Economy
- B Efficiency
- C Effectiveness
- D Expertise

51 Which of the following statements about Value for Money is correct?

- A Economy is minimising the input costs of the organisation
- B Efficiency is minimising the output/input ratio
- C Effectiveness is minimising the costs of meeting the organisation's objectives
- D Enhancement is maximising the input/output ratio

52 A government body uses measures based upon the 'three E's' to measure value for money generated by a publicly funded hospital.

Which of the following relates to efficiency?

- A Cost per successfully treated patient
- B Cost per operation
- C Proportion of patients readmitted after unsuccessful treatment
- D Percentage change in doctors' salaries compared with previous year

Case Scenario Questions

Measuring achievement of corporate objectives

1 A shareholder of QSX Co is concerned about the recent performance of the company and has collected financial information covering the period 20X4 to 20X6.

Year end 31 May	Share price (\$)	Dividend per share (\$)	Earnings per share (\$)
20X4	7.40	0.37	0.607
20X5	8.35	0.385	0.642
20X6	6.48	0.40	0.589

(1) What was the capital gain for the year to 31 May 20X5?

- A 12.8%
- B 11.4%
- C -12.4%
- D -22.4%

(2) What was the total shareholder return for the year to 31 May 20X6?

- A 27.2%
- B -17.6%
- C -22.4%
- D -22.7%

(3) What was the average earnings yield for the period 20X4 to 20X6 (to one decimal place)?

%

(4) Which TWO of the following statements are correct?

- A One of the problems with maximising accounting profit as a financial objective is that accounting profit is affected by accounting policy choices
- B A target for a minimum level of dividend cover is a target for a minimum dividend payout ratio
- C The welfare of employees is a non-financial objective
- D One reason shareholders are interested in earnings per share is that accounting profit takes account of risk

(5) Which of the following statements is NOT correct?

- A Return on capital employed can be defined as operating profit divided by the sum of shareholders' funds and prior charge capital
- B Return on capital employed is the product of operating profit margin and net asset turnover
- C Dividend yield can be defined as dividend per share divided by the ex-dividend share price
- D Return on equity can be defined as operating profit divided by shareholders' funds

2 Summary financial information for ABC Co is given below, covering the last two years.

STATEMENT OF PROFIT OR LOSS (EXTRACT)

	20X8 \$'000	20X7 \$'000
Revenue	74,521	68,000
Cost of sales	28,256	25,772

	20X8 \$'000	20X7 \$'000
Salaries and wages	20,027	19,562
Other costs	11,489	9,160
Profit before interest and tax	14,749	13,506
Interest	1,553	1,863
Tax	4,347	3,726
Profit after interest and tax	8,849	7,917
Dividends payable	4,800	3,100

STATEMENT OF FINANCIAL POSITION (EXTRACT)

	20X8 \$'000	20X7 \$'000
Shareholders' funds	39,900	35,087
Long-term debt	14,000	17,500
Total	53,900	52,587

Other information

	20X8	20X7
Number of shares in issue ('000)	14,000	14,000
P/E ratio (average for year): ABC Co	14.0	13.0
P/E ratio (average for year): Industry	15.2	15.0
EPS	\$0.63	\$0.57
Share price	\$8.82	\$7.41
Dividend per share	\$0.34	\$0.22

(1) What is the percentage increase in return on capital employed (ROCE) for ABC Co between 20X7 and 20X8 (to one decimal place)?

%

(2) What is the operating profit margin for 20X8 (to one decimal place)?

%

(3) What is the total shareholder return?

- A 14.4%
- B 19.0%
- C 19.8%
- D 23.6%

As well as the information above, the following extra data is available:

	20X8	20X7
Gearing (debt/equity)	35.1%	49.9%
Interest cover (PBIT/interest)	9.5	7.2
Inflation	3%	3%

Based on all of the information available, are the following statements true or false?

(4) (i) Employees may be unhappy with their wages in 20X8. (ii) Financial risk for shareholders appears to be a problem area.

- A Statement (i) is true and statement (ii) is false
- B Both statements are true
- C Statement (i) is false and statement (ii) is true
- D Both statements are false

(5) Accounting profits may not be the best measure of a company's performance. Which of the following statements support this theory?

- (i) Profits are affected by accounting policies.
- (ii) Profits take no account of risk.
- (iii) Profits take no account of the level of investment made during the year.
- (iv) Profits are measures of short-term historic performance.

- A (ii) and (iv) only
- B (i), (ii), (iii) and (iv)
- C (ii) and (iii) only
- D (i) only

3 The directors of Fence Co, a listed company, have a remuneration package which includes a financial reward for achieving an annual return on capital employed greater than 15%. The remuneration package does not include a share option scheme.

(1) Which of the following statements about Fence Co is/are correct?

- (i) Managerial reward schemes of listed companies should encourage the achievement of stakeholder objectives
- (ii) Requiring investment projects to be evaluated with return on capital employed is an example of dysfunctional behaviour encouraged by performance-related pay
- (iii) Fence Co has an agency problem, as the directors are not acting to maximise the wealth of shareholders

- A (i) and (ii) only
- B (i) only
- C (ii) and (iii) only
- D (i), (ii) and (iii)

(2) Which of the following statements about Fence Co directors' remuneration package is/are correct?

- (i) Directors' remuneration should be determined by senior executive directors
- (ii) Introducing a share option scheme would help bring directors' objectives in line with shareholders' objectives
- (iii) Linking financial rewards to a target return on capital employed will encourage short-term profitability and discourage capital investment

- A (ii) only
- B (i) and (iii) only
- C (ii) and (iii) only
- D (i), (ii) and (iii)

(ACCA F9 September 2016)

Answers

Objective Test Questions

1 C

The others are secondary objectives that could be used to help achieve the primary objective.

2 B, C and D

Financial management aims to ensure that the money is available to finance profitable projects and to select those projects that the company should undertake. Once profits have been made, the decision then needs to be made about how much to distribute to the owners and how much to re-invest for the future. Income decisions are really sub-divisions of the investment decision. Appraisal decisions would be sub-divisions of the investment decision, and budget decisions would be performed by management accountants.

3 D

The management accountant would produce variance reports comparing actual to budget. All the others are common in the role of a financial manager. A relates to the dividend decision and B and C relate to the financing decision.

4 A

A review of overtime spending and apportioning overheads to cost units is typically carried out by a member of the management accounting team. Depreciation of non-current assets is typically carried out by a member of the financial accounting team.

5 C

Advising on investments in non-current assets is a key role of financial management. Annual accounts are generally produced by financial accountants and management accounts by management accountants. The decision on pay rates varies from business to business but would normally include line managers and HR.

6 D

It is the role of the board, rather than the financial manager, to appoint non-executive directors.

7 A

‘Providing information to management for day-to-day functions...’ is management accounting. ‘Providing information to external users about the historical results...’ is financial accounting. Financial management is relevant to all organisations, not only organisations with shareholders. It aims to achieve efficient acquisition and deployment of financial resources to ensure achievement of the organisation’s objectives. Therefore, option D is incorrect.

8 C

Minimising the company’s cost of capital will maximise the present value of its future operating cash flows, thereby maximising the value of the company. This increase in value belongs to its participating investors (i.e. the value of equity will rise and with it, shareholder wealth).

9 C

If capital markets are efficient, the market price of a share represents the present value of its future dividend stream and hence shareholder wealth.

10 Financial: Earnings growth, Sales revenue growth, Achieving a target level of return on capital employed. Non-financial: Maximisation of market share, Achieving a target level of customer satisfaction.

Achieving market share (a relative measure), or customer satisfaction (a qualitative measure), are non-financial objectives.

11 C and E

C: ROCE is a financial objective. E: this is a financial objective that relates to the level of financial risk that the company is prepared to accept. The other objectives are non-financial.

12 True, True, True**13 B and C**

The three E's in option D should be economy, efficiency and effectiveness. Maximising market share may lead to financial gains, but is not directly a financial objective. It could, for example, be achieved by cutting prices and accepting lower profits.

14 A and B

The sources of shareholder wealth are share price and dividend payments, so increasing both of these would be associated with the objective of shareholder wealth maximisation. Increasing profit and EPS are not directly correlated with shareholder wealth maximisation, although they can be used as secondary goals to help drive share prices up and generate cash for dividend payments. Reducing the WACC, rather than increasing it, would increase business value and therefore shareholder wealth.

15 C

At no time is satisficing (company managers doing just enough to satisfy the shareholders) consistent with the maximisation of shareholder wealth.

16 D

Financing of the company via equity alone is not consistent with maximising shareholder wealth, since it deprives shareholders of the benefits of (generally cheaper) debt finance.

17 B

Cash flows correlate more closely with business value than profit does. If directors' financial rewards are linked to increasing earnings per share, there is an incentive to increase short-term reported profit at the expense of longer-term cash flows and shareholder wealth. By contrast, a decrease in the weighted average cost of capital raises the present value of future cash flows, investing in a positive-NPV project adds to shareholder wealth, and compliance with the corporate governance code tends to support the share price, so A, C and D are all consistent with increasing shareholder wealth.

18 C

The separation of ownership and control creates a situation where managers act as the agents of the owners (shareholders).

19 A and D

Customers, suppliers and finance providers are examples of connected (not internal) stakeholders.

20 C

To facilitate effective management of organisations and to make organisations more visibly accountable to a wider range of stakeholders.

21 C

Debt-holders entrust their money to shareholders through company management, and so shareholders are the agents of debt-holders.

22 B

Invoices are received from suppliers, not any of the other three stakeholders.

23 A

The Code clearly states that the requirement is for independent non-executive directors, but only one member of the committee (as a minimum) needs to have recent and relevant financial experience, not all of them.

24 C

Corporate governance best practice aims to manage risk to desired and controlled levels, not to minimise risk. Running a business implies taking calculated risks in anticipation of a commensurate return.

25 B and D

Statement A is false: profit has a poor correlation with shareholder value, since it ignores the time value of money, risk, and can be manipulated by accounting policy choices, so increasing profit is not always in shareholders' best interests. Statement B is true: a fiduciary duty means the agents (directors) must act in the best interests of the principals (shareholders), which is the basis of the director's stated priority. Statement C is false: closing a factory and making staff redundant may also affect the company's reputation, its relationships with government/regulators, and its future ability to recruit and motivate staff elsewhere, not just its cost base. Statement D is true: in a semi-strong form efficient market, share prices react quickly and correctly to all publicly available information, so the share price will move once the redundancy and new-factory news is made public, not only once the cost savings materialise.

26 A, B and D

Managerial reward schemes should be clearly defined and impossible to manipulate. Their targets should align with the aims of shareholders, so should be focused on both long- and short-term goals.

27 Is a criticism, Is a criticism, Is a criticism

All three statements are typical criticisms of executive share option schemes.

28 C

All the costs are a direct or indirect consequence of the agency problem except the costs of obtaining a listing on the stock exchange, which are not dependent on the relationship between shareholders and managers.

29 C

Restrictive covenants in loan notes are used to reduce the agency problem between shareholders and the providers of debt finance (not between shareholders and managers).

30 C

Statement (1) is incorrect, since share option schemes do not always reward good performance, as share prices are affected by many factors outside managers' control. Statements (2) and (3) are correct.

31 C

WORKING

TSR = $((350 - 360) + 21)/360 = 11/360 = 3.06\%$, i.e. 3.1% to one decimal place.

Question 32

Action	Increase	Decrease
The average cost of capital is increased by a recent financing decision		✓
The company's operating cycle becomes longer		✓
The board of directors accepts a negative net present value project		✓
The annual report declares full compliance with the corporate governance code	✓	

33 D

WORKING

Total shareholder return = $((350 - 310) + 21)/310 = 61/310 = 19.7\%$.

34 False, True, False, True

Value for money means economy, efficiency and effectiveness (not 'engagement'). It is correct that cum dividend means the buyer of the share is entitled to receive the dividend shortly to be paid. The dividend payout ratio compares the dividend to the profit for the year (not the market price per share). It is correct that the agency problem means that shareholder wealth is not being maximised.

35 Answer: \$0.80

WORKING: Profit after tax = \$2,628,000 – \$788,000 = \$1,840,000.

Preference dividend = \$4,000,000 × 6% = \$240,000.

Earnings attributable to ordinary shareholders = \$1,840,000 – \$240,000 = \$1,600,000.

EPS = \$1,600,000/2,000,000 shares = \$0.80.

36 Answer: 6.0

WORKING: The share price must first be adjusted to an ex-dividend basis: $\$3.72 - \$0.12 = \$3.60$. P/E ratio = share price/EPS = $\$3.60/\$0.60 = 6.0$.

37 D

WORKING: Mean growth in EPS = average of $(31.8 - 30.0)/30.0$, $(33.9 - 31.8)/31.8$ and $(35.7 - 33.9)/33.9$ = average of 6.0%, 6.6% and 5.3% $\approx 6\%$ per year.

The dividend payout ratio in Year 3 is $13.3/33.9 = 39.2\%$, which is below 40%, so statement A is incorrect. Mean growth in dividends per share is closer to 5% than 4%, so statement B is incorrect. Total shareholder return for the third year cannot be reliably isolated as 26% from the data given, so statement C is incorrect.

38 Answer: \$3.60

WORKING: Shareholder return = $(P_1 - P_0 + D_1)/P_0$. $0.25 = (P_1 - 3.00 + 0.15)/3.00$, so $0.75 = P_1 - 3.00 + 0.15$, therefore $P_1 = \$3.60$.

39 Answer: 36.4%

WORKING: Shareholder return = $(P_1 - P_0 + D_1)/P_0 = (0.75 + 0.25)/(3.50 - 0.75) = 36.4\%$.

40 True, False, True

Statement 1: the economist's concept of profit is broadly in terms of cash, whereas accounting profits may not equate to cash flows, so accounting profit is not the same as economic profit. Statement 2: profit does not take account of risk. Statement 3: accounting profit can be manipulated to some extent by choices of accounting policies.

41 Answer: 8.0%

WORKING: Dividend yield compares the dividend paid over a year to the current ex-div share price.

Dividend for the year = $\$0.08 + \$0.06 = \$0.14$.

Ex-div share price = $\$1.83 - \$0.08 = \$1.75$.

Dividend yield = $0.14/1.75 \times 100 = 8.0\%$.

42 C

WORKING: ROCE = Profit before interest and taxation/(Non-current liabilities + Equity) = $\$2,500,000/(\$2,500,000 + \$10,000,000) = \$2,500,000/\$12,500,000 = 20\%$.

43 C

Value for money relates to economy (value for money on inputs), efficiency (value for money on converting inputs to outputs) and effectiveness (value for money on outputs). Economy relates to keeping costs down without sacrificing quality, efficiency is about effective resource utilisation (such as decreasing waste levels), and effectiveness relates to the overall objectives of the organisation being achieved.

44 A and C

Efficiency targets aim to make the most efficient use of the resources available. Paying rates for staff of appropriate levels of qualification is an 'economy' target (B). Customer satisfaction rating is an 'effectiveness' target (D).

45 B

Cost per journey to hospital is a measure of efficiency. Percentage of members who re-use the service is a measure of effectiveness. A comparison of actual operating expenses against budget is an economy measure. Number of communities served is an effectiveness measure.

46 B

Economy is the acquisition of resources of appropriate quantity and quality at minimum cost. Total medical staff salary cost is the only answer option that does this. Cost per successfully treated patient is a measure of efficiency (how well inputs are converted into outputs), not economy. Number of patients admitted per year and average waiting time relate to effectiveness.

47 B and E

Effectiveness measures an organisation's success in meeting its objectives. The number of beaches awarded the government's clean beach certification and the amount of litter cleared each year are the only answer options that measure this. Answers A, C and D are measures of efficiency, which looks at how well inputs (hours used, money spent) are converted into outputs.

48 A and D

Although multiple stakeholders with conflicting objectives can also occur in the corporate sector, it is particularly likely in the not-for-profit sector. The key financial objective of not-for-profit organisations is to provide value for money, which embodies economy, efficiency and effectiveness (not 'equality').

49 B

Value for money refers to economy, efficiency and effectiveness. Diversifying sources of finance is not directly associated with any of the '3 E's'.

50 C

Exam success is an objective of a school, so the percentage of students passing five exams or more is a measure of effectiveness.

51 A

Value for money means achieving the '3 E's':

Economy (minimising the input costs of the organisation).

Efficiency (maximising the output/input ratio).

Effectiveness (meeting the organisation's objectives).

52 A

Efficiency measures relate the resources used to the output produced (getting as much as possible for what goes in). 'Proportion of patients readmitted after unsuccessful treatment' relates to effectiveness. 'Cost per operation' relates to economy (spending money frugally), as does 'Percentage change in doctors' salaries compared with previous year'.

Case Scenario Questions

1 (1) A

WORKING: The capital gain is the difference between the opening and closing share prices, expressed as a percentage of the opening share price.

20X5: capital gain = $(8.35 - 7.40)/7.40 = 12.8\%$.

(2) B

WORKING: Total shareholder return is the sum of the dividend paid and the monetary capital gain, expressed as a percentage of the opening share price.

20X6: total shareholder return = $(6.48 - 8.35 + 0.40)/8.35 = -17.6\%$.

(3) Answer: 8.3%

WORKING: Earnings yield = $(\text{earnings per share}/\text{share price}) \times 100$.

Average yield = $(9.1 + 7.7 + 8.2)/3 = 8.3\%$.

Tutorial note: earnings yield is the reciprocal of a P/E ratio.

(4) A and C

(5) D

Return on equity is defined as profit for the year (not operating profit) divided by shareholders' funds.

2 (1) Answer: 6.6%

WORKING: ROCE = $\text{PBIT}/\text{long-term capital}$.

20X8: ROCE = $\$14,749/\$53,900 = 27.4\%$.

20X7: ROCE = $\$13,506/\$52,587 = 25.7\%$.

Percentage increase = $(27.4 - 25.7)/25.7 = 6.6\%$.

(2) Answer: 19.8%

WORKING: Operating profit margin = $\text{PBIT}/\text{Sales} = 14,749/74,521 = 19.8\%$.

(3) D

WORKING: Total shareholder return = $(P_1 - P_0 + D_1)/P_0 = (8.82 - 7.41 + 0.34)/7.41 = 23.6\%$.

(4) A

The shareholders of ABC would probably be reasonably pleased with the performance over the two years (for example, share price has increased by 19%: $(8.82 - 7.41)/7.41 \times 100$). However, salaries and wages have only increased by 2.4% $((20,027 - 19,562)/19,562 \times 100)$, which is below the rate of inflation, so employees may be less pleased with the situation. Statement (i) is true. The financial risk that the shareholders are exposed to does not appear to be a problem area, as gearing has decreased from 49.9% to 35.1% and interest cover is more than sufficient, so statement (ii) is false.

(5) B

Accounting profits can be manipulated to some extent by choices of accounting policies. Profit does not take account of risk; shareholders will be very interested in the level of risk, and maximising profits may be achieved by increasing risk to unacceptable levels. Profits on their own take no account of the volume of investment that it has taken to earn the profit. Profits are reported every year (with half-year interim results for quoted companies) and are measures of short-term historic performance, whereas a

company's performance should ideally be judged over a longer term with future prospects considered as well as past profits. All four statements therefore support the theory.

3 (1) D

All three statements are correct.

- (i) Managerial reward schemes should encourage managers to act in the interests of the company's stakeholders (particularly shareholders), so this is correct as a general principle.
- (ii) Fence Co's directors are rewarded for achieving a target ROCE, which can encourage them to accept only projects that boost the accounting return on capital employed in the short term, and to reject projects that are financially attractive (positive NPV) but would depress ROCE in the short term; this is a form of dysfunctional behaviour caused by performance-related pay.
- (iii) Because the remuneration package rewards a short-term accounting target rather than shareholder wealth (which is better proxied by NPV/cash flow-based measures), the directors may make decisions that are not in shareholders' best interests, meaning Fence Co does have an agency problem.

(2) C

- (i) is incorrect: an independent remuneration committee (of non-executive directors) should determine directors' remuneration, not senior executive directors themselves, in order to avoid a conflict of interest.
- (ii) is correct: a share option scheme gives directors a direct stake in the company's share price, which better aligns their objectives with shareholders' objective of wealth maximisation, rather than rewarding a short-term accounting target.
- (iii) is correct: because ROCE is based on the accounting book value of assets, a target ROCE reward can encourage directors to avoid new capital investment (which increases the asset base and can depress ROCE in early years, even when NPV-positive), favouring short-term profitability over the company's longer-term capital investment needs.

Chapter 2: Basic Investment Appraisal Techniques

Questions

Objective Test Questions

Investment appraisal process

1 Which of the following statements about investment decision making is correct?

- A Opportunity costs are not relevant
- B The return on capital employed considers the time value of money
- C A strength of the payback method is that it is based on profitability
- D Capital budgeting is based on predictions of an uncertain future

ROCE

2 LEFM Co is considering investing in a new 20-year project which will require an initial investment of \$100,000 (with a \$10,000 scrap value) and will have a payback of 10 years. The project has consistent cash flows each year.

What is the ROCE calculated using the average investment?

- A 8.2%
- B 9.1%
- C 10.0%
- D 11.1%

3 A company is considering investing in the following project:

Initial outlay: \$100,000. Residual value: \$0. Project life: 10 years. Annual project cash flow: \$14,500.

What is the accounting rate of return (ARR) for the project using the average investment?

- A 4.5%
- B 9%
- C 14.5%
- D 29%

4 A company is considering an investment of \$340,000 in equipment for a project which will have a life of five years. The equipment can be sold at the end of the five-year period for \$55,000.

What is the average investment to be included in the calculation of ROCE?

- A \$340,000
- B \$197,500
- C \$170,000
- D \$142,500

5 Peach Co's latest results are as follows (\$000):

Item	\$000
Profit before interest and taxation	2,500
Profit before taxation	2,250
Profit after tax	1,400

In addition, extracts from its latest statement of financial position are as follows (\$000):

Item	\$000
Equity	10,000
Non-current liabilities	2,500

What is Peach Co's return on capital employed (ROCE)?

- A 14%
- B 18%
- C 20%
- D 25%

(ACCA FM September 2016)

6 A company undertakes a project that involves purchasing machinery at a cost of \$65,000. The machinery is used on the project for four years, generating operating cash inflows of \$20,000 per year. It is sold at the end of the project for \$10,000. Taxation is charged at a rate of 30%.

Calculate the initial return on capital employed (ROCE) for the project, to the nearest whole percentage.

%

7 Backpay Co is considering investing \$50,000 in a new machine. The machine will have scrap value of \$10,000 at the end of its five-year life. It is expected that 20,000 units will be sold each year at a selling price of \$3.00 per unit. Variable production costs are expected to be \$1.65 per unit, while incremental fixed costs are expected to be \$10,000 per year.

What is the project's return on capital employed, based on average investment?

- A 30%
- B 34%
- C 45%
- D 57%

8 NW Co is considering investing \$46,000 in a new delivery lorry that will last for 4 years, after which time it will be sold for \$7,000. Depreciation is charged on a straight-line basis. Forecast operating profits/(losses) to be generated by the machine are as follows:

Year	1	2	3	4
Operating profit/(loss)	\$16,500	\$23,500	\$13,500	\$(1,500)

What is the return on capital employed (ROCE) for the lorry (using the average investment method, to the nearest %)?

%

9 EE Co is considering investing in a new 40-year project which will require an initial investment of \$50,000 (with zero scrap value) and has a payback period of 20 years. The 40-year project has consistent cash flows each year.

What is the ROCE (using the average investment method, to one decimal place)?

%

Advantages and disadvantages of ROCE

10 Which of the following is NOT an advantage of the ROCE method of investment appraisal?

- A It is based on profits rather than cash flows
- B It is simple to calculate and understand
- C It uses readily available accounting information
- D It is often used by financial analysts to appraise performance

11 In relation to the return on capital employed (ROCE) investment appraisal method, which of the following statements is correct?

- A ROCE leads to better investment decisions since it uses accounting profit rather than estimated cash flows
- B Investment projects with a ROCE greater than the weighted average cost of capital should be accepted
- C Investment projects with a ROCE less than the current ROCE of an organisation should be rejected
- D ROCE takes into account all years of operation of an investment project

Accounting profits and cash flows

12 A project has average estimated cash flows of \$3,000 per year with an initial investment of \$9,000. Depreciation is straight-line with no residual value and the project has a five-year life span. The company has a target return on capital employed (ROCE) of 15% and a target payback period of 2.5 years. ROCE is based on initial investment.

Under which investment appraisal method(s), using the company's targets, will the project be accepted?

(1) ROCE

(2) Payback basis

- A 1 only
- B 2 only
- C Both 1 and 2
- D Neither 1 nor 2

Relevant cash flows

13 A company is trying to decide between keeping an existing machine and replacing it with a new machine. The old machine was purchased just two years ago for \$50,000 and had an expected life of 10 years. It now costs \$1,000 a month for maintenance and repairs due to a mechanical problem. A new machine is being considered to replace it at a cost of \$60,000. The new machine is more efficient and it will only cost \$200 a month for maintenance and repairs. The new machine has an expected life of 10 years.

In deciding to replace the old machine, which of the following should the company NOT consider?

- A Any estimated scrap value on the old machine
- B The original cost of the old machine
- C The estimated useful life of the new machine
- D The lower maintenance cost on the new machine

14 Carter Co paid \$1m for land three years ago. Carter estimates that it could sell the land today for \$1.2m. If the land is not sold, Carter plans to develop the land at an initial cost of \$1.5m. Carter estimates the net operating cash inflow during the first year following development would be \$500,000.

What is Carter's opportunity cost of the development?

- A \$1.5m
- B \$1.2m
- C \$1.0m
- D \$0.5m

15 SW Co has a barrel of chemicals in its warehouse that it purchased for a project a while ago at a cost of \$1,000. It would cost \$400 for a professional disposal company to collect the barrel and dispose of it safely. However, the chemicals could be used in a potential project which is currently being assessed.

What is the relevant cost of using the chemicals in a new project proposal?

- A \$1,000 cost
- B \$400 benefit
- C \$400 cost
- D \$Nil

16 A new project being considered by BLW Co would require 1,000 hours of skilled labour. The current workforce is already fully employed but more workers can be hired in at a cost of \$20 per hour. The current workers are paid \$15 per hour on a project that earns a contribution of \$10 per hour.

What is the relevant cost of labour to be included in the project appraisal?

- A \$10,000
- B \$15,000
- C \$20,000
- D \$25,000

17 LW Co has a half empty factory on which it pays \$5,000 pa rent. If it takes on a new project, it will have to move to a new bigger factory costing \$17,000 pa and it could rent the old factory out for \$3,000 pa until the end of the current lease.

What is the rental cost to be included in the project appraisal?

- A \$14,000
- B \$17,000
- C \$9,000
- D \$19,000

18 An accountant is paid \$30,000 per year and spends 2 weeks working on appraising project Alpha.

Why should the accountant NOT charge half of her month's salary to the project?

- A Because her salary cannot be apportioned
- B Because her salary is not incremental
- C Because her salary is not a cash flow
- D Because her salary is an opportunity cost

Payback method of appraisal

19 Harvey Co is evaluating a capital investment proposal with the following information:

Initial cost: \$500,000. Life: 10 years. Annual operating cash inflow: \$200,000. Scrap value: \$100,000.

The investment will be depreciated using the straight-line method.

What is the payback period for this investment?

- A 3.25 years
- B 2.67 years
- C 2.5 years
- D 2 years

20 A project requires an initial outlay of \$1,000. The forecast cash inflows are as follows:

Year	1	2	3	4
Cash inflow	\$200	\$200	\$400	\$400

What is the project's payback period (to one decimal place)?

21 Kuchman Kookies will invest \$100,000 in new equipment. The company's discount rate is 8% and the operating cash flows from the investment are expected to be as follows:

Year	1	2	3	4	5
Cash flow	\$35,000	\$38,000	\$25,000	\$20,000	\$10,000

What is the investment's payback period?

- A 2.3 years
- B 3.1 years
- C 4.0 years
- D 4.7 years

22 In considering the payback period for three projects, Fly Co gathered the following data about cash flows:

	Year 1	Year 2	Year 3	Year 4	Year 5
Project A	(10,000)	3,000	3,000	3,000	3,000
Project B	(25,000)	15,000	15,000	(10,000)	15,000
Project C	(10,000)	5,000	5,500		

Which of the projects will achieve payback within three years?

- A Projects A, B, and C
- B Projects B and C only
- C Project B only
- D Projects A and C only

23 Backpay Co is considering investing \$50,000 in a new machine. The machine will have scrap value of \$10,000 at the end of its five-year life. It is expected that 20,000 units will be sold each year at a selling price of \$3.00 per unit. Variable production costs are expected to be \$1.65 per unit, while incremental fixed costs are expected to be \$10,000 per year.

What is the project's payback period?

- A 1.0 year
- B 1.9 years
- C 2.5 years
- D 2.9 years

24 NW Co is considering investing \$46,000 in a new delivery lorry that will last for 4 years, after which time it will be sold for \$7,000. Depreciation is charged on a straight-line basis. Forecast operating profits/(losses) to be generated by the machine are as follows:

Year	1	2	3	4
Operating profit/(loss)	\$16,500	\$23,500	\$13,500	\$(1,500)

Assuming operational cash flows arise evenly over the year, what is the payback period for this investment (to the nearest month)?

- A 1 year 7 months
- B 2 years 7 months
- C 1 year 5 months
- D 3 years 2 months

Advantages and disadvantages of payback

25 Which of the following statements concerning the payback method is correct?

- A It does not consider the time value of money
- B It is the time required for profits to recover the initial cost of investment
- C It considers the whole life of a project
- D It is reliable for project selection decisions

26 Which TWO of the following statements about the investment appraisal methods of return on capital employed (ROCE) and payback are true?

- A Both methods are affected by changes in the cost of capital
- B The ROCE does not take account of returns over the entire life of the project
- C The payback method is based on the project's cash flows
- D A requirement for an early payback can increase a company's liquidity

27 Which of the following is a drawback of the payback period method of investment appraisal?

- A It is cash flow based
- B It considers the time value of money
- C It doesn't measure the potential impact on shareholder wealth
- D It is profit based

Case Scenario Questions

1 The following information relates to an investment project which is being evaluated by the directors of Fence Co, a listed company. The initial investment, payable at the start of the first year of operation, is \$3.9 million.

	1	2	3	4
Net operating cash flow (\$000)	1,200	1,500	1,600	1,580
Scrap value (\$000)				100

The directors believe that this investment project will increase shareholder wealth if it achieves a return on capital employed greater than 15%. As a matter of policy, the directors require all investment projects to be evaluated using both the payback and return on capital employed methods. Shareholders have recently criticised the directors for using these investment appraisal methods, claiming that Fence Co ought to be using the academically-preferred net present value method.

The directors have a remuneration package which includes a financial reward for achieving an annual return on capital employed greater than 15%. The remuneration package does not include a share option scheme.

(1) What is the payback period of the investment project?

- A 2.75 years
- B 1.50 years
- C 2.65 years
- D 1.55 years

(2) Based on the average investment method, what is the return on capital employed of the investment project?

- A 13.3%
- B 26.0%
- C 52.0%
- D 73.5%

(3) Which of the following statements about investment appraisal methods is correct?

- A The return on capital employed method considers the time value of money
- B Return on capital employed must be greater than the cost of equity if a project is to be accepted
- C Riskier projects should be evaluated with longer payback periods
- D Payback period ignores the timing of cash flows within the payback period

(4) Which of the following statements about Fence Co is/are correct?

- (i) Managerial reward schemes of listed companies should encourage the achievement of stakeholder objectives
- (ii) Requiring investment projects to be evaluated with return on capital employed is an example of dysfunctional behaviour encouraged by performance-related pay
- (iii) Fence Co has an agency problem as the directors are not acting to maximise the wealth of shareholders

- A (i) and (ii) only
- B (i) only
- C (ii) and (iii) only
- D (i), (ii) and (iii)

(5) Which of the following statements about the Fence Co directors' remuneration package is/are correct?

- (i) Directors' remuneration should be determined by senior executive directors
 - (ii) Introducing a share option scheme would help bring directors' objectives in line with shareholders' objectives
 - (iii) Linking financial rewards to a target return on capital employed will encourage short-term profitability and discourage capital investment
- A (ii) only
 B (i) and (iii) only
 C (ii) and (iii) only
 D (i), (ii) and (iii)

(ACCA FM September 2016)

2 Trecor Co plans to buy a machine costing \$250,000 which will last for 4 years and then be sold for \$5,000.

	Year 1	Year 2	Year 3	Year 4
Net cash flow (\$)	122,000	143,000	187,000	78,000

Depreciation is charged on a straight-line basis over the life of the asset.

(ACCA FM Specimen Exam 2007, amended)

(1) Calculate the before-tax return on capital employed (accounting rate of return) based on the average investment (to the nearest whole percentage).

%

(2) Are the following statements on return on capital employed (ROCE) true or false?

Statement	True	False
If ROCE is less than the target ROCE then the purchase of the machine can be recommended		
ROCE can be used to compare two mutually exclusive projects		

(3) What is the payback period for the machine (to the nearest whole month)?

(ACCA FM Specimen Exam 2007, amended)

Answers

Objective Test Questions

1 D

Capital budgeting requires management's evaluation of an uncertain future since it involves acquisition of long-term assets which will produce uncertain revenues and operating costs.

2 C

WORKING

If payback is 10 years, net annual inflow is \$10,000 (\$100,000/10).

ROCE = Average annual accounting profit/Average investment.

Average annual accounting profit = \$10,000 cash inflow less depreciation.

Depreciation = (\$100,000 – \$10,000)/20 = \$4,500.

So average annual accounting profit = \$10,000 – \$4,500 = \$5,500.

Average investment = \$(100,000 + 10,000)/2 = \$55,000.

Therefore, ROCE = \$5,500/\$55,000 = 10.0%.

3 B

Tutorial note: ARR is calculated as average annual profit/average investment × 100%. ARR can be calculated using initial investment but the question specifically states “average investment”.

WORKING

Average annual profit = Annual cash flow – depreciation = \$14,500 – \$100,000/10 (as the investment has no residual value) = \$4,500.

Average investment = (\$100,000 + \$0)/2 = \$50,000.

Therefore, ARR = \$4,500/\$50,000 × 100% = 9%.

4 B

Average investment is calculated as $\frac{1}{2}$ (initial investment + scrap value) = $\frac{1}{2}$ (\$340,000 + \$55,000) = \$197,500.

5 C

WORKING

ROCE = Operating profit/(Equity + Non-current liabilities) = $100 \times 2,500 / (10,000 + 2,500)$ = 20%.

6 10%

WORKING

Initial ROCE = Average annual operating profit/Initial capital $\times$ 100.

Operating profit = Operating cash flows before taxation – depreciation.

Depreciation over project life = \$65,000 – \$10,000 = \$55,000; depreciation per year = \$55,000/4 = \$13,750.

Average annual profit = \$20,000 – \$13,750 = \$6,250 per year.

Initial ROCE = \$6,250/\$65,000 $\times$ 100 = 9.6%, i.e. 10% to the nearest whole percentage.

Tutorial note: the 30% tax rate given in the question is not used, ROCE is based on operating profit, not a post-tax figure, so it does not require any tax adjustment.

7 A

WORKING

Contribution per unit = 3.00 – 1.65 = \$1.35 per unit.

Annual cash flow = (20,000 $\times$ 1.35) – \$10,000 = \$17,000.

Annual depreciation = (\$50,000 – \$10,000)/5 = \$8,000.

Annual operating profit = \$17,000 – \$8,000 = \$9,000.

Average investment = $\frac{1}{2}$ (\$50,000 + \$10,000) = \$30,000.

ROCE = \$9,000/\$30,000 = 30%.

8 Answer: 49%

WORKING

Average annual accounting profits = (16,500 + 23,500 + 13,500 – 1,500)/4 = \$13,000 p.a. (Note: accounting profits are already after depreciation, so no further adjustment is required.)

Average investment = (initial investment + scrap)/2 = (\$46,000 + \$7,000)/2 = \$26,500.

ROCE = \$13,000/\$26,500 = 49%.

9 Answer: 5.0%

WORKING

A payback of 20 years suggests net annual inflow of \$50,000/20 = \$2,500 per year.

Depreciation = \$50,000/40 = \$1,250 per year.

Average annual accounting profit = \$2,500 – \$1,250 = \$1,250.

Average investment = (\$50,000 + \$0)/2 = \$25,000.

ROCE = \$1,250/\$25,000 = 5.0% per year.

10 A

Profits are affected by accounting policy choices whereas cash flows are not.

11 D

ROCE does take into account all years of operation of an investment project. Discounted cash flow-based decisions are better than profit-based ones (refuting A). The ROCE calculation is compared to a ROCE-based percentage target that is unrelated to the cost of capital (refuting B). A decision under ROCE is compared to a target set by management rather than to existing ROCE levels (refuting C), otherwise, decisions could be made to reject acceptable projects just because they aren't quite as good as current operations.

12 D**WORKING**

Depreciation per year = $\$9,000/5 \text{ years} = \$1,800$.

Profit per year = $\$3,000 - \$1,800 = \$1,200$.

ROCE = Profit/Initial investment = $\$1,200/\$9,000 = 13.33\%$. The target ROCE is 15%; therefore, the project would be rejected under ROCE.

Payback = $\$9,000/\$3,000 = 3 \text{ years}$. Target payback is 2.5 years; therefore, the project would be rejected under payback too.

13 B

Tutorial note: The original cost of the old machine is a sunk cost that will not change regardless of the decision that is made.

14 B

The opportunity cost is the lost disposal proceeds (\$1.2m) if Carter develops the land instead of selling it.

15 B

The \$1,000 original cost is sunk. If the chemicals are used in a new project, SW Co saves the \$400 it would otherwise have to spend disposing of them. This equates to an effective net cash inflow (or, more precisely, the avoidance of an outflow) of \$400, so the project appraisal should show an inflow (benefit) of \$400 in relation to using this chemical.

16 C**WORKING**

Assume BLW Co would choose the cheapest source of labour.

Cost to buy in = $\$20 \times 1,000 \text{ hours} = \$20,000$.

Cost to divert existing labour = lost contribution + labour cost = $(\$10 + \$15) \times 1,000 \text{ hours} = \$25,000$.

The cheapest alternative is to buy in at a cost of \$20,000.

Tutorial note: to calculate how the existing project would suffer as a result of diverting labour, the current labour cost is added back to the lost contribution to give the full impact of diverting labour away from its current role.

17 A

WORKING

The current rental cost is \$5,000. The net new rental cost, should the project proceed, would be $(\$17,000 + \$5,000 - \$3,000) = \$19,000$, so an increment of $\$19,000 - \$5,000 = \$14,000$.

18 B

The cost should not feature in the project appraisal, as the accountant is paid anyway, i.e. her salary is not incremental.

19 C

WORKING

With even cash flows, payback period is calculated as initial cost divided by annual net cash inflows: $\$500,000/\$200,000 = 2.5$ years.

Tutorial note: Depreciation expense is not relevant as it is a non-cash cost.

20 3.5 years

WORKING

As at the end of the third year the project still needs to recover another 200 (cumulative cash flow at the end of Year 3 is $(1,000 - 200 - 200 - 400) = (200)$). This will be generated during the fourth year which itself generates 400 in total.

Payback period = 3 years + $(200/400) = 3.5$ years.

21 B

WORKING

As at the end of the third year, the project still needs to recover another 2,000 (cumulative cash flow: $(100,000) + 35,000 + 38,000 + 25,000 = (2,000)$). This will be generated during the fourth year which itself generates 20,000 in total.

Payback period = 3 years + $(2,000/20,000) = 3.1$ years.

Tutorial note: The 8% discount rate is irrelevant, as the requirement did not ask for discounted payback.

22 B

When the cumulative cash flow (both inflow and outflow) is zero, the project has reached payback. Projects B and C achieve payback by the end of Year 3. The payback period for Project A is somewhere between the end of Year 3 and Year 4.

23 D

WORKING

Contribution per unit = $3.00 - 1.65 = \$1.35$ per unit.

Total annual contribution = $20,000 \times 1.35 = \$27,000$ per year.

Annual cash flow after fixed costs = $\$27,000 - \$10,000 = \$17,000$ per year.

Payback period = $\$50,000/\$17,000 = 2.9$ years.

24 Answer: A**WORKING**

Payback period is the amount of time taken to repay the initial investment. Depreciation = $(\$46,000 - \$7,000)/4 = \$9,750$ per year.

Time	Profit (\$)	Depreciation (\$)	Cash flow (\$)	Cumulative cash flow (\$)
0	Investment		(46,000)	(46,000)
1	16,500	9,750	26,250	(19,750)
2	23,500	9,750	33,250	13,500

Payback period = $1 + (19,750/33,250) = 1.59$ years, i.e. 1 year 7 months to the nearest month.

25 A

The payback method determines the number of years that it will take for an investment's operating cash flows to recover the cost of investment. The payback method does not consider the time value of money.

26 C and D

The payback method is based on cash flows whereas the ROCE method is based on accounting profit. ROCE looks at the project's entire life whereas payback may not. Neither method is related to the cost of capital.

27 C

On the assumption that the basic reason for approving a project is that it will increase shareholder wealth, a major drawback of the payback period is that it does not attempt to measure the impact on shareholder wealth should the project go ahead.

Being cash flow based is a benefit, not a drawback.

The payback period does not take account of the time value of money.

Its calculation is not based on profit.

Case Scenario Questions

1 (1) A

WORKING

Cumulative cash flows:

Time	Calculation	Cumulative cash flow
t0		(3,900)
t1	(3,900) + 1,200	(2,700)
t2	(2,700) + 1,500	(1,200)
t3	(1,200) + 1,600	400

Payback period = 2 years + (1,200/1,600) = 2.75 years.

(2) B

WORKING

Average annual operating profit = (Total operating cash flows – total depreciation)/life of project. Capital investment and scrap value are not operating cash flows; total depreciation = difference between capital investment and scrap value.

= (5,880 – 3,800)/4 = \$520,000 per year.

Average investment = average of capital investment and scrap value = (3,900 + 100)/2 = \$2,000,000.

ROCE = $100 \times 520/2,000 = 26.0\%$.

(3) D

Payback period ignores the timing of cash flows within the payback period is correct. Risk can be taken into account by shortening (not lengthening) the required payback period, so that only cash flows closer to the present, and thus less risky, are considered. ROCE does not incorporate the time value of money, so it cannot be directly compared to the cost of equity; it needs to be compared to a specific ROCE target instead.

(4) D

All the statements are correct. Managerial reward schemes should encourage the achievement of shareholder (stakeholder) objectives to address the principal-agent problem. The use of ROCE means that profitability, instead of shareholder wealth, would be the main decision criterion used to evaluate projects, dysfunctional because it does not necessarily align with improving shareholder wealth. For these reasons, the agency problem exists at Fence Co.

(5) C

Introducing a share option scheme would help bring directors' objectives in line with shareholders' objectives, as directors would become potential shareholders themselves. Linking financial rewards to a target return on capital employed will encourage short-term profitability and discourage capital investment, since capital projects often return low profits in their early years. Executive directors' pay in a listed company should be subject to the recommendations of a remuneration committee made up wholly of independent non-executive directors, not determined by senior executive directors themselves.

2 (1) 56%**WORKING**

Depreciation = \$250,000 – \$5,000 = \$245,000 over the asset's life.

Total accounting profit over the 4 years = total cash inflows – depreciation = (\$122,000 + \$143,000 + \$187,000 + \$78,000) – \$245,000 = \$530,000 – \$245,000 = \$285,000.

Average annual profit = \$285,000/4 = \$71,250. Average investment = (\$250,000 + \$5,000)/2 = \$127,500.

ROCE = \$71,250/\$127,500 = 55.9%, i.e. 56% to the nearest whole percentage.

(2) False, True

If ROCE is less than the target ROCE then the purchase of the machine can be recommended, False: ROCE needs to be higher than the target ROCE (not lower) for the purchase of the machine to be recommended.

ROCE can be used to compare two mutually exclusive projects, True: two or more mutually exclusive projects can be compared using ROCE, with the project offering the highest ROCE normally being selected.

(3) 1 year 11 months**WORKING**

Cumulative cash flow at the end of Year 1 = –\$250,000 + \$122,000 = \$(128,000).

Payback period = 1 year + (128,000/143,000) × 12 months = 1 year + 10.7 months, i.e. 1 year 11 months to the nearest month.

Chapter 3: Investment Appraisal: Discounted Cash Flow Techniques

Questions

Objective Test Questions

The time value of money

1 Which of the following statements about investment appraisal methods is correct?

- A The return on capital employed method considers the time value of money
- B Return on capital employed must be greater than the cost of equity if a project is to be accepted
- C Riskier projects should be evaluated with longer payback periods
- D Payback period ignores the timing of cash flows within the payback period

(ACCA FM September 2016)

2 Money received today is generally considered worth more than the same sum received in the future. Which of the following is NOT one of the reasons for this?

- A The potential to earn interest on the money if it is received now
- B The impact of inflation eroding the future sum's purchasing power
- C The risk that the future sum may not be received at all
- D The rate of corporation tax payable on the sum

Compounding

3 \$5,000 is invested today at a compound interest rate of 6% per year, with no withdrawals.

What will the investment be worth in 4 years' time (to the nearest \$)?

- A \$6,312
- B \$6,000
- C \$5,300
- D \$7,000

4 An investor deposits \$2,000 in an account paying 8% compound interest per year, with no withdrawals.

What is the value of the deposit after 3 years (to the nearest \$)?

\$

Discounting

5 What is the present value of \$10,000 receivable in 5 years' time, if the discount rate is 8% per year?

- A \$6,810
- B \$8,000
- C \$10,000
- D \$14,693

6 A company will receive \$50,000 in 3 years' time. Using a discount rate of 12% per year, what is the present value of this receipt (to the nearest \$)?

\$

The cost of capital

7 Which of the following rates is most commonly compared to a project's internal rate of return to evaluate whether to make an investment?

- A Risk-free rate
- B Return on capital employed
- C Weighted average cost of capital
- D Cost of equity

8 Which of the following statements about the relationship between the cost of capital and a project's internal rate of return (IRR) is correct?

- A A decrease in the cost of capital will increase the IRR
- B A decrease in the cost of capital will decrease the IRR
- C The cost of capital has no effect on the IRR of a project
- D The IRR cannot be calculated unless the cost of capital is known first

The Net Present Value (NPV)

9 An investment in a new product will require an initial outlay of \$20,000. The cash inflow from the project will be \$4,000 each year for the next six years.

Using an 8% discount rate, what is the net present value of the investment?

- A \$(4,876)
- B \$(1,508)
- C \$(29)
- D \$1,508

10 Salem Co is considering a six-year project that produces annual net cash inflows of \$420,000 for the first five years and a net cash inflow of \$100,000 at the end of the sixth year. The project will require an initial investment of \$1,800,000. Salem's cost of capital is 10%.

What is the net present value for this project, to the nearest \$100?

- A \$83,000
- B \$(108,200)
- C \$(151,400)
- D \$(442,000)

11 A company is considering purchasing a machine that costs \$100,000 and has a \$20,000 scrap value. The machine will produce annual operating cash inflows of \$25,000 each year and has a six-year life. The company uses a discount rate of 10%.

What is the net present value of the machine?

- A \$(2,405)
- B \$8,875
- C \$20,155
- D \$28,875

12 Which of the following will normally affect a project's net present value?

- A Estimated scrap value of the asset
- B Net book value of the asset
- C Amount of financial accounting depreciation on the asset
- D Amount of head office costs recharged to the project

13 Net present value in investment decision-making is based on which of the following?

- A Net income
- B Earnings before interest, taxes and depreciation
- C Investing, operating and financing cash flows
- D Investing and operating cash flows

14 A company is considering investing in a two-year project. Machine set-up costs will be \$125,000, payable immediately. Working capital of \$4,000 is required at the beginning of the contract and will be released at the end. The company's cost of capital is 10%.

What is the minimum acceptable contract price (to the nearest thousand dollars), receivable at the end of the two years, for the project to have a net present value of zero?

\$

Advantages and disadvantages of using NPV

15 Indicate whether the following statements about the net present value method are true or false.

Statement	True	False
It is a relative measure		

Statement	True	False
It uses accrual basis, not cash basis, accounting for a project		
It takes into account the time value of money		
It accounts for compounding of returns		

16 Which investment appraisal method is most likely to result in the maximisation of shareholder wealth?

- A Payback period
- B Return on capital employed
- C Internal rate of return
- D Net present value

17 Which of the following actions is LEAST likely to increase shareholder wealth?

- A The weighted average cost of capital is decreased by a recent financing decision
- B The financial rewards of directors are linked to increasing earnings per share
- C The board of directors decides to invest in a project with a positive net present value
- D The annual report declares full compliance with the corporate governance code

(ACCA FM Specimen Exam)

18 Which of the following limitations is common to the calculations of payback period, discounted payback, internal rate of return and net present value?

- A They do not consider the time value of money
- B They require multiple trial and error calculations
- C They require knowledge of a company's cost of capital
- D They rely on the forecasting of future data

Time-saving in the exam (annuities and perpetuities)

19 An investor has a cost of capital of 10%. She is due to receive a 5-year annuity starting in 3 years' time of \$7,000 per year.

What lump sum amount would you need to offer today to make her indifferent between the annuity and your offer (to the nearest \$)?

\$

20 A newspaper reader has won first prize in a national competition and they have a choice as to how they take the prize:

Option 1: Take \$90,000 per year indefinitely starting in 3 years' time (and bequeath this right to their children and so on); or

Option 2: Take a lump sum of \$910,000 in 1 year's time.

Assuming a cost of capital of 10%, which would you advise and why?

- A Option 1 because \$90,000 p.a. indefinitely is an infinite amount of money compared to a one-off payment
- B Option 1 because it is worth more in present value terms
- C Option 2 because it is worth more in present value terms
- D Option 2 because the lump sum has the flexibility to be invested and earn a larger return than \$90,000 p.a.

21 JCW Co is appraising an opportunity to invest in some new machinery that has the following cash flows:

Item	\$
Initial investment	40,000
Net cash inflows for 5 years in advance (per year)	12,000
Decommissioning costs after 5 years	15,000

At a cost of capital of 10%, what is the net present value of this project (to the nearest \$100)?

\$

22 A lease agreement has an NPV of \$(26,496) at a rate of 8%. The lease involves an immediate down payment of \$10,000 followed by 4 equal annual payments.

What is the amount of the annual payment?

- A \$11,020
- B \$4,981
- C \$1,513
- D \$14,039

23 Paulo plans to buy a holiday villa in five years' time for cash. He estimates the cost will be \$1.5m. He plans to set aside the same amount of funds each year for five years, starting immediately, earning a rate of 10% interest per year compound.

To the nearest \$100, how much does he need to set aside each year?

\$

24 An investor has a cost of capital of 6%. She is due to receive a 6-year annuity starting in two years' time of \$5,000 per year.

What lump sum amount would she need to be offered today to make her indifferent between the annuity and the lump sum?

- A \$19,870
- B \$23,195
- C \$24,585
- D \$27,910

25 A company will lease new machinery. The lease term is five years and lease payments of \$10,000 will be made annually in advance.

What is the present value of the lease payments, using a discount rate of 10%?

- A \$4,170
- B \$6,209
- C \$37,910
- D \$41,700

The Internal Rate of Return (IRR)

26 CWE Co is appraising an opportunity to invest in some new machinery that has the following cash flows:

Item	\$
Initial investment	70,000
Net cash inflows for 5 years (per year)	20,000

What is the internal rate of return of the project, calculated using discount factors for 10% and 15% (to one decimal place)?

%

27 Backpay Co is evaluating an investment project with the following forecast cash flows:

Time	0	1	2	3	4
Cash flow (\$m)	(6.5)	2.4	3.1	2.1	1.8

Using discount rates of 15% and 20%, what is the internal rate of return of the investment project (to one decimal place)?

%

28 The net present value (NPV) of an investment with uneven cash flows has been calculated as follows:

Discount rate	NPV \$
15%	50,000
20%	(20,000)

What is the internal rate of return of the investment (to the nearest %)?

- A 11%
- B 17%
- C 19%
- D 22%

29 Which of the following phrases defines the internal rate of return on a project?

- A The number of years it takes to recover the investment
- B The discount rate at which the net present value of the project equals zero
- C The project's average annual operating profit as a percentage of the investment cost
- D The weighted-average cost of capital used to finance the project

30 If a project has a required rate of return of 6%, which of the following statements is correct?

- A The NPV will be positive if the IRR is equal to 5%
- B The project will be rejected if the IRR is equal to 7%
- C The NPV will be negative if the IRR is greater than 6%
- D The project will be accepted if the IRR is greater than 6%

31 Which of the following statements about NPV and IRR is accurate?

- A Two NPV calculations are needed to estimate the IRR using linear interpolation
- B The graphical approach to IRR is only an estimate; linear interpolation using the formula is required for a precise answer
- C The IRR is unique
- D An IRR graph with NPV on the 'Y' axis and discount rate on the 'X' axis will have a negative slope

32 Sudan Co wishes to undertake a project requiring an investment of \$732,000 which will generate equal annual inflows of \$146,400 in perpetuity. The first inflow from the investment is expected one year after the initial investment.

What is the internal rate of return of the project?

- A 20%
- B 25%
- C 400%
- D 500%

33 A company is considering a project that has an initial cash outflow followed by several years of cash inflows, with a cash outflow again in the final year.

How many internal rates of return could there be for this project?

- A Either zero or two
- B Either one or two
- C Zero, one or two
- D Only two

34 A company is considering a project with the following cash flows:

Year	0	1	2
Cash flow	\$(11,000)	\$7,100	\$7,100

Using discount rates of 15% and 20%, what is the internal rate of return (IRR) of the project (to one decimal place)?

- A 18.9%
- B 21.2%
- C 24.2%
- D 44.3%

35 A sixth project, Project F, is causing considerable confusion among board members whose sole means of appraising projects is to find an internal rate of return (IRR) using a spreadsheet function. The summarised cash flows of Project F are as follows:

Time	Action	Amount
0	Invest	\$4.00m
1	Receive	\$8.80m
2	Spend	\$4.83m

Which TWO of the following statements represent likely causes of confusion over Project F?

- A It has a negative NPV when undiscounted
- B It never earns a positive NPV
- C It has no real IRRs
- D It has one positive and one negative IRR
- E It may have more than one IRR

Advantages and disadvantages of IRR

36 Which of the following are correct advantages of the IRR approach to investment appraisal, and which are not?

Statement	Correct	Incorrect
Clear decision rule		
Takes into account the time value of money		
Assumes funds are reinvested at the IRR		
Considers the whole project		

37 Which of the following statements describes a weakness of the internal rate of return (IRR) method?

- A IRR can only be estimated using a financial calculator or spreadsheet
- B Cash flows from the investment are assumed to be reinvested at the IRR
- C The IRR calculation ignores the time value of money
- D The IRR calculation ignores project cash flows occurring after the initial investment is recovered

38 Which THREE of the following are advantages of the internal rate of return (IRR) method of investment appraisal?

- A Considers the whole life of the project
- B Uses cash flows, not accounting profits
- C It is a measure of absolute return
- D It is an accurate calculation
- E It is useful when liquidity is poor
- F It considers the time value of money

NPV versus IRR

39 Four mutually exclusive projects have been appraised using net present value (NPV), internal rate of return (IRR), return on capital employed (ROCE) and payback period (PP). The company objective is to maximise shareholder wealth.

	NPV	IRR	ROCE	PP
Project A	\$1.0m	40%	34%	4 years
Project B	\$1.4m	24%	35%	2.5 years
Project C	\$0.9m	18%	25%	3 years
Project D	\$1.5m	12%	18%	7 years

Which project should be chosen?

- A Project A
- B Project B
- C Project C
- D Project D

40 A project has an initial outflow followed by several years of inflows.

What is the effect of an increase in the cost of capital on (i) the project's net present value (NPV) and (ii) the project's internal rate of return (IRR)?

- A NPV decreases; IRR decreases
- B NPV decreases; IRR is unaffected
- C NPV increases; IRR is unaffected
- D NPV is unaffected; IRR decreases

41 Which of the following statements about the relative merits of NPV and IRR in evaluating mutually exclusive investment projects is correct?

- A NPV assumes reinvestment of surplus cash flows at the IRR, while IRR assumes reinvestment at the cost of capital
 - B With mutually exclusive projects, the IRR method can rank projects in a different order to the NPV method
 - C The IRR method never produces more than one solution, even for projects with unconventional cash flows
 - D NPV cannot be used where the cost of capital changes from year to year, whereas IRR can
- (ACCA FM September/December 2021, amended)

42 Which of the following statements is true if the net present value of a project is \$(4,000) and the required rate of return is 5%?

- A The project's IRR is less than 5%
- B The required rate of return is lower than the IRR
- C The project would increase shareholders' wealth
- D The NPV would be positive if the IRR was equal to 5%

43 Which TWO of the following statements are true of the internal rate of return (IRR) and the net present value (NPV) methods of investment appraisal?

- A IRR ignores the relative sizes of investments
- B IRR is easy to use where there are non-conventional cash flows
- C NPV is widely used in practice
- D IRR is technically superior to NPV

44 Which TWO of the following statements about the internal rate of return (IRR) are true?

- A IRR ignores the relative sizes of investments
- B IRR measures the increase in company value
- C IRR can incorporate discount rate changes during the life of the project
- D IRR and NPV sometimes give conflicting rankings over which project should be prioritised

(ACCA Specimen Exam 2007, amended)

Answers

Objective Test Questions

1 D

Payback period ignores the timing of cash flows within the payback period is correct. Risk can be taken into account by shortening (not lengthening) the required payback period, so that only cash flows closer to the present, and thus less risky, are considered. ROCE does not incorporate the time value of money, so it cannot be directly compared to the cost of equity; it needs to be compared to a specific ROCE target instead.

2 D

Money has a time value for three reasons:

- (1) The potential for earning interest on funds received now (rather than later).
- (2) The impact of inflation eroding the purchasing power of a future sum.
- (3) The effect of risk, since a future receipt is less certain than a present one.

The rate of corporation tax is not one of these three reasons.

3 A

WORKING

$$FV = PV \times (1 + r)^n = \$5,000 \times (1.06)^4 = \$5,000 \times 1.2625 = \$6,312.$$

4 Answer: \$2,519

WORKING

$$FV = PV \times (1 + r)^n = \$2,000 \times (1.08)^3 = \$2,000 \times 1.2597 = \$2,519.$$

5 A

WORKING

$$PV = FV \times (1 + r)^{-n} = \$10,000 \times (1.08)^{-5} = \$10,000 \times 0.681 = \$6,810.$$

6 Answer: \$35,600

WORKING

$$PV = FV \times (1 + r)^{-n} = \$50,000 \times (1.12)^{-3} = \$50,000 \times 0.712 = \$35,600.$$

7 C

The weighted average cost of capital (WACC) is frequently used as the 'hurdle rate' for capital budgeting decisions, i.e. the minimum required rate of return. Investments that provide an IRR that exceeds the WACC should add value to the company.

8 C

The IRR is the discount rate at which the NPV of a project is zero, i.e. it is a function of the project's own cash flows, not of the discount rate used to appraise it. The cost of capital determines a project's NPV (a

higher cost of capital reduces NPV), but changing the cost of capital used to evaluate a project does not change that project's own IRR.

9 B

WORKING

	Cash flow	DF @ 8%	PV
Investment	(20,000)	1.000	(20,000)
Years 1-6 inflows	4,000	4.623	18,492

Net present value = \$(1,508).

10 C

WORKING

Years 1-5: $\$420,000 \times 3.791 = \$1,592,220$.

Year 6: $\$100,000 \times 0.564 = \$56,400$.

Total present value of inflows = \$1,648,620.

Less initial investment \$1,800,000.

Net present value = \$(151,380), i.e. \$(151,400) to the nearest \$100.

11 C

WORKING

	Cash flow	DF @ 10%	PV
Investment	(100,000)	1.000	(100,000)
Years 1-6 inflows	25,000	4.355	108,875
Year 6 scrap value	20,000	0.564	11,280

Net present value = \$20,155 (positive).

12 A

A project's NPV is a function of the future cash flows it will create, including proceeds from the sale of the asset at the end of its useful economic life. Net book value, financial accounting depreciation and recharged head office costs are accounting entries, not cash flows, and do not directly affect NPV.

13 D

NPV is the present value of a project's operating cash flows minus the present value of its investing cash flows. Financing cash flows are not separately included, the cost of financing is implied within the discount rate (the cost of capital) used to calculate the present values.

14 Answer: \$152,000**WORKING**

Time 0 outflows = \$125,000 (set-up) + \$4,000 (working capital) = \$129,000.

Time 2 inflow (working capital released) = \$4,000 × 0.826 = \$3,304.

For NPV = 0, the present value of the contract price must equal: \$129,000 – \$3,304 = \$125,696.

Contract price = \$125,696/0.826 = \$152,174, i.e. \$152,000 to the nearest thousand dollars.

Question 15

Statement	True	False
It is a relative measure		✓
It uses accrual basis, not cash basis, accounting for a project		✓
It takes into account the time value of money	✓	
It accounts for compounding of returns	✓	

NPV is an absolute measure of the wealth a project creates (not a relative/percentage measure), and it uses cash flows, not accrual-basis accounting profit. The NPV method assumes that surplus cash flows are reinvested at the minimum required rate of return (the cost of capital), thereby considering both the time value of money and the compounding of returns.

16 D

NPV estimates the absolute change in the value of equity due to an investment and hence has a direct link with the assumed key financial objective of maximising shareholders' wealth.

17 B

Cash flows better correlate to business value (which can be calculated as the present value of the company's future cash flows) than profit does. If the financial rewards of directors are linked to increasing earnings (profit) per share, for example through a performance-related reward scheme, there is an incentive to increase short-term profit at the expense of longer-term growth in cash flows and hence shareholder wealth. By contrast:

- (1) If the WACC decreases, the present value of the business's future cash flows rises and business value rises.
- (2) Positive NPV projects add directly to business value.
- (3) Declaring full compliance with the corporate governance code would tend to boost the company's share price in a semi-strong efficient market.

18 D

The common disadvantage of all capital budgeting models is their reliance on forecast data. Capital investment decisions relate to longer periods of time that are subject to greater levels of uncertainty than short-term budgeting decisions.

19 Answer: \$21,924

WORKING

The present value of the annuity = $\$7,000 \times AF_{3-7}$, where AF_{3-7} is the 10% annuity factor from years 3-7 inclusive. $AF_{3-7} = AF_{1-7} - AF_{1-2} = 4.868 - 1.736 = 3.132$. Therefore the present value = $\$7,000 \times 3.132 = \$21,924$.

20 C

WORKING

Option 1: the future value of the perpetuity, one year before the first receipt (i.e. at time 2), = $\$90,000/0.1 = \$900,000$. Discounting this back to today at the 10% 2-year discount factor (0.826) gives a present value of $\$900,000 \times 0.826 = \$743,400$.

Option 2: the present value of the lump sum = $\$910,000 \times 0.909$ (the 1-year 10% discount factor) = $\$827,190$.

The lump sum (Option 2) has the higher present value, so it should be chosen.

21 Answer: \$700

WORKING

Cash outlays or receipts occurring at the beginning of a time period are taken to occur at the end of the previous year. An inflow of \$12,000 in advance for 5 years (i.e. starting now) is therefore taken to occur in years 0, 1, 2, 3 and 4.

Time	Item	Cash flow	DF @ 10%	PV
0	Investment	(40,000)	1.000	(40,000)
0-4	Net cash inflows	12,000	$1 + 3.17 = 4.17$	50,040
5	Decommissioning	(15,000)	0.621	(9,315)

Net present value = \$725, i.e. \$700 to the nearest \$100.

22 B

WORKING

$\$26,496 = (a \times AF_{1-4}@8\%) + \$10,000$, where $AF_{1-4}@8\% = 3.312$. So $\$16,496 = a \times 3.312$, therefore $a = \$16,496/3.312 = \$4,981$.

23 Answer: \$223,400

WORKING

Present value of the villa = $\$1.5m \times DF_5@10\% (0.621) = \$931,500$.

Since the deposits start immediately (i.e. at times 0-4), the relevant annuity factor is $AF_{0-4} = 1 + AF_{1-4}@10\% = 1 + 3.17 = 4.170$.

Annual deposit = $\$931,500/4.170 = \$223,381$, i.e. \$223,400 to the nearest \$100.

24 B

WORKING

The 6-year annuity of \$5,000 starts in Year 2, so its value at Time 0 is $\$5,000 \times (AF_{1-7}@6\% - AF_1@6\%) = \$5,000 \times (5.582 - 0.943) = \$5,000 \times 4.639 = \$23,195$.

25 D

WORKING

Since the lease payments are made in advance (i.e. at times 0-4), the relevant annuity factor is $1 + AF_{1-4}@10\% = 1 + 3.17 = 4.170$. Present value = $\$10,000 \times 4.170 = \$41,700$.

26 Answer: 13.3%

WORKING

Time	Cash flow	DF @ 10%	PV	DF @ 15%	PV
0	(70,000)	1.000	(70,000)	1.000	(70,000)
1-5	20,000	3.791	75,820	3.352	67,040
			5,820		(2,960)

$IRR = 10\% + [5,820/(5,820 + 2,960)] \times (15\% - 10\%) = 10\% + 3.3\% = 13.3\%$ (to one decimal place). Cash flows are assumed to arise in arrears unless a question states otherwise, which is why the cash inflows are taken to occur in time periods 1 to 5.

27 Answer: 17.8%

WORKING

Time	Cash flow \$m	DF @ 15%	PV \$m	DF @ 20%	PV \$m
0	(6.5)	1.000	(6.500)	1.000	(6.500)
1	2.4	0.870	2.088	0.833	1.999
2	3.1	0.756	2.344	0.694	2.151
3	2.1	0.658	1.382	0.579	1.216
4	1.8	0.572	1.030	0.482	0.868
			0.344		(0.266)

$IRR = 15\% + [0.344/(0.344 + 0.266)] \times (20\% - 15\%) = 15\% + 2.8\% = 17.8\%$ (to one decimal place).

28 C

WORKING

$IRR = 15\% + [50,000/(50,000 + 20,000)] \times (20\% - 15\%) = 15\% + 3.6\% = 18.6\%$, i.e. 19% to the nearest percent.

29 B

This is the definition of the internal rate of return: the discount rate at which a project's net present value equals zero.

30 D

If the IRR of a project exceeds the 'hurdle rate' (the required rate of return) established by management, the project should be accepted, because it will have a positive NPV.

31 A

Two NPV calculations at two different discount rates give only an approximation of the IRR by linear interpolation, so statement A is accurate. This estimate is not precise, because the true NPV/discount-rate relationship is curved, not a straight line, so statement B is inaccurate, interpolation is still only an estimate, not a precise answer. The IRR is not necessarily unique, projects with unconventional (multiple

sign-change) cash flows can have more than one IRR, so statement C is false. An IRR graph plotting NPV against the discount rate has a negative (not positive) slope, so statement D is also false.

32 A**WORKING**

The IRR is the discount rate at which the present value of the perpetuity equals the initial investment: $\$146,400/r = \$732,000$, so $r = \$146,400/\$732,000 = 20\%$.

33 C

A project can have as many internal rates of return as there are sign changes in its cash flow stream. This project's cash flows change sign twice (outflow to inflow, then inflow to outflow again in the final year), so it could have zero, one or two IRRs, depending on the exact shape of its NPV profile.

34 A**WORKING**

Time	Cash flow	DF @ 15%	PV	DF @ 20%	PV
0	(11,000)	1.000	(11,000)	1.000	(11,000)
1-2	7,100	1.626	11,545	1.527	10,842
			545		(158)

$IRR = 15\% + [545/(545 + 158)] \times (20\% - 15\%) = 15\% + 3.9\% = 18.9\%$ (to one decimal place).

35 A and E

The NPV of Project F at a discount rate of zero (the sum of its undiscounted cash flows) is negative: $\$(4.00m) + \$8.80m - \$4.83m = \$(0.03m)$, which may cause confusion since board members may feel the project could therefore never earn a positive NPV (A is true). The sign of the cash flows changes twice (an outflow followed by an inflow, then another outflow), so the project may have more than one IRR (E is true).

It is not correct that the project never earns a positive NPV (B is false, since NPV can be positive at some discount rates despite the undiscounted total being negative).

It is not correct that it has no real IRRs (C is false).

It is not correct that it has exactly one positive and one negative IRR (D is false, since the number and sign of the IRR(s) cannot be determined without further calculation).

Question 36

Statement	Correct	Incorrect
Clear decision rule		✓
Takes into account the time value of money	✓	
Assumes funds are reinvested at the IRR		✓
Considers the whole project	✓	

Statement 1 is not an advantage: the IRR decision rule depends on the shape of the IRR curve, and there could be several IRRs. Statement 2 is an advantage: IRR is a discounting technique and so takes into account the time value of money. Statement 3 is a disadvantage (a flaw), since there is no reason to

suppose that funds generated early in a project will be reinvested at the IRR after that point, the funds may well be distributed elsewhere. Statement 4 is an advantage: unlike the payback period, IRR considers all of the future incremental cash flows associated with a decision.

37 B

Assuming that funds generated by a project can be reinvested at its own computed IRR may not be realistic, especially if that IRR is particularly high compared to the company's normal rate of return; this reinvestment assumption is a key weakness of the IRR method.

38 A, B and F

IRR considers the whole life of the project (A) and, like NPV, uses cash flows rather than accounting profits (B) and considers the time value of money by discounting (F). IRR is not a measure of absolute return; it is a percentage/relative measure, unlike NPV (refuting C). The IRR is only an estimate, typically found by linear interpolation between two trial discount rates, not an exact calculation (refuting D). IRR is not especially useful when liquidity is poor, since as a percentage return it gives no indication of the absolute cash amounts involved (refuting E).

39 D

The project with the highest NPV will maximise shareholder wealth, as NPV directly measures the absolute impact of a project on shareholder wealth, so Project D should be chosen, even though it has the lowest IRR and ROCE and the longest payback period of the four.

40 B

A higher cost of capital discounts future cash inflows more heavily, reducing a project's NPV. The cost of capital does not feature in the calculation of a project's IRR at all; it only features in the decision rule that is applied after the IRR has been calculated (i.e. comparing the IRR to the cost of capital to decide accept/reject).

41 B

With mutually exclusive projects, the IRR method can (incorrectly) rank projects in a different order to the NPV method. This is due to the reinvestment assumption built into IRR: IRR assumes that net cash inflows generated during a project's life are reinvested at the project's own IRR, whereas NPV assumes reinvestment at the cost of capital, generally the more realistic assumption, which is why NPV ranks projects correctly. (NPV can also be used where the cost of capital changes from year to year, which is difficult for IRR; and unconventional cash flows can produce multiple IRRs, so option C is also incorrect.)

42 A

A negative NPV necessarily implies that a project's IRR is less than the discount rate used to calculate that NPV. Here, since the required rate of return (the discount rate used) is 5%, it must be the case that the IRR is less than 5%.

43 A and C

IRR ignores the relative sizes (absolute scale) of the investments being compared, since it is a percentage measure, a small project with a very high IRR is not necessarily better for shareholder wealth than a large project with a lower IRR (A is true). NPV is indeed widely used in practice, alongside IRR and payback (C is true). IRR is not easy to use where cash flows are non-conventional (i.e. change sign more than once), since this can produce multiple IRRs or no meaningful IRR at all (B is false). IRR is not technically superior to NPV, NPV is generally regarded as the technically superior method, since it gives

an absolute measure of wealth creation and correctly assumes reinvestment at the cost of capital (D is false).

44 A and D

IRR ignores the relative sizes of the investments being compared, since it is expressed as a percentage rather than an absolute amount (A is true). IRR and NPV can indeed give conflicting rankings for mutually exclusive projects, because IRR assumes reinvestment of surplus funds at the project's own IRR while NPV assumes reinvestment at the cost of capital (D is true). IRR does not directly measure the increase in company value; that is what NPV measures; IRR only indicates a percentage return (B is false). IRR cannot incorporate discount rate changes during a project's life, since a single IRR percentage is calculated for the whole project; it is NPV that can readily incorporate a different discount rate for each year (C is false).

Chapter 4: Investment Appraisal: Further Aspects of Discounted Cash Flows

Questions

Objective Test Questions

The impact of inflation on interest rates

1 A company's money (nominal) cost of capital is 21% and the general rate of inflation is expected to average 9% per year.

Using the Fisher formula, what is the company's real cost of capital (to one decimal place)?

- A 10.0%
- B 11.0%
- C 12.0%
- D 30.0%

2 A company's money (nominal) cost of capital is 21% and the general rate of inflation is expected to average 8% per year.

Using the Fisher formula, what is the company's real cost of capital (to one decimal place)?

- A 11.0%
- B 12.0%
- C 13.0%
- D 29.0%

3 Shadowline Co has a money cost of capital of 10%.

If inflation is 4%, what is Shadowline Co's real cost of capital (to one decimal place)?

%

The impact of inflation on cash flows

4 A project is expected to generate a cash flow of \$120,000 in 3 years' time, stated in current (year 0) price terms. General inflation is expected to average 5% per year, and the company's real cost of capital is 7%.

Using the money (nominal) method, what is the present value of this cash flow (to the nearest \$)?

\$

5 AM Co will receive a perpetuity starting in 2 years' time of \$10,000 per year, increasing thereafter by the rate of inflation (which is 2%).

What is the present value of this perpetuity, assuming a money cost of capital of 10.2%?

- A \$90,910
- B \$125,000
- C \$115,740
- D \$74,403

6 FW Co is expecting a receipt of \$10,000 (in real terms) in 1 year's time.

If FW Co expects inflation to increase, and receipts are expected to rise in line with the general rate of inflation, what impact will this have on the present value of that receipt?

- A Nil
- B Reduce
- C Increase
- D Cannot say

7 A cost-saving from new equipment is \$80,000 per year in current (year 0) price terms, arising for the first time in 1 year's time and continuing for 10 years. The saving is expected to inflate at 6% per year, and the money (nominal) discount rate is 15%.

Using the money method, what is the present value of the cost savings over the 10 years (to the nearest \$)?

\$

8 A company is considering investment in new labour-saving equipment costing \$1,000,000. The current wage rate is \$5 per hour but this is expected to increase by 5% each year into the foreseeable future. The equipment is expected to save 20,000 labour hours per year. The company's nominal cost of capital is 15.5%.

What is the present value of the savings in wage costs over a 10-year planning period (to the nearest \$000)?

- A \$385,000
- B \$558,000
- C \$615,000
- D \$676,000

9 A company expects to receive annual cash flows of \$75,000 per year (in current price terms) for a period of five years. The cash flows will inflate at 4% per year and the company's nominal cost of capital is 10%.

What is the present value of the expected cash flows (to the nearest \$000)?

- A \$375,000
- B \$318,000
- C \$296,000
- D \$284,000

Specific and general inflation rates

10 A company's real cost of capital is 8% and the general rate of inflation is expected to average 3.7% per year. Cash flows for a project have each been inflated using their own specific inflation rates, which differ from the general rate.

Which discount rate should be used to discount these money (nominal) cash flows?

- A 8.0%
- B 3.7%
- C 12.0%
- D 11.7%

11 Paisley Co is appraising a two-year project. In current (year 0) price terms, Year 1 sales revenue is \$150,000 and Year 1 labour costs are \$80,000; Year 2 sales revenue is \$160,000 and Year 2 labour costs are \$85,000. Sales revenue is expected to inflate at 6% per year and labour costs at 8% per year. Paisley Co's money (nominal) cost of capital is 12%.

What is the total present value of the net cash flows for Years 1 and 2 combined (to the nearest \$100)?

\$

12 NCW Co is considering investing \$10,000 immediately in a 1-year project with the following cash flows (stated in current terms, arising at the end of the year):

Item	\$
Income	100,000
Expenses	35,000

Income is subject to 10% inflation; expenses will not vary. The real cost of capital is 8% and general inflation is 2%.

Using the money cost of capital, to the nearest whole percentage, what is the net present value of the project?

- A \$68,175
- B \$60,190
- C \$58,175
- D \$78,175

13 Bourton Co is considering investing \$100,000 immediately in a 3-year project. Income of \$60,000 per year (in current price terms) is expected to inflate at 5% per year. A related fixed cost of \$20,000 per year is contractually fixed in money terms and will not increase. Bourton Co's money (nominal) cost of capital is 10%.

What is the net present value of the project (to the nearest \$)?

\$

14 For the real-terms method of NPV appraisal to give a valid result, which of the following statements are true?

Statement	True	False
All of a project's cash flows must be expected to inflate at the same rate		
That common inflation rate must equal the general rate of inflation		
The discount rate used must be the money (nominal) cost of capital		
Working capital must be excluded from the appraisal		

15 The following data is relevant to the evaluation of a particular project:

Item	Rate
Cost of capital in real terms	10% per year
General inflation rate	5% per year
Specific inflation rate of the project's annual cash inflow	6% per year
Specific inflation rate of the project's annual cash outflow	4% per year

Which of the following sets of adjustments will lead to the correct calculation of net present value?

- A Inflate the cash inflow at 4% per year, inflate the cash outflow at 4% per year, and discount at 15.0%
- B Inflate the cash inflow at 5% per year, inflate the cash outflow at 5% per year, and discount at 15.5%
- C Inflate the cash inflow at 6% per year, inflate the cash outflow at 4% per year, and discount at 15.5%

16 Melplash Co is considering an investment in a new, high-performance motorcycle model, with the following forecast production and sales.

Year	1	2	3	4
Motorcycles (units/year)	25,000	45,000	40,000	20,000

Initial investment \$220m; residual value at the end of four years \$30m. Selling price (Year 1 price terms) \$20,000 per motorcycle, expected to inflate at 6% per year. Variable costs (Year 1 price terms) \$14,000 per motorcycle, expected to inflate at 6% per year. Incremental fixed costs (Year 1 price terms) \$120m per year, expected to inflate at 10% per year. Nominal (money) discount rate 14%. Taxation should be ignored.

What is the net present value of the proposed investment (to the nearest \$0.1m)?

(ACCA FM March/June 2022)

\$

17 Paisley Co plans to purchase a machine costing \$13,500. The machine will save labour costs of \$7,000 in the first year. Labour rates in the second year will increase by 10%. The general rate of inflation is 8% and the company's real cost of capital is estimated at 12%. The machine has a two-year life with an estimated scrap value of \$5,000.

What is the NPV of the proposed investment (to the nearest \$10)?

- A \$550
- B \$770
- C \$970
- D \$1,150

Dealing with tax in NPV calculations

18 SW Co has a 31 December year end and pays corporation tax at a rate of 30%, 12 months after the end of the year to which the cash flows relate. It can claim tax-allowable depreciation at a rate of 25% reducing balance. It pays \$1m for a machine on 31 December 20X4. SW Co's cost of capital is 10%.

What is the present value, on 31 December 20X4, of the benefit of the first portion of tax-allowable depreciation?

- A \$250,000
- B \$227,250
- C \$68,175
- D \$75,000

19 A company receives a perpetuity of \$20,000 per year in arrears, and pays 30% corporation tax 12 months after the end of the year to which the cash flows relate.

At a cost of capital of 10%, what is the after-tax present value of the perpetuity?

- A \$140,000
- B \$145,454
- C \$144,000
- D \$127,274

20 Trecor Co plans to buy a machine costing \$250,000 which will last for 4 years and then be sold for \$5,000. Trecor Co can claim tax-allowable depreciation on the machine on a 25% reducing balance basis. It pays tax at an annual rate of 30% one year in arrears.

What amount of tax relief would be received by Trecor Co in time 4 of a net present value (NPV) calculation?

(ACCA Specimen Exam 2007)

\$

21 ABC Co is appraising a project that generates a before-tax operating cash flow of \$60,000 per year. The project requires an investment of \$80,000 in an asset that can be depreciated for tax purposes on a straight-line basis over its 4-year life. Corporation tax is charged at 25% and is payable in the same year as the cash flow that gives rise to it.

What is the after-tax cash flow for the project in each of the 4 years?

- A \$45,000
- B \$50,000
- C \$55,000
- D \$60,000

22 Deakin Co buys equipment costing \$2,000,000. Tax-allowable depreciation is available at 25% on a reducing balance basis, and corporation tax is charged at 30%. The equipment is sold at the end of Year 3 for \$600,000, which triggers a balancing allowance in that year (tax-allowable depreciation is claimed as normal in Years 1 and 2).

What is the tax benefit (cash inflow) arising from the balancing allowance in Year 3?

\$

23 Moore Co buys equipment costing \$500,000. Tax-allowable depreciation is available at 25% on a reducing balance basis, and corporation tax is charged at 25%. The equipment is sold at the end of Year 2 for \$410,000, which triggers a balancing charge in that year (tax-allowable depreciation is claimed as normal in Year 1 only).

What additional amount of tax is payable as a result of the balancing charge?

- A \$8,750
- B \$35,000
- C \$93,750
- D \$102,500

24 Which of the following statements about balancing allowances and balancing charges is correct?

- A A balancing allowance arises when an asset's disposal proceeds exceed its tax written down value
- B A balancing allowance arises when an asset's disposal proceeds are less than its tax written down value
- C A balancing charge reduces the amount of tax a company must pay in the year of disposal
- D Balancing allowances and charges only arise where tax-allowable depreciation is claimed on a straight-line basis

25 Gunning is considering investment in a new machine costing \$190,000, which has a five-year life with zero scrap value. The investment will also require an increase in working capital of \$35,000. Tax-allowable depreciation is available on a straight-line basis. Gunning is subject to a 40% tax rate and has a 10% cost of capital. Tax is paid at the end of the year in which profits are earned.

What is the present value of the tax saving on the first year's tax-allowable depreciation?

- A \$13,817
- B \$15,200
- C \$16,762
- D \$20,725

26 Wendy's Shop acquires an asset for \$100,000 that has no residual value and a 10-year life. Wendy's tax rate is 40%. Tax-allowable depreciation is available on a straight-line basis.

What is Wendy's annual tax saving from the asset?

- A \$10,000
- B \$6,000
- C \$4,000
- D \$2,000

27 ABC Co plans to buy a new machine. The cost of the machine is \$100,000 and it has a five-year life with no disposal value. The machine will be depreciated on a straight-line basis, which matches the policy for tax-allowable depreciation. The machine will increase annual operating cash flows by \$50,000. ABC Co's profit tax rate is 35%.

What is the annual after-tax cash flow generated by the machine?

- A \$19,500
- B \$30,000
- C \$32,500
- D \$39,500

28 Moore Co is considering acquiring a new machine for \$105,000. The machine is estimated to have a 10-year life and a scrap value of \$5,000. Over its life the machine is expected to produce 2,000 units each year with a sales price per unit of \$500 and combined material and labour costs of \$450 per unit. Tax-allowable depreciation is available on a straight-line basis on cost over five years only. Moore Co has a 40% tax rate and tax is paid in the year of the related cash flow.

What is the after-tax cash flow for the tenth year of the project?

- A \$81,000
- B \$68,400
- C \$63,000
- D \$60,000

Incorporating working capital

29 A project has the following projected cash inflows.

Year	Cash inflow
1	\$100,000
2	\$125,000
3	\$105,000

Working capital is required to be in place at the start of each year equal to 10% of the cash inflow for that year. The cost of capital is 10%.

What is the present value of the working capital?

- A \$Nil
- B \$(80,036)
- C \$(2,735)
- D \$33,000

30 AW Co needs to have \$100,000 working capital in place immediately for the start of a 2-year project. The amount will stay constant in real terms. Inflation is running at 10% per year, and AW Co's money cost of capital is 12%.

What is the present value of the cash flows relating to working capital?

- A \$(21,260)
- B \$(20,300)
- C \$(108,730)
- D \$(4,090)

31 Gunning Industries requires \$80,000 of working capital in place at the start of a 4-year project (time 0), with a further \$20,000 required at the start of Year 2 (time 1). All of the working capital invested is recovered in full at the end of the project (time 4).

What is the working capital cash flow at time 4?

\$

32 Which of the following statements about the treatment of working capital in an NPV appraisal is correct?

- A Working capital investment is a tax-allowable expense
- B The recovery of working capital at the end of a project should be included as a cash inflow in the final year
- C Working capital is ignored in NPV appraisals because it does not appear in the income statement
- D Working capital only needs to be provided once, at the start of the project, even where sales grow year on year

33 Gunning Industries is considering investment in a new machine which has a five-year life. The investment in the new machine would also require an immediate increase in working capital of \$35,000, which would be fully recovered at the end of five years. Gunning Industries is subject to a 40% corporate tax rate and has a cost of capital of 10%.

What is the effect of working capital on the net present value of the investment?

- A \$(7,959)
- B \$(10,680)
- C \$(13,265)
- D \$(35,000)

Dealing with questions with both tax and inflation

34 A company is appraising a three-year project which requires an initial outlay on 1 January 20X4 of \$30,000. The project is expected to give the following cash inflows on 31 December of each year:

Year	20X4	20X5	20X6
Cash inflow	\$10,000	\$20,000	\$25,000

All of the above cash flows are before taking account of specific annual inflation of 5% per year. The real cost of capital is 4% and the nominal cost of capital is 14%.

Using a nominal approach, what is the NPV of the project on 1 January 20X4 (to the nearest dollar)?

(ACCA FM September/December 2020)

\$

35 Which of the following is true about the 'inflation' figure that is included in the money cost of capital?

- A It is historic and specific to the business
- B It is historic general inflation suffered by the investors
- C It is expected and specific to the business
- D It is expected general inflation suffered by the investors

Case Scenario Questions

Question 1

BRT Co has developed a new confectionery line that can be sold for \$5.00 per box and that is expected to have continuing popularity for many years. The finance director has proposed that investment in the new product should be evaluated over a four-year time-horizon, even though sales would continue after the fourth year, on the grounds that cash flows after four years are too uncertain to be included.

The variable cost (in current price terms) will depend on sales volume, as follows.

Sales volume (boxes)	Less than 1 million	1-1.9 million	2-2.9 million	3-3.9 million
Variable cost (\$ per box)	2.80	3.00	3.00	3.05

Forecast sales volumes are as follows.

Year	1	2	3	4
Demand (boxes)	0.7 million	1.6 million	2.1 million	3.0 million

Tax: Tax-allowable depreciation on a 25% reducing balance basis could be claimed on the cost of equipment. Profit tax of 30% per year will be payable one year in arrears. A balancing allowance would be claimed in the fourth year of operation.

Inflation: The average general level of inflation is expected to be 3% per year for the selling price and variable costs. BRT Co uses a nominal after-tax cost of capital of 12% to appraise new investment projects.

A trainee accountant at BRT Co has started a spreadsheet to calculate the net present value (NPV) of the proposed new project. An extract, showing the rows and figures completed so far, is reproduced below (all figures \$000):

	A	B	C	D	E	F	G
Row	Item	0	1	2	3	4	5
5	Fixed costs		(1,030)	(1,910)	(3,060)	(4,277)	
6	Net cash flow		556	1,485	1,530	2,308	
9	Working capital	(750)	(23)	(23)	(24)	750	
10	Investment	(2,000)					
12	Discount factor @ 12%	1.000	0.893	0.797	0.712	0.636	0.567

(1) What is the sales figure for Year 2 (cell D3 in the spreadsheet), to the nearest \$'000?

\$

(2) What are the variable costs for Year 3 (cell E4 in the spreadsheet), to the nearest \$'000?

\$

(3) What are the tax benefits generated by the tax-allowable depreciation on the equipment in Year 4 (cell F8), to the nearest \$'000?

\$

(4) Which of the following statements about the project appraisal are true/false?

Statement	True	False
The trainee accountant has used the wrong percentage for the cost of capital		
Ignoring sales after four years underestimates the value of the project		
The working capital figure in Year 4 is wrong		

(5) Consider the following two statements about the project's appraisal using the internal rate of return (IRR):

Statement 1: Because the project has an unconventional pattern of cash flows, it may have more than one IRR, and so the NPV and IRR decision rules may not agree.

Statement 2: The project is financially viable under the IRR decision rule if its IRR is greater than the cost of capital of 12%.

- A Only Statement 1 is true
- B Only Statement 2 is true
- C Both statements are true
- D Neither statement is true

(ACCA F9 June 2011)

Constructed Response Questions

1 Hebac Co is preparing to launch a new product in a new market which is outside its current business operations. The company has undertaken market research and test marketing at a cost of \$500,000, as a result of which it expects the new product to be successful. Hebac Co plans to charge a lower selling price initially and then increase the selling price on the assumption that the new product will establish itself in the new market. Forecast sales volumes, selling prices and variable costs are as follows.

Year	1	2	3	4
Sales volume (units/year)	200,000	800,000	900,000	400,000
Selling price (\$/unit)	15	18	22	22
Variable costs (\$/unit)	9	9	9	9

Selling price and variable cost are given here in current price terms before taking account of forecast selling price inflation of 4% per year and variable cost inflation of 5% per year.

Incremental fixed costs of \$500,000 per year in current price terms would arise as a result of producing the new product. Fixed cost inflation of 8% per year is expected.

The initial investment cost of production equipment for the new product will be \$2.5 million, payable at the start of the first year of operation. Production will cease at the end of four years because the new product is expected to have become obsolete due to new technology. The production equipment would have a scrap value at the end of four years of \$125,000 in future value terms.

Investment in working capital of \$1.5 million will be required at the start of the first year of operation. Working capital inflation of 6% per year is expected and working capital will be recovered in full at the end of four years.

Hebac Co pays corporation tax of 20% per year, with the tax liability being settled in the year in which it arises. The company can claim tax-allowable depreciation on a 25% reducing balance basis on the initial investment cost, adjusted in the final year of operation for a balancing allowance or charge. Hebac Co currently has a nominal after-tax weighted average cost of capital (WACC) of 12% and a real after-tax WACC of 8.5%. The company uses its current WACC as the discount rate for all investment projects.

Required:

- Calculate the net present value of the investment project in nominal terms and comment on its financial acceptability. (12 marks)
- Discuss how the capital asset pricing model can assist Hebac Co in making a better investment decision with respect to its new product launch. (8 marks)

(Total = 20 marks)

(ACCA F9 September 2016)

2 Pinks Co is a large company listed on a major stock exchange. In recent years, the board of Pinks Co has been criticised for weak corporate governance and two of the company's non-executive directors have just resigned. A recent story in the financial media has criticised the performance of Pinks Co and claims that the company is failing to satisfy the objectives of its key stakeholders.

Pinks Co is appraising an investment project which it hopes will boost its performance. The project will cost \$20m, payable in full at the start of the first year of operation. The project life is expected to be four years. Forecast sales volumes, selling price, variable cost and fixed costs are as follows.

Year	1	2	3	4
Sales (units/year)	300,000	410,000	525,000	220,000
Selling price (\$/unit)	125	130	140	120

Year	1	2	3	4
Variable cost (\$/unit)	71	71	71	71
Fixed costs (\$'000/year)	3,000	3,100	3,200	3,000

Selling price and cost information are in current price terms, before applying selling price inflation of 5% per year, variable cost inflation of 3.5% per year and fixed cost inflation of 6% per year.

Pinks Co pays corporation tax of 26%, with the tax liability being settled in the year in which it arises. The company can claim tax-allowable depreciation on the full initial investment of \$20m on a 25% reducing balance basis. The investment project is expected to have zero residual value at the end of four years.

Pinks Co has a nominal after-tax cost of capital of 12% and a real after-tax cost of capital of 8%. The general rate of inflation is expected to be 3.7% per year for the foreseeable future.

Required:

- (a) (i) Calculate the nominal net present value of Pinks Co's investment project. (8 marks)
- (ii) Calculate the real net present value of Pinks Co's investment project and comment on your findings. (4 marks)
- (b) Discuss FOUR ways to encourage managers to achieve stakeholder objectives. (8 marks)

(Total = 20 marks)

(ACCA FM March/June 2019)

3 Uftin Co is a large company which is listed on a major stock market. The company has been evaluating an investment proposal to manufacture Product K3J. The initial investment of \$1,800,000 will be payable at the start of the first year of operation. The following draft evaluation has been prepared by a junior employee.

Year	1	2	3	4
Sales (units/year)	95,000	100,000	150,000	150,000
Selling price (\$/unit)	25	25	26	27
Variable costs (\$/unit)	11	12	12	13

(The above selling prices and variable costs per unit have not been inflated.)

Draft evaluation prepared by the junior employee:

	1 (\$000)	2 (\$000)	3 (\$000)	4 (\$000)
Sales revenue	2,475	2,605	4,064	4,220
Variable costs	(1,097)	(1,260)	(1,890)	(2,048)
Fixed costs	(155)	(155)	(155)	(155)
Interest payments	(150)	(150)	(150)	(150)
Cash flow before tax	1,073	1,040	1,869	1,867
Tax allowable depreciation	(450)	(450)	(450)	(450)
Taxable profit	623	590	1,419	1,417
Taxation	-	(137)	(130)	(312)
Net cash flow	623	453	1,289	1,105
Discount at 12%	0.893	0.797	0.712	0.636
Present values	556	361	918	703

	\$000
Present value of cash inflows	2,538
Cost of machine	(1,800)
NPV	738

The junior employee also provided the following information.

1. Relevant fixed costs are forecast to be \$150,000 per year.
2. Sales and production volumes are the same and no finished goods inventory is held.
3. The corporation tax rate is 22% per year and tax liabilities are payable one year in arrears.
4. Uftin Co can claim tax allowable depreciation of 25% per year on a reducing balance basis on the initial investment.
5. A balancing charge or allowance can be claimed at the end of the fourth year.
6. It is expected that selling price inflation will be 4.2% per year, variable cost inflation will be 5% per year and fixed cost inflation will be 3% per year.
7. The investment has no scrap value.
8. The investment will be partly financed by a \$1,500,000 loan at 10% per year.
9. Uftin Co has a weighted average cost of capital of 12% per year.

Required:

- (a) Prepare a revised draft evaluation of the investment proposal and comment on its financial acceptability. (11 marks)
- (b) Explain any TWO revisions you have made to the draft evaluation in part (a) above. (4 marks)

(Total = 15 marks)

(ACCA F9 December 2014)

4 HDW Co is a listed company which plans to meet increased demand for its products by buying new machinery costing \$5 million. The machinery would last for four years, at the end of which it would be replaced. The scrap value of the machinery is expected to be 5% of the initial cost. Capital allowances would be available on the cost of the machinery on a 25% reducing balance basis, with a balancing allowance or charge claimed in the final year of operation.

This investment will increase production capacity by 9,000 units per year and all of these units are expected to be sold as they are produced. Relevant financial information in current price terms is as follows.

	Current price terms	Forecast inflation
Selling price	\$650 per unit	4.0% per year
Variable cost	\$250 per unit	5.5% per year
Incremental fixed costs	\$250,000 per year	5.0% per year

In addition to the initial cost of the new machinery, initial investment in working capital of \$500,000 will be required. Investment in working capital will be subject to the general rate of inflation, which is expected to be 4.7% per year.

HDW Co pays tax on profits at the rate of 20% per year, one year in arrears. The company has a nominal (money terms) after-tax cost of capital of 12% per year.

Required:

- (a) Calculate the net present value of the planned purchase of the new machinery using a nominal (money terms) approach and comment on its financial acceptability. (14 marks)
- (b) Discuss the difference between a nominal (money terms) approach and a real terms approach to calculating net present value. (5 marks)
- (c) Identify TWO financial objectives of a listed company such as HDW Co and discuss how each of these financial objectives is supported by the planned investment in new machinery. (6 marks)

(Total = 25 marks)*(ACCA F9 June 2013)*

5 Argnil Co is appraising the purchase of a new machine, costing \$1.5 million, to replace an existing machine which is becoming out of date and which has no resale value. The forecast levels of production and sales for the goods produced by the new machine, which has a maximum capacity of 400,000 units per year, are as follows.

Year	1	2	3	4
Sales volume (units/year)	350,000	380,000	400,000	400,000

The new machine will incur fixed annual maintenance costs of \$145,000 per year. Variable costs are expected to be \$3.00 per unit and selling price is expected to be \$5.65 per unit. These costs and selling price estimates are in current price terms and do not take account of general inflation, which is forecast to be 4.7% per year.

It is expected that the new machine will need replacing in four years' time due to advances in technology. The resale value of the new machine is expected to be \$200,000 at that time, in future value terms. The purchase price of the new machine is payable at the start of the first year of the four-year life of the machine.

Working capital investment of \$150,000 will already exist at the start of the four-year period, due to the operation of the existing machine. This investment in working capital is expected to increase in nominal terms in line with the general rate of inflation.

Argnil Co pays corporation tax one year in arrears at an annual rate of 27% and can claim 25% reducing balance tax-allowable depreciation on the purchase price of the new machine. The company has a real after-tax weighted average cost of capital of 6% and a nominal after-tax weighted average cost of capital of 11%.

Required:

- (a) Using a nominal terms net present value approach, evaluate whether purchasing the new machine is financially acceptable. (10 marks)
- (b) Discuss the reasons why investment finance may be limited, even when a company has attractive investment opportunities available to it. (5 marks)

(Total = 15 marks)*(ACCA F9 September/December 2015)*

6 Gini Co is a listed company that manufactures innovative products for the telecommunications industry. Gini Co has just developed a new product called the LP500, which is expected to have a project life of four years.

Gini Co's management has decided that the sales price for the LP500 will be \$90 per unit, as anything higher than this will dramatically reduce the expected annual sales volume which is currently forecast to be 20,000 units per year.

The variable cost of the LP500 includes the cost of a bought-in component, the X22. The price of the X22 has been volatile due to supply problems with the manufacturers and consequently, the variable cost per unit is at present uncertain. The management of Gini Co has made the following estimates of the variable cost per unit, along with its associated probability.

Variable cost per unit (\$)	Probability
40	20%
50	40%
60	20%
70	20%

Incremental fixed costs will be \$300,000 per year. All of the above figures are expressed in current price terms. Sales price inflation is expected to be 5% per year. Variable cost per unit inflation and fixed cost inflation are both expected to be 3% per year.

New plant and equipment costing \$1.5m will need to be purchased at the start of the first year of operation and is expected to have a residual value of \$125,000, in nominal terms, at the end of Year 4. Gini Co pays corporation tax at a rate of 20% and tax is payable at the end of the year following the year to which it relates. The company can claim tax-allowable depreciation at a rate of 25% per year on a reducing balance basis with a balancing adjustment in the year of disposal.

Additional working capital totalling \$100,000 will also be required at the start of the project. Working capital inflation is expected to be 3% per year. Gini Co's nominal after-tax weighted average cost of capital for projects of this nature is 12%.

Required:

(a) Calculate the expected net present value of the LP500 project and comment on its financial acceptability. (12 marks)

(b) (i) Briefly discuss how your decision in part (a) will help Gini Co achieve their primary objective. (2 marks)

(ii) Discuss THREE limitations of using probability analysis in deciding whether or not to proceed with the LP500. (6 marks)

(Total = 20 marks)

(ACCA FM March/June 2023)

7 Elaan Co is a company appraising an investment project to purchase and operate a hotel. The hotel will offer basic accommodation, including its own restaurant, mainly for the business-traveller market. The hotel will be operated for a four-year period and then sold at the end of Year 4.

A suitable property has been located costing \$5.5m but it will require new fittings which are expected to cost an additional \$1.5m. It is expected that once the property is purchased, the fittings can be installed quickly so that the new hotel will be ready for business almost immediately after purchase. The hotel will have 50 rooms and be open for 365 days a year.

Forecasting assumptions are as follows.

Year	1	2	3	4
Room occupancy %	50	60	75	90

The daily room rate will be \$85 for the first year and then increase by 6% per year afterwards. Restaurant revenue will be equivalent to 35% of the accommodation revenue.

Variable costs are expected to be equivalent to 20% of the total revenue. Incremental fixed costs are expected to be \$433,000 for the first year and then increase by 4% per year afterwards.

Elaan Co pays corporation tax at 20% per year and tax is paid in the same year to which it relates. The purchase of the property is not eligible for tax relief, but the fittings will be eligible for tax-allowable depreciation at 25% per year on a reducing balance basis.

Working capital equivalent to 5% of the total revenue will be required at the start of each year. Working capital will be released at the end of Year 4. The expected sale value of the property at the end of Year 4 is expected to be \$7.5m after tax and the fixtures will be sold for an additional \$100,000.

The real after-tax cost of capital for the project is 4.8% and the general rate of inflation is expected to be constant at 5% per year. Elaan Co has never operated a hotel before, which has made the estimation of the above assumptions difficult. Also, inflation has been volatile in the country in which Elaan Co is undertaking the project.

Required:

- Using a nominal-terms approach, calculate the net present value (NPV) of the project and comment on its financial acceptability to Elaan Co. (12 marks)
- Explain how the approach to the NPV appraisal in part (a) would change if a real-terms approach had been used. Note: Further calculations are NOT required. (4 marks)
- Explain how simulation analysis could be used by Elaan Co in its appraisal of the hotel project AND the expected benefits of using simulation analysis. (4 marks)

(Total = 20 marks)

(ACCA FM March/June 2024)

8 Hraxin Co is appraising an investment project which has an expected life of four years and which will not be repeated. The initial investment, payable at the start of the first year of operation, is \$5 million. Scrap value of \$500,000 is expected to arise at the end of four years.

There is some uncertainty about what price can be charged for the units produced by the investment project, as this is expected to depend on the future state of the economy. The following forecast of selling prices and their probabilities has been prepared.

Future economic state	Weak	Medium	Strong
Probability of future economic state	35%	50%	15%
Selling price in current price terms	\$25 per unit	\$30 per unit	\$35 per unit

These selling prices are expected to be subject to annual inflation of 4% per year, regardless of which economic state prevails in the future. Forecast sales and production volumes, and total nominal variable costs, have already been forecast, as follows.

Year	1	2	3	4
Sales and production (units)	150,000	250,000	400,000	300,000
Nominal variable cost (\$000)	2,385	4,200	7,080	5,730

Incremental overheads of \$400,000 per year in current price terms will arise as a result of undertaking the investment project. A large proportion of these overheads relate to energy costs which are expected to increase sharply in the future because of energy supply shortages, so overhead inflation of 10% per year is expected.

The initial investment will attract tax-allowable depreciation on a straight-line basis over the four-year project life. The rate of corporation tax is 30% and tax liabilities are paid in the year in which they arise. Hraxin Co has traditionally used a nominal after-tax discount rate of 11% per year for investment appraisal.

Required:

- (a) Calculate the expected net present value of the investment project and comment on its financial acceptability. (9 marks)
- (b) Critically discuss if sensitivity analysis will assist Hraxin Co in assessing the risk of the investment project. (6 marks)

(Total = 15 marks)

(ACCA F9 June 2015, amended)

9 Duo Co needs to increase production capacity to meet increasing demand for an existing product, 'Quago', which is used in food processing. A new machine, with a useful life of four years and a maximum output of 600,000 kg of Quago per year, could be bought for \$800,000, payable immediately. The scrap value of the machine after four years would be \$30,000. Forecast demand and production of Quago over the next four years is as follows.

Year	1	2	3	4
Demand (kg)	1.4 million	1.5 million	1.6 million	1.7 million

Existing production capacity for Quago is limited to one million kilograms per year and the new machine would only be used for demand additional to this.

The current selling price of Quago is \$8.00 per kilogram and the variable cost of materials is \$5.00 per kilogram. Other variable costs of production are \$1.90 per kilogram. Fixed costs of production associated with the new machine would be \$240,000 in the first year of production, increasing by \$20,000 per year in each subsequent year of operation.

Duo Co pays tax one year in arrears at an annual rate of 30% and can claim tax-allowable depreciation on a 25% reducing balance basis. A balancing allowance is claimed in the final year of operation.

Duo Co uses its after-tax weighted average cost of capital when appraising investment projects. It has a cost of equity of 11% and a before-tax cost of debt of 8.6%. The long-term finance of the company, on a market-value basis, consists of 80% equity and 20% debt.

Required:

(a) Calculate the net present value of buying the new machine and advise on the acceptability of the proposed purchase (work to the nearest \$1,000). (13 marks)

(b) Explain the difference between risk and uncertainty in the context of investment appraisal, and describe how sensitivity analysis and probability analysis can be used to incorporate risk into the investment appraisal process. (7 marks)

(Total = 20 marks)

(ACCA F9 December 2007, amended)

10 Ridag Co is evaluating two investment projects, as follows.

Project 1

This is an investment in new machinery to produce a recently-developed product. The cost of the machinery, which is payable immediately, is \$1.5 million, and the scrap value of the machinery at the end of four years is expected to be \$100,000. Tax-allowable depreciation (capital allowances) can be claimed on this investment on a 25% reducing balance basis. Information on future returns from the investment has been forecast to be as follows.

Year	1	2	3	4
Sales volume (units/year)	50,000	95,000	140,000	75,000
Selling price (\$/unit)	25.00	24.00	23.00	23.00
Variable cost (\$/unit)	10.00	11.00	12.00	12.50
Fixed costs (\$/year)	105,000	115,000	125,000	125,000

This information must be adjusted to allow for selling price inflation of 4% per year and variable cost inflation of 2.5% per year. Fixed costs, which are wholly attributable to the project, have already been adjusted for inflation. Ridag Co pays profit tax of 30% per year one year in arrears.

Project 2

Ridag Co plans to replace an existing machine and must choose between two machines. Machine 1 has an initial cost of \$200,000 and will have a scrap value of \$25,000 after four years. Machine 2 has an initial cost of \$225,000 and will have a scrap value of \$50,000 after three years. Annual maintenance costs of the two machines are as follows.

Year	1	2	3	4
Machine 1 (\$/year)	25,000	29,000	32,000	35,000
Machine 2 (\$/year)	15,000	20,000	25,000	

Where relevant, all information relating to Project 2 has already been adjusted to include expected future inflation. Taxation and tax-allowable depreciation must be ignored in relation to Machine 1 and Machine 2.

Other information

Ridag Co has a nominal before-tax weighted average cost of capital of 12% and a nominal after-tax weighted average cost of capital of 7%.

Required:

- (a) Calculate the net present value of Project 1 and comment on whether this project is financially acceptable to Ridag Co. (12 marks)
- (b) Calculate the equivalent annual costs of Machine 1 and Machine 2, and discuss which machine should be purchased. (6 marks)
- (c) Critically discuss the use of sensitivity analysis and probability analysis as ways of including risk in the investment appraisal process, referring in your answer to the relative effectiveness of each method. (7 marks)

(Total = 25 marks)

(ACCA F9 June 2012)

11 Melplash Co manufactures motorcycles and is considering an investment in a new, high-performance model. The market for high-performance motorcycles is very competitive and Melplash Co's directors want to recover the costs of investment in the model quickly, before it is made obsolete by more advanced models at the end of four years. The directors also wish to show shareholders that the investment will be profitable.

The directors have therefore set a target payback period of two years and a target return on capital employed of 20% for the investment.

The directors have also asked the finance director to provide net present value (NPV) and internal rate of return (IRR) calculations for the investment. They plan to consider the results of these calculations if the investment fulfils the targets for payback and return on capital employed.

Melplash Co has prepared the following forecast of production and sales of the new model.

Year	1	2	3	4
Motorcycles (units/year)	25,000	45,000	40,000	20,000

Melplash Co's finance director has prepared the following information relating to the investment proposal. Taxation should be ignored.

Item	Value
Initial investment	\$220m
Residual value at the end of four years	\$30m
Selling price (Year 1 price terms)	\$20,000 per motorcycle
Expected selling price inflation	6% per year
Variable costs (Year 1 price terms)	\$14,000 per motorcycle
Expected variable cost inflation	6% per year
Incremental fixed costs (Year 1 price terms)	\$120m per year
Expected fixed cost inflation	10% per year
Nominal (money) discount rate	14%

Required:

- (a) For the proposed investment in the new motorcycle, calculate the following:
- (i) Net present value (5 marks)
 - (ii) Internal rate of return (2 marks)
 - (iii) Payback period (1 mark)
 - (iv) Return on capital employed (accounting rate of return) based on average investment. (3 marks)
- (b) Discuss the suitability of the techniques used in (a) for determining whether to undertake an investment and advise whether Melplash Co's proposed investment is financially acceptable. (9 marks)

(Total = 20 marks)*(ACCA FM March/June 2022)*

12 The following draft appraisal of a proposed investment project has been prepared for the finance director of OKM Co by a trainee accountant. The project is consistent with the current business operations of OKM Co.

Year	1	2	3	4	5
Sales (units/year)	250,000	400,000	500,000	250,000	
Contribution (\$000)	1,330	2,128	2,660	1,330	
Fixed costs (\$000)	(530)	(562)	(596)	(631)	
Depreciation (\$000)	(438)	(438)	(437)	(437)	
Interest payments (\$000)	(200)	(200)	(200)	(200)	
Taxable profit (\$000)	162	928	1,427	62	
Taxation (\$000)		(49)	(278)	(428)	(19)
Profit for the year (\$000)	162	879	1,149	(366)	(19)
Scrap value (\$000)				250	
After-tax cash flows (\$000)	162	879	1,149	(116)	(19)
Discount at 10%	0.909	0.826	0.751	0.683	0.621
Present values (\$000)	147	726	863	(79)	(12)

Net present value = \$1,645,000 - \$2,000,000 = \$(355,000), so reject the project.

The following information was included with the draft investment appraisal.

1. The initial investment is \$2 million.
2. Selling price: \$12/unit (current price terms), selling price inflation is 5% per year.
3. Variable cost: \$7/unit (current price terms), variable cost inflation is 4% per year.
4. Fixed overhead costs: \$500,000/year (current price terms), fixed cost inflation is 6% per year.
5. \$200,000/year of the fixed costs are development costs that have already been incurred and are being recovered by an annual charge to the project.
6. Investment financing is by a \$2 million loan at a fixed interest rate of 10% per year.
7. OKM Co can claim 25% reducing balance tax-allowable depreciation on this investment and pays taxation one year in arrears at a rate of 30% per year.
8. The scrap value of machinery at the end of the four-year project is \$250,000.
9. The real weighted average cost of capital of OKM Co is 7% per year.
10. The general rate of inflation is expected to be 4.7% per year.

Required:

- (a) Identify and comment on any errors in the investment appraisal prepared by the trainee accountant. (5 marks)
- (b) Prepare a revised calculation of the net present value of the proposed investment project and comment on the project's acceptability. (10 marks)
- (c) Discuss the problems faced when undertaking investment appraisal in the following areas and comment on how these problems can be overcome:
 - (i) an investment project has several internal rates of return (2 marks)
 - (ii) the business risk of an investment project is significantly different from the business risk of current operations. (3 marks)

(Total = 20 marks)

(ACCA F9/FM June 2010, modified)

13 AgeAid is a charity that provides day-centre activities and hot-meal deliveries for elderly people in the local community. AgeAid does not pay tax on its activities. AgeAid is considering whether to buy or lease a fleet of vehicles to support these services over a four-year period, and its finance team has produced the following information.

Buy option

Year	0	1	2	3	4
Initial cost of vehicles (\$)	(300,000)				
Maintenance costs (\$)		12,000	note (1)	note (1)	note (1)
Running costs (\$)		20,000	note (2)	note (2)	note (2)

Note (1): Maintenance costs are expected to rise by 20% in Year 2, a further 30% in Year 3, and a further 40% in Year 4, compared with the previous year's cost.

Note (2): Running costs are expected to increase by 5% per year in Years 2 to 4, compared with the previous year's cost.

Depreciation is charged at 20% per year on a reducing balance basis. The vehicles are expected to be sold for their written-down value at the end of Year 4. AgeAid's cost of capital for this decision is 9% per year.

Lease option

AgeAid could instead lease the vehicles for annual lease payments of \$80,000, payable in advance, that is, at the start of each of the four years (Year 0, Year 1, Year 2 and Year 3). Running costs under the lease option would be 75% of the running costs which would apply under the buy option in the corresponding year. The lessor would be responsible for maintenance of the vehicles and for their eventual replacement.

Required:

- (a) Evaluate, on financial grounds, whether AgeAid should buy or lease the vehicles. You should show clearly how all of the costs mentioned above are treated. (10 marks)
- (b) Discuss TWO reasons, other than the present value of costs, why AgeAid should lease the vehicles rather than buy them. (4 marks)
- (c) Explain the 3Es framework (economy, efficiency and effectiveness) and how this can be applied by AgeAid to assess value for money in its charitable activities. (6 marks)

(Total = 20 marks)

Answers

Objective Test Questions

1 B

WORKING

$(1 + i) = (1 + r)(1 + h)$, where i = money rate, r = real rate, h = inflation rate. $(1.21) = (1 + r)(1.09)$, so $(1 + r) = 1.21/1.09 = 1.1101$, i.e. $r = 11.0\%$.

2 B

WORKING

$(1.21) = (1 + r)(1.08)$, so $(1 + r) = 1.21/1.08 = 1.1204$, i.e. $r = 12.0\%$.

3 Answer: 5.8%

WORKING

$(1 + r) \times (1.04) = (1.10)$, so $(1 + r) = 1.10/1.04 = 1.058$, i.e. $r = 5.8\%$.

4 Answer: \$97,956

WORKING

Money cash flow in Year 3 = $\$120,000 \times (1.05)^3 = \$138,915$.

Nominal (money) discount rate: $(1 + i) = (1.07)(1.05) = 1.1235$, i.e. $i = 12.35\%$.

Present value = $\$138,915/(1.1235)^3 = \$97,956$. (This equals the result of discounting the real cash flow of $\$120,000$ directly at the real rate of 7% for 3 years, both methods must always agree.)

5 C

WORKING

In order to use the perpetuity factor $(1/r)$, the annual amount must be constant, so the calculation needs to be done in real terms. The money cost of capital is given in the question, so the real rate needs to be calculated using $(1 + r)(1 + h) = (1 + i)$: $(1 + r) \times (1.02) = (1.102)$, so $(1 + r) = 1.102/1.02 = 1.08$, i.e. $r = 8\%$.

The perpetuity factor from T_2 -infinity = $(1/r) - DF_1 = (1/0.08) - 0.926 = 11.574$. Present value = $\$10,000 \times 11.574 = \$115,740$.

6 A

An increased expectation of inflation has two effects: (1) a higher expected nominal (money) cash flow, and (2) a higher nominal discount rate. These two effects cancel each other out exactly, leaving the present value of the real receipt unchanged.

7 Answer: \$525,129

WORKING

Since the saving inflates at exactly the same rate (6%) as the general/money environment implied by the 15% nominal discount rate, the money method (inflating the cash flow at 6% and discounting at 15%) and

the real method (discounting the constant real \$80,000 at the implied real rate of 8.49%) give the same answer: \$525,129 over the 10 years.

8 C

WORKING

Since the nominal cost of capital given is 15.5% and wages inflate at 5% per year, the implied real rate = $(1 + \text{real rate}) = (1 + \text{nominal rate}) / (1 + \text{inflation rate}) = 1.155 / 1.05$, so the real rate = 10%.

The 10-year annuity factor at 10% = 6.145.

Present value of savings = 20,000 hours x \$5/hour x 6.145 = \$614,500, i.e. \$615,000 to the nearest \$000.

9 B

WORKING

Deflating the nominal cost of capital: $(1.10) = (1 + r)(1.04)$, so $r = 1.10 / 1.04 - 1 = 5.8\%$, say 6%.

Five-year annuity factor at 6% = 4.212.

Present value = \$75,000 x 4.212 = \$315,900 (this quick approximation uses a real rate rounded to the nearest whole percent, so it only gets close to the precise answer; this alternative method only works at all because all cash flows inflate at the same, single rate).

Discounting each year's inflated cash flow individually at the nominal rate of 10% gives the precise answer: Year 1 \$70,902 + Year 2 \$67,005 + Year 3 \$63,358 + Year 4 \$59,926 + Year 5 \$56,666 = \$317,857, i.e. \$318,000 to the nearest \$000.

10 C

WORKING

Where cash flows have been inflated using their own specific inflation rates, they must always be discounted using the money (nominal) cost of capital, regardless of what specific rate(s) were used to inflate the individual cash flows. Nominal rate = $(1.08)(1.037) - 1 = 11.996\%$, i.e. 12.0% to one decimal place.

11 Answer: \$129,100

WORKING

Year	Revenue (inflated)	Labour cost (inflated)	Net cash flow	DF @ 12%	PV
1	150,000 x 1.06 = 159,000	80,000 x 1.08 = 86,400	72,600	0.893	64,821
2	160,000 x 1.1236 = 179,776	85,000 x 1.1664 = 99,144	80,632	0.797	64,279

Total present value = \$64,821 + \$64,279 = \$129,100 (to the nearest \$100).

12 C**WORKING**

As not all cash flows inflate at the same rate, cash flows are inflated where necessary and discounted using the money rate. $(1 + \text{money rate}) = (1.08) \times (1.02) = 1.1016$, so the money rate = 10% to the nearest whole percentage.

Nominal income = $\$100,000 \times (1 + 10\%) = \$110,000$.

Nominal expenses = $\$35,000$ (zero inflation).

NPV = $[(\$110,000 - \$35,000) \times DF_1] - \$10,000$, where DF_1 = the 1-year 10% discount factor = 0.909. = $(\$75,000 \times 0.909) - \$10,000 = \$58,175$.

13 Answer: \$14,390**WORKING**

Year	Income (inflated at 5%)	Fixed cost (money terms)	Net cash flow	DF @ 10%	PV
1	63,000	(20,000)	43,000	0.909	39,087
2	66,150	(20,000)	46,150	0.826	38,120
3	69,458	(20,000)	49,458	0.751	37,143

Total present value of net cash flows = $\$39,087 + \$38,120 + \$37,143 = \$114,350$ (minor rounding).

Less initial investment $\$100,000$.

NPV = $\$14,350$ (approximately $\$14,390$ using unrounded discount factors).

Question 14

Statement	True	False
All of a project's cash flows must be expected to inflate at the same rate	✓	
That common inflation rate must equal the general rate of inflation	✓	
The discount rate used must be the money (nominal) cost of capital		✓
Working capital must be excluded from the appraisal		✓

The real-terms method is only valid where all of a project's cash flows are expected to inflate at the same (general) rate. In that case the calculation must use the REAL discount rate, not the money rate, using the money rate to discount real (un-inflated) cash flows would be a mismatch. Working capital, like any other cash flow, must still be included in a real-terms appraisal.

15 C**WORKING**

As the cash inflow and cash outflow are subject to different specific inflation rates, they must be inflated at their own specific rates (6% and 4% respectively) and the resulting nominal cash flows discounted using the nominal (money) discount rate. Using the Fisher formula: $(1 + \text{nominal rate}) = (1 + \text{real rate})(1 + \text{general inflation rate}) = 1.10 \times 1.05 = 1.155$, i.e. a nominal discount rate of 15.5% (NOT the general inflation rate of 5%, nor either specific rate).

16 Answer: approximately \$16.7m (positive)

WORKING

Year	0	1	2	3	4
Revenue (\$m)		500.00	954.00	898.88	476.40
Variable costs (\$m)		(350.00)	(667.80)	(629.20)	(333.48)
Fixed costs (\$m)		(120.00)	(132.00)	(145.20)	(159.72)
Investment/residual value (\$m)	(220.00)				30.00
Net cash flow (\$m)	(220.00)	30.00	154.20	124.48	13.20
Discount at 14%	1.000	0.877	0.769	0.675	0.592
Present value (\$m)	(220.00)	26.31	118.58	84.02	7.81

Net present value = $$(220.00) + 26.31 + 118.58 + 84.02 + 7.81 = \16.72m . The NPV is positive, so the investment is financially acceptable (no tax is charged, so the calculation is entirely pre-tax). Selling price inflated at 6%/year gives \$20,000/\$21,200/\$22,472/\$23,820 per unit; multiplied by the volumes of 25,000/45,000/40,000/20,000 units gives revenue of \$500.00m/\$954.00m/\$898.88m/\$476.40m. A similar approach, using variable cost inflation of 6%/year and fixed cost inflation of 10%/year, gives the variable and fixed cost figures shown.

17 C

WORKING

Nominal cost of capital = $(1.08 \times 1.12) - 1 = 20.96\%$.

Time	0	1	2
Outlay	(13,500)		
Labour savings		7,000	7,700
Scrap value			5,000
Discount factor @ 20.96%	1.000	0.827	0.684
Present value	(13,500)	5,787	8,680

NPV = $$(13,500) + $5,787 + $8,680 = 967 , i.e. \$970 to the nearest \$10. (Labour cost in Year 2 = $\$7,000 \times 1.10 = \$7,700$, as labour rates rise by 10% in Year 2; Year 2 cash flow of $\$7,700 + \$5,000$ scrap = $\$12,700$ is discounted at the Year 2 factor.)

18 C

WORKING

The asset is purchased on 31 December 20X4 (T_0), so the first portion of tax-allowable depreciation is accounted for on that date (as this is the end of the year). The amount of the depreciation is $\$1\text{m} \times 25\% = \$250,000$. Claiming this allowance saves $(\$250,000 \times 30\%) = \$75,000$ tax when it is paid at T_1 (one year later). Present value = $\$75,000 \times DF_1 = \$75,000 \times 0.909 = \$68,175$.

19 B

WORKING

As tax is paid one year in arrears, the \$20,000 and its associated tax are treated separately.

PV of perpetuity = $\$20,000 \times (1/0.1) = \$200,000$.

Less PV of tax: the tax perpetuity runs from T_2 onwards, so $AF_2\text{-infinity} = (1/0.1) - DF_1 = 10 - 0.909 = 9.091$.

PV of tax = $\$20,000 \times 30\% \times 9.091 = \$54,546$.

After-tax present value = $\$200,000 - \$54,546 = \$145,454$.

20 Answer: \$10,547

WORKING

Year	Tax-allowable depreciation		Tax benefit
1	$250,000 \times 0.25 =$	62,500	2: $62,500 \times 0.3 = 18,750$
2	$62,500 \times 0.75 =$	46,875	3: $46,875 \times 0.3 = 14,063$
3	$46,875 \times 0.75 =$	35,156	4: $35,156 \times 0.3 = 10,547$

The tax benefit received by Trecor Co in time 4 is \$10,547.

21 B

WORKING

Tax-allowable depreciation = $\$80,000/4 = \$20,000$ per year.

Taxable profit = $\$60,000 - \$20,000 = \$40,000$.

Tax at 25% = \$10,000.

After-tax cash flow = $\$60,000 - \$10,000 = \$50,000$ (the tax-allowable depreciation reduces taxable profit and hence the tax cash outflow, but, since it is a non-cash item, is not itself deducted from the cash flow).

22 Answer: \$157,500

WORKING

Tax written down value after Year 1 = $\$2,000,000 \times 0.75 = \$1,500,000$.

After Year 2 = $\$1,500,000 \times 0.75 = \$1,125,000$.

Balancing allowance in Year 3 = tax written down value less disposal proceeds = $\$1,125,000 - \$600,000 = \$525,000$.

Tax benefit = $\$525,000 \times 30\% = \$157,500$.

23 A

WORKING

Tax written down value after Year 1 = $\$500,000 \times 0.75 = \$375,000$.

Balancing charge in Year 2 = disposal proceeds less tax written down value = $\$410,000 - \$375,000 = \$35,000$ (proceeds exceed the tax written down value, so a charge, not an allowance, arises).

Additional tax payable = $\$35,000 \times 25\% = \$8,750$.

24 B

A balancing allowance arises when disposal proceeds are LESS than the tax written down value (an extra tax-deductible allowance is given for the shortfall); a balancing charge arises when disposal proceeds EXCEED the tax written down value (too much depreciation has been claimed over the asset's life, so some is clawed back, increasing the tax payable in that year, not reducing it). Balancing allowances/charges arise under reducing-balance TAD (the method used throughout this chapter), not only under straight-line.

25 Answer: \$13,817

WORKING

	\$
Tax depreciation in first year	38,000
Tax rate	x 0.40
Tax saving	15,200
Discount factor @ 10%	x 0.909
Discounted tax saving for first year	13,817

Tax-allowable depreciation = $\$190,000/5 = \$38,000$ in Year 1 (working capital has no tax effect and is not relevant to this calculation).

26 C

WORKING

Tax-allowable depreciation = $\$100,000/10 = \$10,000$ per year. Annual tax saving = $\$10,000 \times 40\% = \$4,000$.

27 D

WORKING

	\$
After-tax operating cash flow ($50,000 \times (1 - 35\%)$)	32,500
Tax saving on tax-allowable depreciation ($((100,000/5) \times 35\%)$)	7,000
Total after-tax cash flow	39,500

28 C

WORKING

	\$
Cash inflow (sales) ($2,000 \times \$500$)	1,000,000
Cash outflow (materials & labour) ($2,000 \times \$450$)	(900,000)
Operating cash inflow	100,000
Tax at 40%	(40,000)
Cash inflow from operations after tax	60,000
Scrap value of equipment	5,000
Tax on balancing charge ($40\% \times \$5,000$)	(2,000)
Cash inflow from sale of machine after tax	3,000
Total cash inflow after tax	63,000

By the end of the fifth year the machine will already have been written down to zero tax value (tax-allowable depreciation is only claimed over five years, on a machine with a ten-year operating life). Therefore, by Year 10 no further tax-allowable depreciation is available, but the entire scrap value of

\$5,000 represents a balancing charge (a taxable gain on disposal, since the tax written down value is already nil).

29 C

WORKING

Time	Working capital required	Incremental cash flow	DF @ 10%	PV
0	10,000	(10,000)	1.000	(10,000)
1	12,500	(2,500)	0.909	(2,273)
2	10,500	2,000	0.826	1,652
3	0	10,500	0.751	7,886

Present value of working capital = \$(2,735).

30 A

WORKING

The working capital required inflates year on year, then the inflated amount is returned in full at the end of the project.

Time	Working capital required (with 10% inflation)	Incremental cash flow	DF @ 12%	PV
0	100,000	(100,000)	1.000	(100,000)
1	110,000	(10,000)	0.893	(8,930)
2	0	110,000	0.797	87,670

Present value of the cash flows relating to working capital = \$(21,260).

31 Answer: +\$100,000

WORKING

All working capital invested over the life of the project (\$80,000 at time 0 plus \$20,000 at time 1, totalling \$100,000) is recovered in full when the project ends, so the working capital cash flow at time 4 is a single inflow of +\$100,000.

32 B

Working capital is a real cash flow (an investment tied up in inventory, receivables less payables) even though it does not appear as a cost in the income statement, so it must be included in an NPV appraisal: as an outflow when invested (including incremental top-ups as sales grow) and as an inflow (its full recovery) when the project ends. It is not a tax-allowable expense, since it is not a revenue or capital cost for tax purposes.

33 C

WORKING

Time	Cash flow	Discount factor	PV	
0	(35,000)	1.000	(35,000)	Initial investment
5	35,000	0.621	21,735	Recovery
Overall effect			(13,265)	

Working capital flows have no tax effect. Only the initial investment of working capital at time 0 and its full recovery at the end of Year 5 are relevant, since the two present values do not offset (because of discounting), there is a net negative effect on NPV of \$(13,265).

34 Answer: \$15,701**WORKING**

Year 1 PV = $10,000 \times 1.05 \times 0.877 = \$9,209$.

Year 2 PV = $20,000 \times 1.05^2 \times 0.769 = \$16,957$.

Year 3 PV = $25,000 \times 1.05^3 \times 0.675 = \$19,535$.

NPV = $9,209 + 16,957 + 19,535 - 30,000 = \$15,701$. (Other answers differing only due to rounding of discount factors would also be correct.)

35 D

The inflation included in the money (nominal) cost of capital is required by investors to compensate them for the loss of general purchasing power their money is expected to suffer in the future as a result of investing in the business, i.e. it reflects EXPECTED GENERAL inflation, not a business-specific or historic rate.

Case Scenario Questions

Question 1

(1) Answer: \$8,487 (or \$8,488)

WORKING

Year 2 sales = 1,600,000 boxes x \$5.00 x $(1.03)^2$ = \$8,487,200 (or \$8,488,000 if the inflated price per box, \$5.305, is rounded first).

(2) Answer: \$6,884,180

WORKING

Year 3 demand of 2.1 million boxes falls within the 2-2.9 million band, so the current-terms variable cost is \$3.00 per box. Variable costs = 2,100,000 x \$3.00 x $(1.03)^3$ = \$6,884,180.

(3) Answer: \$84,375

WORKING

Year	Tax-allowable depreciation		Tax benefit received
1	2,000,000 x 0.25 =	500,000	2: 500,000 x 0.3 = 150,000
2	500,000 x 0.75 =	375,000	3: 375,000 x 0.3 = 112,500
3	375,000 x 0.75 =	281,250	4: 281,250 x 0.3 = 84,375

The tax benefit generated in Year 4 is \$84,375.

(4)

Statement	True	False
The trainee accountant has used the wrong percentage for the cost of capital		✓
Ignoring sales after four years underestimates the value of the project	✓	
The working capital figure in Year 4 is wrong	✓	

Statement 1: As inflated sales and costs have been used, the cost of capital should be the nominal cost of capital (12%), which is what has been used, so this statement is FALSE. Statement 2: cutting off cash flows after four years underestimates the value of the project, as future cash inflows beyond Year 4 will be ignored, TRUE. Statement 3: the final year should recover the TOTAL working capital invested over the life of the project, i.e. \$750k + \$23k + \$23k + \$24k = \$820k, not the \$750k shown, TRUE.

(5) Answer: C, Both statements are true

When there are unconventional cash flow patterns, there may be multiple IRRs, and so the NPV and IRR decisions may not be the same. A project is financially viable under the IRR criterion if the IRR is greater than the cost of capital (12% in this case).

Constructed Response Questions

1 (a)

	Year 1 (\$000)	Year 2 (\$000)	Year 3 (\$000)	Year 4 (\$000)
Sales revenue (W1)	3,120	15,576	22,275	10,296
Variable cost (W2)	(1,890)	(7,936)	(9,378)	(4,376)
Contribution	1,230	7,640	12,897	5,920
Fixed cost	(540)	(583)	(630)	(680)
Taxable cash flow	690	7,057	12,267	5,240
Taxation	(138)	(1,411)	(2,453)	(1,048)
TAD tax benefits (W3)	125	94	70	186
After-tax cash flow	677	5,740	9,884	4,378
Scrap value				125
Working capital (W4)	(90)	(95)	(102)	1,787
Net cash flows	587	5,645	9,782	6,290
Discount at 12%	0.893	0.797	0.712	0.636
Present values	524	4,499	6,965	4,000

PV of future cash flows = \$15,988,000.

Initial investment = \$4,000,000 (\$2.5m + \$1.5m).

NPV = \$11,988,000.

The NPV is strongly positive, so the project is financially acceptable.

WORKINGS

(W1) Sales revenue

Year	1	2	3	4
Selling price (\$/unit)	15.60	19.47	24.75	25.74
Sales volume (units)	200,000	800,000	900,000	400,000
Sales revenue (\$)	3,120,000	15,576,000	22,275,000	10,296,000

Selling prices of \$15/\$18/\$22/\$22 (current terms) inflated at 4%/year.

(W2) Variable cost

Year	1	2	3	4
Variable cost (\$/unit)	9.45	9.92	10.42	10.94
Sales volume (units)	200,000	800,000	900,000	400,000
Variable cost (\$)	1,890,000	7,936,000	9,378,000	4,376,000

Variable cost of \$9.00/unit inflated at 5%/year.

(W3) Tax-allowable depreciation and tax benefits

Year	TA depreciation (\$)	Tax benefit (\$)
1	625,000	125,000
2	469,000	94,000
3	352,000	70,000
4	929,000	186,000
Scrap value	125,000	

Year	TA depreciation (\$)	Tax benefit (\$)
Total	2,500,000	

25% reducing balance on \$2,500,000, with a balancing allowance in Year 4 after deducting the \$125,000 scrap value.

Tax benefits at 20%, settled in the same year the TAD is claimed (Hebac Co's tax liability is settled in the year it arises).

(W4) Working capital

Year	1	2	3	4
Working capital balance (\$)	1,590,000	1,685,000	1,787,000	
Incremental cash flow (\$)	(90,000)	(95,000)	(102,000)	1,787,000

Working capital of \$1.5m (Year 0) inflated at 6%/year; the full balance is recovered as an inflow in Year 4.

(b)

A company can use its weighted average cost of capital (WACC) as the discount rate for an investment project only where the project's business risk and financial risk are similar to those of its existing operations. Where a project's business risk differs significantly, as here, since the new product is outside Hebac Co's current business operations, a project-specific discount rate is needed.

The capital asset pricing model (CAPM) can provide this project-specific discount rate. The equity beta of a proxy company operating in a similar business area reflects the systematic business risk of the project (and, if the proxy company is geared, also its systematic financial risk). The proxy equity beta is first ungeared to strip out the proxy's financial risk, giving an asset beta that reflects only business risk; this asset beta is then regressed using Hebac Co's own gearing to give an equity beta reflecting Hebac Co's financial risk. The regressed equity beta is inserted into the CAPM formula to give a project-specific cost of equity (and hence a project-specific WACC), which, unlike Hebac Co's current WACC, properly reflects the systematic risk of this specific investment, leading to a better-informed accept/reject decision.

Marking scheme	
(a) Inflated selling price	1
Sales revenue	1
Inflated variable cost	1
Inflated fixed costs	1
Tax liabilities	1
TAD benefits Years 1-3	1
TAD benefit Year 4	1
Incremental working capital and recovery	2
PVs	1
Initial investment	1
Comment on acceptability	1
Maximum	12
(b) Business risk, financial risk and WACC	2
Proxy company	1
Systematic business/financial risk	1
Ungearing	1

Marking scheme	
Regearing	1
Project-specific cost of equity and WACC	2
Maximum	8
Total	20 marks

2 (a) (i) Nominal terms appraisal of the investment project

	Year 1 (\$000)	Year 2 (\$000)	Year 3 (\$000)	Year 4 (\$000)
Sales revenue	39,375	58,765	85,087	32,089
Variable cost	(22,047)	(31,185)	(41,328)	(17,923)
Contribution	17,328	27,580	43,759	14,166
Fixed costs	(3,180)	(3,483)	(3,811)	(3,787)
Cash flows before tax	14,148	24,097	39,948	10,379
Tax at 26%	(3,679)	(6,265)	(10,387)	(2,699)
TAD benefits	1,300	975	731	2,194
Cash flows after tax	11,769	18,807	30,292	9,874
Discount at 12%	0.893	0.797	0.712	0.636
Present values	10,510	14,989	21,568	6,280

Sum of PVs of future cash flows = \$53,347,000.

Initial investment = \$20,000,000.

NPV = \$33,347,000.

(ii) Real terms appraisal of the investment project

	Year 1 (\$000)	Year 2 (\$000)	Year 3 (\$000)	Year 4 (\$000)
Nominal cash flows before tax	14,148	24,097	39,948	10,379
Real cash flows before tax	13,643	22,408	35,823	8,975
Tax at 26%	(3,547)	(5,826)	(9,314)	(2,334)
TAD benefits	1,300	975	731	2,194
Cash flows after tax	11,396	17,557	27,240	8,835
Discount at 8%	0.926	0.857	0.794	0.735
Present values	10,553	15,046	21,629	6,494

Sum of PVs of future cash flows = \$53,722,000.

Initial investment = \$20,000,000.

NPV = \$33,722,000.

The real-terms appraisal gives almost the same positive NPV as the nominal-terms appraisal (the small difference is due to applying a different discount rate to the same TAD benefits, which are not adjusted for inflation). As the NPV is positive under both approaches, the investment project is financially acceptable.

(b)

The achievement of stakeholder objectives by managers can be encouraged by managerial reward schemes and by regulatory requirements.

Share option schemes: the agency problem arises from the separation of ownership and control, with managers pursuing their own objectives rather than shareholders' objective of maximising wealth. Turning managers into shareholders through share option schemes increases goal congruence, though the award criteria need careful design.

Performance-related pay: remuneration can be made conditional on specified performance targets consistent with stakeholder objectives (e.g. EPS growth for shareholders, emissions targets for government objectives), though such targets need careful design to avoid dysfunctional behaviour; long-term incentive plans (LTIPs), benchmarked against competitors, are now generally seen as more effective than simple PRP.

Corporate governance codes of best practice: these apply best practice to executive remuneration, risk management, auditing, internal control and board accountability, and support the key role of non-executive directors in exercising independent judgement.

Stock exchange listing regulations: these seek a fair and efficient market, encouraging disclosure of price-sensitive information and decreasing information asymmetry, one of the causes of the agency problem, giving stakeholders a clearer view of whether managers are attending to their objectives.

Marking scheme	
(a) (i) Sales nominal	1
VC nominal	1
FC nominal	1
Tax liability	1
TAD	1
TAD benefits	1
Tax timing	1
PVs and nominal NPV	1
(ii) Real CF before tax	1
Tax treatment	1
PVs and real NPV	1
Comment	1
Maximum	12
(b) 2 marks per way to a maximum of 4 ways	8
Maximum	8
Total	20 marks

3 (a) Revised draft evaluation of investment proposal

	1 (\$000)	2 (\$000)	3 (\$000)	4 (\$000)	5 (\$000)
Sales revenue (W1)	2,475	2,714	4,413	4,775	
Variable costs (W2)	(1,097)	(1,323)	(2,084)	(2,370)	
Fixed costs (W3)	(155)	(159)	(164)	(169)	
Cash flow before tax	1,223	1,232	2,165	2,236	
TA depreciation (W4)	(450)	(338)	(253)	(759)	
Taxable profit	773	894	1,912	1,477	

	1 (\$000)	2 (\$000)	3 (\$000)	4 (\$000)	5 (\$000)
Taxation	-	(170)	(197)	(421)	(325)
After-tax profit	773	724	1,715	1,056	(325)
TA depreciation added back	450	338	253	759	
After-tax cash flow	1,223	1,062	1,968	1,815	(325)
Discount at 12%	0.893	0.797	0.712	0.636	0.567
Present values	1,092	846	1,401	1,154	(184)

	\$000
Present value of cash inflows	4,309
Cost of machine	(1,800)
NPV	2,509

The revised draft evaluation indicates that a positive net present value of \$2,509,000 is expected to be produced. The investment project is therefore financially acceptable and accepting it will increase the wealth of the shareholders of Uftin Co.

WORKINGS

(W1) Sales revenue

Year	1	2	3	4
Selling price (\$/unit)	26.05	27.14	29.42	31.83
Sales volume (units)	95,000	100,000	150,000	150,000
Sales revenue (\$)	2,475,000	2,714,000	4,413,000	4,775,000

Selling prices of \$25/\$25/\$26/\$27 (current terms) inflated at 4.2%/year.

(W2) Variable costs

Year	1	2	3	4
Variable cost (\$/unit)	11.55	13.23	13.89	15.80
Sales volume (units)	95,000	100,000	150,000	150,000
Variable costs (\$)	1,097,000	1,323,000	2,084,000	2,370,000

Variable costs of \$11/\$12/\$12/\$13 (current terms) inflated at 5%/year.

(W3) Fixed costs

Year	1	2	3	4
Fixed costs (\$)	155,000	159,000	164,000	169,000

Fixed costs of \$150,000/year (current terms) inflated at 3%/year.

(W4) Tax-allowable depreciation and tax benefits

Year	TA depreciation (\$)	Tax benefit (\$)
1	450,000	99,000
2	337,500	74,250
3	253,125	55,688
4	759,375	167,063

25% reducing balance on \$1,800,000, with a balancing allowance in Year 4 since scrap value is nil; tax benefits at 22%, received one year in arrears (i.e. in Years 2-5).

(b) Revisions to the original draft evaluation

Inflation: only one year's inflation had been applied in Year 1, with no further inflation applied to Years 2, 3 and 4. The effect of inflation on cash flows is cumulative, and specific inflation rates for sales, variable costs and fixed costs needed to be applied cumulatively in every year.

Interest payments: these should not have been included, because the financing effect of the loan is already reflected in the discount rate (Uftin Co's WACC). In a large company, a loan this size relative to existing finance will not itself affect the WACC.

Tax-allowable depreciation: a constant allowance equal to 25% of the initial investment had been used every year, instead of 25% reducing balance (giving smaller allowances in Years 2 and 3 and a balancing allowance in Year 4). In addition, although TAD had been deducted to produce taxable profit, it had not been added back to produce after-tax cash flow.

Year 5 tax liability: this had been omitted, perhaps because only a four-year evaluation period was used. However, the Year 5 tax cash flow (arising from Year 4's taxable profit, paid one year in arrears) is a relevant cash flow arising as a result of the investment decision and needed to be included.

(Only TWO of the above revisions were required to be explained for full marks.)

Marking scheme	
(a) Sales revenue	1
Inflated sales revenue	1
Inflated variable costs	1
Inflated fixed costs	1
Excluding interest payments	1
TAD	1
Balancing allowance	1
Tax liabilities	1
Timing of tax liabilities	1
NPV	1
Comment on acceptability	1
Maximum	11
(b) Explanation of first revision	1-3
Explanation of second revision	1-3
Maximum	4
Total	15 marks

4 (a) Net present value of investment in new machinery

	1 (\$000)	2 (\$000)	3 (\$000)	4 (\$000)	5 (\$000)
Sales income (W1)	6,084	6,327	6,580	6,844	
Variable cost (W2)	(2,374)	(2,504)	(2,642)	(2,787)	
Contribution	3,710	3,823	3,938	4,057	
Fixed costs (W3)	(263)	(276)	(289)	(304)	
Cash flow	3,447	3,547	3,649	3,753	
Taxation		(689)	(709)	(730)	(751)
CA tax benefits (W4)		250	188	141	372
After-tax cash flow	3,447	3,108	3,128	3,164	(379)

	1 (\$000)	2 (\$000)	3 (\$000)	4 (\$000)	5 (\$000)
Working capital (W5)	(24)	(25)	(26)	(27)	
Scrap value				250	
Net cash flow	3,423	3,083	3,102	3,387	(379)
Discount at 12%	0.893	0.797	0.712	0.636	0.567
Present values	3,057	2,457	2,209	2,154	(215)

PV of future cash flows = \$9,662,000.

Initial investment = \$5,000,000.

Working capital = \$500,000.

NPV = \$4,162,000.

As the net present value of \$4.162 million is positive, the expansion can be recommended as financially acceptable.

WORKINGS

(W1) Sales income

Year	1	2	3	4
Selling price (\$/unit)	676.00	703.04	731.16	760.41
Sales volume (units)	9,000	9,000	9,000	9,000
Sales income (\$)	6,084,000	6,327,000	6,580,000	6,844,000

Selling price of \$650/unit inflated at 4%/year; constant volume of 9,000 units/year.

(W2) Variable cost

Year	1	2	3	4
Variable cost (\$/unit)	263.75	278.26	293.56	309.71
Sales volume (units)	9,000	9,000	9,000	9,000
Variable cost (\$)	2,374,000	2,504,000	2,642,000	2,787,000

Variable cost of \$250/unit inflated at 5.5%/year.

(W3) Fixed costs

Year	1	2	3	4
Fixed costs (\$)	263,000	276,000	289,000	304,000

Fixed costs of \$250,000 inflated at 5%/year.

(W4) Tax-allowable depreciation and tax benefits

Year	TA depreciation (\$)	Tax benefit (\$)
1	1,250,000	250,000
2	937,500	188,000
3	703,125	141,000
4	1,859,375	372,000

25% reducing balance on \$5,000,000, with a balancing allowance in Year 4 after deducting the 5%-of-cost scrap value of \$250,000; tax benefits at 20%, paid one year in arrears.

(W5) Working capital

Year	1	2	3	4
Incremental cash flow (\$)	(24,000)	(25,000)	(26,000)	(27,000)

Working capital of \$500,000 inflates at 4.7%/year; no separate recovery, since it continues indefinitely beyond the 4-year appraisal in this question.

(b)

A nominal (money terms) approach to investment appraisal discounts nominal cash flows with a nominal cost of capital. Nominal cash flows are found by inflating forecast values from current price estimates, for example, using specific inflation rates for different project cash flows in order to generate nominal project cash flows.

A real terms approach to investment appraisal discounts real cash flows with a real cost of capital. Real cash flows are found by deflating nominal cash flows by the general rate of inflation. The real cost of capital is found by deflating the nominal cost of capital by the general rate of inflation, using the Fisher equation: $(1 + \text{real discount rate}) \times (1 + \text{inflation rate}) = (1 + \text{nominal discount rate})$.

The net present value for an investment project does not depend on whether a nominal terms approach or a real terms approach is adopted, since nominal cash flows and the nominal discount rate are both discounted by the general rate of inflation to give real cash flows and the real discount rate, respectively. Both approaches give the same net present value: deflating the net cash flows above by the general rate of inflation of 4.7% and discounting at the real rate of 7% ($1.12/1.047 = 1.07$) gives an NPV of \$4.154 million, which, allowing for rounding, is the same as the \$4.161 million (four significant figures) found under the nominal approach.

(c)

A listed company such as HDW Co is likely to have a range of financial objectives. Maximisation of shareholder wealth is often suggested to be the primary financial objective, and this can be substituted by the objective of maximising the company's share price. Other financial objectives that might be used by HDW Co could relate to earnings per share (for example, a target EPS value for a given period), operating profit (for example, a target level of profit before tax or PBIT), revenue (for example, a desired increase in revenue or sales) and so on. These examples of financial objectives can all be quantified, so that progress towards meeting them can be measured over time.

The investment in the new machine will enable HDW Co to meet increased demand for its products and the company expects to be able to sell all of the increased production at a profit. This will lead to increased revenue and operating profit (profit before interest and tax), so financial objectives relating to these accounting figures will be supported. Whether a financial objective relating to increasing earnings per share (EPS) will be supported will depend on how the investment is financed: raising equity finance by issuing new shares will dilute (decrease) EPS, while raising debt finance will increase interest payments, which will also dilute EPS.

The investment in the new machine has a positive net present value (NPV), so the market value of the company is expected to increase by the amount of the NPV. This increases the wealth of shareholders irrespective of how the investment is financed, since financing costs were accounted for by the discount rate (whether nominal or real). The investment in the new machine will therefore support the objective of shareholder wealth maximisation.

Marking scheme	
(a) Inflated selling price	1

Marking scheme	
Sales income	1
Inflated total variable cost	1
Inflated incremental fixed costs	1
Tax liability	1
Capital allowance Years 1-3	1
Balancing allowance Year 4	1
Capital allowance tax benefits	1
Timing of tax liabilities and benefits	1
Incremental working capital investment	1
Scrap value	1
Discount at 12%	1
Calculated value for NPV	1
Comment on financial acceptability	1
Maximum	14
(b) Discussion of nominal terms approach	2-3
Discussion of real terms approach	2-3
Maximum	5
(c) Identification of two financial objectives	1
Discussion of support for objectives of planned investment	5
Maximum	6
Total	25 marks

5 (a) Evaluation of purchase of new machine using net present value method

	1 (\$000)	2 (\$000)	3 (\$000)	4 (\$000)	5 (\$000)
Sales income (W1)	2,072	2,352	2,596	2,716	
Variable cost (W2)	(1,099)	(1,250)	(1,376)	(1,444)	
Contribution	973	1,102	1,220	1,272	
Maintenance cost	(152)	(159)	(166)	(174)	
Cash flow before tax	821	943	1,054	1,098	
Tax		(222)	(255)	(285)	(297)
Depreciation benefits (W3)		101	76	57	117
Cash flow after tax	821	822	875	870	(180)
Working capital (W4)	(7)	(7)	(8)	(8)	
Resale value				200	
Net cash flow	814	815	867	1,062	(180)
Discount at 11%	0.901	0.812	0.731	0.659	0.593
Present values	733	662	634	700	(107)

PV of future cash flows = \$2,622,000.

Initial investment = \$1,500,000.

Net present value = \$1,122,000.

The investment in the new machine has a positive net present value and is therefore financially acceptable.

WORKINGS

(W1) Sales income

Year	1	2	3	4
Selling price (\$/unit)	5.92	6.19	6.49	6.79
Sales volume (units)	350,000	380,000	400,000	400,000
Sales income (\$)	2,072,000	2,352,000	2,596,000	2,716,000

Selling price of \$5.65/unit inflated at 4.7%/year.

(W2) Variable cost

Year	1	2	3	4
Variable cost (\$/unit)	3.14	3.29	3.44	3.61
Sales volume (units)	350,000	380,000	400,000	400,000
Variable cost (\$)	1,099,000	1,250,000	1,376,000	1,444,000

Variable cost of \$3.00/unit inflated at 4.7%/year.

(W3) Tax-allowable depreciation and tax benefits

Year	TA depreciation (\$)	Tax benefit (\$)
1	375,000	101,000
2	281,000	76,000
3	211,000	57,000
4	433,000	117,000

25% reducing balance on \$1,500,000, with a balancing allowance in Year 4 after deducting the \$200,000 resale value; tax benefits at 27%, paid one year in arrears.

(W4) Working capital

Year	1	2	3	4
Working capital balance (\$)	157,000	164,000	172,000	180,000
Incremental cash flow (\$)	(7,000)	(7,000)	(8,000)	(8,000)

Working capital of \$150,000 inflates at 4.7%/year; no recovery, since the working capital pre-existed and continues to support ongoing operations after the new machine is due for replacement.

(b)

Theoretically, a company should invest in all projects with a positive net present value in order to maximise shareholder wealth. If a company has attractive investment opportunities available to it, with positive net present values, it will not be able to maximise shareholder wealth if it does not invest in them, for example, because investment finance is limited or rationed.

If investment finance is limited for reasons outside a company, it is called 'hard capital rationing'. This may arise because a company is seen as too risky by potential investors, for example, because its level of gearing is so high that it is believed it may struggle to deliver adequate returns on invested funds. Hard capital rationing could also arise if a company wants to raise debt finance for investment purposes, but lacks sufficient assets to offer as security, leading again to a risk-related problem. During a time of financial crisis, investors may seek to reduce risk by limiting the amount of funds they are prepared to invest and by choosing to invest only in low-risk projects. Companies could also struggle to secure investment when capital markets are depressed, or when economic prospects are poor, for example, during a recession.

If investment funds are limited for reasons within a company, the term 'soft capital rationing' is used. Investing in all projects with a positive net present value could mean that a company increases in size quite dramatically, which incumbent managers and directors may wish to avoid in favour of a strategy of controlled growth, limiting the investment finance available as a consequence. Managers and directors may limit investment finance in order to avoid some consequences of external financing, such as an increased commitment to fixed interest payments if new debt finance were raised, or potential dilution of earnings per share if new equity finance were raised, whether from existing or new shareholders. Investment finance may also be limited internally in order to require investment projects to compete with each other for funds, so that only robust investment projects with the strongest net present values gain access to funds, reducing the uncertainty associated with making investment decisions based on financial forecasts.

Marking scheme	
(a) Inflated sales income	1
Inflated variable cost	1
Inflated maintenance cost	0.5
Tax liabilities	1
Timing of tax liabilities	1
Tax-allowable depreciation	1
Tax benefits of tax-allowable depreciation	1
Incremental working capital	1
Resale value	0.5
Present value calculations	1
Evaluation of financial acceptability	1
Maximum	10
(b) Reasons for hard capital rationing	1-4
Reasons for soft capital rationing	1-4
Maximum	5
Total	15 marks

6 (a) Project net present value

	1 (\$000)	2 (\$000)	3 (\$000)	4 (\$000)	5 (\$000)
Sales (W1)	1,890	1,985	2,084	2,188	
Variable costs (W2)	(1,112)	(1,146)	(1,180)	(1,216)	
Fixed costs	(309)	(318)	(328)	(338)	
Taxable cash flow	469	521	576	634	
Taxation at 20%		(94)	(104)	(115)	(127)
Tax savings from TAD (W3)		75	56	42	102
Residual value				125	
Working capital cash flows (W4)	(3)	(3)	(3)	109	
Net cash flows	466	499	525	795	(25)
Discount factors (12%)	0.893	0.797	0.712	0.636	0.567
Present values	416	398	374	506	(14)

Sum of PVs (t1-t4) = \$1,680,000.

Initial investment = \$1,500,000.

Initial working capital = \$100,000.

NPV = \$80,000.

The expected NPV is positive and is therefore financially acceptable.

It should be noted, however, that this assumes that variable cost per unit is \$54 (the expected value), which is not itself a possible value: there is a 60% chance that the variable cost per unit could be less than \$54, which would have a beneficial effect on the NPV obtained, but there is a 40% chance it could be more than \$54, which would reduce the NPV, possibly resulting in a negative NPV.

WORKINGS

Expected variable cost per unit = $(\$40 \times 0.2) + (\$50 \times 0.4) + (\$60 \times 0.2) + (\$70 \times 0.2) = \$54$.

(W1) Sales

Year	1	2	3	4
Selling price (\$/unit)	94.50	99.23	104.19	109.40
Sales volume (units)	20,000	20,000	20,000	20,000
Sales (\$)	1,890,000	1,985,000	2,084,000	2,188,000

Selling price of \$90/unit inflated at 5%/year; constant volume of 20,000 units/year.

(W2) Variable cost

Year	1	2	3	4
Variable cost (\$/unit)	55.62	57.29	59.01	60.78
Sales volume (units)	20,000	20,000	20,000	20,000
Variable cost (\$)	1,112,000	1,146,000	1,180,000	1,216,000

Expected variable cost of \$54/unit inflated at 3%/year.

(W3) Tax-allowable depreciation and tax benefits

Year	TA depreciation (\$)	Tax benefit (\$)
1	375,000	75,000
2	281,000	56,000
3	211,000	42,000
4	508,000	102,000

25% reducing balance on \$1,500,000, with a balancing allowance in Year 4 after deducting the \$125,000 residual value; tax benefits at 20%, paid one year in arrears.

(W4) Working capital

Year	1	2	3	4
Working capital balance (\$)	103,000	106,000	109,000	
Incremental cash flow (\$)	(3,000)	(3,000)	(3,000)	109,000

Working capital of \$100,000 inflates at 3%/year; the full balance is recovered as an inflow in Year 4.

(b) (i)

Gini Co's primary objective is to maximise shareholder wealth. A project's positive NPV represents the amount of value added to the company by undertaking the project. Given the positive NPV of \$80,000 for the LP500 (subject to the actual variable cost per unit turning out close to its expected value), accepting the project will add \$80,000 to the value of Gini Co and consequently increase the shareholders' wealth.

(ii)

Subjective probabilities: probability estimation ideally requires historic data of the event frequency in order to give any reliability to the figures. Even then, past frequency/probability is not necessarily an indication of future frequency/probability. There is little detail regarding the source of the probabilities used to determine the expected variable cost, and if these are little more than an educated guess, this would undermine the credibility of the expected figure calculated.

Variability of the outcome: the variability of the outcomes is being ignored in the decision, as the expected value of the variable cost of \$54/unit gives no indication of the best and worst case figures (\$40 and \$70). There is a 40% probability that the variable cost could actually be more than \$54; if that were to happen, the impact on the NPV could be significant, and could indicate a negative NPV would result.

Attitude to risk: the previously highlighted problem of outcome variability also means that Gini Co's attitude to risk is being ignored in its decision. If Gini Co has a cautious approach to risk, knowing there could be a significant chance of a negative NPV could result in it rejecting the project; a more adventurous approach to risk could see Gini Co being happier to accept the project, given the 60% chance that the variable cost per unit is less than \$54.

Long-run average: an expected value is essentially a long-run average over a sufficiently high number of repetitions and consequently may not itself be a possible outcome, for the LP500 project, \$54 as a unit variable cost is not one of the four possible outcomes at all.

(Only THREE limitations were required for full marks.)

Marking scheme	
(a) Sales revenue	1
Expected variable cost per unit	1

Marking scheme	
Total variable cost	1
Fixed costs	0.5
Tax liability	1
TAD	1
TAD benefit	1
Tax timing	1
Residual value	0.5
Working capital (0-3)	1
Working capital release	1
PVs/NPV	1
Comment	1
Maximum	12
(b) (i) Link to primary objective	2
(ii) First limitation	2
Second limitation	2
Third limitation	2
Maximum	8
Total	20 marks

7 (a) Hotel net present value (NPV)

	1 (\$000)	2 (\$000)	3 (\$000)	4 (\$000)
Accommodation revenue (W1)	776	987	1,307	1,663
Restaurant revenue (35%)	271	345	458	582
Total revenue	1,047	1,332	1,765	2,245
Variable costs (20%)	(209)	(266)	(353)	(449)
Fixed costs	(433)	(450)	(468)	(487)
Taxable cash flow	405	616	944	1,309
Taxation at 20%	(81)	(123)	(189)	(262)
TAD benefit (W2)	75	56	42	107
After-tax cash flow	399	549	797	1,154
Sale value				7,600
Working capital (W3)	(14)	(22)	(24)	112
Net cash flow	385	527	773	8,866
DCF (10%)	0.909	0.826	0.751	0.683
Present value (PV)	350	435	581	6,055

PV of future cash flows = \$7,421,000.

Less: initial investment, property \$5,500,000, fitting-out \$1,500,000, initial working capital \$52,000.

NPV = \$369,000.

NPV is positive and is therefore financially acceptable.

Nominal cost of capital = $(1.048 \times 1.05) - 1 = 10\%$ (this is why tax is charged in the same year as the underlying cash flow, per the scenario, so no one-year lag applies here, unlike most other questions in this chapter).

WORKINGS

(W1) Accommodation revenue

Year	1	2	3	4
Room-nights (50 rooms x 365 days x occupancy %)	9,125	10,950	13,688	16,425
Room rate (\$, inflated at 6%/year)	85.00	90.10	95.51	101.24
Accommodation revenue (\$)	776,000	987,000	1,307,000	1,663,000

Restaurant revenue = 35% of accommodation revenue.

(W2) Tax-allowable depreciation and tax benefits

Year	TA depreciation (\$)	Tax benefit (\$)
1	375,000	75,000
2	281,000	56,000
3	211,000	42,000
4	100,000*	107,000

25% reducing balance on the \$1,500,000 fittings only (the \$5,500,000 property cost is not tax-allowable).

*Year 4's "TA depreciation" figure is not a depreciation charge — it is the fittings' \$100,000 disposal value. WDV brought forward of \$632,812 less this \$100,000 disposal value gives a \$532,812 balancing allowance, which at 20% tax produces the \$107,000 tax benefit shown.

Tax benefits at 20%.

(W3) Working capital

Year	1	2	3	4
Working capital balance (5% of total revenue) (\$)	67,000	88,000	112,000	
Incremental cash flow (\$)	(14,000)	(22,000)	(24,000)	112,000

Initial working capital of \$52,000 invested at Year 0 (5% of Year 1 revenue); the balance is recovered in full in Year 4, alongside the Year 4 sale value of the property (\$7,500,000 after tax) and fixtures (\$100,000), totalling \$7,600,000.

(b)

The NPV calculation in part (a) included the effects of inflation. Under a real-terms approach, the effects of general inflation are excluded, which affects both the cash flows and the cost of capital used for the NPV.

Cash flows: cash flows which already suffer only general inflation would be left uninflated at today's prices (current terms). For Elaan Co, no cash flows fall into this category.

Where cash flows suffer specific rates of inflation rather than the general rate, the cash flows would first need to be inflated at their specific rate (as in part (a)). These nominal cash flows would then be deflated using the general (not specific) rate of inflation to give them in real terms.

The real-terms figures would differ from the current-price figures due to the difference between the specific and general rates of inflation.

Cost of capital: the real cash flows would need to be discounted using the real cost of capital (4.8%), not the nominal cost of capital (10%) used in part (a).

Result: in the absence of tax-allowable depreciation, which suffers no inflation, the NPV obtained under both methods would be identical. In the case of the NPV for Elaan Co, the effects of tax-allowable depreciation would result in a slight difference between the two approaches, but it is likely to be insignificant.

(c)

Where there is risk within a proposed project, tools such as simulation analysis can assist; given the many risks within Elaan Co's forecasting assumptions, simulation analysis would be useful in deciding whether to proceed. The process would involve:

- (1) Specifying the variables.
- (2) Specifying the relationship between the variables.
- (3) Simulating the environment.
- (4) Analysing the results.

Stages 1 and 2: Elaan Co would need to identify all the major variables and, for each variable, the range of possible values and their associated probabilities, then build these into a mathematical model input into a computer. Stage 3: the computer would use random number generation to change multiple variables at a time and recalculate the NPV each time, repeating this routine many times over (possibly millions). Stage 4: the end result would be a mathematical distribution of all envisaged possible NPV outcomes along with a likelihood/probability of occurrence.

Benefits: basic sensitivity analysis can only assess the impact of one variable changing at a time, whereas simulation looks at many (possibly all) variables changing at the same time, so is more 'real-world', since it is unlikely a single variable will change in isolation. Simulation can also help Elaan Co's decision-making by showing the full range of envisaged possible outcomes as a statistical distribution, enabling further mathematical analysis, such as determining the probability of a negative NPV, given the relatively small NPV for the hotel project compared with the investment required, knowing the probability of a negative NPV could be extremely useful to Elaan Co.

Marking scheme	
(a) Total room-nights	1
Room rate and revenue	1
Restaurant revenue	0.5
Variable costs	0.5
Fixed costs	1
Tax on cash flow	1

Marking scheme	
TAD	1
TAD benefit	1
Working capital balance	1
Working capital cash flow	1
Nominal cost of capital	1
NPV	1
Comment	1
Maximum	12
(b) Real-terms approach	4
Maximum	4
(c) Simulation analysis	4
Maximum	4
Total	20 marks

8 (a) Calculation of expected net present value

	1 (\$000)	2 (\$000)	3 (\$000)	4 (\$000)
Revenue (W1)	4,524	7,843	13,048	10,179
Variable cost	(2,385)	(4,200)	(7,080)	(5,730)
Contribution	2,139	3,643	5,968	4,449
Overhead (W2)	(440)	(484)	(532)	(586)
Cash flow before tax	1,699	3,159	5,436	3,863
Tax	(510)	(948)	(1,631)	(1,159)
Depreciation benefits	338	338	338	338
Cash flow after tax	1,527	2,549	4,143	3,042
Scrap value				500
Project cash flow	1,527	2,549	4,143	3,542
Discount at 11%	0.901	0.812	0.731	0.659
Present values	1,376	2,070	3,029	2,334

Note: tax on each year's cash flow before tax, and the related tax-allowable depreciation benefit, are both charged/received in the same year (no one-year lag in this question, since Hraxin Co pays tax in the year the liability arises), and the scrap value arises together with the final year's cash flow.

PV of future cash flows = \$1,376,000 + \$2,070,000 + \$3,029,000 + \$2,334,000 = \$8,809,000.

Initial investment = \$5,000,000.

Expected net present value (ENPV) = \$3,809,000.

The investment project has a positive ENPV of \$3,809,000. This is a mean or average NPV which will result from the project being repeated many times. However, as the project is not being repeated, the NPVs associated with each future economic state must be calculated, as it is one of these NPVs which is expected to occur; management's decision on financial acceptability should be based on these individual NPVs and the risk associated with each one.

WORKINGS

Mean/average selling price = $(25 \times 0.35) + (30 \times 0.5) + (35 \times 0.15) = \$29/\text{unit}$.

(W1) Revenue

Year	1	2	3	4
Selling price (\$/unit)	30.16	31.37	32.62	33.93
Sales volume (units)	150,000	250,000	400,000	300,000
Revenue (\$)	4,524,000	7,843,000	13,048,000	10,179,000

Mean selling price of \$29/unit inflated at 4%/year.

(W2) Overhead

Year	1	2	3	4
Overhead (\$)	440,000	484,000	532,000	586,000

Overhead of \$400,000 inflated at 10%/year.

As a check on the depreciation tax benefit of \$338,000:

Total tax-allowable depreciation = \$5,000,000 - \$500,000 (scrap value) = \$4,500,000, claimed on a straight-line basis over the four-year project life, giving annual tax-allowable depreciation of $\$4,500,000/4 = \$1,125,000$ per year.

The annual cash flow benefit from this = $\$1,125,000 \times 30\%$ (the corporation tax rate) = \$337,500 per year, i.e. \$338,000 to the nearest \$000.

This reconciles exactly with the \$338,000 shown in each of Years 1-4 above, giving a total ENPV of \$3,809,000 (to the nearest \$000).

(b)

Sensitivity analysis assesses the extent to which the net present value (NPV) of an investment project responds to changes in project variables. Two methods are commonly used: one method determines the percentage change in a project variable which results in a negative NPV, while the other determines the percentage change in NPV which results from a fixed percentage change (for example, 5%) in each project variable in turn. The key or critical project variables are identified as those to which the NPV is most sensitive.

Unlike uncertainty, risk can be quantified and measured: the probabilities of the occurrence of particular future outcomes can be assessed, for example, and used to evaluate the volatility of future cash flows. The probabilities of the future economic states in the appraisal of Hraxin Co's investment project are an example of probability analysis, and these probabilities can lead to an assessment of project risk.

While sensitivity analysis can indicate the critical variables of an investment project, it does not give any indication of the probability of a change in any critical variable. Selling price may be a critical variable, for example, but sensitivity analysis is not able to say whether a change in selling price is likely to occur, in the appraisal of Hraxin Co's investment project, the probabilities of different selling prices arising with related economic states have come from probability analysis, not from sensitivity analysis.

Sensitivity analysis will not therefore directly assist Hraxin Co in assessing the risk of the investment project. However, it does provide useful information which helps management gain a deeper understanding of the investment project and focuses management attention on aspects of the investment project where problems may arise.

Marking scheme	
(a) Mean selling price per unit	0.5
Inflated selling price per unit	1
Inflated revenue	1
Inflated overhead	1
Tax liabilities	1
Timing of tax liabilities	1
Tax-allowable depreciation benefits	1
Scrap value	0.5
Present values of future cash flows	1
Comment on financial acceptability	1
Maximum	9
(b) Explanation of sensitivity analysis	1-3
Explanation of risk in investment appraisal	1-2
Discussion of sensitivity analysis and risk	1-3
Maximum	6
Total	15 marks

9 (a) Net present value evaluation of investment

After-tax weighted average cost of capital = $(11\% \times 0.8) + (8.6\% \times (1 - 0.3) \times 0.2) = 8.8\% + 1.204\% = 10\%$ (approximately).

	1 (\$000)	2 (\$000)	3 (\$000)	4 (\$000)	5 (\$000)
Contribution (W1)	440	550	660	660	
Fixed costs	(240)	(260)	(280)	(300)	

	1 (\$000)	2 (\$000)	3 (\$000)	4 (\$000)	5 (\$000)
Taxable cash flow	200	290	380	360	
Taxation		(60)	(87)	(114)	(108)
CA tax benefits (W2)		60	45	34	92
Scrap value				30	
After-tax cash flows	200	290	338	310	(16)
Discount at 10%	0.909	0.826	0.751	0.683	0.621
Present values	182	240	254	212	(10)

Present value of benefits = \$878,000.

Initial investment = \$800,000.

Net present value = \$78,000.

The net present value is positive and so the investment is financially acceptable. However, demand becomes greater than production capacity in the fourth year of operation, so further investment in new machinery may be needed after three years; the new machine will itself need replacing after four years if production capacity is to be maintained at an increased level, a more complete appraisal of the proposed investment could address these expansion and replacement considerations, as well as the assumption of constant selling price and variable cost per kilogram, the absence of any consideration of general inflation, and the linear increase in fixed costs and demand over time.

WORKINGS

(W1) Annual contribution

Year	1	2	3	4
Excess demand over existing capacity (kg)	400,000	500,000	600,000	700,000
Incremental production, capped at 600,000 kg/yr (kg)	400,000	500,000	600,000	600,000
Contribution (\$/kg)	1.10	1.10	1.10	1.10
Contribution (\$)	440,000	550,000	660,000	660,000

Contribution per kg = \$8.00 selling price - \$5.00 material cost - \$1.90 other variable cost = \$1.10/kg.

(W2) Tax-allowable depreciation and tax benefits

Year	TA depreciation (\$)	WDV c/f (\$)	Tax benefit (\$), one year in arrears
1	200,000	600,000	
2	150,000	450,000	60,000 (Year 2, from Year 1's TAD)
3	112,500	337,500	45,000 (Year 3, from Year 2's TAD)
4 (balancing allowance)	307,500	0	34,000 (Year 4, from Year 3's TAD, rounded from \$33,750)
5			92,000 (Year 5, from the Year 4 balancing allowance, rounded from \$92,250)

25% reducing balance on \$800,000, with a balancing allowance in Year 4 after deducting the \$30,000 scrap value (Year 4 = \$337,500 WDV - \$30,000 = \$307,500); tax benefits at 30%.

As a check: taxable cash flow less tax (one year in arrears at 30%) plus the CA tax benefit (also one year in arrears) reconciles exactly to the after-tax cash flow shown above in every year, e.g. Year 3 = \$380,000 taxable cash flow - \$87,000 tax (30% of Year 2's \$290,000) + \$45,000 CA benefit (30% of Year

2's \$150,000 TAD) = \$338,000, matching the table. Discounting at 10% and summing gives \$878,000, less the \$800,000 initial cost, giving the NPV of \$78,000.

(b)

Risk refers to the situation where an investment project has several possible outcomes, all of which are known and to which probabilities can be attached, for example on the basis of past experience; risk can therefore be quantified and measured by the variability of returns of an investment project. Uncertainty refers to the situation where an investment project has several possible outcomes but it is not possible to assign probabilities to their occurrence, so it is not possible to say which outcomes are likely to occur. The difference between risk and uncertainty, therefore, is that risk can be quantified whereas uncertainty cannot; risk increases with the variability of returns, while uncertainty increases with project life.

Sensitivity analysis assesses how the net present value of an investment project is affected by changes in project variables: considering each variable in turn, either the change in the variable required to make the NPV zero is determined, or the change in NPV arising from a fixed change in a given project variable. In this way, the key or critical project variables are determined. However, sensitivity analysis does not assess the probability of changes in project variables occurring, and so is often dismissed as a way of incorporating risk into investment appraisal.

Probability analysis refers to the assessment of the separate probabilities of a number of specified outcomes of an investment project. For example, a range of expected market conditions could be formulated and the probability of each arising in each of several future years assessed; the net present values arising from combinations of future conditions could then be assessed and linked to the joint probabilities of those combinations. The expected net present value (ENPV) could be calculated, together with the probability of the worst-case scenario and the probability of a negative net present value, so that the downside risk of the investment could be determined and incorporated into the investment decision.

Marking scheme	
(a) After-tax weighted average cost of capital	2
Annual contribution	2
Fixed costs	1
Taxation	1
TAD tax benefits	3
Scrap value	1
Discount factors	1
Net present value	1
Comment	1-2
Maximum	13
(b) Risk and uncertainty	2-3
Discussion of sensitivity analysis	2-3
Discussion of probability analysis	2-3
Maximum	7
Total	20 marks

10 (a) Calculation of net present value (NPV) of Project 1

As nominal after-tax cash flows are to be discounted, the nominal after-tax weighted average cost of capital of 7% must be used.

	1 (\$000)	2 (\$000)	3 (\$000)	4 (\$000)	5 (\$000)
Sales revenue (W1)	1,300	2,466	3,622	2,018	
Variable costs (W2)	(513)	(1,098)	(1,809)	(1,035)	
Contribution	787	1,368	1,813	983	
Fixed costs	(105)	(115)	(125)	(125)	
Taxable cash flow	682	1,253	1,688	858	
Tax liabilities		(205)	(376)	(506)	(257)
CA tax benefits (W3)		113	84	63	160
After-tax cash flow	682	1,161	1,396	415	(97)
Scrap value				100	
Net cash flow	682	1,161	1,396	515	(97)
Discount at 7%	0.935	0.873	0.816	0.763	0.713
Present values	638	1,014	1,139	393	(69)

Present value of cash inflows = \$3,115,000.

Cost of machine = \$(1,500,000).

NPV = \$1,615,000.

Project 1 has a positive NPV of \$1,615,000 and so it is financially acceptable to Ridag Co. However, the discount rate used here is the current weighted average after-tax cost of capital. As this is a recently-developed product, it may be appropriate to use a project-specific discount rate that reflects the risk of the new product launch.

WORKINGS**(W1) Sales revenue**

Year	1	2	3	4
Selling price (\$/unit)	26.00	25.96	25.87	26.91
Sales volume (units)	50,000	95,000	140,000	75,000
Sales revenue (\$)	1,300,000	2,466,200	3,621,800	2,018,250

Selling prices of \$25.00/\$24.00/\$23.00/\$23.00 (current terms) inflated at 4%/year.

(W2) Variable costs

Year	1	2	3	4
Variable cost (\$/unit)	10.25	11.56	12.92	13.80
Sales volume (units)	50,000	95,000	140,000	75,000
Variable costs (\$)	512,500	1,098,200	1,808,800	1,035,000

Variable costs of \$10.00/\$11.00/\$12.00/\$12.50 (current terms) inflated at 2.5%/year.

(W3) Capital allowances (tax-allowable depreciation) and tax benefits

Year	CA (\$)	Tax benefit (\$)
1	375,000	112,500
2	281,250	84,375
3	210,938	63,281
4	532,812	159,844

25% reducing balance on \$1,500,000, with a balancing allowance in Year 4 after deducting the \$100,000 scrap value (Year 4 = $843,750 - 210,938 - 100,000 = \$532,812$); tax benefits received one year in arrears.

(b) Calculation of equivalent annual cost (EAC)

Since taxation and tax-allowable depreciation are to be ignored, and Project 2's cash flows have already been adjusted for inflation, the correct discount rate to use is the nominal before-tax weighted average cost of capital of 12%.

Machine 1	0 (\$)	1 (\$)	2 (\$)	3 (\$)	4 (\$)
Maintenance costs		(25,000)	(29,000)	(32,000)	(35,000)
Investment and scrap	(200,000)				25,000
Net cash flow	(200,000)	(25,000)	(29,000)	(32,000)	(10,000)
Discount at 12%	1.000	0.893	0.797	0.712	0.636
Present values	(200,000)	(22,325)	(23,113)	(22,784)	(6,360)

Present value of cash flows = \$(274,582).

Cumulative present value factor (4 years, 12%) = 3.037.

Equivalent annual cost = $274,582 / 3.037 = \$90,412$.

Machine 2	0 (\$)	1 (\$)	2 (\$)	3 (\$)
Maintenance costs		(15,000)	(20,000)	(25,000)
Investment and scrap	(225,000)			50,000
Net cash flow	(225,000)	(15,000)	(20,000)	25,000
Discount at 12%	1.000	0.893	0.797	0.712
Present values	(225,000)	(13,395)	(15,940)	17,800

Present value of cash flows = \$(236,535).

Cumulative present value factor (3 years, 12%) = 2.402.

Equivalent annual cost = $236,535 / 2.402 = \$98,474$.

The machine with the lowest equivalent annual cost should be purchased, and calculation shows this to be Machine 1. If the present value of future cash flows had been considered alone, Machine 2 (cost of \$236,535) would have been preferred to Machine 1 (cost of \$274,582); however, the lives of the two machines are different, and the equivalent annual cost method allows this to be taken into consideration.

(c)

Within the context of investment appraisal, risk relates to the variability of returns and so it can be quantified, for example by forecasting the probabilities related to future cash flows. From this point of view, risk can be differentiated from uncertainty, which cannot be quantified. Uncertainty can be said to increase with project life, while risk increases with the variability of returns.

It is commonly said that risk can be included in the investment appraisal process by using sensitivity analysis, which determines the effect on project net present value of a change in individual project variables. The analysis highlights the project variable to which the project net present value is most sensitive in relative terms. However, since sensitivity analysis changes only one variable at a time, it ignores interrelationships between project variables.

While sensitivity analysis can indicate the key or critical variable, it does not indicate the likelihood of a change in the future value of this variable, i.e. sensitivity analysis does not indicate the probability of a change in the future value of the key or critical variable. For this reason, given the earlier comments on risk and uncertainty, it can be said that sensitivity analysis is not itself a method of including risk in the investment appraisal process.

Probability analysis, as its name implies, attaches probabilities to the expected future cash flows of an investment project and uses these to calculate the expected net present value (ENPV). The ENPV is the average NPV that would be expected to occur if an investment project could be repeated a large number of times, and can also be seen as the mean or expected value of an NPV probability distribution. Given the earlier discussion of risk and uncertainty, probability analysis is therefore a way of including a consideration of risk in the investment appraisal process, and is certainly a more effective way of considering the risk of investment projects than sensitivity analysis.

A weakness of probability analysis, however, lies in the difficulty of estimating the probabilities that are to be attached to expected future cash flows. While these probabilities can be based on expert judgement and previous experience of similar investment projects, there remains an element of subjectivity which cannot be escaped.

Marking scheme	
(a) Sales revenue	1
Inflated selling price	1
Variable costs	1
Inflated variable cost	1
Fixed costs	1
Tax liabilities	1
Timing of tax	1
CA tax benefits	2
Scrap value	1
PVs and NPV	1
Comment	1
Maximum	12
(b) Discount rate used	1
Machine 1 net cash flows and PV	1
Machine 1 EAC	1
Machine 2 net cash flows and PV	1
Machine 2 EAC	1
Recommendation	1

Marking scheme	
Maximum	6
(c) Risk vs uncertainty	1
Discussion of sensitivity analysis	1-2
Discussion of probability analysis	1-2
Relative effectiveness	1-2
Maximum	7
Total	25 marks

11 (a) (i) Calculation of net present value (NPV)

	0 (\$m)	1 (\$m)	2 (\$m)	3 (\$m)	4 (\$m)
Revenue (W1)		500.00	954.00	898.88	476.40
Variable costs (W2)		(350.00)	(667.80)	(629.20)	(333.48)
Fixed costs (W3)		(120.00)	(132.00)	(145.20)	(159.72)
Investment/residual value	(220.00)				30.00
Net cash flow	(220.00)	30.00	154.20	124.48	13.20
Discount at 14%	1.000	0.877	0.769	0.675	0.592
Present values	(220.00)	26.31	118.58	84.02	7.81

Net present value = \$(220.00m) + \$26.31m + \$118.58m + \$84.02m + \$7.81m = \$16.72m.

WORKINGS**(W1) Revenue**

Year	1	2	3	4
Selling price (\$/unit)	20,000	21,200	22,472	23,820
Sales volume (units)	25,000	45,000	40,000	20,000
Revenue (\$m)	500.00	954.00	898.88	476.40

Selling price of \$20,000/unit inflated at 6%/year.

(W2) Variable cost

Year	1	2	3	4
Variable cost (\$/unit)	14,000	14,840	15,730	16,674
Sales volume (units)	25,000	45,000	40,000	20,000
Variable cost (\$m)	350.00	667.80	629.20	333.48

Variable cost of \$14,000/unit inflated at 6%/year.

(W3) Fixed costs

Year	1	2	3	4
Fixed costs (\$m)	120.00	132.00	145.20	159.72

Fixed costs of \$120m inflated at 10%/year.

(ii) Calculation of internal rate of return (IRR)

	0 (\$m)	1 (\$m)	2 (\$m)	3 (\$m)	4 (\$m)
Net cash flow	(220.00)	30.00	154.20	124.48	13.20
Discount at 20%	1.000	0.833	0.694	0.579	0.482
Present values	(220.00)	24.99	107.01	72.07	6.36

Net present value at 20% = \$(9.57m).

$IRR = L + ((H - L) \times (NL / (NL - NH))) = 14 + ((20 - 14) \times (16.72 / (16.72 + 9.57))) = 17.82\%$.

(iii) Calculation of payback period

	0 (\$m)	1 (\$m)	2 (\$m)	3 (\$m)
Net cash flow	(220.00)	30.00	154.20	124.48
Cumulative cash flow	(220.00)	(190.00)	(35.80)	88.68

Payback period = $2 + (12 \times (35.80 / (35.80 + 88.68))) = 2 \text{ years } 3 \text{ months}$.

(iv) Calculation of return on capital employed (ROCE)

Total cash inflow = \$30.00m + \$154.20m + \$124.48m + \$(13.20 - 30.00)m = \$291.88m.

Total depreciation = \$220m - \$30m = \$190m.

Total accounting profit = \$291.88m - \$190m = \$101.88m.

Average accounting profit = \$101.88m/4 = \$25.47m.

Average investment = (\$220m + \$30m)/2 = \$125m.

ROCE = (\$25.47m/\$125m) × 100% = 20.4%.

(b) Discussion

Net present value (NPV): NPV takes account of the time value of money and the whole investment life. It is an absolute measure, enabling better business planning.

The cost of capital it uses takes into account the risk of the investment and its financing; however, the results of the NPV calculation will depend on whether the cost of capital has been accurately estimated.

If the options being considered are mutually exclusive, NPV can be used to rank them, as it indicates the effect on shareholder wealth; however, if shareholders have a short time horizon, they may not be prepared to wait for an investment to generate returns over its later life, even if it does have the best NPV.

Internal rate of return (IRR): IRR can be calculated without knowing the company's cost of capital, but a benchmark is needed to determine whether the IRR is acceptable. If IRR is used for comparison, it does not indicate which investment will generate more wealth for shareholders; it may give a different result to NPV because, rather than being an indicator of wealth creation, it shows how sensitive the investment is to changes in the cost of capital. In some situations there is more than one IRR.

Payback: payback has the advantage of using cash flows. It indicates when the investment will break even and whether this is before projected cash flows become very uncertain.

However, payback ignores the cash flows after the payback period, which for Melplash Co's Years 3-4 are a significant proportion of total cash flows; it also ignores the total return on the investment, and the simple payback figure ignores the time value of money (although a discounted payback figure can be calculated).

The payback target is subjective. Using payback to compare investments may result in choosing shorter-term investments giving immediate returns rather than investments offering greater returns in the longer term; however, investments with a shorter period for recouping the amount invested may be preferred if Melplash Co has liquidity problems.

Return on capital employed (ROCE): ROCE may reflect the wishes of shareholders that the company generates sufficient profits, and does consider the entire life of the investment. However, the choice of an acceptable rate is subjective, here the justification for choosing the 20% target is unclear. ROCE does not take into account the time value of money, and it is possible that an investment which increases company value will be rejected if its ROCE is below the target. The acceptability of the ROCE may also depend on the assumptions about realisable value at the end of the project. As ROCE is a percentage measure it can be used to compare this investment with other options, but it does not indicate the absolute change in shareholder wealth generated by different investments.

Advice: the investment is acceptable using three of the four methods, but fails the payback test which the directors have set. However, the directors should choose to go ahead with the investment, on the basis that it has a positive NPV and will therefore increase long-term shareholder wealth. If the directors are concerned about the uncertainties of the income in later years, they should carry out sensitivity analysis.

Marking scheme	
(a) (i) Income	1
Variable costs	1
Fixed costs	1
Initial investment	0.5
Residual value	0.5
PVs and NPV	1
(ii) IRR setup	1
IRR calculation	1
(iii) Payback	1

Marking scheme	
(iv) Average accounting profit	1
Average investment	1
ROCE	1
Maximum	11
(b) NPV	2
IRR	2
Payback	2
ROCE	2
Advice	1
Maximum	9
Total	20 marks

12 (a) Errors in the original investment appraisal

Inflation was incorrectly applied to selling prices and variable costs in calculating contribution, since only one year's inflation was allowed for in each year of operation, rather than inflation being applied cumulatively year on year.

The fixed costs were correctly inflated, but included \$200,000 per year (before inflation) of development costs that have already been incurred and are not a relevant cash flow; only relevant costs should be included in investment appraisal.

Straight-line accounting depreciation had been used in the calculation, but this depreciation method is not acceptable to the tax authorities; the approved 25% reducing balance tax-allowable depreciation should be used instead.

Interest payments have been included in the investment appraisal, but these are already allowed for by the discount rate used in calculating the net present value, so including them as well double-counts the cost of debt finance.

The interest rate on the debt finance (10%) has been used as the discount rate, when the nominal weighted average cost of capital should have been used instead to discount the nominal after-tax cash flows.

(b) Revised calculation of net present value

Nominal weighted average cost of capital = $1.07 \times 1.047 = 1.12$, i.e. 12% per year.

	1 (\$000)	2 (\$000)	3 (\$000)	4 (\$000)	5 (\$000)
Contribution (W1)	1,330	2,264	3,010	1,600	
Fixed costs (W2)	(318)	(337)	(357)	(379)	
Taxable cash flow	1,012	1,927	2,653	1,221	
Taxation		(304)	(578)	(796)	(366)
CA tax benefits (W3)		150	112	84	178
After-tax cash flow	1,012	1,773	2,187	509	(188)
Scrap value				250	
Net cash flow	1,012	1,773	2,187	759	(188)
Discount at 12%	0.893	0.797	0.712	0.635	0.567
Present values	904	1,413	1,557	482	(107)

Present value of future cash flows = \$4,249,000.

Initial investment = \$2,000,000.

Net present value = \$2,249,000.

The net present value is positive and so the investment is financially acceptable.

WORKINGS

(W1) Annual contribution

Year	1	2	3	4
Sales volume (units/yr)	250,000	400,000	500,000	250,000
Selling price (\$/unit)	12.60	13.23	13.89	14.59
Variable cost (\$/unit)	7.28	7.57	7.87	8.19
Contribution (\$/unit)	5.32	5.66	6.02	6.40
Contribution (\$/yr)	1,330,000	2,264,000	3,010,000	1,600,000

Selling price \$12.00/unit inflated at 5%/year; variable cost \$7.00/unit inflated at 4%/year.

(W2) Fixed costs

Year	1	2	3	4
Fixed costs (\$)	318,000	337,000	357,000	379,000

Relevant fixed costs of \$300,000 (\$500,000 less the \$200,000 non-relevant development cost) inflated at 6%/year.

(W3) Tax-allowable depreciation and tax benefits

Year	TA depreciation (\$)	Tax benefit (\$)
1	500,000	150,000
2	375,000	112,500
3	281,250	84,375
4	593,750	178,125
Scrap value	250,000	
Total	2,000,000	

25% reducing balance on \$2,000,000, with a balancing allowance in Year 4 after deducting the \$250,000 scrap value (Year 4 = $843,750 - 250,000 = \$593,750$); tax benefits at 30%, paid one year in arrears.

As a check: an alternative calculation via taxable profit (taxable cash flow less tax-allowable depreciation, less tax, plus tax-allowable depreciation added back) reconciles exactly to the same after-tax cash flow in every year, giving the same net present value of \$2,249,000.

(c) (i) Multiple internal rates of return

An investment project may have multiple internal rates of return if it has unconventional cash flows, that is, cash flows that change sign more than once over the life of the project. A mining operation, for example, may have initial investment (a cash outflow) followed by many years of successful operation (cash inflows) before decommissioning and environmental repair (a further cash outflow). This technical difficulty makes it difficult to use the internal rate of return (IRR) method to offer investment advice. One solution is to use the net present value (NPV) method instead of IRR, since unconventional cash flows are easily accommodated by NPV; this is one area where NPV is considered superior to IRR.

(ii) Projects with significantly different business risk to current operations

Where a proposed investment project has business risk that is significantly different from current operations, it is no longer appropriate to use the weighted average cost of capital (WACC) as the discount rate in calculating the project's net present value. WACC can only be used as a discount rate where business risk and financial risk are not significantly affected by undertaking the investment project. Where business risk changes significantly, the capital asset pricing model (CAPM) should instead be used to calculate a project-specific discount rate that takes account of the systematic risk of the proposed investment project, typically by selecting a proxy company in a similar line of business, ungearing its equity beta to an asset beta, and regearing it to reflect the investing company's own financial risk.

Marking scheme	
(a) Identification of errors in the evaluation	5
Maximum	5
(b) Nominal weighted average cost of capital	1
Inflated selling prices	1
Inflated variable costs	1
Inflated contribution	1
Inflated fixed costs	1
TA depreciation and/or related tax benefits	3
Scrap value	1
Discount factors	1
Net present value	1
Comment	1-2
Maximum	10
(c) (i) Discussion of projects with several IRR	2
(ii) Discussion of projects with different business risk	3
Maximum	5
Total	20 marks

13 (a) Evaluation of buy versus lease

Buy option

Year	0 (\$)	1 (\$)	2 (\$)	3 (\$)	4 (\$)
Initial cost	(300,000)				
Maintenance costs		(12,000)	(14,400)	(18,720)	(26,208)
Running costs		(20,000)	(21,000)	(22,050)	(23,153)
Residual value					122,880
Net cash flow	(300,000)	(32,000)	(35,400)	(40,770)	73,519
Discount factors at 9%	1.000	0.917	0.842	0.772	0.708
Present values	(300,000)	(29,344)	(29,807)	(31,474)	52,051

Present value of cost of buying = \$(300,000) + \$(29,344) + \$(29,807) + \$(31,474) + \$52,051 = \$(338,574) (small rounding differences of a few thousand dollars may arise depending on the number of decimal places used for discount factors and intermediate figures).

Note that depreciation is not itself a cash flow and is not included as a cost; it is used only to establish the vehicles' written-down (residual) value at the end of Year 4, which is the relevant cash inflow on disposal.

WORKINGS

Maintenance and running costs

Year	1	2	3	4
Maintenance costs (\$)	12,000	14,400	18,720	26,208
Running costs (\$)	20,000	21,000	22,050	23,153

Maintenance costs rise by 20%/30%/40% in Years 2/3/4 respectively; running costs increase by 5%/year.

The written-down value of the vehicles after four years of 20% reducing balance depreciation is \$300,000 x 0.8 x 0.8 x 0.8 x 0.8 = \$122,880, which is received as the residual value in Year 4.

Lease option

Year	0 (\$)	1 (\$)	2 (\$)	3 (\$)	4 (\$)
Lease payments	(80,000)	(80,000)	(80,000)	(80,000)	
Running costs		(15,000)	(15,750)	(16,538)	(17,365)
Net cash flow	(80,000)	(95,000)	(95,750)	(96,538)	(17,365)
Discount factors at 9%	1.000	0.917	0.842	0.772	0.708
Present values	(80,000)	(87,115)	(80,622)	(74,527)	(12,294)

Present value of lease payments = \$80,000 x (1 + 0.917 + 0.842 + 0.772) = \$80,000 x 3.531 = \$(282,480).

Running costs (lease option)

Year	1	2	3	4
Running costs (75% of buy option) (\$)	15,000	15,750	16,538	17,365
Discount factor @ 9%	0.917	0.842	0.772	0.708
Present value (\$)	13,755	13,262	12,767	12,294

Present value of running costs = \$(13,755) + \$(13,262) + \$(12,767) + \$(12,294) = \$(52,078). Total present value of cost of leasing = \$(282,480) + \$(52,078) = \$(334,558).

Conclusion: leasing has the lower present value of cost (\$334,558 compared with \$338,574 for buying), a saving of approximately \$4,000, and so on financial grounds AgeAid should lease the vehicles rather than buy them.

(b) Non-financial reasons to lease rather than buy

Leasing gives AgeAid fixed, regular, budgetable annual payments of \$80,000, rather than requiring a single large upfront outflow of \$300,000. As a charity, AgeAid is likely to be reliant on donations and grants, which can be volatile and unpredictable, and it may have limited access to loan finance because it lacks security to offer a lender. Avoiding a large one-off cash outflow protects AgeAid's liquidity and reduces the risk that other charitable activities would have to be postponed or scaled back to fund the purchase.

Under the lease option, the lessor is responsible for maintaining the vehicles and for eventually replacing them. This removes the administrative burden of arranging maintenance from AgeAid, and transfers the risk of unexpected repair costs and vehicle obsolescence to the lessor, while giving AgeAid greater certainty that roadworthy vehicles will be available throughout the lease term to support its day-centre and meal-delivery services.

(c) The 3Es: economy, efficiency and effectiveness

Economy means obtaining the appropriate quantity and quality of resources at the lowest cost. For AgeAid, this could involve sourcing vehicles, fuel and meal ingredients at appropriately low prices, and making use of volunteer labour where possible to minimise staff costs, while still ensuring resources are of a suitable quality for the charity's activities.

Efficiency means maximising the output achieved from the resources used, that is, the relationship between inputs and outputs. For AgeAid, this could be measured by, for example, the number of hot meals delivered or elderly people transported per vehicle per day, or the cost per day-centre activity provided.

Effectiveness means the extent to which AgeAid achieves its stated objectives, that is, a measure of outputs or outcomes against goals, rather than of resources used. For AgeAid, this could be measured by, for example, the number of day-centre activities run or the number of elderly beneficiaries actually served, assessed against the charity's own objectives for supporting elderly people in the community.

Together, the 3Es allow AgeAid's trustees and management to assess value for money, since a charity does not have a profit motive against which to judge performance. Answers that give only bare textbook definitions of economy, efficiency and effectiveness, without applying them to AgeAid's own operations, cannot be awarded full marks.

Marking scheme	
(a) Buy option cash flows and present value	5
Lease option cash flows and present value	4
Conclusion	1
Maximum	10
(b) One mark for identifying each reason, up to 1 further mark for developing each reason	4
Maximum	4
(c) Explanation and application of economy	2
efficiency	2
effectiveness	2
Maximum	6
Total	20 marks

Chapter 5: Asset Investment Decisions and Capital Rationing

Questions

Objective Test Questions

Lease versus buy

1 A company is comparing the cost of leasing an asset against the cost of borrowing to buy it.

Which discount rate should normally be used to evaluate both options?

- A The company's weighted average cost of capital
- B The company's cost of equity
- C The after-tax cost of borrowing
- D The risk-free rate

2 A company is deciding whether to buy or lease an asset with a 10-year economic life and a zero residual value. It can be purchased for \$80,000 payable immediately. Alternatively, it can be leased with 10 lease rentals of \$12,000 per year, payable annually in advance. The appropriate discount rate is 10% per year. Ignore taxation.

Should the company buy or lease the asset?

- A Buy, because its present value cost (\$80,000) is lower than leasing (\$81,108)
- B Lease, because its present value cost (\$81,108) is lower than buying (\$80,000)
- C Buy, because the total cash paid (\$80,000) is lower than the total lease rentals (\$120,000)
- D It makes no difference, because the asset is used either way

3 A company is deciding whether to buy or lease an asset with a five-year economic life. It can be purchased for \$81,000 payable immediately and will have a residual value of \$40,000 after five years. Alternatively, it can be leased with five lease rentals of \$14,000 per year, payable annually in arrears. The appropriate discount rate is 10% per year. Ignore taxation.

What is the present value cost of buying the asset (to the nearest \$)?

\$

4 Which of the following statements about the lease-versus-buy decision are true or false?

Statement	True	False
The lease-versus-buy decision should be evaluated separately from the decision to invest in the asset itself		
Leasing is always the cheaper financing option once tax is taken into account		
If the asset is purchased, tax-allowable depreciation is a relevant cash flow in the appraisal		

Statement	True	False
The evaluation compares the present value COST of each financing option, not its NPV benefit		

5 The cost of purchasing a machine is \$100,000 payable immediately. Its disposal value is expected to be \$10,000 in five years' time. The same asset can be leased for a period of five years with rentals of \$25,000 payable annually in advance. The appropriate discount rate is 10% per year.

What is the net present value to the LESSOR if it purchases the machine and then leases it to the user on the above terms?

- A \$10,460 positive
- B \$990 positive
- C \$1,960 negative
- D \$11,440 negative

6 Which of the following is potentially a benefit to a lessee, as opposed to a company that purchases an asset outright?

- A Avoiding tax exhaustion
- B Attracting lease customers that may not have been otherwise possible
- C Exploiting a low cost of capital
- D Potential future scrap proceeds

Replacement decisions using equivalent annual cost and equivalent annual benefit

7 A professional kitchen is attempting to choose between gas and electricity for its main heat source. Once a choice is made, the kitchen intends to keep to that source indefinitely. Each gas oven has an NPV of \$50,000 over its useful life of 5 years. Each electric oven has an NPV of \$68,000 over its useful life of 7 years. The cost of capital is 8%.

Which should the kitchen choose, and why?

- A Gas, because its average NPV per year is higher than electric
- B Electric, because its NPV is higher than gas
- C Electric, because its equivalent annual benefit is higher
- D Electric, because it lasts longer than gas

8 Which TWO of the following are typically benefits of a shorter replacement cycle for an asset?

- A Higher scrap value
- B Better company image and efficiency
- C Lower annual depreciation
- D Less time to benefit from owning the asset
- E Lower total maintenance costs over the asset's whole working life

9 Which of the following statements about the use of the equivalent annual cost (EAC) method are true or false?

Statement	True	False
The use of equivalent annual cost is most appropriate in periods of high inflation		
The EAC method assumes that the machine can be replaced by exactly the same machine in perpetuity		

Equivalent annual costs (EACs)

10 Guilder Co decides to replace an existing machine with a Super machine, which will last four years. The following present value table has been prepared.

	0	1	2	3	4
Maintenance costs		(20,000)	(29,000)	(32,000)	(35,000)
Investment and scrap	(250,000)				25,000
Net cash flow	(250,000)	(20,000)	(29,000)	(32,000)	(10,000)
Discount at 12%	1.000	0.893	0.797	0.712	0.636
Present values	(250,000)	(17,860)	(23,113)	(22,784)	(6,360)

What is the equivalent annual cost (EAC) of the Super machine (to the nearest whole number)?

\$

11 A machine costing \$450,000 has a useful life of six years, after which time its estimated present value of resale will be \$25,800. The present value of the total running costs over the next six years is \$82,280. The company's cost of capital is 10%.

What is the equivalent annual cost of using the machine if it were bought and replaced every six years in perpetuity (to the nearest dollar)?

\$

12 Togue Co is investigating how often it should replace its company car fleet. It is considering replacement every two, three, four or five years. Cash flow estimates have been compiled, giving the following negative NPVs, based on Togue Co's cost of capital of 11%.

Years to replacement	Negative NPV
2	\$149,308
3	\$209,142
4	\$254,893
5	\$310,860

If Togue Co adopts the optimal replacement cycle, what will be the equivalent annual cost (to the nearest dollar, rounded down)?

\$

13 A company buys a machine for \$10,000 and sells it for \$2,000 at time 3. Running costs of the machine are as follows:

Time	1	2	3
Running cost	\$3,000	\$5,000	\$7,000

If a series of machines are bought, run and sold on an infinite cycle of replacements, what is the equivalent annual cost of the machine if the discount rate is 10%?

- A \$22,114
- B \$8,892
- C \$8,288
- D \$7,371

14 A company is deciding how often to replace an item of equipment costing \$20,000. If the equipment is replaced every year, it can be traded in for \$10,000 at the end of Year 1. If instead it is kept for two years, running costs of \$5,000 will be incurred at the end of Year 2, but the equipment can then be traded in for \$8,000. The cost of capital is 10%.

Using the annuity factors $AF1 = 0.909$ and $AF2 = 1.736$, what is the equivalent annual cost of the optimal replacement cycle (to the nearest dollar)?

- A \$10,093 (2-year cycle)
- B \$12,002 (1-year cycle)
- C \$17,522 (2-year cycle)
- D \$10,910 (1-year cycle)

15 Care Co needs to replace a major piece of office equipment that is in constant use, for which there is expected to continue to be a need for the foreseeable future. Two types of machine are available, with different capital costs, useful lives, scrap values and annual running costs.

Machine 1 will initially cost \$480,000, have a life of four years, a scrap value of \$60,000, and annual running costs of \$72,000. Machine 2 will initially cost \$540,000, have a life of three years, a scrap value of \$120,000, and annual running costs of \$47,000.

Care Co's cost of capital is 10%. Assume that all cash flows, except the initial capital cost, occur at the end of the relevant year, and that taxation and inflation can be ignored.

What is the equivalent annual cost of Machine 2 (to the nearest \$000)?

\$

Are the following statements about Care Co's use of the equivalent annual cost method true or false?

Statement	True	False
The equivalent annual cost calculation assumes that the same type of machine will be used into the foreseeable future		
The equivalent annual cost calculation assumes that the capabilities of Machine 1 and Machine 2 are identical		

16 A machine costs \$150,000 and has a useful life of eight years, after which time its estimated resale value will be \$25,000. Annual running costs will be \$5,000 for each of the first three years of use, and then \$8,000 for each of the next five years. All running costs are payable on the last day of the year to which they relate. The appropriate discount rate is 20% per year.

Working to the nearest \$100, what is the equivalent annual cost of the machine?

- A \$46,600
- B \$43,900
- C \$43,300
- D \$21,100

Capital rationing

17 Which of the following statements is correct?

- A Tax-allowable depreciation is a relevant cash flow when evaluating borrowing to buy compared with leasing as a financing choice
- B Asset replacement decisions require relevant cash flows to be discounted by the after-tax cost of debt
- C If capital is rationed, divisible investment projects can be ranked by the profitability index when determining the optimum investment schedule
- D Government restrictions on bank lending are associated with soft capital rationing

(ACCA F9 December 2014)

18 Are the following statements about capital rationing true or false?

Statement	True	False
The profitability index is suitable for handling multi-period capital rationing problems if projects are divisible		
Assuming that projects are divisible is generally an unrealistic assumption		

19 Which TWO of the following are limitations of using the profitability index to solve a capital rationing problem?

- A It uses accounting profits rather than cash flows
- B It ignores the time value of money
- C It cannot deal with non-divisible (indivisible) projects
- D It cannot deal with multi-period capital rationing
- E It cannot be used where projects are mutually exclusive

20 NB Co is faced with an immediate capital constraint of \$100m available to invest. It is considering investing in four divisible projects.

Project	Initial cost \$m	NPV \$m
1	40	4
2	30	5
3	50	6
4	60	5

What is the NPV generated from the optimum investment programme?

\$

21 NB Co is faced with an immediate capital constraint of \$100m available to invest, in the same four projects as the previous question, but the projects are now indivisible.

Project	Initial cost \$m	NPV \$m
1	40	4
2	30	5
3	50	6
4	60	5

What is the NPV generated from the optimum investment programme?

- A \$13m
- B \$11m
- C \$9m
- D \$15m

22 Progress Co has the following non-divisible projects available.

Project	Initial investment \$	PV of cash inflows \$
1	200,000	300,000
2	450,000	725,000
3	350,000	425,000
4	900,000	1,300,000
5	600,000	1,000,000

Assuming the company has \$1m to invest in capital projects, which combination of projects should Progress Co accept?

- A 1 and 5
- B 2 and 4
- C 3 and 5
- D 1, 2 and 3

23 Gaimett Co has \$1.5 million of surplus cash which it wishes to invest in long-term projects. All of the following projects are divisible and must be started immediately.

Project	NPV \$000	Investment value \$000
A	125	375
B	240	800
C	80	400
D	40	50
E	150	600

If the surplus cash available is \$1.5 million, what is the optimum combination of projects that Gaimett Co should undertake?

- A B, E and 26.7% of A
- B C, E and 62.5% of B
- C A, D, E, C and 9.4% of B
- D A, B, D and 45.8% of E

24 Before Gaimett Co's decision on the projects above is made, Project A becomes unavailable, and it is realised that the remaining projects (B, C, D and E, with the same figures as the previous question) are actually indivisible.

Under the new circumstances, what is the optimum combination of projects that Gaimett Co should undertake?

- A B, C and D
- B C, D and E
- C B, D and E
- D B, C, D and E

25 A company has four independent, divisible projects available, as follows.

Project	Capital needed at time 0 \$	NPV \$
1	10,000	30,000
2	8,000	25,000
3	12,000	30,000
4	16,000	36,000

If the company has \$32,000 to invest at time 0, and none of the projects can be delayed, what is the maximum NPV that can be earned?

- A \$85,000
- B \$89,500
- C \$102,250
- D \$103,000

Case Scenario Questions

1 A company is trying to determine the optimal replacement cycle for the laptops used by its sales team. The following information is relevant to the decision. The cost of each laptop is \$2,400. Maintenance costs are payable at the end of each full year of ownership, but not in the year of replacement (e.g. if a laptop is owned for three years, then the maintenance cost is payable at the end of Year 1 and at the end of Year 2).

Interval between replacement (years)	Trade-in value \$	Maintenance cost \$
1	1,200	Nil
2	800	75 (payable at end of Year 1)
3	To be determined	150 (payable at end of Year 2)

The company uses a relatively high discount rate of 14% as it faces capital rationing.

(1) What is the annual equivalent cost of replacing a laptop every year?

- A \$1,348
- B \$1,413
- C \$1,537
- D \$1,611

(2) What is the Year 3 trade-in value, given that the annual equivalent cost of a 3-year replacement cycle is \$1,024?

- A \$268
- B \$301
- C \$412
- D \$727

(3) Which of the following are potential tax implications of the asset replacement decision?

- (i) Tax savings on tax-allowable depreciation
- (ii) Tax payment on a balancing charge
- (iii) Tax savings on maintenance costs

- A (i) only
- B (i) and (ii) only
- C (ii) and (iii) only
- D (i), (ii) and (iii)

(4) Which of the following factors may encourage the company to replace the laptops relatively often?

- (i) Changing technology
- (ii) Supplier discounts
- (iii) Environmental impacts

- A (i) only
- B (i) and (ii) only
- C (ii) and (iii) only
- D (i), (ii) and (iii)

(5) Which of the following would NOT explain why the company faces capital rationing?

- A Asymmetry of information
- B Poor credit rating
- C Strict government monetary policy
- D Large proportion of tangible assets

2 Easter Co is about to hold its annual strategic planning meeting, and a number of capital investment projects will be discussed. A summary of the projects' cash flows is shown below.

Project	NPV \$000	Investment \$000
A	4,900	3,200
B	7,400	9,300
C	5,900	7,300
D	7,500	5,200
E	9,000	5,600

The investments will need to be made at the start of the coming year; no projects can be delayed and none are divisible. The funds available for these projects are limited to \$10,000,000.

(1) Are the following statements about capital rationing true or false?

Statement	True	False
Both hard and soft capital rationing are the result of external factors		
Both hard and soft capital rationing are the result of internal policies		
Hard rationing is the result of external factors; soft rationing is the result of internal policies		
Soft rationing is the result of external factors; hard rationing is the result of internal policies		

(2) Which TWO of the following projects should be accepted, based on the circumstances described?

- A Project A
- B Project B
- C Project C
- D Project D
- E Project E

(3) What projects would be accepted if it were found that projects A and E were mutually exclusive?

- A A and D
- B A and E
- C A, D and E
- D D and E

(4) What projects, and fractions of projects, would be accepted if a partner could be found to invest in a proportion of one of the projects, making them effectively infinitely divisible (ignoring the mutually exclusive limitation)?

- A A, D and 2/7 E
- B A, E and 3/13 D
- C A, D and E
- D D and E

3 Guilder Co is appraising four different projects but is experiencing capital rationing in Year 0. No capital rationing is expected in future periods, but none of the four projects that Guilder Co is considering can be postponed, so a decision must be made now. Guilder Co's cost of capital is 12%.

Project	Outlay in Year 0 \$	PV \$	NPV \$
Amster	100,000	111,400	11,400
Eind	56,000	62,580	6,580
Utrec	60,000	68,760	8,760
Tilbur	90,000	102,400	12,400

(1) Arrange the projects in order of their preference to Guilder Co using the profitability index, with the most attractive first.

- A Utrec, Tilbur, Eind, Amster
- B Amster, Tilbur, Utrec, Eind
- C Tilbur, Utrec, Amster, Eind
- D Eind, Utrec, Tilbur, Amster

(2) Which of the following statements about Guilder Co's decision to use the profitability index is true?

- A The PI takes account of the absolute size of the individual projects
- B PI highlights the projects which are slowest in generating returns
- C PI can only be used if projects are divisible
- D PI allows for uncertainty about the outcome of each project

Constructed Response Questions

1 Melanie Co is considering the acquisition of a new machine with an operating life of three years. The new machine could be leased for three payments of \$55,000, payable annually in advance. Alternatively, the machine could be purchased for \$160,000 using a bank loan at a cost of 8% per year. If the machine is purchased, Melanie Co will incur maintenance costs of \$8,000 per year, payable at the end of each year of operation. The machine would have a residual value of \$40,000 at the end of its three-year life.

Melanie Co's production manager estimates that if maintenance routines were upgraded, the new machine could be operated for a period of four years, with maintenance costs increasing to \$12,000 per year, payable at the end of each year of operation. If operated for four years, the machine's residual value would fall to \$11,000.

Taxation should be ignored.

Required:

(a) (i) Assuming that the new machine is operated for a three-year period, evaluate whether Melanie Co should use leasing or borrowing as a source of finance. (6 marks)

(ii) Using a discount rate of 10%, calculate the equivalent annual cost of purchasing and operating the machine for both three years and four years, and recommend which replacement interval should be adopted. (6 marks)

(b) Critically discuss FOUR reasons why NPV is regarded as superior to IRR as an investment appraisal technique. (8 marks)

(Total = 20 marks)

(ACCA FM September/December 2018)

2 Dink Co is a small company that is finding it difficult to raise funds to acquire a new machine costing \$750,000. Dink Co would ideally like a four-year loan for the full purchase price at a before-interest tax rate of 8.6% per year.

The machine would have an expected life of four years. At the end of this period the machine would have a residual value of \$50,000. Tax-allowable servicing costs for the machine would be \$23,000 per year. Tax-allowable depreciation on the full purchase price would be available on a 25% reducing balance basis.

A leasing company has offered a contract whereby Dink Co could have use of the new machine for four years in exchange for an annual lease rental payment of \$200,000, payable at the start of each year. The contract states that the leasing company would undertake maintenance of the machine at no additional cost to Dink Co. At the end of four years the leasing company would remove the machine from the manufacturing facility of Dink Co.

Dink Co pays corporation tax of 30% one year in arrears.

Required:

- (a) (i) Calculate the present value of the cost of borrowing to buy. (6 marks)
- (ii) Calculate the present value of the cost of leasing. (3 marks)
- (iii) Recommend which option is more attractive in financial terms to Dink Co. (1 mark)
- (b) (i) Discuss general reasons why investment capital may be rationed. (6 marks)
- (ii) Discuss ways in which the external capital rationing experienced by Dink Co might be overcome. (4 marks)

(Total = 20 marks)*(ACCA FM September/December 2019)*

3 Cabrerias Co is a construction company. It uses a large earth-moving vehicle called the Beast to prepare foundations for buildings. It needs to decide whether the cheapest replacement interval for the Beast is three or four years.

Cabrerias Co purchases the Beast from a manufacturer for \$800,000, payable one year after delivery. Its resale value will fall by 40% of the purchase price at the end of its first year of operation. The resale value will then reduce by 25% of its previous year's resale value for each further year of operation.

Yearly maintenance costs are \$20,000 at the end of its first year of operation, rising by 5% per year. Maintenance must be provided in the year of sale. Yearly fuel costs are \$28,000 in the first year, rising by \$5,000 for each extra year it is operated.

If the Beast is operated beyond three years, it is subject to a government safety and carbon emissions test. The test would be paid for and would take place at the beginning of the fourth year of operation. Correction of any faults discovered by this test is mandatory. There is an 80% chance that the test and remedial work will cost Cabrerias Co \$50,000, and a 20% chance it will cost \$120,000.

Cabrerias Co's cost of capital is 8%. Ignore taxation.

Required:

- (a) Calculate the equivalent annual cost of the three-year and four-year replacement intervals for the Beast, and advise Cabrerias Co which replacement interval to adopt. (11 marks)
- (b) Discuss why replacement interval decisions should be based upon equivalent annual cost (EAC). (4 marks)
- (c) Discuss why discounted cash flow methods of investment appraisal are considered superior to non-discounted cash flow methods. (5 marks)

(Total = 20 marks)*(ACCA FM March/June 2021)*

4 The Board of OAP Co has decided to limit investment funds to \$10 million for the next year and is preparing its capital budget. The company is considering five projects, as follows.

	Initial investment	Net present value
Project A	\$2,500,000	\$1,000,000
Project B	\$2,200,000	\$1,550,000
Project C	\$2,600,000	\$1,350,000
Project D	\$1,900,000	\$1,500,000
Project E	\$5,000,000	To be calculated

All five projects have a project life of four years. Projects A, B, C and D are divisible, and Projects B and D are mutually exclusive. All net present values are in nominal, after-tax terms.

Project E is a strategically important project which the Board of OAP Co has decided must be undertaken in order for the company to remain competitive, regardless of its financial acceptability. Information relating to the future cash flows of this project is as follows.

Year	1	2	3	4
Sales volume (units)	12,000	13,000	10,000	10,000
Selling price (\$/unit)	450	475	500	570
Variable cost (\$/unit)	260	280	295	320
Fixed costs (\$000)	750	750	750	750

These forecasts are before taking account of selling price inflation of 5.0% per year, variable cost inflation of 6.0% per year and fixed cost inflation of 3.5% per year. The fixed costs are incremental fixed costs associated with Project E. At the end of four years, machinery from the project will be sold for scrap with a value of \$400,000. Tax-allowable depreciation on the initial investment cost of Project E is available on a 25% reducing balance basis, and OAP Co pays corporation tax of 28% per year, one year in arrears. A balancing charge or allowance is available at the end of the fourth year of operation. OAP Co has a nominal after-tax cost of capital of 13% per year.

Required:

- Calculate the nominal after-tax net present value of Project E, and comment on the financial acceptability of this project. (14 marks)
- Calculate the maximum net present value which can be obtained from investing the fund of \$10 million, assuming here that the nominal after-tax NPV of Project E is zero. (5 marks)
- Discuss the reasons why the Board of OAP Co may have decided to limit investment funds for the next year. (6 marks)

(Total = 25 marks)

(ACCA F9 June 2014)

5 Hawker Co is about to replace its existing delivery vehicle with a new design of vehicle that offers greater fuel economy. It estimates that replacing the existing vehicle will save running costs of \$2,000 per year. There are two financing options.

Option 1

The vehicle could be purchased for \$34,000 using a bank loan with an after-tax cost of borrowing of 4% per year. The vehicle would have a useful life of four years and would have a residual value of \$14,000 at the end of that period. Straight-line tax-allowable depreciation is available on the vehicle. The vehicle would be subject to a government CO₂ emissions tax of \$600 at the end of each year of operation. Emissions tax expenses are corporation tax deductible.

Option 2

The vehicle could be leased for a period of four years for a payment of \$6,000 per year, payable at the start of each year. The lessor will pay the CO₂ emissions tax. Lease payments are a corporation tax deductible expense.

Hawker Co's after-tax weighted average cost of capital is 8%. It pays corporation tax at a rate of 20% one year in arrears.

Required:

- (a) Evaluate whether Hawker Co should use leasing or borrowing as a source of finance for the new vehicle. (10 marks)
- (b) Discuss TWO reasons (other than possible after-tax cost advantages) why Hawker Co may choose to lease rather than buy the new delivery vehicle. (4 marks)
- (c) Discuss THREE advantages of using NPV rather than IRR in investment appraisal. (6 marks)

(Total = 20 marks)*(ACCA FM September/December 2021)*

6 Clover Co must replace a machine which is essential to its production process. The company has identified two machines of identical capacity, but different technology, which would be suitable. Only one machine would be purchased in this mutually exclusive investment decision.

Machine 1

This machine would cost \$2,000,000 and would need replacing after three years, when its residual value would be \$200,000. Maintenance costs would be \$52,000 per year in the first year of operation, increasing by 8% per year in each subsequent year. Annual operating costs, expected to be \$300,000 in the first year of operation, would increase by 10% per year in each subsequent year.

Machine 2

This machine would cost \$2,500,000 and would need replacing after five years, when its residual value would be \$150,000. Maintenance costs are forecast to be \$39,000 per year in the first year of operation, increasing by 6% per year in each subsequent year. Annual operating costs, expected to be \$248,000 in the first year of operation, would increase by 12% per year in each subsequent year.

Other information

Clover Co believes that whether Machine 1 or Machine 2 is selected, future replacement machines would, in each case, be more advanced versions of the machines, with identical life cycles. Buying Machine 1 would therefore commit the company to a three-year replacement cycle, while buying Machine 2 would commit the company to a five-year replacement cycle.

The residual value of Machine 1 is a buyback price from the machine's seller that would be included in the contract of sale. The residual value of Machine 2 is an estimate made by Clover Co based on existing market information.

The technology used in each machine is expected to continue to develop in the future. While Clover Co thinks it is unlikely that Machine 1 would become obsolescent during its three-year life, the company thinks it is likely that Machine 2 could be affected by obsolescence towards the end of its life.

For both machines, all increases in costs are due to increasing age.

Clover Co uses a discount rate of 13% to appraise investment projects. Ignore taxation.

Required:

- (a) (i) Calculate the equivalent annual cost of each machine's replacement cycle. (10 marks)
- (ii) Discuss which machine should be selected by Clover Co. (4 marks)
- (b) Discuss why discounted cash flow (DCF) investment appraisal methods are considered to be superior to non-DCF methods. (6 marks)

(Total = 20 marks)*(ACCA FM September/December 2022)*

7 Crocket Co is a manufacturing company that has three investment decisions for the coming year.

Investment decision 1

Six investment projects are being considered, with the following details.

Project	Initial outlay \$000	Net present value \$000
A	1,000	390
B	1,500	Not yet known
C	750	325
D	1,125	590
E	1,850	840
F	1,300	635

Project B is expected to generate the following cash flows.

Year	1	2	3	4
Sales income \$000	725	765	885	612
Costs \$000	145	168	202	94

Project B cash flows are before allowing for inflation of 4% per year for sales income and 5% per year for costs. Crocket Co has a nominal cost of capital of 10%.

Due to management reluctance to raise new finance, capital for investment in the above projects is currently restricted to \$5m. Projects A, B, D and F are all independent, but Projects C and E are mutually exclusive. All of the above projects are divisible and none can be delayed or repeated.

Investment decision 2

A number of Crocket Co's employees have a company car. The entire company car fleet is now due for renewal and, in the past, it has been replaced every four years. Management are not sure if this is the optimum length of time, and feel that other fleet replacement cycles, such as every three or five years, should also be considered.

Investment decision 3

The management of Crocket Co are considering the financial viability of another project, but as yet no detailed financial information is available to perform an NPV appraisal. One of the reasons for this is that the various cash flows will be subject to a number of different rates of inflation that are very uncertain at present. For example, the selling price inflation may be no more than 2% per year, whereas material cost inflation could be anything from 4% to 6% per year. The general rate of inflation is expected to differ from both of these. Management are not sure whether the appraisal could be performed by simply ignoring inflation altogether.

Note: the \$5m capital constraint outlined with investment decision 1 applies to that investment decision only and not to investment decisions 2 and 3.

Required:

- (a) (i) For investment decision 1, calculate the net present value of Project B; and (4 marks)
- (ii) Given the capital constraint, calculate the optimum investment combination and the resulting net present value. (6 marks)
- (b) For investment decision 2, explain the approach Crocket Co should use to determine the optimum replacement cycle for the company car fleet. (4 marks)
- (c) In relation to investment decision 3, describe the two approaches for dealing with inflation AND provide a reasoned recommendation as to which approach Crocket's management should follow. (6 marks)

(Total = 20 marks)*(ACCA FM September/December 2020)*

8 Dysxa Co is looking to expand the capacity of an existing factory in its Alpha Division by 850,000 units per year in order to meet increased demand for one of its products. The expansion will cost \$3.2 million.

The selling price of the unit is \$3.10 per unit and variable costs of production are \$1.10 per unit, both in current price terms. Selling price inflation of 3% per year and variable cost inflation of 6% per year are expected. Nominal fixed costs of production have been forecast as follows.

Year	1	2	3	4
Fixed costs \$	110,000	205,000	330,000	330,000

Dysxa Co has a nominal after-tax weighted average cost of capital of 10% and pays corporation tax of 20% per year one year in arrears. The company can claim 25% reducing balance tax-allowable depreciation on the full cost of the expansion, which should be assumed to be paid at the start of the first year of operation.

Dysxa Co evaluates all investment projects as though they have a project life of four years and assumes zero scrap value at the end of four years.

Dysxa Co has limited the capital investment funds in its Delta division to \$7 million. The division has identified five possible investment projects, as follows. All the projects are divisible and cannot be deferred or repeated; Projects C and E are mutually exclusive.

Project	Initial investment \$	Net present value \$
A	3,000,000	6,000,000
B	2,000,000	3,200,000
C	1,000,000	1,700,000
D	1,000,000	2,100,000
E	2,000,000	3,600,000

Required:

- (a) Calculate the net present value of the investment project and comment on its financial acceptability. (8 marks)
- (b) Determine the net present value of the optimum investment schedule for Delta division. (3 marks)
- (c) Discuss the reasons why hard and soft capital rationing occur. (5 marks)
- (d) Discuss TWO ways in which the risk of a project can be assessed. (4 marks)

(Total = 20 marks)*(ACCA F9 December 2016)*

9 Sulu Co manufactures sports equipment. It is considering launching a new stopwatch product, requiring the purchase of machinery costing \$600,000 now (Year 0). \$200,000 has already been spent on research and development for the stopwatch, and this cost is being amortised over 4 years (\$50,000 per year) in the company's management accounts, which forecast the following operating results for the project (in \$'000):

Year	1	2	3	4
Operating profit/(loss) per management accounts, after deducting \$50,000/year R&D amortisation and \$150,000/year machinery depreciation	(75)	40	110	160

The machinery is eligible for a 100% initial (first-year) tax allowance. Sulu Co pays corporation tax at 20%, paid in the same year the related cash flow arises. Sulu Co's cost of capital is 8%.

Sulu Co is also considering a separate portfolio of five other investment projects (Projects A, B, C, D and E), each requiring initial outlay in Year 0 only. Each project is divisible and cannot be repeated. Project A and Project B are mutually exclusive, so only one of the two can be undertaken.

Project	Initial outlay \$000	NPV \$000
A	900	300
B	850	350
C	500	(50)
D	950	400
E	600	215

Sulu Co has only \$2,100,000 of capital available for investment in this portfolio in Year 0 (a hard capital rationing constraint), separate from the funding required for the stopwatch project above.

Required:

- Calculate the net present value (NPV) of the stopwatch project and comment on your result. (6 marks)
- Determine the optimal investment plan for Sulu Co's \$2.1m of available capital among Projects A-E, and calculate the resulting NPV of that plan. (5 marks)
- Discuss TWO reasons why Sulu Co may be subject to hard capital rationing. (4 marks)
- Discuss the usefulness of return on capital employed (ROCE) to Sulu Co as an investment appraisal method. (5 marks)

(Total = 20 marks)

Answers

Objective Test Questions

1 C

The financing decision (lease vs borrow-and-buy) should be evaluated using the after-tax cost of borrowing, since both options are financing methods for the same underlying investment and the comparison is about the relative cost of the two sources of finance, not the return on the investment itself.

2 A

WORKING

Buy: present value cost = \$80,000 (paid immediately, no further cash flows).

Lease: 10 payments of \$12,000 in advance can be discounted using a 9-year annuity factor plus 1 (for the immediate payment): $(AF_{1-9} @ 10\% + 1) \times \$12,000 = (5.759 + 1) \times \$12,000 = \$81,108$.

As buying has the lower present value cost, the asset should be bought.

3 Answer: \$56,160

WORKING

Buy: $-\$81,000 + (\$40,000 \times DF_5 @ 10\%) = -\$81,000 + (\$40,000 \times 0.621) = -\$81,000 + \$24,840 = -\$56,160$, i.e. a present value cost of \$56,160.

(For comparison: Lease = $AF_{1-5} @ 10\% \times \$14,000 = 3.791 \times \$14,000 = \$53,074$. As leasing has the lower present value cost here, it, not buying, would be the recommended option for this asset.)

Question 4

Statement	True	False
The lease-versus-buy decision should be evaluated separately from the decision to invest in the asset itself	✓	
Leasing is always the cheaper financing option once tax is taken into account		✓
If the asset is purchased, tax-allowable depreciation is a relevant cash flow in the appraisal	✓	
The evaluation compares the present value COST of each financing option, not its NPV benefit	✓	

The finance decision (how to pay for an asset) is kept separate from the investment decision (whether the asset itself is worth acquiring), the operating cash flows generated by the asset are the same regardless of how it is financed. Leasing is not always cheaper: which option wins depends on the specific interest rate, lease terms and tax position in each case (see Questions 2 and 3 above, where each financing method wins once). Where an asset is purchased, tax-allowable depreciation affects the tax cash flows and so is relevant. Because this is purely a financing comparison, the two options are compared on a present value COST basis (the lower cost wins), not by calculating an NPV.

5 A**WORKING**

NPV to the lessor = $-\$100,000 + (\$10,000 \times 0.621) + \$25,000 \times (AF_{1-4} @ 10\% + 1) = -\$100,000 + \$6,210 + \$25,000 \times (3.17 + 1) = -\$100,000 + \$6,210 + \$104,250 = \$10,460$ positive.

(As the lease payments are received in advance, they can be discounted using a 4-year annuity factor (AF) + 1 for the initial receipt.) The lessor earns a positive NPV of \$10,460 from purchasing the machine and leasing it out on these terms.

6 A

'Avoiding tax exhaustion' is a potential benefit to a lessee. Tax exhaustion is when a business has insufficient taxable profit to make full use of tax relief such as tax-allowable depreciation. In this situation, it may be beneficial to lease the asset from a business that can benefit from the tax-allowable depreciation and share in that benefit via lower lease payments. 'Attracting lease customers that may not have been otherwise possible' is a potential benefit to a LESSOR, not a lessee. 'Exploiting a low cost of capital' is a potential benefit for the purchaser, not the lessee. 'Potential future scrap proceeds' is a potential benefit for the purchaser, not the lessee, as the lessee is not entitled to scrap proceeds.

7 C**WORKING**

EAB gas = $\$50,000/AF_5 = 50,000/3.993 = \$12,522$ per year. EAB electric = $\$68,000/AF_7 = 68,000/5.206 = \$13,062$ per year. The NPVs cannot be compared directly as they relate to different time periods; equivalent annual benefits must be compared instead. Electric should be chosen as its EAB is higher.

8 A and B

A shorter replacement cycle tends to produce a higher scrap/trade-in value (since the asset is sold while still relatively new) and can support a better company image and greater operating efficiency (newer equipment tends to be more reliable and efficient). It does not reduce annual depreciation (a shorter life generally means HIGHER annual depreciation) and gives LESS, not more, time to benefit from owning the asset; total maintenance costs are not necessarily lower either, since more frequent replacement means the purchase/set-up cost is incurred more often.

Question 9

Statement	True	False
The use of equivalent annual cost is most appropriate in periods of high inflation		✓
The EAC method assumes that the machine can be replaced by exactly the same machine in perpetuity	✓	

The equivalent annual cost method is most convenient to use in a period of NO (or low) inflation, because in times of high inflation the EAC itself would keep rising each cycle, undermining the simple year-on-year comparison, so statement 1 is false. The EAC method does assume that a machine can be replaced by exactly the same machine in perpetuity, which is a genuine weakness of the method, since assets (particularly technology) are constantly developing and it is not usually possible to replace something with exactly the same thing, so statement 2 is true.

10 Answer: \$105,406

WORKING

Present value of cash flows = $(250,000 + 17,860 + 23,113 + 22,784 + 6,360) = \$320,117$.

Cumulative present value factor (4-year annuity factor at 12%) = 3.037.

Equivalent annual cost = $\$320,117 / 3.037 = \$105,406$.

11 Answer: \$116,299

WORKING

Present value of one 6-year cycle: outlay \$(450,000), running costs \$(82,280), resale \$25,800 = present value cost \$(506,480). Using the 6-year annuity factor at 10% (4.355), the equivalent annual cost = $\$506,480 / 4.355 = \$116,299$ (to the nearest dollar).

12 Answer: \$82,170

WORKING

Years to replacement	Negative NPV	Annuity factor at 11%	EAC
2	149,308	1.713	87,162
3	209,142	2.444	85,574
4	254,893	3.102	82,170
5	310,860	3.696	84,107

The lowest equivalent annual cost is the optimal replacement cycle, the 4-year cycle, at \$82,170.

13 C

WORKING

Time	Cash flow	DF @ 10%	PV
0	(10,000)	1.000	(10,000)
1	(3,000)	0.909	(2,727)
2	(5,000)	0.826	(4,130)
3	(5,000)*	0.751	(3,755)

*Time 3 figure is the \$7,000 running cost net of the \$2,000 scrap proceeds = \$5,000.

NPV = \$(20,612). Equivalent annual cost = PV of costs/annuity factor for the length of the replacement cycle = $20,612 / 2.487 = \$8,288$.

14 A

WORKING

Net present cost of the 1-year cycle = $\$20,000 - (\$10,000 \times 0.909) = \$10,910$. EAC (1-year cycle) = $\$10,910 / 0.909 = \$12,002$.

Net present cost of the 2-year cycle = $\$20,000 - [(\$8,000 - \$5,000) \times 0.826] = \$20,000 - \$2,478 = \$17,522$. EAC (2-year cycle) = $\$17,522 / 1.736 = \$10,093$.

The 2-year cycle should be chosen, with the lower equivalent annual cost of \$10,093.

15 Answer: \$228,000**WORKING**

Net present cost of Machine 2 = -\$540,000 (initial cost) - \$116,889 (maintenance costs, \$47,000 x 2.487) + \$90,120 (scrap proceeds, \$120,000 x 0.751) = \$(566,769).

Equivalent annual cost = \$566,769/2.487 = \$227,893, i.e. \$228,000 to the nearest \$000.

Statement	True	False
The equivalent annual cost calculation assumes that the same type of machine will be used into the foreseeable future	✓	
The equivalent annual cost calculation assumes that the capabilities of Machine 1 and Machine 2 are identical	✓	

Both statements are true. The EAC method assumes the assets will be replaced in perpetuity, or at least for the foreseeable future, by an identical machine, and it also assumes that the operating efficiency/capabilities of the machines being compared are similar, if one machine were more efficient or productive than the other, a simple EAC comparison alone would not capture that difference.

16 B**WORKING**

Present value cost = \$150,000 (initial cost) + (\$5,000 x AF₁₋₃ @ 20%) + [\$8,000 x (AF₁₋₈ @ 20% - AF₁₋₃ @ 20%)] - (\$25,000 x DF₈ @ 20%)

= \$150,000 + (\$5,000 x 2.106) + [\$8,000 x (3.837 - 2.106)] - (\$25,000 x 0.233)

= \$150,000 + \$10,530 + \$13,848 - \$5,825 = \$168,553

Equivalent annual cost = \$168,553/3.837 = \$43,930, i.e. \$43,900 to the nearest \$100.

17 C

The profitability index can only be used to rank DIVISIBLE projects under SINGLE-period capital rationing.

Tax-allowable depreciation is not itself a cash flow relevant to the LEASE-VS-BUY financing choice (it affects the buy option's tax cash flows either way, so it does not distinguish between the two financing methods).

Asset replacement decisions use the EAC method with the same discount rate as the wider appraisal, not specifically the after-tax cost of debt.

Government restrictions on bank lending are an external factor and so are associated with HARD, not soft, capital rationing.

(ACCA F9 December 2014)

Question 18

Statement	True	False
The profitability index is suitable for handling multi-period capital rationing problems if projects are divisible		✓
Assuming that projects are divisible is generally an unrealistic assumption		✓

The profitability index is only suitable for handling SINGLE-period capital rationing problems, even if projects are divisible, multi-period rationing requires a linear programming approach. Whether a project

may be considered divisible depends on the specific project: investing in a single machine is unlikely to be divisible (half a machine will not generate half the return), but buying a chain of shops, for example, could reasonably be divisible.

19 C and D

If projects are non-divisible, only trial-and-error testing of feasible combinations can find the optimal investment plan (the profitability index ranking alone cannot be used to allocate fractional amounts of capital). Multi-period capital rationing can only be solved using a linear programming model, not a simple profitability index ranking. The profitability index does use discounted cash flows (so it does incorporate the time value of money) and is not itself an accounting-profit-based measure.

20 Answer: \$13m

WORKING

Project	Initial cost \$m	NPV \$m	Profitability index (NPV/cost)	Rank
1	40	4	0.100	3
2	30	5	0.167	1
3	50	6	0.120	2
4	60	5	0.083	4

Investment plan: 100% of Project 2 (\$30m cost, \$5m NPV) + 100% of Project 3 (\$50m cost, \$6m NPV) = \$80m committed, leaving \$20m. This funds 50% of Project 1 (\$20m cost, \$2m NPV, being half of its \$40m cost and \$4m NPV). Total cost = \$100m; total NPV = \$5m + \$6m + \$2m = \$13m.

21 B

WORKING

As the projects are now indivisible, the possible combinations within the \$100m budget must be tested directly (the profitability index can only rank projects, it cannot itself solve an indivisible-project problem). Projects 2 and 3 together cost \$30m + \$50m = \$80m (within budget) and give NPV \$5m + \$6m = \$11m. No other feasible pair or triple within the \$100m limit beats this:

Combination	Cost	NPV
Projects 1, 2 and 4	\$130m (exceeds budget)	-
Projects 1 and 3	\$90m	\$10m
Projects 1 and 4	\$100m	\$9m

The optimum indivisible combination is Projects 2 and 3, giving \$11m.

22 A

WORKING

Combination	Calculation	NPV \$000
1 and 5	(1,000 + 300) - (200 + 600)	500
3 and 5	(1,000 + 425) - (600 + 350)	475
1, 2 and 3	(300 + 725 + 425) - (200 + 450 + 350)	450

(Combination 2 and 4 is not feasible, as it exceeds the \$1m available capital.) Projects 1 and 5 produce the highest total NPV, \$500,000, on a total investment of \$800,000.

23 D

WORKING

Project	NPV \$000	Investment value \$000	Profitability index	Ranking
A	125	375	0.33	2
B	240	800	0.30	3
C	80	400	0.20	5
D	40	50	0.80	1
E	150	600	0.25	4

As the projects are divisible, the profitability index (NPV per \$ invested) is used to rank the projects and cash is allocated in order of that ranking. Of the \$1.5 million, cash can be allocated to Projects D, A and B in full (a combined cost of \$1,225,000), leaving \$275,000 for the fourth-ranked project, E. This funds $275/600 = 45.8\%$ of Project E.

24 C

WORKING

The projects are now indivisible, so cannot be done in part, the optimum combination is found by testing all feasible combinations and choosing the one with the highest total NPV. B, C and D would generate \$360,000. C, D and E would generate \$270,000. B, D and E would generate \$430,000 (the highest). B, C, D and E together would require more investment cash than is available, so is not a feasible combination. The optimum indivisible combination is therefore B, D and E.

25 B

WORKING

Project	Capital \$	NPV \$	PI (NPV/capital)	Rank
1	10,000	30,000	3.000	2nd
2	8,000	25,000	3.125	1st
3	12,000	30,000	2.500	3rd
4	16,000	36,000	2.250	4th

Investment plan: 100% of Project 2 (cost \$8,000, NPV \$25,000) + 100% of Project 1 (cost \$10,000, NPV \$30,000) + 100% of Project 3 (cost \$12,000, NPV \$30,000) = \$30,000 committed, leaving \$2,000. This funds $\$2,000/\$16,000 = 12.5\%$ of Project 4, giving NPV = $12.5\% \times \$36,000 = \$4,500$. Total NPV = $\$25,000 + \$30,000 + \$30,000 + \$4,500 = \$89,500$.

Case Scenario Questions

1 (1) C

WORKING

PV of the 1-year replacement cycle = $$(2,400) + ($1,200 \times 0.877) = $(1,348)$. Equivalent annual cost = $\$1,348/0.877 = \$1,537$.

(2) B

WORKING

PV of costs over a 3-year cycle = $\$1,024 \times 2.322 = \$2,378$. This equals $(2,400 + (75 \times 0.877)) + (150 \times 0.769) - (\text{trade-in value} \times 0.675) = 2,466 + 115 - (\text{trade-in value} \times 0.675)$. So $\text{trade-in value} \times 0.675 = \203 , giving $\text{trade-in value} = \$203/0.675 = \$301$.

(3) D

All three are potential tax implications. A balancing charge refers to a taxable gain on disposal, and would arise if the trade-in value exceeded the asset's tax written down value.

(4) B

Rapidly changing technology may encourage a shorter replacement cycle, and replacing the laptops more often could also lead to discounts being offered by the supplier. However, replacing laptops more often may lead to GREATER, not less, environmental damage (from manufacturing and disposing of more units), so this would discourage, not encourage, frequent replacement.

(5) D

Tangible assets provide good potential security for providers of debt finance, so a large proportion of tangible assets would tend to make it EASIER, not harder, to raise finance; this would not explain why the company faces capital rationing. The other three options are all genuine causes of capital rationing.

2 (1) False, False, True, False

Hard rationing is the limit on the amount of finance imposed by external lending institutions/capital markets, so it comes from external factors. Soft rationing arises from internal management decisions, for instance, the decision to hold back some cash for a later project, or an unwillingness to tie up all available cash for many years.

(2) A and E

WORKING

Project	NPV \$000	Investment \$000	NPV/\$ invested	Rank
A	4,900	3,200	1.53	2nd
B	7,400	9,300	0.80	5th
C	5,900	7,300	0.81	4th
D	7,500	5,200	1.44	3rd
E	9,000	5,600	1.61	1st

Total investment required for all five projects is \$30.6m, far exceeding the \$10m available, and the projects are indivisible, so feasible combinations must be tested directly. Projects A and D would give NPV $\$4,900\text{k} + \$7,500\text{k} = \$12,400\text{k}$. Projects A and E would give NPV $\$4,900\text{k} + \$9,000\text{k} = \$13,900\text{k}$, the highest of any feasible pair, and higher than any single project alone. Projects A and E should be accepted.

(3) A

Since A and E can no longer both be accepted (they are mutually exclusive), the next-best feasible combination from the ranking above is A and D (NPV \$12,400k), which becomes the optimal choice.

(4) B

Using the profitability index ranking above, the combination accepted becomes A plus E plus as much of the next-ranked project (D) as remaining capital allows. Projects A and E together require \$3,200k + \$5,600k = \$8,800k, leaving \$1,200k of the \$10,000k available. Since Project D requires a total investment of \$5,200k, the proportion of Project D that can be undertaken with the remaining capital is $\$1,200k/\$5,200k = 3/13$ (approximately 23.1%).

3 (1) A**WORKING**

Project	Outlay \$	PV \$	NPV \$	PI (PV/outlay)	Rank
Amster	100,000	111,400	11,400	1.114	4th
Eind	56,000	62,580	6,580	1.118	3rd
Utrech	60,000	68,760	8,760	1.146	1st
Tilbur	90,000	102,400	12,400	1.138	2nd

Ranked by profitability index (PV of future cash flows divided by outlay), from most to least attractive: Utrecht (1.146), Tilbur (1.138), Eind (1.118), Amster (1.114).

(2) C

The profitability index can only be used to rank projects, and select an optimum combination, when projects are DIVISIBLE, if projects are indivisible, feasible combinations must instead be tested directly by trial and error. The PI does NOT take account of the absolute size of individual projects (a small project can have a high PI while generating only a small NPV), does NOT highlight which projects are slowest to generate their returns, and does NOT allow for uncertainty about a project's outcome; it assumes complete certainty about each project's forecast cash flows.

Constructed Response Questions

1 (a) (i) Evaluation of leasing versus borrowing (3-year period)

Lease	0	1	2
Lease payment	(55,000)	(55,000)	(55,000)
Discount at 8%	1.000	0.926	0.857
Present value	(55,000)	(50,930)	(47,135)

Borrow and buy	0	1	2	3
Initial cost	(160,000)			
Residual value				40,000
Maintenance		(8,000)	(8,000)	(8,000)
Total	(160,000)	(8,000)	(8,000)	32,000
Discount at 8%	1.000	0.926	0.857	0.794
Present value	(160,000)	(7,408)	(6,856)	25,408

Present value cost of leasing = \$(153,065). Present value cost of borrowing and buying = \$(148,856). As borrowing and buying offers the cheaper present value cost, the machine should be financed by borrowing.

(a) (ii) Equivalent annual cost of purchasing and operating the machine

3-year replacement cycle	0	1	2	3
Initial cost	(160,000)			
Residual value				40,000
Maintenance		(8,000)	(8,000)	(8,000)
Total	(160,000)	(8,000)	(8,000)	32,000
Discount at 10%	1.000	0.909	0.826	0.751
Present value	(160,000)	(7,272)	(6,608)	24,032

Present value cost = \$(149,848). EAC (3-year cycle) = $149,848 / 2.487$ (3-year annuity factor at 10%) = \$60,253.

4-year replacement cycle	0	1	2	3	4
Initial cost	(160,000)				
Residual value					11,000
Maintenance		(12,000)	(12,000)	(12,000)	(12,000)
Total	(160,000)	(12,000)	(12,000)	(12,000)	(1,000)
Discount at 10%	1.000	0.909	0.826	0.751	0.683
Present value	(160,000)	(10,908)	(9,912)	(9,012)	(683)

Present value cost = \$(190,515). EAC (4-year cycle) = $190,515 / 3.170$ (4-year annuity factor at 10%) = \$60,099.

As the 4-year cycle has the (marginally) lower equivalent annual cost, the machine should be replaced every four years.

(b)

NPV is regarded as superior to IRR as an investment appraisal technique for several reasons. First, NPV directly measures the absolute increase in shareholder wealth that a project is expected to create, since it is expressed in monetary terms, whereas IRR is a relative (percentage) measure that says nothing about the scale of value created. Second, NPV assumes that intermediate cash flows are reinvested at the company's cost of capital, a realistic assumption, whereas IRR assumes reinvestment at the project's own IRR, which, especially for a high-IRR project, is often unrealistic. Third, unconventional cash flows (for example, a project with a large cash outflow partway through its life) can produce multiple IRRs, making the IRR decision rule ambiguous, whereas NPV always gives a single, unambiguous figure. Fourth, NPV can easily accommodate a cost of capital that changes from year to year (simply by discounting each year's cash flow at that year's specific rate), whereas IRR cannot readily be compared against a changing hurdle rate. Finally, for mutually exclusive projects, ranking by IRR can give a different (and wealth-reducing) result compared with ranking by NPV, particularly when projects differ significantly in scale or timing of cash flows, NPV ranking always identifies the wealth-maximising choice.

(ACCA FM September/December 2018)

2 (a) (i) Present value of the cost of borrowing to buy

After-tax cost of borrowing = $8.6\% \times (1 - 0.3) = 8.6\% \times 0.7 = 6\%$.

	0	1	2	3	4	5
Purchase	(750,000)					
Residual value					50,000	
Service costs, net of one-year-arrears tax relief		(23,000)	(16,100)	(16,100)	(16,100)	6,900
Tax-allowable depreciation benefit (one year in arrears)			56,250	42,188	31,641	79,922
Net cash flow	(750,000)	(23,000)	40,150	26,088	65,541	86,822
Discount at 6%	1.000	0.943	0.890	0.840	0.792	0.747
Present value	(750,000)	(21,689)	35,734	21,914	51,908	64,856

WORKINGS

Tax-allowable depreciation and tax benefits

Year	TA depreciation (\$)	Tax benefit (\$)
1	187,500	56,250
2	140,625	42,188
3	105,469	31,641
4	266,406	79,922

25% reducing balance on \$750,000, with a balancing allowance in Year 4 (= \$750,000 - \$187,500 - \$140,625 - \$105,469 - \$50,000 residual value); tax benefits at 30%, received one year in arrears. The \$23,000/year service cost is tax-deductible at 30%, giving tax relief of \$6,900/year, also received one year in arrears (so Year 1's service cost is not yet relieved, while Year 5, after servicing has ended, still receives the relief on Year 4's cost).

Present value of the cost of borrowing to buy = \$(597,277) (using the standard discount factors above; a spreadsheet NPV calculation using unrounded factors gives \$(597,268)).

(a) (ii) Present value of the cost of leasing

	0	1	2	3	4	5
Lease rentals	(200,000)	(200,000)	(200,000)	(200,000)		
Tax benefits			60,000	60,000	60,000	60,000
Net cash flow	(200,000)	(200,000)	(140,000)	(140,000)	60,000	60,000
Discount at 6%	1.000	0.943	0.890	0.840	0.792	0.747
Present value	(200,000)	(188,600)	(124,600)	(117,600)	47,520	44,820

Present value of the cost of leasing = \$(538,460).

(a) (iii)

Financial benefit of leasing = \$597,277 - \$538,460 = \$58,817. Leasing is recommended as the option that is more attractive in financial terms to Dink Co.

(b) (i)

Investment capital may be rationed for several reasons. Hard (external) capital rationing can arise because capital markets are depressed, making it difficult or expensive to raise new finance, or because lenders regard a company (particularly a smaller company such as Dink Co) as too high-risk to lend to on favourable terms, or because the company lacks sufficient security/collateral to support further borrowing. Soft (internal) capital rationing can arise because management is reluctant to raise new equity finance (fearing dilution of control or earnings per share), because management wishes to avoid the risk associated with high levels of gearing and fixed interest commitments, or because management prefers a policy of steady, organic growth funded from retained earnings rather than rapid, externally-financed expansion.

(b) (ii)

The external (hard) capital rationing experienced by Dink Co might be overcome in several ways:

- (1) Seeking finance from alternative sources such as venture capital providers, business angels, peer-to-peer lending platforms, or government-backed small business loan schemes, which may be more willing to lend to a smaller, higher-risk company.
- (2) Offering additional security or a personal guarantee to reduce the lender's perceived risk and so improve loan terms.
- (3) Using asset-based finance such as leasing (as evaluated in part (a)) or invoice discounting/factoring, which is often more accessible to smaller companies than a general-purpose bank loan.
- (4) Building a stronger trading track record and financial history over time to improve the company's creditworthiness and future access to finance.

(ACCA FM September/December 2019)

3 (a) Equivalent annual cost of the three-year and four-year replacement intervals**WORKINGS****Maintenance costs, fuel costs and resale value**

Year	1	2	3	4
Maintenance costs (\$)	20,000	21,000	22,050	23,153
Fuel costs (\$)	28,000	33,000	38,000	43,000
Resale value (\$)	480,000	360,000	270,000	202,500

Maintenance costs rise by 5% per year from \$20,000. Fuel costs rise by \$5,000 per year from \$28,000. Resale value falls by 40% in Year 1, then 25% per year thereafter, from \$800,000 (end of Year 3 for the 3-year cycle; end of Year 4 for the 4-year cycle).

Government test cost (4-year cycle only, expected value): $(80\% \times \$50,000) + (20\% \times \$120,000) = \$40,000 + \$24,000 = \$64,000$, payable at the start of Year 4 (i.e. discounted as a Year 3 cash flow).

3-year cycle	1	2	3
Purchase cost	(800,000)		
Maintenance	(20,000)	(21,000)	(22,050)
Fuel	(28,000)	(33,000)	(38,000)
Resale value			270,000
Net cash flow	(848,000)	(54,000)	209,950
Discount at 8%	0.926	0.857	0.794
Present value	(785,248)	(46,278)	166,700

Present value cost of the 3-year cycle = \$(664,626), i.e. \$(664,816) after rounding of the underlying maintenance, fuel and resale figures (own-figure basis).

4-year cycle	1	2	3	4
Purchase cost	(800,000)			
Maintenance	(20,000)	(21,000)	(22,050)	(23,153)
Fuel	(28,000)	(33,000)	(38,000)	(43,000)
Government test (expected value)			(64,000)	
Resale value				202,500
Net cash flow	(848,000)	(54,000)	(124,050)	136,347
Discount at 8%	0.926	0.857	0.794	0.735
Present value	(785,248)	(46,278)	(98,496)	100,215

Present value cost of the 4-year cycle = \$(829,807), i.e. \$(829,800) after rounding of the underlying maintenance, fuel and resale figures (own-figure basis).

EAC of the 3-year cycle = NPV of the 3-year cycle / 3-year annuity factor at 8% (2.577) = $\$664,816 / 2.577 = \$257,981$.

EAC of the 4-year cycle = NPV of the 4-year cycle / 4-year annuity factor at 8% (3.312) = $\$829,800 / 3.312 = \$250,500$.

Recommendation: on an expected-cost basis, the four-year replacement interval has the lower equivalent annual cost, so Cabrerias Co should replace the Beast every four years. However, it is a close decision, and there is a 20% chance that the government test will cost \$120,000 rather than \$50,000, which would make the four-year interval more expensive than shown. Cabrerias Co should also consider replacement intervals other than three and four years, as a cheaper option may exist outside this range.

(b)

In simple situations, choosing between one-off projects of different lengths is straightforward: the NPV technique evaluates the costs and benefits of a project over its life, and the project with the largest NPV is selected. However, when assets of different lives can be endlessly replaced, and form part of an infinite chain of identical replacement decisions, project NPVs (or present value costs) cannot be meaningfully compared directly, because they relate to different lengths of time, a three-year cycle's present value cost cannot simply be compared with a four-year cycle's.

To allow for the different lengths of the replacement intervals, each cycle's present value cost is divided by the annuity factor appropriate to its length (a three-year annuity factor for a three-year cycle, and so on). The resulting figure is the equivalent annual cost: the annual cost, payable at the end of each year of the cycle, that is equivalent in present value terms to the actual cash flows of that cycle. Once expressed on this common, per-year basis, the EAC of each possible replacement interval can be compared meaningfully, and the interval with the lowest EAC represents the optimal (cheapest) choice, assuming continual like-for-like replacement into the foreseeable future.

(c)

Discounted cash flow (DCF) methods, such as NPV and IRR, are considered superior to non-discounted cash flow methods, such as payback period and accounting rate of return (ROCE), for several reasons. First, DCF methods take account of the time value of money, the principle that a sum of cash received today is worth more than the same sum received in the future, because of its earning potential, the effect of inflation, and risk, whereas payback and ROCE do not discount cash flows at all. Second, DCF methods are based on cash flows, not accounting profits; cash flow is a more objective measure than profit, which can be affected by the choice of accounting policies (e.g. depreciation method), whereas ROCE is profit-based. Third, NPV in particular considers the whole of a project's life and directly measures the absolute increase in shareholder wealth a project is expected to generate, whereas payback ignores all cash flows occurring after the payback period is reached, potentially rejecting projects with strong long-term returns. Non-DCF methods remain popular in practice mainly for their simplicity and ease of communication (particularly payback, as a rough guide to liquidity/risk), but they are theoretically inferior to DCF methods.

(ACCA FM March/June 2021)

4 (a) Nominal after-tax NPV of Project E**WORKING**

Year	1	2	3	4
Selling price (\$/unit)	450	475	500	570
Inflated selling price (\$/unit)	472.50	523.69	578.81	692.84
Sales volume (units)	12,000	13,000	10,000	10,000
Sales income (\$000)	5,670	6,808	5,788	6,928

Year	1	2	3	4
Variable cost (\$/unit)	260	280	295	320
Inflated variable cost (\$/unit)	275.60	314.61	351.35	403.99
Sales volume (units)	12,000	13,000	10,000	10,000
Variable cost (\$000)	3,307	4,090	3,514	4,040

Fixed costs

Year	1	2	3	4
Fixed costs (\$)	776,000	803,000	832,000	861,000

\$750,000 inflated at 3.5% per year (all rounded to the nearest \$000).

Tax-allowable depreciation and tax benefits

Year	TA depreciation (\$)	Tax benefit (\$)
1	1,250,000	350,000
2	937,500	262,500
3	703,125	196,875
4	1,709,375	478,625

25% reducing balance on the \$5,000,000 investment.

Year 4 balancing allowance = \$2,109,375 minus \$400,000 scrap = \$1,709,375.

Tax benefits at 28%.

Year	1	2	3	4	5
Sales income (\$000)	5,670	6,808	5,788	6,928	
Variable cost (\$000)	(3,307)	(4,090)	(3,514)	(4,040)	
Fixed cost (\$000)	(776)	(803)	(832)	(861)	
Cash flow before tax (\$000)	1,587	1,915	1,442	2,027	
Tax at 28% (\$000)		(444)	(536)	(404)	(568)
Depreciation tax benefit (\$000)		350	263	197	479
Scrap value (\$000)				400	
Net cash flow (\$000)	1,587	1,821	1,169	2,220	(89)
Discount at 13%	0.885	0.783	0.693	0.613	0.543
Present value (\$000)	1,405	1,426	810	1,361	(48)

Sum of present values = \$4,954,000.

Initial investment = \$5,000,000.

Net present value = \$4,954,000 – \$5,000,000 = \$(46,000), i.e. a negative NPV of approximately \$46,000.

Although the NPV of Project E is negative and so the project is not financially acceptable on a standalone basis, the Board of OAP Co has decided that it must be undertaken regardless, since it is regarded as strategically important to the company remaining competitive.

(b) Calculation of maximum NPV

WORKING

Project	A	B	C	D	E
Investment (\$000)	2,500	2,200	2,600	1,900	5,000
NPV (\$000)	1,000	1,550	1,350	1,500	nil
PV of future cash flows (\$000)	3,500	3,750	3,950	3,400	5,000
Profitability index	1.400	1.705	1.519	1.789	1.000
Ranking	4	-	3	2	1

Project E has been ranked first as it must be undertaken regardless of its financial merit. Project B cannot be undertaken alongside Project D, since the two are mutually exclusive; Project D has the higher profitability index of the two, so it is preferred.

Project	Investment (\$000)	NPV (\$000)
E	5,000	nil
D	1,900	1,500

Project	Investment (\$000)	NPV (\$000)
C	2,600	1,350
A	500	200
Total	10,000	3,050

As Project A is divisible and only \$500,000 (20% of its \$2,500,000 initial cost) of capital remains available after cumulative investment in Projects E, D and C, only 20% of Project A's NPV can be earned, $20\% \times \$1,000,000 = \$200,000$. The maximum NPV obtainable from the \$10 million fund is \$3,050,000, comprising Projects E (in full, for strategic reasons), D and C (in full) and 20% of Project A.

(c)

When a company restricts or limits its investment funds in this way, it is imposing 'soft' or internal capital rationing on itself. Capital rationing means that a company is unable to invest in all projects with a positive net present value, and so, strictly, is not acting to maximise shareholder wealth in the short term. There are several reasons why the Board of OAP Co may have decided to limit investment funds for the next year. It may not wish to issue new equity finance, in order to avoid diluting existing shareholders' earnings per share and control. Issuing new equity finance can also increase the risk of a company's shares being acquired by an outside party, potentially leading to an unwelcome future takeover bid. The Board may not wish to take on new debt finance if it wishes to avoid increasing its commitment to fixed interest payments; this could be because economic prospects are seen as poor or uncertain, or because existing debt obligations are already high and the Board does not wish to increase financial risk further. Alternatively, the Board may simply prefer a strategy of organic growth, financing capital investment from retained earnings rather than seeking additional external finance, in order to maintain full control over the pace and direction of the company's expansion.

(ACCA F9 June 2014)

5 (a) Evaluation of leasing versus borrowing

Option 1, Borrow and buy	0	1	2	3	4	5
Purchase price	(34,000)					
Residual value					14,000	
TAD benefits			1,000	1,000	1,000	1,000
Emissions tax		(600)	(600)	(600)	(600)	
Emissions tax benefits			120	120	120	120
Net cash flow	(34,000)	(600)	520	520	14,520	1,120
Discount at 4%	1.000	0.962	0.925	0.889	0.855	0.822
Present value	(34,000)	(577)	481	462	12,412	921

Present value cost of borrowing = \$(20,301).

Option 2, Lease	0	1	2	3	4	5
Lease payment	(6,000)	(6,000)	(6,000)	(6,000)		
Lease payment tax benefits			1,200	1,200	1,200	1,200
Net cash flow	(6,000)	(6,000)	(4,800)	(4,800)	1,200	1,200
Discount at 4%	1.000	0.962	0.925	0.889	0.855	0.822
Present value	(6,000)	(5,769)	(4,438)	(4,267)	1,026	986

Present value cost of leasing = \$(18,462). Hawker Co should lease the new vehicle, because this option has the lower present value cost.

(b) Reasons for leasing (other than after-tax cost)

Leasing can offer more flexibility than borrowing and buying: lease periods can be for less than an asset's useful life, or break clauses may be offered by the lessor, so if technology changes and the asset becomes out of date before the end of its expected life, the lessee does not have to keep on using it. Also, under the 'borrow and buy' option, Hawker Co would carry the risk of the vehicle failing to achieve its expected disposal value of \$14,000; if leased, this risk would instead be transferred to the lessor. (A further acceptable reason: the lessee may not have enough cash to pay for the vehicle and might have difficulty obtaining a bank loan, in which case leasing may be the only way of getting use of the asset. Only two reasons are required.)

(c) Advantages of NPV over IRR

In most simple accept-or-reject decisions, IRR and NPV select the same project, but NPV has several advantages over IRR:

- (1) The NPV of a project, calculated at an appropriate cost of capital, equals the increase in shareholder wealth the project offers, directly linking it to the objective of maximising shareholder wealth; IRR, being a percentage rate of return, does not measure this absolute increase in wealth.
- (2) NPV looks at absolute increases in wealth and so can be used to compare projects of different sizes, whereas IRR looks at relative rates of return and so ignores the scale of the investment, a small project may have a very high IRR but bring only a small increase in wealth.
- (3) NPV is not subject to the technical difficulties that limit IRR: where cash flows reverse direction more than once, IRR can produce multiple internal rates of return, and for mutually exclusive projects IRR can rank projects differently (and incorrectly) compared with NPV, because IRR assumes reinvestment of interim cash flows at the project's own IRR (often unrealistic) rather than at the cost of capital as NPV assumes.
- (4) NPV can easily be used where the cost of capital changes from year to year, whereas it is difficult to make an accept/reject decision using IRR in these circumstances.

(ACCA FM September/December 2021)

6 (a) (i) Equivalent annual cost of each machine's replacement cycle

Machine 1 (3-year cycle)	1	2	3
Residual value			200,000
Maintenance cost	(52,000)	(56,160)	(60,653)
Operating cost	(300,000)	(330,000)	(363,000)
Net cash flow	(352,000)	(386,160)	(223,653)
Discount at 13%	0.885	0.783	0.693
Present value	(311,504)	(302,363)	(154,991)

Sum of present values = \$(768,858).

Initial investment = \$(2,000,000).

NPV = \$(2,768,858).

EAC = \$2,768,858/2.361 = \$1,172,748.

Machine 2 (5-year cycle)	1	2	3	4	5
Residual value					150,000
Maintenance cost	(39,000)	(41,340)	(43,820)	(46,450)	(49,237)
Operating cost	(248,000)	(277,760)	(311,091)	(348,422)	(390,233)
Net cash flow	(287,000)	(319,100)	(354,912)	(394,872)	(289,469)
Discount at 13%	0.885	0.783	0.693	0.613	0.543
Present value	(253,995)	(249,855)	(245,954)	(242,056)	(157,182)

Sum of present values = \$(1,149,042).

Initial investment = \$(2,500,000).

NPV = \$(3,649,042).

EAC = $\$3,649,042 / 3.517 = \$1,037,544$.

(a) (ii) Discussion of which machine should be selected

Choosing between the two machines on financial grounds cannot be achieved using NPV alone, since they have different replacement cycles reflecting their unequal lives; the two decisions must instead be compared using equivalent annual cost, which shows that Machine 2 has the lower EAC and so should be preferred on financial grounds.

However, the residual value of Machine 1 is a contractual buyback price and so is relatively certain, whereas the residual value of Machine 2 is merely an estimate and so is subject to uncertainty; Machine 1's three-year cycle might therefore be chosen despite its higher EAC. Purchasing Machine 1 also carries little risk of obsolescence within its three-year life, whereas Machine 2's five-year cycle carries significant obsolescence risk late in its life, which again might favour Machine 1 despite its higher EAC. Machine 1's shorter replacement cycle may also give Clover Co greater flexibility to respond to future changes in technology. On financial grounds alone, however, Machine 2 with its lower EAC should be recommended, and even if Machine 2's residual value were zero it would still have the lower EAC.

(b) DCF versus non-DCF methods

Discounted cash flow methods, such as NPV and IRR, are considered superior to non-DCF methods, such as payback and ROCE, for several reasons. DCF methods take account of the time value of money by discounting future cash flows to their present value, whereas payback and ROCE ignore this. DCF methods use cash flows, which are not distorted by company-specific accounting policy choices such as depreciation method, making them more objective than profit-based measures like ROCE. DCF methods also consider cash flows over the whole life of a project, whereas payback ignores cash flows after the payback period, so a project with a higher NPV might be rejected under payback in favour of one with a lower NPV but a shorter payback period, which is inconsistent with the objective of maximising shareholder wealth.

(ACCA FM September/December 2022)

7 (a) (i) Net present value of Project B

Year	1	2	3	4
Sales income (inflated at 4%) \$000	754	827	996	716
Costs (inflated at 5%) \$000	(152)	(185)	(234)	(114)
Net cash flow \$000	602	642	762	602
Discount at 10%	0.909	0.826	0.751	0.683
Present value \$000	547	530	572	411

Total present value = \$2,061,000. Less initial outlay \$1,500,000. NPV of Project B = \$561,000.

(a) (ii) Optimum investment combination

Project	Outlay \$000	NPV \$000	Profitability index (NPV/outlay)	Rank
A	1,000	390	0.390	5th
B	1,500	561	0.374	6th
C	750	325	0.433	Excluded
D	1,125	590	0.524	1st
E	1,850	840	0.454	3rd

Project	Outlay \$000	NPV \$000	Profitability index (NPV/outlay)	Rank
F	1,300	635	0.488	2nd

As the projects are divisible, the project with the higher profitability index is chosen out of the mutually exclusive pair C and E; this is Project E (0.454) rather than Project C (0.433).

Rank	Project	%	Capital \$000	NPV \$000
1	D	100	1,125	590
2	F	100	1,300	635
3	E	100	1,850	840
4	A	72.5	725	283
Total			5,000	2,348

After selecting Projects D, F and E in full, \$725,000 of the \$5,000,000 capital remains, funding 72.5% (725/1,000) of Project A and so 72.5% of its NPV ($0.725 \times \$390,000 = \$283,000$). The optimum investment combination generates a total NPV of \$2,348,000.

(b) Approach to determine the optimum replacement cycle

First, the cash flows must be determined for each replacement cycle under consideration (three, four and five years). These cash flows will include the initial outlay of the car fleet, maintenance and running costs, and the residual value of the fleet at the end of the cycle. These cash flows are then discounted at a suitable discount rate to determine the present value of each cycle; all these present values will be negative, as they represent the cost of running the fleet with no relevant income attributable.

Given the continually repeating nature of the decision, a common timescale must then be used to allow the cycles to be compared meaningfully: while the present value of the three-year cycle will appear cheapest, it recurs more frequently than the four- or five-year cycles going forward indefinitely. To allow for the differing timescales, each cycle's present value cost is converted to an equivalent annual cost (EAC), calculated by dividing that cycle's present value by the annuity factor for its length (e.g. the three-year present value is divided by a three-year annuity factor). Once each cycle's EAC has been calculated, the lowest (cheapest) figure indicates the optimum replacement cycle for the fleet, since all cycles are now expressed on the same, comparable, one-year basis.

(c) Approaches to dealing with inflation, and recommendation

When appraising an investment, the treatment of inflation must be considered, as it affects both the cash flows and the required rate of return used as the discount rate. Real-terms and nominal-terms approaches differ in how the effects of inflation are incorporated.

Nominal-terms approach: both the cash flows and the discount rate incorporate the effects of inflation. The cost of capital includes the effect of general inflation on investors' required rate of return, and specific rates of inflation are applied to sales, material costs and other cash flows. The uncertainty surrounding Crocket Co's inflation rates will make a nominal-terms appraisal more difficult to prepare accurately, and this should be considered when reviewing the results.

Real-terms approach: the effects of general inflation are excluded. Nominal cash flows (incorporating specific inflation rates) are deflated by the general rate of inflation to give real-terms cash flows, and a real discount rate, representing investors' base level of required return before inflation, is used.

Recommendation: a real-terms approach is only suitable if two conditions are met: a single rate of inflation must affect all of a project's cash flows, and that single rate must also equal the general rate of inflation affecting the discount rate. Neither condition is met here, selling price inflation (2%) differs from material cost inflation (4%-6%), and the general rate of inflation is expected to differ from both. As neither condition is satisfied, a real-terms approach is not suitable for Crocket Co to appraise this project; a nominal-terms approach should be used instead.

(ACCA FM September/December 2020)

8 (a) Net present value of the investment project

WORKING, sales revenue

Year	1	2	3	4
Selling price (\$/unit)	3.10	3.10	3.10	3.10
Inflated at 3% per year	3.19	3.29	3.39	3.49
Sales volume (000 units/year)	850	850	850	850
Sales revenue (\$000/year)	2,712	2,797	2,882	2,967

WORKING, variable cost

Year	1	2	3	4
Variable cost (\$/unit)	1.10	1.10	1.10	1.10
Inflated at 6% per year	1.17	1.24	1.31	1.39
Sales volume (000 units/year)	850	850	850	850
Variable cost (\$000/year)	995	1,054	1,114	1,182

WORKING, tax-allowable depreciation (TAD)

Year	TA depreciation (\$)	Tax benefit (\$)
1	800,000	160,000
2	600,000	120,000
3	450,000	90,000
4	1,350,000	270,000

25% reducing balance on the \$3,200,000 expansion cost, with a balancing allowance in Year 4 (assumed zero scrap value; written-down value brought forward into Year 4 = \$3,200,000 – \$800,000 – \$600,000 – \$450,000 = \$1,350,000, and the balancing allowance is this \$1,350,000 written-down value less nil scrap value = \$1,350,000); tax benefits at 20%, received one year in arrears.

	1	2	3	4	5
Sales revenue \$000	2,712	2,797	2,882	2,967	
Variable costs \$000	(995)	(1,054)	(1,114)	(1,182)	
Contribution \$000	1,717	1,743	1,768	1,785	
Fixed costs \$000	(110)	(205)	(330)	(330)	
Taxable cash flow \$000	1,607	1,538	1,438	1,455	
Taxation at 20% \$000		(321)	(308)	(288)	(291)
TAD tax benefits \$000		160	120	90	270
After-tax cash flow \$000	1,607	1,377	1,250	1,257	(21)
Discount at 10%	0.909	0.826	0.751	0.683	0.621
Present value \$000	1,461	1,137	939	859	(13)

PV of future cash flows = \$4,383,000. Less initial investment \$3,200,000. Net present value = \$1,183,000. The NPV is positive and so the investment project is financially acceptable.

(b) Net present value of the optimum investment schedule for Delta division

WORKING

Project	Initial investment \$	Net present value \$	Profitability index (NPV/investment)	Rank
A	3,000,000	6,000,000	2.0	2nd
B	2,000,000	3,200,000	1.6	4th

Project	Initial investment \$	Net present value \$	Profitability index (NPV/investment)	Rank
C	1,000,000	1,700,000	1.7	Excluded (mutually exclusive with E)
D	1,000,000	2,100,000	2.1	1st
E	2,000,000	3,600,000	1.8	3rd

Project E is preferred to the mutually exclusive Project C, as it has the higher profitability index (1.8 vs 1.7).

Project	Initial investment \$	Rank	Net present value \$
D	1,000,000	1st	2,100,000
A	3,000,000	2nd	6,000,000
E	2,000,000	3rd	3,600,000
B	1,000,000 (of \$2,000,000)	4th	1,600,000 (\$3.2m x \$1m/\$2m)
Total	7,000,000		13,300,000

Projects D, A and E are undertaken in full, using \$6,000,000 of the \$7,000,000 budget and generating \$11,700,000 of NPV. The remaining \$1,000,000 funds half (50%) of Project B, generating a further \$1,600,000 of NPV (50% x \$3,200,000). The NPV of the optimum investment schedule for Delta division is \$13,300,000.

(c) Reasons why hard and soft capital rationing occur

Capital rationing can be divided into hard capital rationing, which is externally imposed, and soft capital rationing, which is internally imposed.

Soft capital rationing: investment capital may be limited internally because a company does not want to take on a commitment to increased fixed interest payments, for example, if it expects future profitability to be poor. A company may wish to avoid diluting existing earnings per share or changing existing patterns of ownership and control by issuing new equity. A company may limit investment funds because it wishes to pursue controlled growth rather than rapid growth. Given the uncertainty associated with forecasting future cash flows, a company may limit investment funds in order to create an internal market where investment projects compete for finance, with only the best investment projects being granted approval.

Hard capital rationing: external reasons for capital rationing can be related to risk and to availability of finance. Providers of finance may see a company as too risky to invest in, perhaps because it is highly geared or because it has a poor record or poor prospects in terms of profitability or cash flow. Long-term finance for capital investment may have limited availability because of the poor economic state of the economy, or because there is a banking crisis.

(d) Two ways of assessing project risk

The risk of an investment project could be assessed by using probability analysis or by using the capital asset pricing model (CAPM).

Probability analysis: project risk can be assessed or quantified by attaching probabilities to expected investment project outcomes. At an overall level, this could be as simple as attaching probabilities to two or more expected scenarios, for example, associated with different economic states. Key project variables might then take different values depending on the economic state. At the level of individual project variables, probability distributions of values could be found through expert analysis, and the probability distributions and relationships between variables then built into a simulation model. This model could then be used to generate a probability distribution of expected project outcomes in terms of net present values. Project risk could then be measured by the standard deviation of the expected net present value.

CAPM: the CAPM uses a beta that represents the systematic risk of a project and of the company undertaking the project to find a risk-adjusted cost of equity for use in discounting. The systematic business risk of an investment project can be assessed by identifying a proxy company in a similar line of business. The equity beta of the proxy company can then be ungeared to give the asset beta of the company, which reflects systematic business risk alone as the effect of the systematic financial risk of the proxy company is removed by the ungearing process. The asset beta can then be regearred to reflect the systematic financial risk of the investing company, giving an equity beta that reflects the systematic risk of the investment project.

(ACCA F9 December 2016)

9 (a) Net present value of the stopwatch project

The forecast operating profit/(loss) figures in the management accounts are accounting figures, not cash flows, and must be adjusted before they can be discounted. Two non-cash items have been deducted in arriving at these figures and must be added back: the \$50,000/year R&D amortisation charge, and the \$150,000/year machinery depreciation charge. The \$200,000 of R&D expenditure is a sunk cost, already spent before the decision is made, and is irrelevant to the investment decision, it must not be deducted again as a cash flow.

Tax must then be calculated on the resulting operating cash flows at 20%, paid in the same year: this gives a tax inflow (relief) in loss-making years and a tax outflow (cost) in profit-making years. Because the machinery is eligible for a 100% initial (first-year) tax allowance, the whole \$600,000 cost is relieved against tax at once, giving a tax saving of $600,000 \times 20\% = \$120,000$ in Year 1 only, with no further capital allowance adjustments needed in Years 2-4.

WORKING, cash flows (\$000)

	0	1	2	3	4
Machinery	(600)				
Operating profit/(loss) per accounts		(75)	40	110	160
Add back: R&D amortisation		50	50	50	50
Add back: depreciation		150	150	150	150
Operating cash flow		125	240	310	360
Tax at 20% on operating cash flow		(25)	(48)	(62)	(72)
Tax saving, 100% initial allowance (600 x 20%)		120			
Net cash flow	(600)	220	192	248	288
Discount factor at 8%	1.000	0.926	0.857	0.794	0.735
Present value	(600)	204	165	197	212

Net present value = $(600) + 204 + 165 + 197 + 212 = \$178,000$ (positive, on an own-figure basis from the illustrative operating profit/(loss) figures above). As the NPV is positive, the stopwatch project is forecast

to increase shareholder wealth and, on financial grounds, should be accepted, subject to the usual caveats about the reliability of the sales and cost forecasts underlying the management accounts figures.

(b) Optimal investment plan for the \$2.1m portfolio

Because Projects A-E are each divisible, cannot be repeated, and Sulu Co faces a single-period hard capital rationing constraint, the optimal plan is found by ranking projects using the profitability index ($PI = NPV / \text{initial outlay}$), not by NPV alone. Project C is rejected immediately as it has a negative NPV.

Project	Outlay \$000	NPV \$000	PI (NPV/outlay)	Rank
A	900	300	0.333	4th (excluded, see below)
B	850	350	0.412	2nd
C	500	(50)	n/a	Rejected, negative NPV
D	950	400	0.421	1st
E	600	215	0.358	3rd

Projects A and B are mutually exclusive: since B ranks above A, B is selected and A is excluded from the ranking entirely.

Project	Rank	Outlay \$000	Cumulative outlay \$000	NPV \$000
D	1st	950	950	400
B	2nd	850	1,800	350
E, 50%	3rd	300	2,100	107.5
Total		2,100		857.5

Projects D and B are undertaken in full, using \$1,800,000 of the \$2,100,000 available. The remaining \$300,000 funds 50% of divisible Project E, contributing a further $50\% \times \$215,000 = \$107,500$ of NPV. The optimal investment plan is therefore D, B and 50% of E, fully utilising the \$2.1m of available capital and generating a total NPV of \$857,500.

(c) Reasons why Sulu Co may be subject to hard capital rationing

Hard capital rationing arises where the limit on available investment funds is imposed externally by the capital markets, rather than by the company's own management. Two reasons why Sulu Co may face hard capital rationing are as follows.

Lenders may perceive Sulu Co as a higher-risk borrower, for example, if its gearing ratio is above the industry average, and may therefore be unwilling to advance further funds. Higher gearing increases concern over Sulu Co's ability to meet interest payments and repay principal, particularly if profits or cash flows are volatile, so providers of debt finance may cap or refuse additional lending until gearing falls.

Sulu Co may lack sufficient security or collateral to offer against new borrowing, for example, if its existing assets are already pledged against current loans. Without adequate unencumbered asset backing, banks and other lenders are likely to be reluctant to extend further debt finance, restricting the funds available for investment regardless of the underlying quality of Sulu Co's projects.

(d) Usefulness of ROCE to Sulu Co

Return on capital employed (ROCE) has some advantages as an investment appraisal method for Sulu Co. It is easy for non-financial managers and other stakeholders to understand, since it expresses a project's return as a simple percentage. As a relative measure, it allows comparison between projects of different sizes and against industry or company-wide benchmarks. It is also a measure widely used by external analysts and investors, so it may be relevant to how outside parties assess Sulu Co's overall performance.

However, ROCE also has significant disadvantages for Sulu Co. It is based on accounting profits, which are affected by subjective accounting policy choices such as depreciation method, rather than on cash flows, so it does not measure the real economic return of a project as directly as discounted cash flow methods. Critically, ROCE ignores the time value of money, unlike the NPV approach that Sulu Co already applies to its other investment decisions, such as the stopwatch project and the A-E portfolio above, which is a significant weakness given that Sulu Co's projects span several years. ROCE can also be manipulated through accounting choices and does not directly indicate whether shareholder wealth is being maximised.

Overall, because Sulu Co already uses NPV, a technique that incorporates the time value of money and is based on cash flows, for its investment decisions, ROCE's practical usefulness to Sulu Co as a primary appraisal method is limited. It may nonetheless retain a secondary role in performance measurement or in communicating results to non-specialist stakeholders.

Chapter 6: Investment Appraisal Under Uncertainty

Questions

Objective Test Questions

Risk and uncertainty

1 Which of the following limitations is common to the calculations of payback period, discounted payback, internal rate of return and net present value?

- A They do not consider the time value of money
- B They require multiple trial and error calculations
- C They require knowledge of a company's cost of capital
- D They rely on the forecasting of future data

2 Ridag Co is appraising a project with a positive NPV. It is concerned about the risk and uncertainty associated with the project.

Which of the following statements about risk, uncertainty and the project is true?

- A Sensitivity analysis takes into account the interrelationship between project variables
- B Probability analysis can be used to assess the uncertainty associated with the project
- C Uncertainty can be said to increase with project life, while risk increases with the variability of returns
- D A discount rate of 5% could be used to lessen the effect of later cash flows on the decision

Sensitivity analysis

3 The NPV of a project is \$56,000. The PV of revenue is \$380,000, the PV of variable costs is \$240,000 and the PV of fixed costs is \$80,000.

What is the sensitivity of the project to changes in sales volume?

- A 14.7%
- B 23.3%
- C 40.0%
- D 93.3%

4 A project has the following cash flows. At the company's cost of capital of 10% the NPV of the project is \$16,800.

Item	\$
Time 0 investment	110,000
Times 1-4 net cash inflow (per year)	40,000

Applying sensitivity analysis to the cost of capital, what percentage change in the cost of capital would cause the project NPV to fall to zero?

- A 70%
- B 17%
- C 5%
- D 4%

5 A company has calculated the NPV of a new project as follows (\$000):

Item	\$000
Sales revenue	4,000
Variable costs	(2,000)
Fixed costs	(500)
Corporation tax at 20%	(300)
Initial outlay	(1,000)
NPV	200

What is the sensitivity of the project decision to a change in sales volume?

- A 12.5%
- B 6.3%
- C 10.0%
- D 5.0%

6 The following financial information relates to an investment project:

Item	\$
PV of sales revenue	50,025,000
PV of variable costs	25,475,000
PV of contribution	24,550,000
PV of fixed costs	18,250,000
PV of operating income	6,300,000
Initial investment	5,000,000
NPV	1,300,000

What is the sensitivity of the NPV to a change in sales volume?

- A 7.1%
- B 2.6%
- C 5.1%
- D 5.3%

7 The following financial information relates to an investment project:

Item	\$
PV of sales revenue	50,025,000
PV of variable costs	25,475,000
PV of contribution	24,550,000
PV of fixed costs	18,250,000
PV of operating income	6,300,000
Initial investment	5,000,000
NPV	1,300,000

What is the sensitivity of the net present value of the investment project to a change in fixed costs?

- A 7.1%
- B 5.3%
- C 5.1%
- D 2.6%

8 A company is considering a project with the following cash flows.

Year	Initial investment \$000	Variable costs \$000	Cash inflows \$000	Net cash flows \$000
0	(11,000)			(11,000)
1		(3,200)	10,300	7,100
2		(3,200)	10,300	7,100

Cash flows arise from selling 1,030,000 units at \$10 per unit. The company has a cost of capital of 9%. The net present value (NPV) of the project is \$1,490,000.

What is the sensitivity of the project to changes in sales volume (to one decimal place)?

- A 8.2%
- B 8.4%
- C 11.9%
- D 26.5%

9 Link Co has decided to expand its existing business operations. Financial information relating to the expansion project is as follows (project life four years, discount rate 10%):

Item	\$
PV of contribution	3,880,000
Initial capital investment	1,800,000
Net present value	511,000

Which of the following statements relating to sensitivity analysis is/are correct?

- (1) Although critical factors may be identified, the management of Link Co may have no control over them.
- (2) A weakness of sensitivity analysis is that it ignores interdependency between project variables.
- (3) Sensitivity analysis can be used by Link Co to assess the risk of an investment project.

- A 1 and 2 only
- B 1 only
- C 2 and 3 only
- D 1, 2 and 3

10 Which of the following statements about sensitivity analysis are true or false?

Statement	True	False
Sensitivity analysis considers each project variable in isolation, one at a time		
Sensitivity analysis assesses the probability that a critical variable will actually change		

Statement	True	False
Sensitivity analysis can identify the key or critical variables in an investment project		

Probability analysis

11 Excalibur Co has developed a model to predict sales levels for its beachwear based on long-range weather forecasts. The probability of various temperatures and related sales units are as follows.

Unit sales	Temperature	Probability
10,000	Below 20°	5%
30,000	20-24°	25%
50,000	24-28°	50%
40,000	28-32°	15%
25,000	Over 32°	5%

What is Excalibur Co's expected level of sales?

- A 31,000 units
- B 40,250 units
- C 50,000 units
- D 155,000 units

12 Dough Co has decided to increase its daily purchases of doughnuts by 100 boxes. A box of doughnuts costs \$2 and sells for \$3 in normal shops. Any boxes not sold in normal shops are sold through Dough Co's economy shop for \$1. Dough Co estimates the following probabilities of selling the additional boxes.

Normal shop sales	Economy shop sales	Probability
60	40	0.6
100	0	0.4

What is the expected value of Dough Co's decision to buy 100 additional boxes of doughnuts?

- A \$28
- B \$40
- C \$52
- D \$68

13 The annual sales volume and the contribution per unit of a new product are uncertain. Estimates for these two variables are as follows.

Sales volume (units)	Probability	Contribution	Probability
80,000	0.1	\$2.00	0.5
75,000	0.6	\$1.50	0.5
50,000	0.3		

The sales volume and contribution per unit can be assumed to be independent. If the annual fixed costs are \$130,000, what is the probability that the company will make a loss?

- A 0.30
- B 0.50
- C 0.65
- D 1.00

14 An investment project has a cost of \$12,000, payable at the start of the first year of operation. The possible future cash flows arising from the investment project have the following present values and associated probabilities.

PV of Year 1 cash flow \$	Probability	PV of Year 2 cash flow \$	Probability
16,000	0.15	20,000	0.75
12,000	0.60	(2,000)	0.25
(4,000)	0.25		

What is the expected value of the net present value of the investment project?

- A \$11,100
- B \$11,850
- C \$28,700
- D \$76,300

15 Care Co is replacing office equipment. Machine 2 has a scrap value of \$120,000 as used in the original investment appraisal. It is now felt that the final scrap value of the machine depends on two factors: whether a new supplier enters the market (which would reduce the likely scrap value) and the strength of the dollar against other currencies (since sales of used machines will be made abroad). Adverse effects will each reduce the scrap value by 10% of the figure used in the original appraisal.

New supplier	Probability	Strong \$	Probability
Yes	0.4	Yes	0.3
No	0.6	No	0.7

What is the expected value of the scrap proceeds from Machine 2?

- A \$106,800
- B \$109,000
- C \$111,600
- D \$113,000

Further techniques for adjusting for risk and uncertainty

16 Indicate whether the following statements about a simulation exercise are true or false.

Statement	True	False
It considers the effect of changing one variable at a time		
It considers the impact of many variables changing at the same time		
It points directly to the correct investment decision		
It assesses the likelihood of a variable changing		

17 When considering risk in project appraisal, what is the main advantage of using simulation to assist the appraisal?

- A A clear decision rule
- B More than one variable can change at a time
- C Statistically more accurate than other methods
- D Being diagrammatic, it is easier to understand

18 To overcome the difficulties of incorporating probabilities into the investment appraisal calculations, one of Care Co's directors suggests performing a simulation exercise to help reach a decision.

Indicate whether the following statements about simulation are true or false.

Statement	True	False
It eliminates the effects of risk associated with various estimates		
It requires probabilities of estimates subject to risk to be known		

19 Which of the following methods can deal adequately with project risk?

- (1) Sensitivity analysis
 - (2) Simulation models
 - (3) Probability analysis
- A 1 only
 - B 1 and 2 only
 - C 2 and 3 only
 - D 1, 2 and 3

20 Which of the following statements is true?

- A The sensitivity of NPV to a change in sales volume can be calculated as NPV divided by the present value of future sales income
- B The certainty equivalent approach converts risky cash flows into riskless equivalent amounts which are discounted by a CAPM-derived project-specific cost of capital
- C Using random numbers to generate possible values of project variables, a simulation model can generate a standard deviation of expected project outcomes
- D The problem with risk and uncertainty in investment appraisal is that neither can be quantified or measured

21 A company has a cost of capital of 8% and is appraising a project with the following cash flows:

Item	\$
Time 0 investment	100,000
Times 1-5 net cash inflow (per year)	40,000

What is the discounted payback period for this project?

- A 2.5 years
- B Just under 3 years
- C 2 years
- D Just over 4 years

22 Blane Co purchases a new machine for \$340,000. The machine is expected to increase annual cash flows by \$110,000 per year for the next four years. The appropriate discount rate is 4%.

Using the discounted payback period method, approximately how many years will it take for Blane Co to recover its investment?

- A 3.09 years
- B 3.16 years
- C 3.37 years
- D 3.58 years

23 A company with an 8% cost of capital purchases a machine for \$43,000. The forecast operating cash flows generated by the machine are as follows:

Year	1	2	3	4
Cash flow	\$10,000	\$15,000	\$20,000	\$27,000

What is the machine's discounted payback period?

- A 2.90 years
- B 3.10 years
- C 3.14 years
- D 3.25 years

24 A company has the following pattern of cash flow for a project. The company uses a discount rate of 10%.

Time/Year	0	1	2	3	4	5
Cash flow	\$(100,000)	\$40,000	\$20,000	\$30,000	\$5,000	\$40,000

In what year does discounted payback occur?

- A Year 3
- B Year 4
- C Year 5
- D Payback does not occur

25 A company is considering a project with the following cash flows.

Year	Initial investment \$000	Variable costs \$000	Cash inflows \$000	Net cash flows \$000
0	(11,000)			(11,000)
1		(3,200)	10,300	7,100
2		(3,200)	10,300	7,100

Cash flows arise from selling 1,030,000 units at \$10 per unit. The company has a cost of capital of 9%.

What is the discounted payback of the project?

- A 1.18 years
- B 1.25 years
- C 1.55 years
- D 1.75 years

26 Leah Co is an all-equity financed company which wishes to appraise a project in a new area of business. Leah Co's existing equity beta is 1.2. The average equity beta for companies in the new business area is 2.0, with an average debt/(debt plus equity) ratio of 25%. The risk-free rate of return is 5% and the market risk premium is 4%.

Ignoring taxation and using the capital asset pricing model, what is Leah Co's risk-adjusted cost of equity for the new project?

- A 8.6%
- B 9.8%
- C 11.0%
- D 13.0%

27 Shyma Co manufactures ships and has an equity beta of 1.6 and a debt:equity ratio of 1:3. It is considering a new project to manufacture farm vehicles. Trant Co is a manufacturer of farm vehicles and has an asset beta of 1.1 and a debt:equity ratio of 2:3. The risk-free rate of return is 5%, the market risk premium is 3% and the corporation tax rate is 40%.

Using the capital asset pricing model, what would be the suitable cost of equity for Shyma Co to use in its appraisal of the farm vehicle project (to one decimal place)?

%

28 Fresh Co operates a chain of supermarkets. It has an equity beta of 0.85 and is financed 30:70 debt to equity by market values. It is considering a project to offer financial services to its customers. The average equity beta of financial services companies is 1.65 and their average gearing ratio is 45:55 debt to equity by market values. The risk-free rate is 3%, the market premium is 5% and the corporate tax rate is 30%.

What cost of equity should Fresh Co use when calculating a discount rate for the financial services project?

- A 7.250%
- B 11.255%
- C 8.250%
- D 9.825%

29 Are the following statements true or false?

Statement	True	False
The sensitivity of a project variable can be calculated by dividing the project net present value by the present value of the cash flows relating to that project variable		
The expected net present value is the value expected to occur if an investment project with several possible outcomes is undertaken once		

Statement	True	False
The discounted payback period is the time taken for the cumulative net present value to change from negative to positive		

Constructed Response Questions

1 Vyxyn Co is evaluating a planned investment in a new product costing \$20m, payable at the start of the first year of operation. The product will be produced for four years, at the end of which production will cease. The investment project will have a terminal value of zero. Financial information relating to the investment project is as follows.

Year	1	2	3	4
Sales volume (units/year)	440,000	550,000	720,000	400,000
Selling price (\$/unit)	26.50	28.50	30.00	26.00
Fixed cost (\$/year)	1,100,000	1,121,000	1,155,000	1,200,000

These selling prices have not yet been adjusted for selling price inflation, expected to be 3.5% per year. The annual fixed costs given above are already in nominal terms. Variable cost per unit depends on whether competition is maintained between suppliers of a key component.

Competition	Strong	Moderate	Weak
Probability	45%	35%	20%
Variable cost (\$/unit)	10.80	12.00	14.70

The variable costs above are before variable cost inflation of 4.0% per year. Vyxyn Co can claim tax-allowable depreciation on a 25% per year reducing-balance basis on the full \$20m investment cost, and pays corporation tax of 28% per year one year in arrears. The investment is financed by 8% loan notes, redeemable in ten years. Vyxyn Co has a nominal after-tax weighted average cost of capital of 10%, a real after-tax weighted average cost of capital of 7%, and a cost of equity of 11%.

Required:

- Discuss the difference between risk and uncertainty in relation to investment appraisal. (3 marks)
- Calculate the expected net present value of the investment project and comment on its financial acceptability and on the risk relating to variable cost. (9 marks)
- Critically discuss how risk can be considered in the investment appraisal process. (8 marks)

(Total = 20 marks)

(ACCA FM March/June 2017)

2 Copper Co is concerned about the risk associated with a proposed investment and is looking for ways to incorporate risk into its investment appraisal process. The company has heard that probability analysis may be useful in this respect, and so the following information relating to the proposed investment has been prepared.

Year 1 cash flow \$	Probability	Year 2 cash flow \$	Probability
1,000,000	0.1	2,000,000	0.3
2,000,000	0.5	3,000,000	0.6
3,000,000	0.4	5,000,000	0.1

However, the company is not sure how to interpret the results of an investment appraisal based on probability analysis. The proposed investment will cost \$3.5m, payable in full at the start of the first year of operation. Copper Co uses a discount rate of 12% in investment appraisal.

Required:

(a) Using a joint probability table:

- (i) Calculate the mean (expected) NPV of the proposed investment; (8 marks)
- (ii) Calculate the probability of the investment having a negative NPV; (1 mark)
- (iii) Calculate the NPV of the most likely outcome; (1 mark)
- (iv) Comment on the financial acceptability of the proposed investment. (2 marks)

(b) Discuss TWO of the following methods of adjusting for risk and uncertainty in investment appraisal: (8 marks)

- (i) Simulation.
- (ii) Adjusted payback.
- (iii) Risk-adjusted discount rates.

(Total = 20 marks)*(ACCA FM March/June 2018)*

3 The directors of Pelta Co are considering a planned investment project costing \$25m, payable at the start of the first year of operation. The following information relates to the investment project.

Year	1	2	3	4
Sales volume (units/year)	520,000	624,000	717,000	788,000
Selling price (\$/unit)	30.00	30.00	30.00	30.00
Variable cost (\$/unit)	10.00	10.20	10.61	10.93
Fixed cost (\$/year)	700,000	735,000	779,000	841,000

This information needs adjusting to take account of selling price inflation of 4% per year and variable cost inflation of 3% per year. The fixed costs, which are incremental and related to the investment project, are already in nominal terms. The Year 4 sales volume is expected to continue for the foreseeable future.

Pelta Co pays corporation tax of 30% one year in arrears. The company can claim tax-allowable depreciation on a 25% reducing-balance basis.

The directors of Pelta Co believe that all investment projects must be evaluated over four years of operations, with an assumed terminal value at the end of the fourth year of 5% of the initial investment cost. Both net present value and discounted payback must be used, with a maximum discounted payback period of two years. The real after-tax cost of capital of Pelta Co is 7% and its nominal after-tax cost of capital is 12%.

Required:

(a) (i) Calculate the net present value of the planned investment project. (9 marks)

(ii) Calculate the discounted payback period of the planned investment project. (2 marks)

(b) Discuss the financial acceptability of the investment project. (3 marks)

(c) Critically discuss the views of the directors of Pelta Co on the company's investment appraisal. (6 marks)

(Total = 20 marks)*(ACCA FM September/December 2017)*

4 Degniss Co is a company which installs kitchens and bathrooms to customer specifications. It is planning to invest \$4,000,000 in a new facility to convert vans and trucks into motorhomes. Each motorhome will be designed and built according to customer requirements. Degniss Co expects motorhome production and sales in the first four years of operation to be as follows.

Year	1	2	3	4
Motorhomes produced and sold	250	300	450	450

The selling price for a motorhome depends on the van or truck which is converted, the quality of the units installed and the extent of conversion work required. Degniss Co has undertaken research into the likely sales and costs of the different kinds of motorhome which could be selected by customers, as follows.

Motorhome type	Basic	Standard	Deluxe
Probability of selection	20%	45%	35%
Selling price (\$/unit)	30,000	42,000	72,000
Conversion cost (\$/unit)	23,000	29,000	40,000

Fixed costs of the production facility are expected to depend on the volume of motorhome production.

Production volume (units/year)	200-299	300-399	400-499
Fixed costs (\$000/year)	4,000	5,000	5,500

Degniss Co pays corporation tax of 28% per year, settled in the year in which it arises. Tax-allowable depreciation on the cost of the investment is available on a straight-line basis over ten years. Degniss Co has an after-tax discount rate of 11% per year.

Required:

- Calculate the expected net present value of the planned investment for the first four years of operation. (7 marks)
- After the fourth year of operation, Degniss Co expects to continue to produce and sell 450 motorhomes per year for the foreseeable future. Calculate the effect on the expected net present value of the planned investment of continuing to produce and sell motorhomes beyond the first four years, and comment on the financial acceptability of the planned investment. (3 marks)
- Critically discuss the use of probability analysis in incorporating risk into investment appraisal. (5 marks)

(Total = 15 marks)

(ACCA FM March/June 2016)

5 Pulorough Co produces household electrical goods for its home country market and for markets in neighbouring countries. Pulorough Co is planning to make a major investment in new production facilities in order to produce an upgraded model of its cooker. Pulorough Co's finance director has assigned probabilities to a number of possible sales volumes of this new model, based on Pulorough Co's previous experience of launching product upgrades. He has calculated annual expected values for sales for the four-year time period over which the investment will be assessed.

However, after a number of years of strong economic growth, Pulorough Co's home country and its neighbours appear likely to enter a period of recession, meaning that consumers may be less willing to spend money on new kitchen goods.

The initial investment cost will be \$39m. The finance director has assumed that there will be zero realisable value at the end of the four years. Working capital can be ignored. The expected volumes of sales are:

Year	1	2	3	4
Expected sales volume (units)	60,000	150,000	140,000	80,000

The expected contribution from the upgraded cooker throughout the four years is \$200 per cooker. This will not be affected by the rate of inflation.

Incremental fixed costs, excluding depreciation, are expected to be \$2.5m in Year 1. They will increase by the rate of inflation in each of Years 2 to 4, which are predicted to be as follows:

Year	2	3	4
Inflation rate	6%	4%	3%

Pulorough Co pays corporation tax of 25% per year one year in arrears. No tax-allowable depreciation is available on the investment expenditure. Pulorough Co currently uses 11% as the after-tax discount rate to appraise investments like this.

One of the other directors has commented that he thinks the level of contribution that the finance director has assumed is too high. He also wondered that, with the economic uncertainty, the cost of capital should be higher. The finance director has said that he will prepare sensitivity calculations, showing in relative terms the level of change in contribution and discount rate that would produce a nil net present value. He will assume that the contribution will be lower by the same percentage each year.

Another director has stated that she has heard that probability analysis can be helpful in decision-making, and wonders if this will be the same as the sensitivity analysis that the finance director is going to prepare.

Required:

- (a) Discuss the difference between risk and uncertainty in the context of investment decisions. (3 marks)
- (b) (i) Calculate the nominal net present value of the investment project and comment on its financial acceptability. (5 marks)
- (ii) Calculate the sensitivity of the project's net present value to a change in the level of contribution per unit; and (3 marks)
- (iii) Calculate the sensitivity of the project's net present value to a change in the discount rate. (3 marks)
- (c) Compare and contrast sensitivity analysis and probability analysis as methods of assisting investment decisions when there is more than one possible outcome. (6 marks)

(Total = 20 marks)*(ACCA FM September/December 2023)*

6 Galle Co is a company which manufactures car batteries. Its latest project is a new battery called the G-Power Excel (GPE) which is a variation on an existing battery but uses less harmful materials in its construction. Due to technological advances, the GPE project is expected to have a life of only four years.

Sales for the first year are expected to be 10,000 units, but the remaining three years are more uncertain due to the risk of similar products being launched by competitors. Galle Co expects that sales volume will remain constant at 10,000 units per year for the final three years, but there is a 30% probability that volume could drop to only 5,000 units per year for Years 2, 3 and 4.

The GPE has a selling price in current-price terms of \$70 per unit, but sales price inflation is expected to be 4.5% per year. The variable cost per unit is \$45 in current-price terms but variable cost inflation is expected to be 3% per year. Fixed costs will remain constant at \$10,000 per year.

Manufacturing equipment costing \$500,000 will be purchased at the start of the first year of production. The equipment will qualify for tax-allowable depreciation at 25% on a reducing balance basis and it is anticipated that the equipment will have zero residual value at the end of four years.

Working capital of \$80,000 will be required at the start of the first year of production and working capital inflation is expected to be 3% per year. Working capital needs to be in place at the start of the year to which it relates and will be released at the end of the project.

Galle Co pays corporation tax at a rate of 20% and tax is paid at the end of the year to which it relates. Galle Co is profitable and can utilise any unused tax-allowable depreciation. Galle Co uses a nominal after-tax cost of capital of 13% for appraising projects of this nature.

Required:

- (a) (i) Calculate the expected net present value; and (10 marks)
- (ii) Comment on its financial acceptability. (2 marks)
- (b) Discuss FOUR techniques for dealing with risk and uncertainty in an investment appraisal. (8 marks)

(Total = 20 marks)*(ACCA FM September/December 2024)*

Answers

Objective Test Questions

1 D

The common disadvantage of all capital budgeting models is their reliance on forecast data. Capital investment decisions relate to longer periods of time that are subject to greater levels of uncertainty than short-term budgeting decisions.

2 C

Uncertainty can be said to increase with project life, while risk increases with the variability of returns; this is the standard distinction between the two terms.

Notes on incorrect answers: simulation (not sensitivity analysis) takes into account the interrelationship between project variables, since sensitivity analysis only changes one variable at a time. Probability analysis can be used to assess the risk (not the uncertainty) associated with a project, since uncertainty by definition cannot be quantified. A lower discount rate of 5% would increase the present value of costs incurred in later years and would therefore increase, not lessen, their effect on the decision.

3 C

WORKING

The sensitivity of a variable = $(NPV/PV \text{ of the relevant cash flow}) \times 100$. The relevant cash flow for sales volume is contribution (revenue less variable costs). $PV \text{ of contribution} = \$380,000 - \$240,000 = \$140,000$. Sensitivity = $\$56,000/\$140,000 \times 100 = 40\%$.

4 A

WORKING

To force the NPV to zero, the 4-year annuity factor needed is $AF_4 = \$110,000/\$40,000 = 2.75$. From annuity tables, the 4-year annuity factor closest to 2.75 is 2.743, corresponding to a discount rate of 17%. The sensitivity of the decision to the cost of capital = $(17 - 10)/10 \times 100 = 70\%$. The cost of capital can therefore increase by 70% (from 10% to 17%) before the project's NPV becomes negative.

5 A

WORKING

A change in sales volume affects both sales revenue and variable costs, together, contribution. Because the tax liability also changes with sales volume, the relevant cash flow is after-tax contribution: $(\$4,000,000 - \$2,000,000) \times (1 - 0.20) = \$1,600,000$. Sensitivity = $\$200,000/\$1,600,000 \times 100 = 12.5\%$.

6 D

WORKING

A change in sales volume affects both the sales revenue and variable cost cash flows, which together make up contribution: $\$50,025,000 - \$25,475,000 = \$24,550,000$ (as already given). Sensitivity to sales volume = $\$1,300,000/\$24,550,000 \times 100 = 5.3\%$.

7 A

WORKING

Net present value (\$6,300,000 - \$5,000,000) = \$1,300,000. The sensitivity to a change in fixed costs = \$1,300,000/\$18,250,000 = 7.1%.

8 C

WORKING

Year	Contribution \$000	Discount factor at 9%	PV \$000
1-2	7,100	1.759	12,489

Contribution = \$10,300,000 - \$3,200,000 = \$7,100,000 per year. Sensitivity of the project to sales volume = \$1,490,000/\$12,489,000 x 100% = 11.9%.

9 A

Statements 1 and 2 are correct. A critical factor may be identified by sensitivity analysis, such as the level of sales volume, but management may have no control over it. Sensitivity analysis only looks at the effect of changing one variable at a time, so it necessarily ignores that a change in one variable may be linked to changes in others; this is a genuine weakness of the technique. Statement 3 is incorrect: sensitivity analysis does not assess the risk of a project (probability analysis does); it only identifies which variables are critical, without saying how likely a change in those variables actually is.

Question 10

Statement	True	False
Sensitivity analysis considers each project variable in isolation, one at a time	✓	
Sensitivity analysis assesses the probability that a critical variable will actually change		✓
Sensitivity analysis can identify the key or critical variables in an investment project	✓	

Sensitivity analysis calculates, for each project variable in turn, the change needed to make the NPV zero (or the change in NPV from a given change in the variable), so it does consider variables one at a time, and it does identify the key or critical variables (those needing the smallest change to turn the decision). However, it gives no indication of how likely such a change actually is: assessing probability is the role of probability analysis, not sensitivity analysis.

11 B

WORKING

Unit sales	Probability	Expected value
10,000	0.05	500
30,000	0.25	7,500
50,000	0.50	25,000
40,000	0.15	6,000
25,000	0.05	1,250
Total		40,250

12 C**WORKING**

There is a 60% probability of a profit of \$20 (60 boxes at a \$1 profit through the normal shops, less 40 boxes at a \$1 loss through the economy shop). There is a 40% probability of a profit of \$100 (100 boxes at a \$1 profit through the normal shops). Expected value = $(0.6 \times \$20) + (0.4 \times \$100) = \$12 + \$40 = \$52$.

13 C**WORKING**

Sales volume	Contribution	Profit/(loss) \$000
80,000	2.00	30.0
80,000	1.50	(10.0)
75,000	2.00	20.0
75,000	1.50	(17.5)
50,000	2.00	(30.0)
50,000	1.50	(55.0)

Probability of a loss = $P(80,000 \text{ units, } \$1.50 \text{ contribution}) + P(75,000 \text{ units, } \$1.50 \text{ contribution}) + P(50,000 \text{ units, either contribution}) = (0.1 \times 0.5) + (0.6 \times 0.5) + 0.3 = 0.05 + 0.30 + 0.30 = 0.65$.

14 A**WORKING**

EV of PV of Year 1 cash flow = $(\$16,000 \times 0.15) + (\$12,000 \times 0.60) - (\$4,000 \times 0.25) = \$2,400 + \$7,200 - \$1,000 = \$8,600$.

EV of PV of Year 2 cash flow = $(\$20,000 \times 0.75) - (\$2,000 \times 0.25) = \$15,000 - \$500 = \$14,500$.

EV of NPV = $-\$12,000 + \$8,600 + \$14,500 = \$11,100$.

15 C**WORKING**

Care Co can expect to receive the full \$120,000 if neither adverse factor occurs (probability $0.6 \times 0.7 = 0.42$). It can expect to receive \$108,000 (a single 10% reduction) if exactly one adverse factor occurs (probability = $(0.4 \times 0.7) + (0.6 \times 0.3) = 0.28 + 0.18 = 0.46$). It can expect to receive only \$96,000 (a 20% reduction) if both adverse factors occur (probability $0.4 \times 0.3 = 0.12$).

Scrap value	Probability	Expected value
\$120,000	0.42	50,400
\$108,000	0.46	49,680
\$96,000	0.12	11,520
Total	1.00	111,600

Question 16

Statement	True	False
It considers the effect of changing one variable at a time		✓
It considers the impact of many variables changing at the same time	✓	
It points directly to the correct investment decision		✓

Statement	True	False
It assesses the likelihood of a variable changing		✓

It is sensitivity analysis, not simulation, that considers the effect of changing one variable at a time, simulation's advantage over sensitivity analysis is that it considers many variables changing together. Simulation gives management more information about the range and likelihood of possible outcomes, but it does not point directly to the correct investment decision, nor does it itself assess the likelihood of any one variable changing (the probability distributions of the individual variables must be supplied as an input to the model, not generated by it).

17 B

A clear decision rule is not provided by simulation. Simulation's real advantage over sensitivity analysis is that more than one variable can change at a time, giving a more realistic picture of a project's risk. Simulation is not statistically more accurate than other methods, the input variables and their probability distributions are still only estimates. Simulation output can be presented diagrammatically, but this is a secondary benefit, not its main advantage.

Question 18

Statement	True	False
It eliminates the effects of risk associated with various estimates		✓
It requires probabilities of estimates subject to risk to be known	✓	

Simulation does not eliminate the effects of risk; it merely helps quantify those effects, by producing a range of possible NPVs and an indication of their probability distribution. To do this, it does require the probabilities of the risky estimates to be known (or at least estimated) in advance, as an input to the model.

19 C

Simulation models and probability analysis can both deal adequately with project risk, because both incorporate the probability of a variable changing. Sensitivity analysis cannot deal adequately with project risk on its own, because it gives no indication of the probability that a project variable will actually change by a significant amount; it only shows the effect if it did.

20 C

Using random numbers to generate possible values of project variables, a simulation model can generate a standard deviation of expected project outcomes, this is true. The sensitivity of NPV to a change in sales volume should be calculated as NPV divided by the present value of contribution, not future sales income (a change in sales volume affects variable costs as well as revenue). Under the certainty equivalent approach, the riskless equivalent cash flows are discounted at the risk-free rate of return, not a CAPM-derived project-specific rate (risk is removed from the cash flows themselves, rather than reflected in the discount rate). Risk is distinguishable from uncertainty precisely because risk CAN be quantified, even though uncertainty cannot.

21 B

WORKING

Year	Cash flow	DF at 8%	Discounted cash flow	Cumulative discounted cash flow
0	(100,000)	1.000	(100,000)	(100,000)
1	40,000	0.926	37,040	(62,960)
2	40,000	0.857	34,280	(28,680)
3	40,000	0.794	31,760	3,080

Discounted payback occurs during Year 3, since the cumulative discounted cash flow turns positive that year ($28,680/31,760 = 90\%$ of the way through it), i.e. just under 3 years.

22 C

WORKING

Year	Cash flow	DF at 4%	PV	Cumulative PV
1	110,000	0.962	105,820	105,820
2	110,000	0.925	101,750	207,570
3	110,000	0.889	97,790	305,360
4	110,000	0.855	94,050	399,410

After three years, Blane Co has recovered \$305,360 of its \$340,000 investment, leaving \$34,640 to recover. In Year 4 it generates \$94,050, so it needs $\$34,640/\$94,050 = 0.37$ of that year. Discounted payback period = $3 + 0.37 = 3.37$ years.

23 D

WORKING

Year	Cash flow	DF at 8%	PV	Cumulative PV
1	10,000	0.926	9,260	9,260
2	15,000	0.857	12,855	22,115
3	20,000	0.794	15,880	37,995
4	27,000	0.735	19,845	57,840

At the end of Year 3, \$5,005 of the \$43,000 investment remains to be recovered ($\$43,000 - \$37,995$). Year 4 generates \$19,845, so the remaining amount is recovered $\$5,005/\$19,845 = 0.25$ of the way through Year 4. Discounted payback period = $3 + 0.25 = 3.25$ years.

24 C

WORKING

Year	Cash flow	Discount factor at 10%	Present value	Cumulative present value
0	(100,000)	1.000	(100,000)	(100,000)
1	40,000	0.909	36,360	(63,640)
2	20,000	0.826	16,520	(47,120)
3	30,000	0.751	22,530	(24,590)
4	5,000	0.683	3,415	(21,175)
5	40,000	0.621	24,840	3,665

The cumulative present value only becomes positive during Year 5, so discounted payback occurs in Year 5.

25 D

WORKING

Year	Net cash flow \$000	Discount factor at 9%	PV \$000	Cumulative PV \$000
0	(11,000)	1.000	(11,000.00)	(11,000.00)
1	7,100	0.917	6,510.70	(4,489.30)
2	7,100	0.842	5,978.20	

$4,489.30/5,978.20 = 0.75$. Therefore the discounted payback = $1 + 0.75 = 1.75$ years.

26 C

WORKING

Leah Co's own equity beta (1.2) should be ignored, since it reflects the risk of Leah Co's EXISTING business, not the new business area. Instead, the industry average equity beta for the new area (2.0) is used as a proxy, and ungeared to remove the effect of the proxy companies' financial risk: asset beta = $2.0 \times (75/100) = 1.5$ (tax is ignored, so the ungearing formula simplifies to equity beta $\times [E/(D+E)]$). Since Leah Co is all-equity financed, this asset beta does not need to be regearing. Using the CAPM: cost of equity = $5\% + (1.5 \times 4\%) = 5\% + 6\% = 11.0\%$.

27 Answer: 9.0%

WORKING

To reflect the business risk of the new project, Shyma Co's own equity beta (1.6) is ignored, and the proxy beta of Trant Co (1.1) is used instead. Since Trant Co's beta is already an asset beta, it does not need to be ungeared, but it does need to be regearing to reflect Shyma Co's own capital structure (debt:equity 1:3, tax 40%):

$$\text{Equity beta} = 1.1 \times [3 + 1(1 - 0.4)]/3 = 1.1 \times (3.6/3) = 1.1 \times 1.2 = 1.32.$$

Using the CAPM: cost of equity = $5\% + (1.32 \times 3\%) = 5\% + 3.96\% = 8.96\%$, i.e. 9.0% to one decimal place.

28 Answer: 9.825%

WORKING

Fresh Co's own equity beta (0.85) reflects the risk of its existing supermarket business, not the new financial services project, so it is ignored. The average equity beta of financial services companies (1.65) is used instead, as a proxy, and first ungeared to remove the effect of the proxy companies' own financial risk (gearing 45:55 debt:equity, tax 30%):

$$\text{Asset beta} = 1.65 \times [55/(55 + 45(1 - 0.3))] = 1.65 \times (55/86.5) = 1.65 \times 0.636 = 1.05 \text{ (to 2 decimal places).}$$

This asset beta is then regearing to reflect Fresh Co's own capital structure (30:70 debt:equity):

$$\text{Equity beta} = 1.05 \times [(70 + 30(1 - 0.3))/70] = 1.05 \times (91/70) = 1.05 \times 1.3 = 1.365.$$

Using the CAPM: cost of equity = $3\% + (1.365 \times 5\%) = 3\% + 6.825\% = 9.825\%$.

Question 29

Statement	True	False
The sensitivity of a project variable can be calculated by dividing the project net present value by the present value of the cash flows relating to that project variable	✓	

Statement	True	False
The expected net present value is the value expected to occur if an investment project with several possible outcomes is undertaken once		✓
The discounted payback period is the time taken for the cumulative net present value to change from negative to positive	✓	

Statement 2 is false: the expected net present value is the value expected to occur, on average, if an investment project with several possible outcomes is undertaken many times, not once, if the project is only ever undertaken once, only one of the possible individual outcomes will actually occur.

Constructed Response Questions

1 (a) Risk and uncertainty

The terms risk and uncertainty are often used interchangeably in everyday discussion; however, there is a clear difference between them in relation to investment appraisal.

Risk refers to the situation where an investment project has several possible outcomes, all of which are known, and to which probabilities can be attached (for example, on the basis of past experience). Risk can therefore be quantified and measured by the variability of returns of an investment project.

Uncertainty refers to the situation where an investment project has several possible outcomes, but it is not possible to assign probabilities to their occurrence, so it is not possible to say which outcomes are likely to occur.

The difference between risk and uncertainty is therefore that risk can be quantified whereas uncertainty cannot. Risk increases with the variability of returns, while uncertainty increases with project life.

(b) Expected net present value

WORKING

Mean variable cost per unit = $(0.45 \times \$10.80) + (0.35 \times \$12.00) + (0.20 \times \$14.70) = \$4.86 + \$4.20 + \$2.94 = \$12.00$ per unit.

Year	1	2	3	4	5
Sales income (\$000)	12,069	16,791	23,947	11,936	
Variable cost (\$000)	(5,491)	(7,139)	(9,720)	(5,616)	
Contribution (\$000)	6,578	9,652	14,227	6,320	
Fixed cost (\$000)	(1,100)	(1,121)	(1,155)	(1,200)	
Taxable cash flow	5,478	8,531	13,072	5,120	
Taxation at 28%		(1,534)	(2,389)	(3,660)	(1,434)
TAD tax benefits		1,400	1,050	788	2,362
After-tax cash flow	5,478	8,397	11,733	2,248	928
Discount at 10%	0.909	0.826	0.751	0.683	0.621
Present values	4,980	6,936	8,812	1,535	576

PV of future cash flows = \$22,839,000.

Initial investment = \$(20,000,000).

Expected NPV = \$2,839,000.

The expected NPV is positive, so the investment project is likely to be financially acceptable. The probability that variable cost per unit will be \$12.00 per unit or less is 80% (45% + 35%), so the probability of a positive NPV is at least 80%. However, the effect on the NPV of variable cost rising to \$14.70 per unit (a 20% probability) must still be investigated, since this may result in a negative NPV.

(c) How risk can be considered in the investment appraisal process**Sensitivity analysis**

This technique looks at the effect on the NPV of an investment project of changes in project variables, such as selling price per unit, variable cost per unit and sales volume. There are two approaches which are used. The first calculates the relative (percentage) change in a given project variable which is needed to make the NPV zero. The second calculates the relative (percentage) change in project NPV which results from a given change in the value of a project variable (e.g. 5%).

Sensitivity analysis considers each project variable individually. Once the sensitivities for each project variable have been calculated, the next step is to identify the key or critical variables; those where the smallest relative change makes the NPV zero, or where the biggest change in NPV results from a given change in a project variable. The key or critical project variables indicate where underlying assumptions may need to be checked, or where managers may need to focus their attention in order to make an investment project successful. However, as sensitivity analysis does not consider risk as measured by probabilities, it can be argued that it is not really a way of considering risk in investment appraisal at all, even though it is often described as such.

Probability analysis

This technique requires that probabilities for each project outcome (or for different values of individual project variables) be assessed and assigned. A range of project NPVs can then be calculated, along with the mean NPV (the expected NPV, or ENPV) associated with repeating the investment project many times. The worst and best outcomes and their probabilities, the most likely outcome and its probability, and the probability of a negative NPV, can also be calculated. Investment decisions can then be based on the risk profile of the investment project, rather than simply on the NPV decision rule alone.

Risk-adjusted discount rate

The risk associated with an investment project can be incorporated into the discount rate as a risk premium over the risk-free rate of return. The risk premium can be determined subjectively, for example, by recognising that launching a new product is intrinsically riskier than replacing an existing machine or making a small expansion of existing operations. It can also be determined more theoretically, using the capital asset pricing model: a proxy company's equity beta can be ungeared and the resulting asset beta regearred to reflect the financing risk of the investing company, giving a project-specific equity beta which can be used to find a project-specific cost of equity or discount rate.

Adjusted payback

If risk and uncertainty are treated as being the same, payback can be used to adjust for them by shortening the payback period required as project risk increases, since this places more emphasis on the earlier cash flows, which are less uncertain. Discounted payback can also be seen as an adjusted payback method, since it reflects risk through the discount rate used to discount the project's cash flows before the payback period is calculated.

(ACCA FM March/June 2017)

2 (a) (i) Mean (expected) NPV

WORKING

PV of Year 1 cash flows at 12% (DF 0.893):

Year 1 cash flow	PV
\$1,000,000	\$893,000
\$2,000,000	\$1,786,000
\$3,000,000	\$2,679,000

PV of Year 2 cash flows at 12% (DF 0.797):

Year 2 cash flow	PV
\$2,000,000	\$1,594,000
\$3,000,000	\$2,391,000
\$5,000,000	\$3,985,000

PV Yr 1 (\$000)	PV Yr 2 (\$000)	Joint probability	Total PV (\$000)	NPV (\$000)	EV of NPV (\$000)
893 (0.1)	1,594 (0.3)	0.03	2,487	(1,013)	(30.4)
893 (0.1)	2,391 (0.6)	0.06	3,284	(216)	(13.0)
893 (0.1)	3,985 (0.1)	0.01	4,878	1,378	13.8
1,786 (0.5)	1,594 (0.3)	0.15	3,380	(120)	(18.0)
1,786 (0.5)	2,391 (0.6)	0.30	4,177	677	203.1
1,786 (0.5)	3,985 (0.1)	0.05	5,771	2,271	113.6
2,679 (0.4)	1,594 (0.3)	0.12	4,273	773	92.8
2,679 (0.4)	2,391 (0.6)	0.24	5,070	1,570	376.8
2,679 (0.4)	3,985 (0.1)	0.04	6,664	3,164	126.6

Mean (expected) NPV = sum of (joint probability x NPV) = \$865,000 (approximately).

(a) (ii) Probability of a negative NPV

The three combinations above with a negative NPV have joint probabilities of 0.03, 0.06 and 0.15.

Probability of a negative NPV = $0.03 + 0.06 + 0.15 = 24\%$.

(a) (iii) NPV of the most likely outcome

The combination with the highest joint probability (0.30) is Year 1 cash flow \$2,000,000 and Year 2 cash flow \$3,000,000, giving an NPV of \$677,000.

(a) (iv) Financial acceptability

The mean (expected) NPV is positive, which might suggest the proposed investment is financially acceptable. However, the mean (expected) NPV is not a value expected to occur from undertaking the investment once, but a mean value that would arise from undertaking it many times, and there is no clear decision rule attached to it on its own. A decision on financial acceptability must also consider the risk of a negative NPV: at 24%, this may be considered too high a risk to be acceptable. The final decision will depend on the risk preferences of Copper Co's directors; there is no single decision rule to follow here.

(b) Methods of adjusting for risk and uncertainty (any TWO required)**Simulation**

Simulation is a computer-based method of evaluating an investment project whereby the probability distributions associated with individual project variables, and the interdependencies between them, are incorporated. Random numbers are assigned to a range of different values of a project variable to reflect its probability distribution. Each simulation run randomly selects values of the project variables using random numbers and calculates a mean (expected) NPV. A picture of the probability distribution of the mean (expected) NPV is built up from the results of many repeated simulation runs, from which project risk can be assessed, via the standard deviation of expected returns, the most likely outcome, and the probability of a negative NPV.

Adjusted payback

If risk and uncertainty are treated as the same, payback can be used to adjust for them in investment appraisal. As uncertainty (risk) increases, the payback period required can be shortened, to increase the emphasis on cash flows nearer to the present (and hence less uncertain); as uncertainty decreases, the required payback period can be lengthened. Discounted payback adjusts for risk in a similar way, since risk is reflected in the discount rate used to discount the cash flows before payback is assessed, so discounted payback can be seen as an adjusted payback method. A weakness of this approach is that it does not consider cash flows that lie outside the payback period.

Risk-adjusted discount rates

The risk associated with an investment project can be incorporated into the discount rate as a risk premium over the risk-free rate of return. This premium can be determined subjectively, for example, recognising that launching a new product is intrinsically riskier than replacing an existing machine. It can also be determined more theoretically using the capital asset pricing model: a proxy company's equity beta can be ungeared and the resulting asset beta regearred to reflect the financial risk of the investing company, giving a project-specific equity beta which can be used to find a project-specific cost of equity or discount rate.

(ACCA FM March/June 2018)

3 (a) (i) Net present value**WORKING**

Year	1	2	3	4	5
Sales income (\$000)	16,224	20,248	24,196	27,655	
Variable costs	(5,356)	(6,752)	(8,313)	(9,694)	
Contribution	10,868	13,495	15,883	17,962	
Fixed costs	(700)	(735)	(779)	(841)	
Cash flow before tax	10,168	12,760	15,104	17,121	
Corporation tax		(3,050)	(3,828)	(4,531)	(5,136)
TAD tax benefits		1,875	1,406	1,055	2,789
After-tax cash flow	10,168	11,585	12,682	13,644	(2,347)
Terminal value				1,250	
Project cash flow	10,168	11,585	12,682	14,894	(2,347)
Discount at 12%	0.893	0.797	0.712	0.636	0.567
Present values	9,080	9,233	9,030	9,473	(1,331)

PV of future cash flows = \$35,485,000.

Initial investment = \$(25,000,000).

NPV = \$10,485,000.

(a) (ii) Discounted payback period

Year	1	2	3	4	5
Present values (\$000)	9,080	9,233	9,030	9,473	(1,331)
Cumulative NPV (\$000)	(15,920)	(6,687)	2,343	11,815	10,485

Discounted payback occurs during Year 3, $6,687/9,030 = 74\%$ of the way through the year. Discounted payback period = $2 + 0.74 =$ approximately 2.7 years.

(b) Financial acceptability

The investment project is financially acceptable under the NPV decision rule, since it has a substantial positive NPV. The discounted payback period of 2.7 years is greater than the directors' maximum target of two years, so from that perspective the project is not financially acceptable. However, the correct advice is given by the NPV method, and so the investment project is financially acceptable.

(c) Critical discussion of the directors' views**Evaluation period**

Sales are expected to continue beyond Year 4, so the directors' view that all investment projects must be evaluated over four years of operations does not seem sensible; it ignores relevant cash flows that would otherwise contribute positively to the project's value, and may lead to profitable projects with longer lives being rejected.

Assumed terminal value

The assumption that the project has a terminal value at the end of Year 4 of 5% of the initial investment cost has no factual or analytical basis. If a terminal value is to be used at all, it should be based on a realistic estimate of the project's actual resale or scrap value at that time, not an arbitrary percentage of the original cost.

Discounted payback method

The directors need to explain their view that a project's discounted payback must be no greater than two years. They may believe that an earlier payback indicates a lower-risk project. Although discounted payback does overcome simple payback's failure to account for the time value of money, it still fails to consider cash flows that arise outside the payback period. Theoretically, Pelta Co should rely on the NPV method as its primary investment appraisal tool.

(ACCA FM September/December 2017)

4 (a) Expected net present value (first four years)**WORKING**

Average selling price = $(0.20 \times \$30,000) + (0.45 \times \$42,000) + (0.35 \times \$72,000) = \$6,000 + \$18,900 + \$25,200 = \$50,100$ per unit.

Average conversion cost = $(0.20 \times \$23,000) + (0.45 \times \$29,000) + (0.35 \times \$40,000) = \$4,600 + \$13,050 + \$14,000 = \$31,650$ per unit.

Average contribution = $\$50,100 - \$31,650 = \$18,450$ per unit. Fixed costs each year are read off the production-volume table:

Year	Units	Band	Fixed costs
1	250	200-299	\$4,000,000
2	300	300-399	\$5,000,000
3-4	450	400-499	\$5,500,000 each

Year	1	2	3	4
Sales income (\$000)	12,525	15,030	22,545	22,545
Conversion cost (\$000)	(7,913)	(9,495)	(14,243)	(14,243)
Contribution (\$000)	4,613	5,535	8,303	8,303
Fixed costs (\$000)	(4,000)	(5,000)	(5,500)	(5,500)
Before-tax cash flow	613	535	2,803	2,803
Corporation tax at 28%	(172)	(150)	(785)	(785)
TAD tax benefits	112	112	112	112
After-tax cash flow	553	497	2,129	2,129
Discount at 11%	0.901	0.812	0.731	0.659
Present values	498	404	1,556	1,403

Tax-allowable depreciation = \$4,000,000/10 years = \$400,000 per year; tax benefit = \$400,000 x 0.28 = \$112,000 per year (tax is settled in the year it arises, so both the tax charge and the TAD benefit fall in the same year as the related cash flow).

Sum of present values = \$3,861,000.

Initial investment = \$(4,000,000).

NPV (first four years only) = \$(139,000).

(b) Effect of continuing beyond Year 4

If, ignoring tax-allowable depreciation, sales and costs continue at the Year 4 level indefinitely, the after-tax cash flow from Year 5 onwards is \$2,803,000 - \$785,000 = \$2,018,000 per year (approximately \$2,017,000 using the rounded before-tax figure from the table above).

PV of this perpetuity, valued at the start of Year 5 and discounted back to the present = (\$2,017,000/0.11) x 0.659 (the Year 4 discount factor) = \$12,083,664.

A further six years of tax-allowable depreciation benefits (to complete the ten-year straight-line write-off) also continue: PV = \$112,000 x 4.231 (6-year annuity factor at 11%) x 0.659 = \$312,282.

Total increase in NPV from continuing beyond Year 4 = \$12,083,664 + \$312,282 = approximately \$12.4 million.

Even continuing production for just one further year (Year 5 only) would be enough to make the project acceptable: NPV = \$(139,000) + (\$2,129,000 x 0.593) = \$(139,000) + \$1,262,497 = \$1,123,497, which is already positive.

If the evaluation is restricted to four years only, the project has a negative NPV and is not financially acceptable. However, once production and sales beyond Year 4 are taken into account, the NPV becomes strongly positive, and the investment project is clearly financially acceptable.

(c) Critical discussion of probability analysis

Risk in investment appraisal refers to a range of outcomes whose probability of occurrence can be quantified. Risk can therefore be distinguished from uncertainty in investment appraisal, where the likelihood of particular outcomes occurring cannot be quantified.

As regards incorporating risk into investment appraisal, probability analysis can be used to calculate the values of possible outcomes and their probability distribution, the value of the worst possible outcome and its probability, the probability that an investment will generate a positive NPV, the standard deviation of the possible outcomes, and the expected value (mean value) of the NPV. Standard deviation is a measure of risk in financial management.

One difficulty with probability analysis is its assumption that an investment can be repeated a large number of times. The expected value of the NPV is a mean or average value of a number of possible NPVs, while standard deviation is a measure of the dispersal of possible NPVs about that expected (mean) NPV. In reality, many investment projects cannot be repeated, so only one of the possible outcomes will actually occur, the expected (mean) value itself may never actually happen, causing difficulties in applying and interpreting the NPV decision rule when using probability analysis.

Another difficulty with probability analysis is the question of how the probabilities of possible outcomes are assessed and calculated. One method is to consider and analyse the outcomes of similar investment projects from the past, but this relies on the assumption that the past is an acceptable guide to the future. Assessing probabilities in this way is also likely to be a subjective process.

(ACCA FM March/June 2016)

5 (a) Risk and uncertainty

Both risk and uncertainty deal with future situations where there are several possible outcomes arising from a decision to invest.

Analysis of risk is possible in situations where probabilities can be assigned to different outcomes and mathematical techniques can be used for analysis. The probability of each outcome is often determined on the basis of past experience, which is the approach Pulorough Co's finance director has taken when calculating the expected values of sales. Risk increases as the variability of returns increases.

Analysis of uncertainty applies to situations where it is not possible to assign probabilities to different outcomes. The position may be new or very different to what has happened before, so past experience cannot be used to determine probabilities. Uncertainty increases over time.

Because economic conditions may change significantly and uncertainty therefore exists, Pulorough Co's finance director may have taken an incorrect approach when calculating the expected values of sales. Past results may not be relevant as a basis for the probabilities of future sales.

(b) (i) Net present value

WORKING

	1 (\$000)	2 (\$000)	3 (\$000)	4 (\$000)	5 (\$000)
Contribution	12,000	30,000	28,000	16,000	
Fixed costs	(2,500)	(2,650)	(2,756)	(2,839)	
Before-tax cash flow	9,500	27,350	25,244	13,161	
Tax liability at 25%		(2,375)	(6,838)	(6,311)	(3,290)
After-tax cash flow	9,500	24,975	18,406	6,850	(3,290)
Discount at 11%	0.901	0.812	0.731	0.659	0.593
Present values	8,560	20,280	13,455	4,514	(1,951)

Sum of present values = \$44,858,000.

Initial investment = \$39,000,000.

NPV = \$5,858,000.

NPV is positive, therefore the project should be accepted.

(b) (ii) Sensitivity to a change in contribution per unit

	1 (\$000)	2 (\$000)	3 (\$000)	4 (\$000)	5 (\$000)
Contribution	12,000	30,000	28,000	16,000	
Tax liability at 25%		(3,000)	(7,500)	(7,000)	(4,000)
Post-tax contribution	12,000	27,000	20,500	9,000	(4,000)
Discount at 11%	0.901	0.812	0.731	0.659	0.593
Present values	10,812	21,924	14,986	5,931	(2,372)

Discounted post-tax contribution = \$51,281,000. Sensitivity = $(\$5,858,000/\$51,281,000) \times 100\% = 11.4\%$.

(b) (iii) Sensitivity to a change in the discount rate

By interpolation:

	1 (\$000)	2 (\$000)	3 (\$000)	4 (\$000)	5 (\$000)
After-tax cash flow	9,500	24,975	18,406	6,850	(3,290)
Discount at 20%	0.833	0.694	0.579	0.482	0.402
Present values	7,914	17,333	10,657	3,302	(1,323)

Sum of present values at 20% = \$37,883,000.

Initial investment = \$39,000,000.

NPV = \$(1,117,000).

IRR = $11 + [5,858/(5,858 - (1,117))] \times (20 - 11) = 18.6\%$. Sensitivity = $[(18.6 - 11)/11] \times 100\% = 69.1\%$.

(By spreadsheet, calculating the IRR of the after-tax cash flows for Years 0-5 in (b)(i) directly, IRR = 18.4% and sensitivity = $[(18.4 - 11)/11] \times 100\% = 67.3\%$. The small difference between the two figures arises from the rounding involved in the two-point interpolation method above.)

(c) Sensitivity analysis versus probability analysis**Description of methods**

Sensitivity analysis identifies the percentage change in a variable, or combination of variables, that would be needed for outcomes to change the decision, for example, for a decision to proceed with an investment project to turn into a decision to reject it, or vice versa. Probability analysis instead involves looking at the likelihood that different outcomes will occur.

Likelihood of outcomes

Sensitivity analysis does not take into account the probabilities of outcomes occurring. Probability analysis shows how likely it is that outcomes will occur which will change the decision, such as the probability that an investment which is expected to have a positive net present value (NPV) will have a negative one.

Decision-making

Sensitivity analysis does not provide a simple indication of whether to undertake a project which is not expected to, but could, have a negative NPV. It indicates by how much variables will have to change for a project to have a negative NPV, but that figure does not tell the decision-maker whether the project should go ahead; the decision-maker must subjectively assess whether the level of change needed is too small given the level of uncertainty.

Probability analysis can be used to calculate the expected NPV of the project, which can provide the decision-maker with a simple decision rule to go ahead if the expected NPV is positive. However, if the decision-maker is also concerned about the dispersal of results and the likelihood of a negative NPV, probability analysis does not provide a simple decision rule either, the decision-maker will still have to decide whether the probability of making a loss is too large for their risk appetite.

Number of variables

Simple sensitivity analysis involves isolating the impact of each variable on the outcome, one at a time. Probability analysis, by contrast, can allow for more than one variable changing by combining their probabilities, and can also allow for the probabilities of different variables changing being interdependent.

(ACCA FM September/December 2023)

6 (a) (i) Project cash flows

	1 (\$)	2 (\$)	3 (\$)	4 (\$)
Revenue (W1)	731,500	649,740	678,980	709,580
Variable costs (W1)	(463,500)	(405,790)	(417,945)	(430,525)
Fixed costs	(10,000)	(10,000)	(10,000)	(10,000)
Taxable cash flow	258,000	233,950	251,035	269,055
Taxation at 20%	(51,600)	(46,790)	(50,207)	(53,811)
TAD benefit (W2)	25,000	18,750	14,063	42,187
After-tax cash flow	231,400	205,910	214,891	257,431
Working capital (W3)	(2,400)	(2,472)	(2,546)	87,418
Net cash flow	229,000	203,438	212,345	344,849
Discount factors at 13%	0.885	0.783	0.693	0.613
Present value (PV)	202,665	159,292	147,155	211,392

PV of future cash flows = \$720,504.

Initial investment = \$500,000.

Working capital = \$80,000.

Expected NPV = \$140,504.

WORKINGS

(W1) Sales revenue and variable costs

Year	1	2	3	4
Sales volume (units)	10,000	8,500	8,500	8,500
Selling price (\$/unit)	73.15	76.44	79.88	83.48
Revenue (\$)	731,500	649,740	678,980	709,580

Year	1	2	3	4
Sales volume (units)	10,000	8,500	8,500	8,500
Variable cost (\$/unit)	46.35	47.74	49.17	50.65
Variable cost (\$)	463,500	405,790	417,945	430,525

Expected volume: 10,000 units in Year 1.

For Years 2-4, expected volume = $(0.3 \times 5,000) + (0.7 \times 10,000) = 8,500$ units/year.

Selling price of \$70/unit inflated at 4.5%/year.

Variable cost of \$45/unit inflated at 3%/year.

(W2) Tax-allowable depreciation and tax benefits

Year	1	2	3	4
Tax-allowable depreciation (\$)	125,000	93,750	70,313	210,937
Tax benefit at 20% (\$)	25,000	18,750	14,063	42,187

25% reducing balance on \$500,000, with a balancing allowance in Year 4 since scrap value is nil: Year 4
TAD = $500,000 - 125,000 - 93,750 - 70,313 = \$210,937$.

(W3) Working capital

Year	1	2	3	4
Working capital balance (\$)	82,400	84,872	87,418	
Incremental cash flow (\$)	(2,400)	(2,472)	(2,546)	87,418

Working capital of \$80,000 inflating at 3%/year, with the balance recovered in full at the end of Year 4.

(ii)

Expected NPV is positive and is therefore financially acceptable. However, the expected NPV is based upon an expected volume of 8,500 units for Years 2-4; there is a 30% probability that the sales volume for these years could be as low as 5,000 units, which would reduce the NPV, possibly resulting in a negative NPV.

(b)

Expected values (EV) / probability analysis: an EV is a long-run, weighted average outcome using the probability of occurrence for each outcome as the weights. An EV simplifies the appraisal by incorporating the variation in outcomes into a single figure, but requires knowledge not only of the individual outcomes but also their probability of occurrence, which can be difficult in practice unless past data is available, and only has any real meaning for repetitive events.

Sensitivity analysis: a form of 'what if?' analysis where the impact on a project's NPV of a change in a key variable can be quantified, allowing key variables to be identified for further investigation. It gives no indication of the likelihood of the changes occurring, nor does it allow more than one variable to be changed at a time, so interdependence between variables is ignored.

Simulation: a more sophisticated version of sensitivity analysis which allows more than one variable to change at a time, usually performed with computer modelling due to its complexity; it gives a good indication of potential outcomes but its main disadvantages are its complexity, the time it takes and the fact that probabilities are still required for the variable changes.

Adjusted payback: payback measures how long, in years, until a project has broken even in its cash flows. A longer payback period means greater reliance on later, more uncertain cash flows; reducing the maximum payback benchmark will tend to reject longer-payback, more uncertain projects. Choice of the maximum payback is subjective, however, and no account is taken of the project's return, as cash flows after payback are ignored.

Risk-adjusted discount rate: for higher risk projects, the discount rate should be increased to ensure future present values are reduced sufficiently to allow for the increased return required by finance providers. This could be performed arbitrarily, by adding a return premium to the existing rate, or using more sophisticated techniques such as the capital asset pricing model (CAPM).

(Only FOUR techniques were required for full marks.)

(ACCA FM September/December 2024)

Marking scheme	
(a) (i) Expected volume	1
Sales revenue	1
Variable cost	1
Tax liability	1
Tax timing	1
TAD calculation	1
TAD benefit	1
Working capital investment	0.5
Working capital increments	1
Working capital recovery	0.5
PVs/NPV	1
(ii) Comments	2
Maximum	12
(b) First technique	2
Second technique	2
Third technique	2
Fourth technique	2
Maximum	8

Marking scheme	
Total	20 marks

Chapter 7: Working Capital Management

Questions

Objective Test Questions

The elements of working capital

1 Which TWO of the following are correct descriptions of net working capital?

- A Current assets minus current liabilities
- B Inventory days plus accounts receivable days minus accounts payable days
- C Current assets divided by current liabilities
- D The long-term capital invested in net current assets

2 For the coming year, a company has budgeted sales of \$2 million per month, 80% of which will be on credit. It expects its accounts receivable payment period to be three months. Forecast average inventory and average accounts payable for the coming year are \$10 million and \$4 million respectively.

What is the company's working capital requirement for the coming year (to one decimal place)?

\$

3 Which TWO of the following would increase the net working capital of a company?

- A Collection of accounts receivable
- B Refinancing of accounts payable with a two-year bank loan
- C Using the proceeds of a rights issue to repay an overdraft
- D Payment of a dividend

The objectives of working capital management

4 PKA Co is a European company that sells goods solely within Europe. The recently appointed financial manager of PKA Co has been investigating working capital management objectives and the working capital management of the company.

What are the main objectives of working capital management at PKA Co?

- (1) To ensure that PKA Co has sufficient liquid resources
 - (2) To increase PKA Co's profitability
 - (3) To ensure that PKA Co's assets give the highest possible returns
- A 1 only
 - B 1 and 2 only
 - C 2 and 3 only
 - D 1, 2 and 3

5 Cat Co is reviewing its working capital management.

Which TWO of the following statements concerning working capital management are correct?

- A The twin objectives of working capital management are profitability and liquidity
- B A conservative approach to working capital investment will increase profitability
- C Working capital management is a key factor in a company's long-term success
- D Liquid assets give the highest returns, leading to conflicts of objectives

6 Andrew Co is a large listed company financed by both equity and debt.

In which of the following areas of financial management will the impact of working capital management be smallest?

- A Liquidity management
- B Interest rate management
- C Management of relationship with the bank
- D Dividend policy

Policies regarding working capital management

7 Which TWO of the following statements about overcapitalisation and overtrading are correct?

- A Overtrading often arises from a rapid increase in sales revenue
- B Overcapitalisation results in a relatively low current ratio
- C Overtrading may result in a relatively high accounts payable turnover period
- D Overcapitalisation is the result of too much short-term capital

8 Which of the following is LEAST likely to characterise overtrading?

- A Increased short-term borrowing
- B Increased cash balances
- C Increased revenue
- D Reduced working capital

9 If Plot Co were overtrading, which TWO of the following could be symptoms?

- A Decreasing levels of trade receivables
- B Increasing levels of inventory
- C Increasing levels of long-term borrowings
- D Increasing levels of current liabilities

10 Match the characteristic below with the appropriate working capital strategy.

Characteristic	Working capital strategy
Relatively high level of current assets	
Relatively low level of current assets	
Relatively large amounts of short-term finance	
Relatively small amounts of short-term finance	

Choose from: *Conservative investment strategy / Aggressive investment strategy / Conservative financing strategy / Aggressive financing strategy*

11 Two companies, Acacia and Birch, have the following average levels of working capital.

Working capital level (\$m)	Acacia	Birch
Maximum	10	12
Minimum	6	9

Acacia's working capital is financed with \$4 million of long-term debt and Birch's working capital is financed with \$9 million of long-term debt. The balance of finance is from short-term sources.

Identify which type of working capital funding strategy each company is employing (Aggressive, Matching, or Conservative).

Company	Funding strategy
Acacia's working capital funding strategy	
Birch's working capital funding strategy	

12 Which of the following is/are true?

- (1) A conservative working capital investment policy implies a higher proportion of permanent current assets to fluctuating current assets
 - (2) Long-term finance is generally cheaper than short-term finance
- A 1 only is correct
 B 2 only is correct
 C 1 and 2 are correct
 D 1 and 2 are incorrect

13 Which of the following indicates that a company is becoming more conservative in its working capital funding policy?

- A Increase in the ratio of current liabilities to non-current liabilities
 B Decrease in the operating cycle
 C Decrease in the current ratio
 D Increase in the ratio of long-term finance to current liabilities

14 Which of the following statements concerning working capital management are correct?

- (1) Working capital should increase as sales increase
 - (2) An increase in the cash operating cycle will decrease profitability
 - (3) Overtrading is also known as under-capitalisation
- A 1 and 2 only
 B 1 and 3 only
 C 2 and 3 only
 D 1, 2 and 3

15 Which of the following is NOT usually associated with overtrading?

- A An increase in the current ratio
- B A rapid increase in revenue
- C A rapid increase in the volume of current assets
- D Most of the increase in current assets being financed by credit

16 Which of the following is NOT a feature of overtrading?

- A Rapid increase in the level of current assets
- B Increase in shareholders' funds
- C Increase in accounts receivable collection period
- D Rapid increase in revenue

17 Which of the following statements is consistent with an aggressive working capital funding policy?

- A Most current assets are financed with long-term funding rather than short-term funding, despite long-term funding being more expensive
- B Most current assets are funded with debt rather than equity, as it is cheaper
- C Most current assets are financed with short-term funding rather than long-term funding, despite the higher risk involved
- D Most current assets are funded by secured funding to enable the financial manager to focus on managing the working capital balances

18 Which of the following is/are true?

(i) An aggressive working capital funding policy implies a lower value of current assets and a higher value of current liabilities than a conservative working capital funding policy

(ii) Long-term debt finance is riskier for the lender than short-term debt finance

- A i only is correct
- B ii only is correct
- C i and ii are correct
- D i and ii are incorrect

19 Pop Co is switching from using mainly long-term fixed rate finance to fund its working capital to using mainly short-term variable rate finance.

Which of the following statements about the change in Pop Co's working capital financing policy is true?

- A Finance costs will increase
- B Re-financing risk will increase
- C Interest rate risk will decrease
- D Overcapitalisation risk will decrease

Working capital ratios: liquidity

20 Which of the following transactions would increase the current ratio and decrease operating profit?

- A Tax due from the previous year is paid
- B A long-term loan note is redeemed
- C A scrip dividend is paid
- D Land is sold for less than its carrying amount

21 Which of the following ratios would most likely be used by management to evaluate short-term liquidity?

- A Return on total assets
- B Financial gearing
- C Accounts receivable turnover
- D Acid test (quick) ratio

22 A company has a current ratio of 2.3 and a quick ratio of 0.8. It increases its overdraft in order to buy more inventory as a cash purchase.

What will happen to the company's current ratio and quick ratio as a result of this transaction (Increase or Decrease)?

Ratio	Increase	Decrease
Current ratio		
Quick ratio		

23 A company sells inventory for cash to a customer, at a selling price which is below the cost of the inventory items.

How will this transaction affect the current ratio and the quick ratio immediately after the transaction (Increase or Decrease)?

Ratio	Increase	Decrease
Current ratio		
Quick ratio		

The cash operating cycle

24 The following information has been calculated for D Co.

Trade receivables collection period	10 weeks
Raw material inventory turnover period	6 weeks
Work in progress inventory turnover period	2 weeks
Trade payables payment period	7 weeks
Finished goods inventory turnover period	6 weeks

What is the length of the working capital cycle?

25 Which of the following is NOT included in the calculation of the operating cycle?

- A Inventory holding period
- B Payables payment period
- C Receivables collection period
- D Settlement discount period

26 The operating cycle of an organisation would shorten under which of the following combinations of changes?

	Inventory holding period	Receivables collection period	Payables payment period
A	Increase	Increase	Increase
B	Decrease	Decrease	Increase
C	Increase	Decrease	Decrease
D	Decrease	Decrease	Decrease

(Operating cycle = inventory holding period + receivables collection period – payables payment period.)

27 The following information has been calculated for AAA Co.

Trade receivables collection period	25 days
Raw material holding period	24 days
Length of the production cycle	30 days
Credit taken from suppliers	56 days
Finished goods holding period	34 days

What is the length of the working capital cycle?

- A 3 days
- B 27 days
- C 57 days
- D 169 days

28 Which of the following situations will produce the most favourable operating cycle?

	Inventory holding period	Receivables collection period	Payables payment period
A	Short	Short	Short
B	Long	Long	Short
C	Short	Short	Long
D	Long	Short	Long

29 What is the operating cycle of company Z, given the following?

Inventory holding period	30 days
Receivables collection period	38 days
Trade payables payment period	44 days

- A 112 days
- B 68 days
- C 36 days
- D 24 days

Working capital ratios: operating cycle

30 The following are extracts from the financial statements of Inci Co.

	\$m
Revenue	2,780
Cost of sales	1,612
Inventories	226
Trade receivables	417
Trade payables	198

10% of Inci Co's sales, by value, are on cash terms. Purchases for the year totalled \$1,451m. Assume a 365-day year.

Calculate Inci Co's inventory holding period, trade receivables collection period and trade payables payment period (all calculations should be to the nearest day).

Ratio	Days
Inventory holding period	
Trade receivables collection period	
Trade payables payment period	

31 Goldstar has an accounts receivable turnover of 10.5 times, an inventory turnover of 4 times and a payables turnover of 8 times.

What is Goldstar's cash operating cycle, to the nearest day (assume 365 days in a year)?

32 A company has a current ratio of 2. Due to having significant surplus cash balances, it has decided to pay its trade payables after 30 days in future, rather than after 50 days as it has in the past.

What will be the effect of this change on the company's current ratio and its operating cycle?

- A Current ratio: Increase; Operating cycle: Increase
- B Current ratio: Increase; Operating cycle: Decrease
- C Current ratio: Decrease; Operating cycle: Increase
- D Current ratio: Decrease; Operating cycle: Decrease

Working capital investment levels

33 A company has annual credit sales of \$27 million and related cost of sales of \$15 million. The company has the following targets for the next year (assume there are 360 days in the year):

Item	Days
Trade receivables collection period	50
Inventory holding period	60
Trade payables payment period	45

What is the net investment in working capital required for the next year?

- A \$8,125,000
- B \$4,375,000
- C \$2,875,000
- D \$6,375,000

34 Crag Co has sales of \$200m per year and the gross profit margin is 40%. The finished goods inventory holding period varies throughout the year within the following range.

	Maximum	Minimum
Inventory holding period (days)	120	90

All purchases and sales are made on a cash basis and no inventory of raw materials or work in progress is carried. Crag Co intends to finance permanent current assets with equity and fluctuating current assets with its overdraft.

In relation to finished goods inventory, and assuming a 360-day year, how much finance will be needed from the overdraft?

\$

35 Mile Co is looking to change its working capital policy to match the rest of the industry. The following results are expected for the coming year.

\$000	
Revenue	20,500
Cost of sales	(12,800)
Gross profit	7,700

Revenue and cost of sales can be assumed to be spread evenly throughout the year. The working capital ratios of Mile Co, compared with the industry, are as follows (assume there are 365 days in each year).

	Mile Co	Industry
Accounts receivable collection period	50	42
Inventory holding period	45	35
Accounts payable payment period	40	35

If Mile Co matches its working capital cycle with the industry, what will be the decrease in its net working capital?

- A \$624,600
- B \$730,100
- C \$835,600
- D \$975,300

36 A company has the following summarised statement of financial position at 31 December 20X4.

	\$000	\$000
Non-current assets		1,000
Current assets		
Inventories	200	
Receivables	150	
Cash	100	
	450	
Current liabilities		
Payables	(200)	
Net current assets		250
		1,250

Over the next year the company should double its sales. The company does not plan to invest in any new non-current assets, but inventories, receivables and payables should all move in line with sales.

What cash balance in one year's time would this imply if the non-current assets were all land, no new capital was raised and all profits were paid out as dividends?

- A \$100,000 cash in hand
- B \$200,000 cash in hand
- C \$50,000 overdraft
- D \$100,000 overdraft

Constructed Response Questions

1 TGA Co, a multinational company, has annual credit sales of \$5.4 million and related cost of sales of \$2.16 million. Approximately half of all credit sales are exports to a European country, which are invoiced in euros. Financial information relating to TGA Co is as follows.

	\$000	\$000
Inventory	473.4	
Trade receivables	1,331.5	1,804.9
Trade payables	177.5	
Overdraft	1,326.6	1,504.1
Net working capital		300.8

TGA Co plans to change working capital policy in order to improve its profitability. This policy change will not affect the current levels of credit sales, cost of sales or net working capital. As a result of the policy change, the following working capital ratio values are expected (assume 365 days per year):

Item	Value
Inventory days	50
Trade receivables days	62
Trade payables days	45
Short-term dollar borrowing rate	5% per year
Short-term dollar deposit rate	4% per year

Required:

- For the change in working capital policy, calculate the change in the operating cycle, the effect on the current ratio and the finance cost saving. Comment on your findings. (8 marks)
- Discuss the key elements of a trade receivables management policy. (7 marks)
- Explain the different types of foreign currency risk faced by a multinational company. (6 marks)
- TGA Co expects to receive €500,000 from export sales at the end of three months. A forward rate of €1.687 per \$1 has been offered by the company's bank and the spot rate is €1.675 per \$1. TGA Co can borrow short term in the euro at 9% per year. Calculate the dollar income from a forward market hedge and a money market hedge, and indicate which hedge would be financially preferred by TGA Co. (4 marks)

(Total = 25 marks)

(ACCA F9 June 2013)

2 Crago Co is concerned that it may be overtrading. Financial information relating to the company is as follows.

	20X5 \$000	20X4 \$000
Credit sales income	17,100	12,000
Cost of sales	8,550	7,500
Inventory	2,500	2,100
Trade receivables	2,000	1,000
Current assets total	4,500	3,100
Trade payables	1,900	1,250
Overdraft	2,400	850
Current liabilities total	4,300	2,100
Net working capital	200	1,000
Long-term debt	3,000	3,000

Sector averages for 20X5 are as follows (assume 360 days per year):

Item	Value
Inventory days	65
Receivables days	30
Payables days	50
Current ratio	1.7
Quick ratio	0.8

Required:

Evaluate whether Crago Co can be considered to be overtrading and discuss how overtrading can be overcome. (Up to 4 marks are available for calculations.) (10 marks)

(ACCA F9 March/June 2016)

3 The current assets and current liabilities of CSZ Co at the end of March 2014 were as follows.

	\$000	\$000
Inventory	5,700	
Trade receivables	6,575	
Current assets total		12,275
Trade payables	2,137	
Overdraft	4,682	
Current liabilities total		6,819
Net current assets		5,456

For the year to March 2014, sales were \$40 million (all credit) and cost of sales was \$26 million. For the year to March 2015, credit sales are forecast to remain at \$40 million, while cost of sales is forecast to fall to 60% of sales. Target ratios for March 2015 are as follows (assume 365 days per year):

Item	Value
Inventory days	60
Receivables days	75
Payables days	55
Current ratio	1.4 times

Required:

- Calculate the working capital cycle of CSZ Co at the end of March 2014, and discuss whether a working capital cycle should be positive or negative. (6 marks)
- Calculate the target quick ratio and target sales/net working capital ratio at the end of March 2015. (5 marks)
- Analyse and compare the current asset and current liability positions for March 2014 and March 2015, and discuss how the working capital financing policy of CSZ Co would have changed. (8 marks)
- Briefly discuss THREE internal methods which could be used by CSZ Co to manage foreign currency transaction risk arising from its continuing business activities. (6 marks)

(Total = 25 marks)

(ACCA F9 June 2014)

4 It is the middle of December 20X6 and Pangli Co is looking at working capital management for January 20X7. Forecast financial information at the start of January 20X7 is as follows:

Item	\$
Inventory	455,000
Trade receivables	408,350
Trade payables	186,700
Overdraft	240,250

All sales are on credit and are expected to be \$3.5m for 20X6. Monthly sales are as follows:

Month	Sales
November 20X6 (actual)	\$270,875
December 20X6 (forecast)	\$300,000
January 20X7 (forecast)	\$350,000

Pangli Co has a gross profit margin of 40%. Although Pangli Co offers 30 days credit, only 60% of customers pay in the month following purchase, while the remaining customers take an additional month of credit.

Inventory is expected to increase by \$52,250 during January 20X7. Pangli Co plans to pay 70% of trade payables in January 20X7 and defer paying the remaining 30% until the end of February 20X7. All suppliers require payment within 30 days. Credit purchases from suppliers during January 20X7 are expected to be \$250,000.

Interest of \$70,000 is due to be paid in January 20X7 on fixed rate bank debt. Operating cash outflows are expected to be \$146,500 in January 20X7. Pangli Co has no cash and relies on its overdraft to finance daily operations. The company has no plans to raise long-term finance during January 20X7. Assume that each year has 360 days.

Required:

- (a) (i) Calculate the cash operating cycle of Pangli Co at the start of January 20X7. (2 marks)
- (ii) Calculate the overdraft expected at the end of January 20X7. (4 marks)
- (iii) Calculate the current ratios at the start and end of January 20X7. (4 marks)
- (b) Discuss FIVE techniques that Pangli Co could use in managing trade receivables. (10 marks)

(Total = 20 marks)

(ACCA F9 March/June 2017 Sample)

5 Pumice Co plans to expand its business operations by opening several new outlets at a cost of \$8m, financed by an issue of loan notes. The company currently generates credit sales of \$80.768m before cost of sales of \$27.7m. All sales are on credit. The current statement of financial position of Pumice Co is as follows.

Assets	\$000
Non-current assets	54,070
Inventories	4,000
Trade receivables	12,320
Cash and cash equivalents	800
Current assets total	17,120
Total assets	71,190
Equity and liabilities	\$000
Equity	6,000

Equity and liabilities	\$000
Reserves	34,000
Total equity	40,000
Non-current liabilities	18,000
Trade payables	9,690
Overdraft	3,500
Current liabilities total	13,190
Total equity and liabilities	71,190

Pumice Co expects that the expansion will increase credit sales by 18.7%, with cost of sales being 33% of credit sales and profit after tax being \$6.818m. Non-current assets will increase by 11%. The bank has demanded that Pumice Co's overdraft be reduced to \$3m and the company expects its cash balance to be \$0.7m after the expansion.

Pumice Co has been receiving complaints from its suppliers about late payment and the company plans to improve its working capital management as part of the expansion. It expects the following working capital ratios to result:

Item	Days
Inventory holding period	50
Trade receivables payment period	60
Trade payables payment period	60

The finance director of Pumice Co wishes to investigate how the expansion will change the following ratios:

- (i) Trade payables payment period.
- (ii) Current ratio.
- (iii) Revenue/net working capital ratio (defining net working capital as inventory plus trade receivables less trade payables).

Assume that there are 360 days in a year.

Required:

- (a) (i) Prepare a forecast statement of financial position for Pumice Co. (6 marks)
- (ii) Calculate the effect of the proposed expansion on the working capital ratios listed by the finance director. (4 marks)
- (b) Discuss the ways in which implementing the proposed changes in working capital represent:
 - (i) Changes in working capital investment policy for Pumice Co; and (5 marks)
 - (ii) Changes in working capital funding policy for Pumice Co. (5 marks)

(Total = 20 marks)

(ACCA F9 March/June 2020)

6 The following financial information relates to HGR Co: statement of financial position at the current date (extracts).

	\$000	\$000
Non-current assets		48,965
Inventory	8,160	
Accounts receivable	8,775	
Current assets total		16,935
Overdraft	3,800	
Accounts payable	10,200	

	\$000	\$000
Current liabilities total		14,000
Net current assets		2,935
Total assets less current liabilities		51,900

Cash flow forecasts from the current date are as follows.

	Month 1 \$000	Month 2 \$000	Month 3 \$000
Cash operating receipts	4,220	4,350	3,808
Cash operating payments	3,950	4,100	3,750
Six-monthly interest on traded bonds		200	
Capital investment			2,000

The finance director has completed a review of accounts receivable management and has proposed staff training and operating procedure improvements, which he/she believes will reduce the accounts receivable collection period to the average sector value of 53 days. This reduction would take six months to achieve from the current date, with an equal reduction in each month. He/she has also proposed changes to inventory management methods, which he/she hopes will reduce the inventory holding period by two days per month each month over a three-month period from the current date. He/she does not expect any change in the current level of accounts payable.

HGR Co has an overdraft limit of \$4,000,000. Overdraft interest is payable at an annual rate of 6.17% per year, with payments being made each month based on the opening balance at the start of that month. Credit sales for the year to the current date were \$49,275,000 and cost of sales was \$37,230,000. These levels of credit sales and cost of sales are expected to be maintained in the coming year. Assume that there are 365 working days in each year.

Required:

- Discuss the working capital financing strategy of HGR Co. (6 marks)
- For HGR Co, calculate: (i) the bank balance in three months' time if no action is taken; and (ii) the bank balance in three months' time if the finance director's proposals are implemented. Comment on the forecast cash flow position of HGR Co and recommend a suitable course of action. (9 marks)
- Discuss how risks arising from granting credit to foreign customers can be managed and reduced. (5 marks)

(Total = 20 marks)

(ACCA F9 June 2009, modified)

7 The following financial information relates to Wobnig Co.

Statement of profit or loss extracts	20X6 \$000	20X5 \$000
Revenue	14,525	10,375
Gross profit	7,990	5,707
Operating profit	4,067	3,735
Interest expense	355	292
Profit before income taxes	3,712	3,443
Taxation	1,485	1,278
Profit for the year	2,227	2,165

Statement of financial position extracts	20X6 \$000	20X5 \$000
Non-current assets	15,284	14,602
Inventory	2,149	1,092
Trade receivables	3,200	1,734
Current assets total	5,349	2,826

Statement of financial position extracts	20X6 \$000	20X5 \$000
Total assets	20,633	17,428
Ordinary shares	8,000	8,000
Reserves	4,268	3,541
Total equity	12,268	11,541
7% Loan notes	4,000	4,000
Trade payables	2,865	1,637
Overdraft	1,500	250
Current liabilities total	4,365	1,887
Total equity and liabilities	20,633	17,428

Average ratios for the last two years for companies with similar business operations to Wobnig Co are as follows:

Item	Value
Current ratio	1.7 times
Quick ratio	1.1 times
Inventory holding period	55 days
Trade receivables collection period	60 days
Trade payables payment period	85 days
Sales revenue/net working capital	10 times

Required:

- (a) Using suitable working capital ratios and analysis of the financial information provided, evaluate whether Wobnig Co can be described as overtrading (undercapitalised). (11 marks)
- (b) Discuss the factors to be considered in formulating a trade receivables management policy. (5 marks)
- (c) Wobnig Co is considering using the Miller-Orr model to manage its cash flows. The minimum cash balance would be \$200,000 and the spread is expected to be \$75,000. Calculate the Miller-Orr model upper limit and return point, and explain how these would be used to manage the cash balances of Wobnig Co. (4 marks)

(Total = 20 marks)

(ACCA F9 June 2012, amended)

8 Gro Co is a fast-growing company that has seen its revenue increase by more than 70% over the last financial year. Despite this strong sales growth and a \$40m increase in profit after tax over the year, the company's cash position has deteriorated sharply, moving from a \$44m positive cash balance at the start of the year to an \$11m bank overdraft by the year end, a swing of \$55m. No new long-term finance (debt or equity) was raised during the year to support this expansion. Extracts from Gro Co's financial statements for the current year and the prior year are provided below.

Statement of profit or loss extracts	Year 2 \$m	Year 1 \$m
Revenue	685	400
Cost of sales	(520)	(280)
Gross profit	165	120
Profit after tax	70	30

Statement of financial position extracts	Year 2 \$m	Year 1 \$m
Inventory	95	45
Trade receivables	115	55
Trade payables	110	40
Cash/(overdraft)	(11)	44

Required:

- (a) Using appropriate working capital ratios and financial analysis, evaluate whether Gro Co is exhibiting signs of overtrading. (12 marks)
- (b) Explain the advantages and disadvantages of a conservative approach and an aggressive approach to financing working capital, and discuss which approach Gro Co appears to have adopted. (8 marks)

(Total = 20 marks)

Answers

Objective Test Questions

1 A and D

Net working capital = current assets minus current liabilities, which is also described as the long-term capital invested in net current assets (since net current assets, unless entirely financed by short-term sources, must be supported by long-term capital). Inventory days + receivables days – payables days is the cash operating cycle, not net working capital. Current assets divided by current liabilities is the current ratio.

2 \$10.8m

WORKING

This question tests the elements of working capital: working capital = inventory + receivables – payables. Accounts receivable = $\$2\text{m} \times 12 \times 80\% \times 3/12 = \4.8m . Working capital requirement = $\$4.8\text{m} + \$10\text{m} - \$4\text{m} = \10.8m .

3 B and C

Net working capital (current assets minus current liabilities) increases only if current assets increase or current liabilities decrease. Exchanging accounts payable or an overdraft (a current liability) for a two-year bank loan (option B) or for the proceeds of a rights issue (option C, which are equity, both long-term finance) would decrease current liabilities and so increase working capital. Collecting accounts receivable (option A) simply converts one current asset into another (receivables into cash), leaving current assets and net working capital unchanged, and paying a dividend (option D) reduces cash without affecting any other current asset or current liability, so it decreases net working capital rather than increasing it.

4 B

The two main objectives of working capital management are to ensure the business has sufficient liquid resources to continue in business, and to increase its profitability. These two objectives will often conflict, because liquid assets give the lowest returns. Statement 3 is therefore not correct: assets giving the highest possible returns is not itself an objective of working capital management (indeed, it can conflict with the liquidity objective).

5 A and C

Statement A is correct: sufficient working capital should be maintained to ensure bills can be paid on time, but working capital (receivables, inventory, payables) does not itself earn a return, so excessive working capital is undesirable, spare cash, for example, should be temporarily invested to earn a return, provided risk is low. Statement C is correct: too much or too little working capital leads to poor business performance, so managing it to an appropriate level is important to a business's long-term success.

Statement B is incorrect: a conservative approach to working capital investment implies aiming to keep relatively high levels of working capital, generally to reduce risk, but this is expensive, money tied up not directly earning a return, so it will decrease, not increase, profitability. Statement D is incorrect: liquid assets give the lowest returns (not the highest), which is precisely why the liquidity and profitability objectives of working capital management often conflict.

6 D

Working capital management may have some impact on dividend policy, for example, a shortage of liquid resources could constrain the level of dividend a company is able to pay in the short term, but this impact is indirect and relatively small compared with the other areas listed. Liquidity management (option A) is directly and substantially affected by working capital decisions, since working capital IS the company's short-term liquidity position. Interest rate management (option B) is closely linked to working capital, since short-term financing of working capital (e.g. via an overdraft) is directly exposed to interest rate movements. Management of the relationship with the bank (option C) is also heavily influenced by working capital management, since the bank is typically the main provider of short-term working capital finance and will closely monitor a company's working capital position. Dividend policy is therefore the area where the impact of working capital management is smallest.

7 A and C

Overtrading often arises from a rapid increase in sales revenue, and it may result in a relatively high (not low) accounts payable turnover period, since a business that is overtrading increasingly relies on delaying payment to suppliers as a source of finance. Overcapitalisation results in a relatively high (not low) current ratio, since it is the result of a company having too much long-term (not short-term) capital tied up in working capital.

8 B

Overtrading is associated with falling liquidity, i.e. a deteriorating cash position, not increased cash balances. Increased short-term borrowing, increased revenue and reduced (net) working capital are all typical characteristics of overtrading.

9 B and D

The two symptoms of overtrading are increasing levels of inventory and increasing levels of current liabilities. Trade receivables increase during overtrading, not decrease, so decreasing receivables is not a symptom. Most of the increase in assets during overtrading is financed by credit (current liabilities) rather than long-term borrowings, so increasing long-term borrowings is not a symptom either.

Question 10

Characteristic	Working capital strategy
Relatively high level of current assets	Conservative investment strategy
Relatively low level of current assets	Aggressive investment strategy
Relatively large amounts of short-term finance	Aggressive financing strategy
Relatively small amounts of short-term finance	Conservative financing strategy

A conservative investment policy involves carrying higher amounts of current assets, while a conservative financing policy involves using more long-term finance than short-term finance. Conservative strategies are lower-risk strategies:

- (1) Holding high levels of current assets is less risky than holding low levels, since there is a lower chance of running out of inventory or of upsetting customers by chasing receivables too aggressively.
- (2) Using low levels of short-term finance implies that larger levels of (less risky) long-term finance have been used instead.

Question 11

Company	Funding strategy
Acacia's working capital funding strategy	Aggressive
Birch's working capital funding strategy	Matching

Acacia has \$4 million of long-term working capital finance, but permanent working capital (its minimum level) of \$6 million. It is therefore using short-term finance for both fluctuating and some of its permanent working capital, so its funding strategy is aggressive. Birch has \$9 million of long-term working capital finance and permanent working capital of \$9 million too; it is using short-term finance only for fluctuating working capital and long-term finance for permanent working capital, so this is a matching strategy. Neither company is using long-term finance for any of its fluctuating working capital, which would be required for a conservative approach.

12 D

A conservative working capital investment policy refers to a higher amount invested in working capital; it does not relate to the proportions of permanent to fluctuating current assets, so statement 1 is incorrect. Long-term finance is not generally cheaper than short-term finance; it is more expensive, since it is riskier for the lender (who therefore demands a higher return), so statement 2 is also incorrect.

13 D

An increase in the ratio of long-term finance to current liabilities means a greater proportion of working capital is being financed from long-term sources; this is what makes a funding policy more conservative. A decrease in the operating cycle and a decrease in the current ratio are unrelated to the financing mix as such, and an increase in the ratio of current liabilities to non-current liabilities would indicate a more aggressive (not conservative) funding policy.

14 D

All three statements are correct. Working capital will naturally increase with sales, as more inventory is needed to satisfy demand and more credit is offered to more customers, statement 1 is correct. Lengthening of the cash operating cycle means cash is out of the business for longer, requiring more funding to cover it, which increases funding costs and so decreases profitability, statement 2 is correct. Overtrading (undercapitalisation) is where a business does not have enough long-term capital funding to sustain its level of trading activity, so overtrading and undercapitalisation are indeed alternative names for the same phenomenon, statement 3 is correct. This is a real ACCA OTQ (December 2014).

15 A

Overtrading is stimulated by a rapid increase in revenue. With higher sales levels, current asset balances for inventory and receivables, as well as the payables level, should rise in proportion to the rise in revenue, with most of this increase in current assets typically financed by credit (payables and overdraft) rather than by long-term capital. This deteriorating liquidity position means the current ratio falls, not rises, as overtrading develops, so an increase in the current ratio is not associated with overtrading.

16 B

Overtrading is characterised by a rapid increase in the level of current assets and in revenue (both financed largely through credit rather than long-term capital), together with a lengthening accounts receivable collection period as the business's credit control comes under strain. An increase in shareholders' funds is not a feature of overtrading, indeed, overtrading arises precisely because a

company's capital base (including shareholders' funds) has not kept pace with its rapidly growing volume of trade.

17 C

With an aggressive working capital funding policy, most current assets are financed with short-term funding rather than long-term funding, despite the higher risk involved. It is an aggressive policy because it aims to keep funding costs low (short-term finance is generally cheaper than long-term finance), despite exposing the business to higher risk (short-term finance may be subject to renewal issues and changing costs).

18 B

An aggressive working capital funding policy refers to an organisation using short-term funding to finance most current assets, including those that permanently exist, such as buffer inventory; it does not relate to holding low values of current assets and high values of current liabilities, which is instead a feature of an aggressive working capital investment policy, not a funding policy. Statement (i) is therefore false. Lenders who lend for longer face higher risk, as their money is exposed to the fortunes of the borrowing company for longer, so long-term debt finance is indeed riskier for the lender than short-term debt finance, statement (ii) is true.

19 B

Switching from long-term fixed rate finance to short-term variable rate finance increases refinancing risk, since short-term finance must be renewed more frequently, and there is no certainty that it will be available on similarly favourable terms (or at all) each time it needs to be rolled over. Finance costs are more likely to decrease than increase, since short-term finance is generally cheaper than long-term finance (per the normal yield curve). Interest rate risk increases rather than decreases, since variable rate finance exposes the company to fluctuating interest costs, unlike the certainty of a fixed rate.

Overcapitalisation risk (having too much long-term capital tied up in working capital) is reduced, if anything, by moving to a more aggressive, short-term financing mix, but this is the opposite of the type of risk that increases here, and in any event the greater concern with this move is the increased liquidity/refinancing risk, not a change in overcapitalisation risk. This is a real ACCA OTQ (September 2016).

20 D

The current ratio is current assets divided by current liabilities. The sale of land would increase cash, and therefore current assets, without increasing current liabilities; this would increase the current ratio. The sale of land at less than its carrying amount would also generate a loss, decreasing operating profit. Paying tax due from the previous year would reduce both cash and the tax payable, so both current assets and current liabilities fall together; this increases (not necessarily decreases) the current ratio and has no effect on operating profit. Redeeming a long-term loan note reduces cash and a non-current liability, so it has no effect on the current ratio, and no effect on operating profit. Paying a scrip dividend involves no cash or profit effect at all.

21 D

The acid test (quick) ratio evaluates short-term liquidity by comparing liquid assets (current assets excluding inventory, which may not be readily convertible to cash) to current liabilities. Return on total assets and financial gearing are not liquidity ratios, and accounts receivable turnover, while related to working capital efficiency, does not by itself evaluate overall short-term liquidity in the way the quick ratio does.

Question 22

Ratio	Increase	Decrease
Current ratio		✓
Quick ratio		✓

WORKING

Suppose current assets (including inventory) are \$230, current liabilities are \$100 and inventory is \$150. Current ratio = $\$230/\$100 = 2.3$. Quick ratio = $(\$230 - \$150)/\$100 = 0.8$.

Now suppose \$30 of inventory is purchased using the overdraft: current assets = \$260, quick assets = \$80 (unchanged, since inventory is excluded), current liabilities = \$130. Current ratio = $\$260/\$130 = 2.0$ (a decrease). Quick ratio = $\$80/\$130 = 0.62$ (a decrease). Both ratios fall because current liabilities increase (via the overdraft) while quick assets are unaffected by an inventory purchase, and current assets increase by proportionately less than current liabilities.

Question 23

Ratio	Increase	Decrease
Current ratio		✓
Quick ratio	✓	

WORKING

The current liabilities figure (the denominator of both ratios) stays the same in both cases, since the transaction is a cash sale for cash. The total of current assets decreases overall, because the reduction in the inventory value (removed at cost) is greater than the increase in cash (added at the lower selling price), so the numerator of the current ratio falls, and the current ratio decreases.

The total of liquid assets (current assets excluding inventory), which is the numerator of the quick ratio, increases because of the higher cash balance, inventory is excluded from this figure both before and after the sale, so only the cash increase is reflected, and the quick ratio therefore increases.

24 17 weeks**WORKING**

The working capital cycle = the asset periods (receivables collection + raw material inventory turnover + work in progress turnover + finished goods inventory turnover) less the liability period (payables payment).

	Weeks
Raw materials	6
Payables	(7)
Production (WIP)	2
Finished goods	6
Receivables	10
Total	17

25 D

The operating cycle = inventory holding period + receivables collection period – payables payment period. The effect of customers paying earlier to take an early settlement discount would already be reflected within the receivables collection period itself (and, symmetrically, within the payables payment period for a company taking a discount from its own suppliers); there is no separate 'settlement discount period' component of the cycle.

26 B

Operating cycle = inventory holding period + receivables collection period – payables payment period. A decrease in both the inventory holding period and the receivables collection period, combined with an increase in the payables payment period, will therefore shorten the operating cycle: the two positive components fall while the component subtracted rises.

27 C**WORKING**

The length of the operating cycle = 25 + 24 + 30 – 56 + 34 = 57 days (receivables collection + raw material holding + production cycle – credit taken from suppliers + finished goods holding).

28 C

A short operating cycle is most favourable, i.e. a low inventory holding period, a low receivables collection period, and a high (long) payables payment period. This minimises the gap between paying suppliers and receiving cash from customers, so option C (short inventory holding, short receivables collection, long payables payment) is the most favourable combination.

29 D

Operating cycle = inventory holding period + receivables collection period – payables payment period = 30 + 38 – 44 = 24 days.

Question 30

Ratio	Days
Inventory holding period	51
Trade receivables collection period	61
Trade payables payment period	50

WORKING

Inventory holding period = inventory/cost of sales x 365 = (\$226m/\$1,612m) x 365 = 51 days.

Trade receivables collection period = trade receivables/credit sales x 365. Credit sales = 90% x \$2,780m = \$2,502m. Trade receivables collection period = (\$417m/\$2,502m) x 365 = 61 days.

Trade payables payment period = trade payables/credit purchases x 365 = (\$198m/\$1,451m) x 365 = 50 days.

31 80 days

WORKING

Receivables collection period = $365/10.5 = 34.76$ days.

Inventory holding period = $365/4 = 91.25$ days.

Payables payment period = $365/8 = 45.63$ days.

Cash operating cycle = $34.76 + 91.25 - 45.63 = 80.38$ days, i.e. 80 days to the nearest day.

32 A

WORKING

The existing current ratio is 2:1, for example, \$2m of current assets and \$1m of current liabilities. Paying suppliers more quickly (after 30 days rather than 50) will reduce cash (a current asset) and reduce payables (a current liability) by the same amount, say \$0.5m: new current ratio = $\$1.5m/\$0.5m = 3$, an increase. Paying suppliers more quickly will also reduce the trade payables payment period, and since the operating cycle = inventory holding period + receivables collection period – payables payment period, a smaller amount being subtracted means the operating cycle increases.

33 B

WORKING

Inventory = $\$15,000,000 \times 60/360 = \$2,500,000$.

Trade receivables = $\$27,000,000 \times 50/360 = \$3,750,000$.

Trade payables = $\$15,000,000 \times 45/360 = \$1,875,000$.

Net investment required = $\$2,500,000 + \$3,750,000 - \$1,875,000 = \$4,375,000$.

34 \$10m

WORKING

As there is a minimum of 90 days' worth of inventory held throughout the year, this element is classed as a permanent current asset and is financed with equity. Only the fluctuating element of the inventory balance, the difference between the maximum (120 days) and minimum (90 days) levels, i.e. 30 days' worth, needs to be financed with the overdraft.

Annual cost of finished goods = $\$200m \times (1 - 40\%) = \$120m$ (cost of sales, at a 40% gross profit margin).

Overdraft finance required = $\$120m \times 30/360 = \$10m$.

35 A**WORKING**

For every day's reduction in a working capital period, $1/365$ th less working capital is needed. Matching the industry would reduce Mile Co's receivables collection period by 8 days (50 to 42), its inventory holding period by 10 days (45 to 35), and its payables payment period would increase by 5 days (40 to 35, a decrease in the number of days Mile Co takes to pay, which reduces the free finance payables provide).

Reduced receivables = $8/365 \times \$20,500,000 = \$449,315$.

Reduced inventory = $10/365 \times \$12,800,000 = \$350,685$.

Reduced payables (finance) = $5/365 \times \$12,800,000 = \$175,342$ (this offsets the other two reductions, since payables is a liability).

Net decrease in working capital = $\$449,315 + \$350,685 - \$175,342 = \$624,658$, i.e. approximately \$624,600.

36 C**WORKING**

As there is no new capital raised and all profits are paid out as dividends (so no retained earnings), the total of the statement of financial position remains \$1,250,000. Since the non-current assets are land, they are not depreciated, so their value stays at \$1,000,000, meaning net current assets must remain \$250,000.

	Now \$000	In one year \$000
Non-current assets (no change)	1,000	1,000
Inventories (double)	200	400
Receivables (double)	150	300
Cash (balancing figure)	100	Nil
Current assets total	450	700
Payables (double)	(200)	(400)
Overdraft (balancing figure)		(50)
Net current assets	250	250
Total	1,250	1,250

Current assets excluding cash (inventory + receivables, both doubled with sales) rise to \$700,000, while current liabilities (payables, also doubled) rise to \$400,000. Since net current assets must stay at \$250,000, the balancing figure needed on the liabilities side is a \$50,000 overdraft, i.e. cash moves from a \$100,000 asset to a \$50,000 overdraft (a \$150,000 net cash outflow), because the growth in inventory and receivables from the doubling of sales outstrips the free financing that doubled payables can provide.

Constructed Response Questions

1 (a) Effect of the working capital policy change

WORKING

(i) Change in operating cycle

Current inventory days = $(\$473,400/\$2,160,000) \times 365 = 80$ days.

Current receivables days = $(\$1,331,500/\$5,400,000) \times 365 = 90$ days.

Current payables days = $(\$177,500/\$2,160,000) \times 365 = 30$ days.

Current operating cycle = $80 + 90 - 30 = 140$ days. Operating cycle after the policy change = $50 + 62 - 45 = 67$ days. The operating cycle decreases by 73 days.

(ii) Effect on the current ratio

Current ratio = $\$1,804,900/\$1,504,100 = 1.20$ times.

Revised inventory = $\$2,160,000 \times 50/365 = \$295,890$.

Revised receivables = $\$5,400,000 \times 62/365 = \$917,260$.

Revised payables = $\$2,160,000 \times 45/365 = \$266,301$.

Since net working capital is unchanged at \$300,800, the revised overdraft is a balancing figure: revised current assets ($\$295,890 + \$917,260 = \$1,213,150$) – revised current liabilities (payables \$266,301 + overdraft) = \$300,800, so overdraft = $\$1,213,150 - \$266,301 - \$300,800 = \$646,049$.

Revised current liabilities = $\$266,301 + \$646,049 = \$912,350$. Revised current ratio = $\$1,213,150/\$912,350 = 1.33$ times, an increase from 1.20 to 1.33.

(iii) Finance cost saving

The overdraft falls from \$1,326,600 to \$646,049, a reduction of \$680,551. At a borrowing rate of 5% per year, this saves $5\% \times \$680,551 = \$34,028$ per year.

Comment: the working capital policy change reduces the operating cycle substantially (by 73 days), reflecting tighter inventory and receivables management only partly offset by a longer payables period. This releases cash that is used to pay down the overdraft, improving the current ratio from 1.20 to 1.33 (better short-term liquidity) and saving over \$34,000 per year in finance costs, directly improving profitability, which was the stated aim of the policy change.

(b) Key elements of a trade receivables management policy

Assessing creditworthiness: before granting credit to new customers, a company should assess their creditworthiness using trade references, bank references, credit reference agencies and published accounts, potentially supported by an internal credit-scoring process. This determines how much credit to offer, and on what terms.

Managing accounts receivable: once credit has been granted, the company needs to ensure customers keep to the agreed credit terms and credit limits. An aged receivables analysis helps identify overdue accounts and how late they are, and customers should be reminded of amounts due via regular statements of account.

Collecting amounts owing: a clear policy is needed for chasing overdue accounts promptly, assessing whether payment is likely to be forthcoming, and escalating through further steps such as legal action or a debt collection agency where necessary.

Offering early settlement discounts: a discount can be offered for payment before the normal due date, in exchange for a small reduction in the amount received. The benefit (e.g. reduced overdraft interest, or matching a competitor's terms) must be weighed against the cost of the discount itself.

Using factoring and invoice discounting: a factor can assist with credit assessment, administration and debt collection, and can advance a percentage of outstanding invoice values for a fee, on a recourse or non-recourse basis. Invoice discounting similarly advances funds against a company's own invoices while it retains control of collection.

Managing foreign accounts receivable: exporting increases the risk of non-payment and the amount tied up in receivables due to longer payment terms. This can be managed through export credit insurance, export factoring, documentary credits, and techniques that reduce the investment in foreign receivables, such as discounting bills of exchange.

(c) Types of foreign currency risk

Foreign currency risk can be divided into transaction risk, translation risk and economic risk.

Transaction risk is the foreign currency risk associated with short-term transactions, such as receiving money from customers in settlement of foreign currency accounts receivable. The risk is that the actual profit or cost associated with a future transaction may differ from the expected or forecast profit or cost, for example, the expected profit on goods invoiced in a foreign currency could be decreased by an adverse exchange rate movement. Transaction risk is therefore cash exposure, since cash transactions are affected by it, and this type of foreign currency risk is usually hedged.

Translation risk is the foreign currency risk associated with the consolidation of foreign currency denominated assets and liabilities. Movements in exchange rates can change the value of such assets and liabilities, resulting in unrealised foreign currency losses or gains when financial statements are consolidated. These gains and losses exist only on paper and do not have a cash effect, translation exposure is often referred to as accounting exposure. It can be hedged using asset and liability management, but hedging this type of risk may be deemed unnecessary.

Economic risk is the foreign currency risk associated with longer-term movements in exchange rates. It refers to the possibility that the present value of a company's future cash flows may be affected by future exchange rate movements, or that a company's competitive position may be affected by such movements. Transaction exposure can be viewed as short-term economic exposure. All companies face economic exposure and it is difficult to hedge against.

(d) Forward market hedge vs money market hedge**WORKING**

Income from a forward market hedge = €500,000/1.687 = \$296,384.

Money market hedge: three-month euro borrowing rate = $9\%/4 = 2.25\%$. Three-month dollar deposit rate = TGA Co's short-term dollar deposit rate of 4% per year given earlier in the question, so $4\%/4 = 1\%$ for three months.

Euros borrowed now = €500,000/1.0225 = €488,998.

Dollar value of this borrowing at spot = €488,998/1.675 = \$291,939.

Dollar income on this sum once deposited for three months = \$291,939 x 1.01 = \$294,858.

The forward market hedge gives \$296,384 – \$294,858 = \$1,526 more income than the money market hedge, and so the forward hedge would be preferred financially by TGA Co.

(ACCA F9 June 2013)

2 WORKING

	20X5	20X4	Sector 20X5
Inventory days (x360/COS)	105	101	65
Receivables days (x360/credit sales)	42	30	30
Payables days (x360/COS)	80	60	50
Current ratio	1.05	1.48	1.7
Quick ratio	0.47	0.48	0.8
Sales/net working capital	86 times	12 times	

Evaluation of overtrading

Rapid increase in sales revenue: credit sales rose by 42.5% (from \$12,000,000 to \$17,100,000) year on year, a classic early warning sign of overtrading, the business is expanding faster than its capital base can comfortably support.

Increased reliance on short-term finance: the overdraft has risen from \$850,000 to \$2,400,000 (a 182% increase) and trade payables from \$1,250,000 to \$1,900,000 (a 52% increase), while long-term debt has stayed flat at \$3,000,000. The sales/net working capital ratio has risen dramatically from 12 times to 86 times, showing net working capital is being stretched far thinner relative to the volume of trade being supported.

Rapid increase in current assets financed by credit rather than long-term capital: receivables have doubled (from \$1,000,000 to \$2,000,000) and inventory has risen by 19% (from \$2,100,000 to \$2,500,000), both well ahead of the underlying sales growth in some respects, and both financed almost entirely through the growth in current liabilities rather than new long-term finance.

Declining solvency and liquidity: the current ratio has fallen from 1.48 to 1.05, moving further below the sector average of 1.7, and the quick ratio has remained broadly flat around 0.47-0.48, well below the sector average of 0.8. Inventory days (105, up from 101) and payables days (80, up from 60) are both far above sector norms (65 and 50 respectively), and receivables days (42, up from 30) has also moved above the sector average of 30.

Taken together, these indicators, rapid sales growth, a soaring reliance on short-term/overdraft finance, current assets growing largely on the back of credit, and liquidity ratios falling well below sector averages, provide clear evidence that Crago Co is overtrading.

Overcoming overtrading

Overtrading arises from an imbalance between the volume of trade being supported and the amount of long-term capital available to finance it. It can therefore be overcome by raising additional long-term finance, either new equity or long-term debt, to restore working capital to a sustainable level relative to the scale of operations. Alternatively, the company could restrain the pace of its sales growth, effectively halting further expansion until its financial position (and in particular its working capital and liquidity ratios) has been consolidated and brought back into line with a sustainable, adequately financed level of trading.

(ACCA F9 March/June 2016)

3 (a) Working capital cycle at end of March 2014**WORKING**

Inventory days = $(\$5,700,000/\$26,000,000) \times 365 = 80$ days.

Receivables days = $(\$6,575,000/\$40,000,000) \times 365 = 60$ days.

Payables days = $(\$2,137,000/\$26,000,000) \times 365 = 30$ days.

Working capital cycle = $80 + 60 - 30 = 110$ days (positive).

Discussion: a positive working capital cycle means a company pays its suppliers before it receives cash from its customers; it therefore has a financing need for that period, typically met through an overdraft or other short-term (or long-term) finance.

A negative cycle, more typical of businesses such as supermarkets that hold low inventory and sell largely for cash while taking extended credit from suppliers, means the company is paid by its customers before it has to pay its own suppliers, generating free financing rather than requiring it.

Whether a cycle is positive or negative is not really a matter of choice for most businesses; it is largely determined by the nature of the industry and the trading relationships within it, rather than being a target management can simply select.

(b) Target quick ratio and target sales/net working capital ratio**WORKING**

Cost of sales for the year to March 2015 = $60\% \times \$40,000,000 = \$24,000,000$.

Target inventory = $\$24,000,000 \times 60/365 = \$3,945,205$. Target receivables = $\$40,000,000 \times 75/365 = \$8,219,178$.

Target current assets = $\$3,945,205 + \$8,219,178 = \$12,164,384$. At a target current ratio of 1.4 times, target current liabilities = $\$12,164,384/1.4 = \$8,688,846$.

Target quick ratio = $(\text{current assets} - \text{inventory})/\text{current liabilities} = \$8,219,178/\$8,688,846 = 0.95$ times.

Target net working capital = $\$12,164,384 - \$8,688,846 = \$3,475,538$. Target sales/net working capital ratio = $\$40,000,000/\$3,475,538 = 11.5$ times.

(c) Comparison of March 2014 and March 2015 positions, and change in financing policy**WORKING**

Target payables = $\$24,000,000 \times 55/365 = \$3,616,438$. Target overdraft (balancing figure) = target current liabilities – target payables = $\$8,688,846 - \$3,616,438 = \$5,072,408$.

	March 2014 \$000	March 2015 \$000
Inventory	5,700	3,945
Receivables	6,575	8,219
Current assets total	12,275	12,164
Trade payables	2,137	3,616

	March 2014 \$000	March 2015 \$000
Overdraft	4,682	5,072
Current liabilities total	6,819	8,689

Overall current assets are little changed (down slightly, from \$12,275,000 to \$12,164,384, as a tighter inventory position roughly offsets a longer receivables period), but current liabilities rise substantially, from \$6,819,000 to \$8,688,846 (a 27.4% increase), driven mainly by trade payables, which rise 69.2% as the payables period lengthens from 30 to 55 days, alongside a further rise in the overdraft.

As a result, current liabilities as a proportion of current assets rise from 56% ($6,819/12,275$) to 71% ($8,689/12,164$), meaning the proportion of current assets financed by long-term sources falls from 44% to 29%. This increasing reliance on short-term finance, both trade credit and overdraft, to fund a similar level of current assets implies an aggressive shift in CSZ Co's working capital financing policy.

(d) Internal methods of managing foreign currency transaction risk

Transaction risk relates to foreign currency transactions which are short-term in nature, such as payments expected from foreign trade receivables, payments made in settling foreign trade payables, and interest payments on foreign currency-denominated debt. It can be managed by several internal methods.

Currency of invoice: one internal hedging method is for CSZ Co to invoice foreign customers in its domestic currency, thereby transferring the foreign currency risk to the foreign customers. This method is usually not commercially viable, however, as foreign customers will transfer their business to competitors who do invoice in the foreign currency, thereby avoiding the foreign currency risk themselves.

Matching: the risk arising from foreign currency receipts and payments can be managed by matching. Receipts and payments in the same foreign currency can be matched, for example by using a foreign currency bank account, so that there is no need to buy the foreign currency. Taking a longer-term view, assets and liabilities can also be matched to hedge foreign currency risk, for example, a company expecting regular foreign currency income can use debt in the same currency to meet a financing need, so that the foreign currency interest on the debt can be met from the foreign currency income.

Leading and lagging: transaction risk can be managed by leading and lagging, where foreign currency payments are made in advance (leading) or in arrears (lagging), depending on the view of the paying company as to whether the currency of payment is expected to appreciate or depreciate against the domestic currency. Lagged payments to accounts payable should not exceed the credit period agreed with the supplier, however.

(ACCA F9 June 2014)

4 (a) (i) Cash operating cycle at the start of January 20X7

WORKING

The cash operating cycle is calculated by adding inventory days and receivables days, and subtracting payables days. Cost of sales = $\$3,500,000 \times (1 - 0.4) = \$2,100,000$.

Inventory days = $360 \times \$455,000/\$2,100,000 = 78$ days.

Trade receivables days = $360 \times \$408,350/\$3,500,000 = 42$ days.

Trade payables days = $360 \times \$186,700/\$2,100,000 = 32$ days.

Cash operating cycle of Pangli Co = $78 + 42 - 32 = 88$ days.

(ii) Overdraft expected at the end of January 20X7

WORKING

Inventory at end of January 20X7 = \$455,000 + \$52,250 = \$507,250.

At the start of January 20X7, 100% of December 20X6 sales (\$300,000) remain outstanding, together with 40% of November 20X6 sales (40% x \$270,875 = \$108,350), a total of \$408,350 as given.

Trade receivables	\$
At start of January 20X7	408,350
Outstanding November 20X6 receivables paid	(108,350)
December 20X6 receivables, 60% paid	(180,000)
January 20X7 credit sales	350,000
At end of January 20X7	470,000

Trade payables	\$
At start of January 20X7	186,700
Payment of 70% of trade payables	(130,690)
January 20X7 credit purchases	250,000
At end of January 20X7	306,010

Overdraft	\$
At start of January 20X7	240,250
Cash received from customers	(288,350)
Cash paid to suppliers	130,690
Interest payment	70,000
Operating cash outflows	146,500
Expected at end of January 20X7	299,090

(iii) Current ratios at start and end of January 20X7

WORKING

Current assets at start of January 20X7 = \$455,000 + \$408,350 = \$863,350.

Current liabilities at start of January 20X7 = \$186,700 + \$240,250 = \$426,950.

Current ratio at start of January 20X7 = \$863,350/\$426,950 = 2.03 times.

Current assets at end of January 20X7 = \$507,250 + \$470,000 = \$977,250.

Current liabilities at end of January 20X7 = \$306,010 + \$299,090 = \$605,100.

Current ratio at end of January 20X7 = \$977,250/\$605,100 = 1.62 times.

(b) Techniques for managing trade receivables

Assessing creditworthiness: Pangli Co can reduce its exposure to bad debt and late payment risk by assessing the creditworthiness of new customers using trade references, bank references, credit reference agencies and published accounts, potentially supported by its own credit-scoring process, before deciding how much credit to offer and on what terms.

Managing accounts receivable: Pangli Co needs to make sure credit customers abide by the agreed terms of trade and credit limits. An aged receivables analysis provides key information on overdue accounts and how late they are, and the company should remind customers of outstanding invoices through regular statements of account.

Collecting amounts owing: ideally customers pay on time, but where an account becomes overdue, Pangli Co must follow it up quickly, assess whether payment is likely to be forthcoming, and have a clear policy on further steps such as legal action or a debt collection agency where it is not.

Offering early settlement discounts: Pangli Co can encourage prompt settlement by offering a discount for payment before the normal due date, for example, a 2% discount for settling after 25 days rather than the full 40 days allowed. The benefit (e.g. reduced overdraft interest, or matching a competitor's terms) must be weighed against the cost of the discount.

Using factoring and invoice discounting: Pangli Co could use a factor, on a recourse or non-recourse basis, to assist with credit assessment, administration and debt collection, and to advance a percentage of outstanding invoice values for a fee. Invoice discounting similarly allows a third party to advance funds against high-value invoices, with the balance paid over (less a fee) once the invoices are settled.

Managing foreign accounts receivable: where relevant, foreign receivables carry increased non-payment risk and tend to be outstanding for longer. This can be managed through creditworthiness assessment, export factoring, export credit insurance, documentary credits, countertrade agreements, and techniques such as advances against collections or discounting bills of exchange to reduce the investment tied up in foreign receivables.

(ACCA F9 March/June 2017 Sample)

5 (a) (i) Forecast statement of financial position**WORKING**

Revenue = \$80.768m x 118.7% = \$95,871,616. Cost of sales = \$95,871,616 x 33% = \$31,637,633.

Assets	\$000
Non-current assets (54,070 x 111%)	60,018
Inventory (31,637,633 x 50/360)	4,394
Trade receivables (95,871,616 x 60/360)	15,979
Cash	700
Current assets total	21,073
Total assets	81,091

Equity and liabilities	\$000
Equity	6,000
Reserves (34,000 + 6,818)	40,818
Total equity	46,818
Non-current liabilities (18,000 + 8,000)	26,000
Trade payables (31,637,633 x 60/360)	5,273
Overdraft	3,000
Current liabilities total	8,273
Total equity and liabilities	81,091

(ii) Effect on working capital ratios**WORKING**

Trade payables payment period: currently $(9,690/27,700) \times 360 = 126$ days. Pumice Co is targeting 60 days, a reduction of 66 days. This looks to be ambitious.

Current ratio: currently $17,120/13,190 = 1.30$ times. After the expansion this becomes $21,073/8,273 = 2.55$ times. This suggests that management is looking to increase solvency.

Revenue/net working capital: net working capital currently $= 4,000 + 12,320 - 9,690 = \$6,630,000$.

Revised net working capital $= 4,394 + 15,979 - 5,273 = \$15,100,000$.

Revenue/NWC currently $= 80,768/6,630 = 12.18$ times.

Revised revenue/NWC $= 95,872/15,100 = 6.35$ times. There would be a substantial increase in the capital supporting revenue.

(b) (i) Changes in working capital investment policy

Working capital investment policy considers the level of current assets used to support revenue generation in relation to different companies. A company adopts an aggressive working capital investment policy relative to another company if it uses a lower level of current assets to support a similar level of revenue generation; conversely, the other company adopts a conservative policy relative to the first.

While there are no other companies here with which to compare Pumice Co, the effect of implementing the proposed changes can be measured by the revenue/current assets ratio: this shows no significant change, moving only from 4.72 times to 4.55 times ($80,768/17,120 = 4.72$ times; $95,872/21,073 = 4.55$ times).

Revenue/current assets appears a less sensitive measure of working capital investment policy than revenue/net working capital, which changes from 12.18 times to 6.35 times as a result of the decreased reliance on short-term working capital funding implicit in the proposed changes.

(ii) Changes in working capital funding policy

Working capital funding policy can be characterised as conservative, matching or aggressive, depending on the extent to which fluctuating and permanent current assets are financed from short-term or long-term sources. A conservative policy uses long-term funds for permanent current assets and part of fluctuating current assets (lower risk, lower profitability, since long-term funds are more expensive). An aggressive policy uses short-term funds for fluctuating current assets and part of permanent current assets (higher risk, higher profitability, since short-term funds are cheaper). A matching policy applies the matching principle exactly.

While there is insufficient information to determine the relative levels of permanent and fluctuating current assets, implementing the proposed changes shows a substantial movement to using long-term funds rather than short-term funds. Before the expansion, 77% of current assets are financed from short-term funds (trade payables plus overdraft) ($100 \times 13,190/17,120 = 77\%$). After the expansion, only 39% of current assets would be financed from short-term funds ($100 \times 8,273/21,073 = 39\%$) and 61% from long-term funds. This change is also apparent from the increase in the current ratio from 1.30 times to 2.55 times.

Therefore, the proposed changes in working capital suggest a movement by Pumice Co from an aggressive working capital funding policy to a conservative working capital funding policy. This is also evidenced by the \$4,917,000 decrease in short-term funds ($13,190 - 8,273$) relative to the \$3,953,000 increase in current assets ($21,073 - 17,120$) and the \$14,818,000 increase in long-term funds ($(46,818 + 26,000) - (40,000 + 18,000)$): the company's current reliance on short-term funds has been reversed.

(ACCA F9 March/June 2020)

6 (a) Working capital financing strategy of HGR Co

When considering the financing of working capital, it is useful to divide current assets into fluctuating current assets and permanent current assets. Fluctuating current assets represent changes in the level of current assets due to the unpredictability of business activity. Permanent current assets represent the core level of investment in current assets needed to support a given level of revenue or business activity; as revenue increases, the level of permanent current assets will also increase. This relationship can be measured by the ratio of revenue to net current assets.

The financing choice as far as working capital is concerned is between short-term and long-term finance. Short-term finance is more flexible than long-term finance, an overdraft, for example, is used as the need arises, with variable interest charged on the outstanding balance, but it is also more risky, since an overdraft facility may be withdrawn or a short-term loan renewed on less favourable terms. In terms of cost, the term structure of interest rates suggests short-term debt finance has a lower cost than long-term debt finance.

The matching principle suggests long-term finance should be used for long-term investment: applied to working capital financing, long-term finance should be matched with permanent current assets and non-current assets, a financing policy called a 'matching policy'. HGR Co is not using this policy, since of the \$16,935,000 of current assets, \$14,000,000 (83%) is financed from short-term sources (overdraft and trade payables) and only \$2,935,000 (17%) is financed from a long-term source (equity or traded bonds).

The financing approach taken by HGR Co, where short-term finance is preferred, is called an aggressive policy. Reliance on short-term finance makes this riskier than a matching approach, but more profitable due to the lower cost of short-term finance. Following an aggressive approach to financing can lead to overtrading (undercapitalisation) and the possibility of liquidity problems.

(b) Bank balance forecasts and recommendation**WORKING**

Monthly overdraft interest rate = $(1.0617)^{1/12} = 1.005$, i.e. 0.5%.

Reduction in accounts receivable collection period: current period = $(8,775/49,275) \times 365 = 65$ days.

Reduction over six months = $65 - 53 = 12$ days.

Monthly reduction = 2 days.

Each receivables collection day is equivalent to $\$49,275,000/365 = \$135,000$, so the monthly reduction in accounts receivable = $2 \times \$135,000 = \$270,000$.

Reduction in inventory holding period: current period = $(8,160/37,230) \times 365 = 80$ days. Each inventory holding day is equivalent to $\$37,230,000/365 = \$102,000$, so the monthly reduction in inventory = $2 \times \$102,000 = \$204,000$.

If no action is taken:

	Month 1 \$000	Month 2 \$000	Month 3 \$000
Receipts	4,220	4,350	3,808
Payments	(3,950)	(4,100)	(3,750)
Interest on bonds		(200)	
Overdraft interest	(19)	(18)	(18)
Capital investment			(2,000)
Net cash flow	251	32	(1,960)
Opening balance	(3,800)	(3,549)	(3,517)
Closing balance	(3,549)	(3,517)	(5,477)

If the finance director's proposals are implemented:

	Month 1 \$000	Month 2 \$000	Month 3 \$000
Receipts	4,220	4,350	3,808
Payments	(3,950)	(4,100)	(3,750)
Interest on bonds		(200)	
Overdraft interest	(19)	(15)	(13)
Capital investment			(2,000)
Accounts receivable reduction	270	270	270
Inventory reduction	204	204	204
Net cash flow	725	509	(1,481)
Opening balance	(3,800)	(3,075)	(2,566)
Closing balance	(3,075)	(2,566)	(4,047)

Discussion: if no action is taken, the cash flow forecast shows that HGR Co will exceed its overdraft limit of \$4 million by \$1.48 million in three months' time. If the finance director's proposals are implemented, there is a positive effect on the bank balance, but the overdraft limit is still exceeded in three months' time, although only by \$47,000 rather than by \$1.47 million. In each subsequent month, the continuing reduction in accounts receivable collection days will improve the bank balance by \$270,000 per month, though without further information on operating receipts and payments it cannot be forecast whether the bank balance will return to within the limit.

Purchase of non-current assets should not be financed by an overdraft, but by a long-term source of finance such as equity or bonds. If the \$2,000,000 capital expenditure were removed from working capital management, the overdraft balance at the end of three months would be \$3.48 million if no action were taken and \$2.05 million if the finance director's proposals were implemented. Given that HGR Co has almost \$50 million of non-current assets that could possibly be used as security, raising long-term debt through either a bank loan or a bond issue appears sensible. Recommendation: implement the finance director's proposals and finance the capital expenditure from a long-term source; consideration could also be given to using some long-term debt finance to reduce the overdraft and to reduce the level of accounts payable, currently standing at 100 days.

(c) Managing risks of granting credit to foreign customers

When credit is granted to foreign customers, two problems become especially significant. First, the longer distances and more complex nature of foreign trade transactions mean foreign accounts receivable need more investment than domestic receivables, longer transaction times increase accounts receivable balances and hence financing costs. Second, the risk of bad debts is higher with foreign accounts receivable than with domestic ones. Exporters therefore seek both to reduce the risk of bad debts and to reduce the level of investment in foreign accounts receivable.

Many foreign transactions are on 'open account', an agreement to settle the amount outstanding on a predetermined date; this reflects a good business relationship between importer and exporter, but also carries the highest risk of non-payment. One way to reduce investment in foreign accounts receivable is to agree early payment with an importer (payment in advance, payment on shipment, or cash on delivery), though such terms are unlikely to be competitive.

Bill finance can be used to accelerate cash receipts: bills of exchange, supported by a documentary letter of credit, can be discounted by a bank to give immediate funds, without recourse if countersigned by the importer's bank. Documentary letters of credit are a payment guarantee backed by one or more banks and carry almost no risk, provided the exporter complies with the terms and conditions and presents the documents specified (bills of lading, shipping documents, bills of exchange, and so on).

Companies can also manage and reduce risk by assessing the creditworthiness of new customers using bank references and credit reports, and by using insurance to cover some of the risks of giving credit to foreign customers, avoiding the cost of pursuing recovery through a foreign legal system where the exporter may be at a disadvantage. Export factoring can also be used, transferring administration and collection to a factor with local expertise, reducing both the investment in foreign accounts receivable and the incidence of bad debts.

(ACCA F9 June 2009, modified)

7 (a) Overtrading

Overtrading arises when a company does not have enough long-term finance to support its level of trading activity. There are a number of signs of overtrading, which are referred to in the following discussion.

Rapid increase in revenue compared to long-term finance: revenue has increased by 40%, from \$10,375,000 to \$14,525,000, while long-term finance has increased by only 4.7% (\$16,268,000/\$15,541,000).

Increase in trade receivables collection period: a rapid increase in revenue may be due to offering more generous credit terms to customers, in which case the trade receivables ratio would be expected to increase. The trade receivables collection period has in fact increased from 61 days to 80 days, an increase of 31%. In 20X5, the collection period was close to the 60-day average for similar companies, but is now 33% more than this. While revenue has increased by 40%, trade receivables have increased by 85% (\$3,200,000/\$1,734,000). It appears that Wobnig has offered more generous credit terms to its customers, although another explanation could be that the company's customers are struggling to settle their accounts on time due to a downturn in economic activity (e.g. a recession), leading to an increase in overdue payments and outstanding invoices.

Decrease in profitability: a rapid increase in revenue may also be due to offering lower prices on products sold, affecting gross profit margin or operating profit margin. The operating profit margin of Wobnig has decreased from 36% in 20X5 to 28% in 20X6. While revenue increased by 40%, operating profit increased by only 8.9% (\$4,067,000/\$3,735,000). While this decrease in profitability supports the possibility that Wobnig has decreased selling prices in order to increase sales volume, such a decrease in profitability may also be caused by an increase in cost of sales or other operating costs.

Rapid increase in current assets: the increase in trade receivables has already been discussed. Inventory increased by 97% (\$2,149,000/\$1,092,000) compared to the revenue increase of 40%, indicating perhaps that further increases in sales volume are being planned by Wobnig. The inventory holding period also increased from 85 days in 20X5 to 120 days in 20X6, well above the 55-day average for similar companies. Therefore, there has been a rapid increase in current assets of 89% (\$5,349,000/\$2,826,000), compared to the increase in long-term finance of only 4.7%.

An increased dependence on short-term finance: Wobnig has certainly increased its dependence on short-term finance and this can be shown in several ways. The sales revenue/net working capital ratio has increased from 11 times in 20X5 to 15 times in 20X6, compared to the average for similar companies of 10 times. There has been a 500% increase in the company's overdraft (\$1,500,000/\$250,000) and a 75% increase in trade payables (\$2,865,000/\$1,637,000). Furthermore, trade payables days rose from 128 days in 20X5 to 160 days in 20X6, higher than the average for similar companies of 85 days. Short-term debt as a proportion of total debt increased from 6% in 20X5 (\$250,000/\$4,250,000) to 27% in 20X6 (\$1,500,000/\$5,500,000). This analysis supports the view that Wobnig is more dependent on short-term finance in 20X6 than in 20X5.

A decrease in liquidity: a key problem arising from overtrading is a decrease in liquidity and a shortage of cash. The current ratio of Wobnig has fallen from 1.5 times in 20X5 to 1.2 times in 20X6, compared to an average for similar companies of 1.7 times. The quick ratio or acid test ratio, which is a more sensitive measure of liquidity, has fallen from 0.9 times in 20X5 to 0.7 times in 20X6, compared to an average for similar companies of 1.1 times. Therefore, there are clear indications that liquidity has fallen over the period and that Wobnig has a weaker liquidity position than similar companies on an average basis. However, the current assets of the company do still exceed its current liabilities, so it does not yet have a liquid deficit.

Conclusion: overall, it can be concluded that there are several indications that Wobnig is moving, or has moved, into an overtrading (undercapitalisation) position.

WORKINGS

	20X6	20X5
Operating profit margin (operating profit/revenue)	$4,067/14,525 = 28\%$	$3,735/10,375 = 36\%$
Current ratio	$5,349/4,365 = 1.2$ times	$2,826/1,887 = 1.5$ times
Quick ratio	$3,200/4,365 = 0.7$ times	$1,734/1,887 = 0.9$ times
Inventory holding period	$365 \times 2,149/6,535 = 120$ days	$365 \times 1,092/4,668 = 85$ days
Receivables collection period	$365 \times 3,200/14,525 = 80$ days	$365 \times 1,734/10,375 = 61$ days
Payables payment period	$365 \times 2,865/6,535 = 160$ days	$365 \times 1,637/4,668 = 128$ days
Net working capital	$5,349 - 4,365 = \$984,000$	$2,826 - 1,887 = \$939,000$
Sales/net working capital	$14,525/984 = 15$ times	$10,375/939 = 11$ times

(b) Receivables policy

Credit analysis: in offering credit, a company must consider that it will be exposed to the risk of late payment and the risk of bad debts. To reduce these risks, the company will assess the creditworthiness of its potential customers. In order to do this, the company needs information, which can come from a variety of sources, such as trade references, bank references, credit reference agencies, published accounts and so on. As a result of assessing the creditworthiness of customers, a company can decide on the amount of credit to offer, the credit terms to offer, or whether to offer credit at all.

Credit control: having extended credit to customers, a company needs to consider ways to ensure that the terms under which credit was granted are followed. It is important that customers settle outstanding accounts on time and keep to agreed credit limits. Factors to consider here are, therefore, the number of overdue accounts and the amount of outstanding cash. This information can be provided by an aged receivables analysis. Another factor to consider is that customers need to be made aware of the amounts outstanding on their accounts and reminded when payment is due. This can be done by providing regular statements of account and by sending reminder letters when payment is due.

Receivables collection: cash received needs to be banked quickly if payment is not made electronically by credit transfer. Overdue accounts must be followed up in order to assess the likelihood of payment and to determine what further action is needed. In the worst cases, legal steps may need to be taken in order to recover outstanding amounts. A key factor to consider here is that the costs incurred chasing overdue amounts must not exceed the benefit gained.

(c) Miller-Orr model**WORKING**

Calculation of upper limit: the upper limit is the sum of the lower limit and the spread. Using the minimum cash balance as the lower limit, upper limit = $\$200,000 + \$75,000 = \$275,000$.

Calculation of return point: the return point is the sum of the lower limit and one-third of the spread. Return point = $\$200,000 + (\$75,000/3) = \$200,000 + \$25,000 = \$225,000$.

Use in managing cash balances: the Miller-Orr model provides decision rules about when to invest surplus cash (if a cash balance increases to a high level), and about when to sell short-term investments (if a cash balance falls to a low level). By using these decision rules, the cash balance is kept between the upper and lower limits set by the Miller-Orr model. When the cash balance reaches the upper limit, $\$50,000$ is invested in short-term securities. This is equal to the upper limit minus the return point ($\$275,000 - \$225,000$). When the cash balance falls to the lower limit, short-term securities worth $\$25,000$ are sold for cash. This is equal to the return point minus the lower limit ($\$225,000 - \$200,000$).

(ACCA F9 June 2012, amended)

8 (a) Overtrading

Overtrading occurs when a company expands its operations faster than it can support with long-term finance, leading to liquidity strain and financial pressure. There are a number of signs of overtrading, which are referred to in the following discussion.

Rapid increase in revenue without matching long-term finance: revenue has increased by 71.3% (\$685m/\$400m), a very rapid rate of growth, yet no new long-term finance (debt or equity) was raised during the year to support this expansion. This mismatch between the rate of growth and the financing base used to support it is a classic overtrading trigger.

Deteriorating cash position despite rising profit: profit after tax has increased by \$40m, from \$30m to \$70m, yet the cash position has swung from a \$44m positive balance to an \$11m overdraft, a deterioration of \$55m. This confirms that profit is not the same as cash: the cash generated by trading has been more than absorbed by the growth in working capital needed to support the higher level of sales (rising inventory and receivables), leaving Gro Co reliant on an overdraft rather than the cash it has actually earned.

Rapid increase in current assets: trade receivables have increased by 109.1% (\$115m/\$55m) and inventory by 111.1% (\$95m/\$45m), both well in excess of the 71.3% increase in revenue. The receivables collection period has risen from 50 days to 61 days and the inventory holding period from 59 days to 67 days, suggesting Gro Co may be offering more generous credit terms and/or building up inventory to support further sales growth, both of which increase the investment tied up in working capital.

An increased dependence on short-term finance: trade payables have grown even faster than revenue, increasing by 175% (\$110m/\$40m), and the payables payment period has lengthened from 52 days to 77 days. This indicates Gro Co has been stretching its supplier credit terms, using trade payables (and now the overdraft) as an informal source of short-term finance to help fund its expansion, itself a symptom of overtrading.

A decrease in liquidity: the current ratio has fallen from 3.6 times to 1.7 times and the quick ratio from 2.5 times to 0.95 times, both indicating a marked tightening of Gro Co's liquidity position. The sales revenue/net working capital ratio has risen sharply from 3.8 times to 7.7 times, showing that a much greater volume of sales is now being supported by a relatively thin margin of net working capital, leaving the company increasingly exposed to any further growth or disruption to trading.

Conclusion: Gro Co displays several classic signs of overtrading, rapid sales growth, a deteriorating cash position, rising working capital requirements, and reliance on short-term/supplier finance rather than new long-term finance. Unless additional long-term finance is raised or growth is moderated, this position is unlikely to be sustainable.

WORKINGS

	Year 2	Year 1
Revenue growth	685/400 = 71.3% increase	
Inventory holding period	365 x 95/520 = 67 days	365 x 45/280 = 59 days
Receivables collection period	365 x 115/685 = 61 days	365 x 55/400 = 50 days
Payables payment period	365 x 110/520 = 77 days	365 x 40/280 = 52 days
Current ratio	210/121 = 1.7 times	144/40 = 3.6 times
Quick ratio	115/121 = 0.95 times	99/40 = 2.5 times
Net working capital	210 - 121 = \$89m	144 - 40 = \$104m
Sales/net working capital	685/89 = 7.7 times	400/104 = 3.8 times

Note: current assets = inventory + receivables + cash (Year 1 only, as Gro Co has an overdraft rather than cash in Year 2) = \$144m (Year 1) and \$95m + \$115m = \$210m (Year 2). Current liabilities = payables + overdraft (Year 2 only) = \$40m (Year 1) and \$110m + \$11m = \$121m (Year 2).

(b) Working capital financing strategy

The question concerns working capital financing strategy, that is, which sources of finance (long-term or short-term) are used to fund current assets, rather than working capital management, which concerns how the individual elements of working capital (such as inventory levels or the speed of receivables collection) are managed on a day-to-day basis.

A conservative approach to financing working capital uses long-term finance (equity or long-term debt) to fund all of the permanent current assets and part of the fluctuating current assets. The main advantage is greater stability: the company is less exposed to the risk of a liquidity crisis or the risk that short-term finance cannot be renewed or is only available on unfavourable terms, and financing costs are more predictable. The main disadvantage is cost: long-term finance is generally more expensive than short-term finance, and because some funding is committed even when it is not needed, funds can be underutilised during periods of lower demand.

An aggressive approach to financing working capital uses short-term finance (such as an overdraft or trade credit) to fund all of the fluctuating current assets and part of the permanent current assets. The main advantages are that short-term finance is usually cheaper than long-term finance and is more flexible, allowing the company to respond quickly to changing working capital needs. The main disadvantages are higher liquidity and refinancing risk, since short-term finance must be regularly renewed and may not always be available on acceptable terms, particularly in a credit market downturn, and greater exposure to cash flow pressure.

Applied to Gro Co: the fact that Gro Co has funded revenue growth of over 70% entirely without raising new long-term finance, and has moved from a cash surplus into a bank overdraft while also significantly stretching its trade payables, strongly suggests that Gro Co has been following an aggressive financing approach. Its rapidly expanding current assets have been funded predominantly by short-term sources (the overdraft and extended supplier credit) rather than by long-term capital, which is consistent with, and has likely contributed to, the liquidity strain identified in part (a).

Chapter 8: Working Capital Management: Inventory Control

Questions

Objective Test Questions

The objectives of inventory management

1 A company's typical inventory holding period at any time is as follows:

Item	Days
Raw materials	15
Work in progress	35
Finished goods	40

Annual cost of goods sold as per the financial statements is \$100m, of which raw material purchases account for 50% of the total.

The company has implemented plans to reduce the level of inventory held, the effects of which are expected to be as follows:

- (1) Raw material holding time to be reduced by 5 days.
- (2) Production time to be reduced by 4 days.
- (3) Finished goods holding time to be reduced by 5 days.

Assuming a 365-day year, what will be the reduction in inventory held?

- A \$1.918m
- B \$2.603m
- C \$3.151m
- D \$3.836m

2 Which TWO of the following would affect the optimal level of inventory?

- A Holding cost per unit of inventory
- B Current level of inventory
- C Cost of placing an order for inventory
- D The re-order level

Costs of low inventory levels

3 In inventory management, which of the following will tend to increase the level of safety stock?

- A Holding cost increases
- B Cost of running out of inventory decreases
- C Variability of lead-time increases
- D Reliability of demand forecasting increases

4 Lyle Co's inventory reorder level is 500 units, the lead time is three weeks, and the sales volume is estimated at 50 units per week.

What quantity of safety (buffer) inventory should Lyle Co hold?

- A 150
- B 350
- C 500
- D 650

5 An increase in which of the following should cause management to reduce the level of inventory?

- A The cost of placing an order
- B The cost of holding inventory
- C The annual demand for the product
- D The lead time needed to acquire inventory

Economic order quantity (EOQ)

6 Which TWO of the following assumptions are associated with the economic order quantity model?

- A The holding cost per unit will vary with quantity ordered
- B The cost of placing an order will vary with quantity ordered
- C Holding costs depend on the average level of inventory
- D The purchase cost per unit will vary based on quantity discounts
- E Constant demand

7 Which of the following statements about the economic order quantity (EOQ) and the re-order level (ROL) is true?

- A The EOQ determines the ROL
- B The ROL determines the EOQ
- C Both are influenced by demand
- D Both are influenced by lead time

8 Which of the following is LEAST relevant to the economic order quantity model for inventory?

- A Safety stock
- B Annual demand
- C Holding costs
- D Order costs

9 Stewart Co uses the Economic Order Quantity (EOQ) model for inventory management.

A decrease in which of the following variables would increase the EOQ?

- A Cost per order
- B Level of buffer inventory
- C Holding cost
- D Quantity demanded

10 A company determines its order quantity for a raw material using the Economic Order Quantity (EOQ) model.

What would be the effects on the EOQ and the total annual ordering cost of a decrease in the cost of ordering a batch of raw material?

	EOQ	Total annual ordering cost
A	Higher	Lower
B	Higher	Higher
C	Lower	Higher
D	Lower	Lower

11 A company uses 9,000 units of a component a year. The component has a purchase price of \$40 per unit and the cost of placing an order is \$160. The annual holding cost of one component is 8% of its purchase price.

What is the economic order quantity of the component (to the nearest unit)?

- A 530
- B 671
- C 949
- D 1,342

12 Will Co incurs costs of \$65 every time it places an order with its raw materials supplier. It orders 300,000 units of product each year. The cost of holding one unit of inventory for a month is \$2.50. Will Co keeps a buffer inventory at all times of 25,000 units of material.

Calculate the economic order quantity (EOQ) that will optimise the business's inventory costs, to the nearest 10 units.

13 PKA Co orders 100,000 units of a raw material when the inventory level falls to 35,000 units. Forecast demand to meet production requirements during the next year is 625,000 units. The cost of placing and processing an order is \$250, while the cost of holding a unit in stores is \$0.50 per unit per year. Both costs are expected to be constant during the next year. You should assume a 50-week year and that demand is constant throughout the year.

What is the economic order quantity?

- A 250
- B 3,536
- C 17,678
- D 25,000

14 PKA Co (same data as above): orders 100,000 units of a raw material when the inventory level falls to 35,000 units. Forecast demand to meet production requirements during the next year is 625,000 units, and orders are received two weeks after being placed with the supplier. You should assume a 50-week year and that demand is constant throughout the year.

What is the current minimum inventory level at PKA Co?

- A 10,000
- B 12,500
- C 22,500
- D 35,000

15 TS Co has daily demand for ball bearings of 40 a day for each of the 250 working days (50 weeks) of the year. The ball bearings are purchased from a local supplier for \$2 each. The cost of placing an order is \$64 per order, regardless of the size of the order. The inventory holding costs, expressed as a percentage of inventory purchase price, is 25% per year.

What is the economic order quantity (EOQ)?

16 EE Co has calculated the following in relation to its inventories:

Item	Value
Buffer inventory level	50 units
Reorder size	250 items
Fixed order costs	\$50 per order
Cost of holding one item per year	\$1.25
Annual demand	10,000 items
Purchase price	\$2 per item

What are the total inventory related costs for a year (to the nearest whole \$)?

\$

17 Plot Co sells Product P, with sales occurring evenly throughout the year. The annual demand for Product P is 300,000 units and an order for new inventory is placed each month. Each order costs \$267 to place. The cost of holding Product P in inventory is 10 cents per unit per year. Buffer inventory equal to 40% of one month's sales is maintained.

What is the total annual cost of the current purchasing policy (to the nearest whole number)?

\$

18 Plot Co (same data as above, Product P, annual demand 300,000 units, order cost \$267, holding cost 10 cents per unit per year, buffer inventory equal to 40% of one month's sales).

What is the total annual cost of a policy based on using the economic order quantity (EOQ) (to the nearest \$100)?

\$

19 Plot Co sells both Product P and Product Q, with sales of both products occurring evenly throughout the year. Plot Co currently places one order per month for 25,000 units of Product P (annual demand 300,000 units). Each order costs \$267 to process. The cost of holding Product P in inventory is \$0.10 per unit per year. Plot Co maintains no buffer inventory of Product P.

What is the total annual ordering cost and holding cost if Plot Co uses the economic order quantity for Product P (to the nearest \$100)?

- A \$6,000
- B \$4,454
- C \$4,000
- D \$2,300

20 Amax Co is a multinational company which has been reviewing its working capital management. Amax Co has received an offer of a discount for bulk purchase from Bard Co, one of its major suppliers. Bard Co has offered a 0.5% discount on orders of 60,000 units or more of Component X. Amax Co currently consumes 240,000 units of Component X each year (purchase price \$25 per unit) and places orders of 20,000 units at the end of each month. Holding cost for Component X is \$1 per unit per year and ordering costs are \$250 per order. Amax Co does not maintain any buffer inventory of Component X.

Using months as a basis, what is the financial consequence of accepting the bulk purchase discount?

- A Benefit of \$8,000 per year
- B Benefit of \$12,000 per year
- C Cost of \$13,000 per year
- D Cost of \$8,000 per year

(ACCA FM March/June 2022)

21 The production manager at Green Co has established the following information about a major inventory item:

Item	Value
Purchase price per unit	\$480
Annual demand	4,000 units
Supplier's delivery cost per order	\$10
Chief buyer's salary per year (relates to all product lines, not just this one)	\$30,000
Annual storage cost per unit	\$2
Cost of capital	10% per year

What is the economic order quantity for this inventory item?

22 Green Co adopts the EOQ as its order quantity for the inventory item above, and it takes one week for an order to be delivered. Green Co does not wish to hold any buffer inventory.

How much inventory will Green Co have on hand when the order is placed? Assume there are 52 weeks in a year.

Calculating the re-order level (ROL)

23 The following information relates to Blanc Co's inventory:

Item	Value
Annual demand	400,000 units
Order quantity	70,000 units
Reorder level	30,000 units
Lead time	3 weeks

Assume 50 weeks in the year.

What is the buffer (safety) level of inventory?

- A 6,000 units
- B 40,000 units
- C 30,000 units
- D 24,000 units

24 The inventory ordering policy of ZAR Co is to order 100,000 units when the inventory level falls to 20,000 units. The cost of placing and processing an order is \$200, while the cost of holding inventory is \$0.50 per unit per year. Orders are received one week after being placed with the supplier. The production requirement for the next year (50 weeks) is 600,000 units.

What is the cost of ZAR Co's inventory ordering policy?

\$

25 Indicate whether the following are or are not assumptions behind the traditional economic order quantity (EOQ) formula, $EOQ = \sqrt{2 \times Co \times D \div Ch}$.

Statement	Is an assumption	Is not an assumption
Constant demand		
Zero lead time		

Inventory management systems

26 Which of the following is/are usually seen as benefits of the just-in-time approach to inventory management?

- (1) Reduced risk of stock-outs
 - (2) Reduced holding costs
 - (3) Reduced dependence on suppliers
- A 2 only
B 1 and 3 only
C 2 and 3 only
D 1, 2 and 3

27 Which TWO of the following are aims of a Just in Time system of inventory control?

- A Reduction in capital tied up in inventory
- B Creation of an inflexible production process
- C Elimination of all activities performed that do not add value
- D Lowering of inventory ordering costs

28 Which of the following is an advantage of implementing just-in-time inventory management?

- A Quality control costs will be eliminated
- B Monthly finance costs incurred in holding inventory will be kept constant
- C The frequency of raw material deliveries is reduced
- D The amount of obsolete inventory will be minimised

29 Cat Co places monthly orders with a supplier for 10,000 components which are used in its manufacturing processes. Annual demand is 120,000 components. The cost per component is \$7.50. The cost of ordering is \$200 per order, while the cost of holding components in inventory is \$1.00 per component per year.

What is the current total annual cost of inventory?

\$

30 Cat Co currently places one order per month for 10,000 components which are used in its manufacturing processes. The cost per component is \$7.50, the cost of ordering is \$200 per order and the cost of holding components in inventory is \$1.00 per component per year.

Which of the following are potential benefits to Cat Co of implementing just-in-time inventory management?

- (1) Less dependence on suppliers
- (2) Shorter operating cycle
- (3) Reduced risk of stock-outs

- A 1, 2 and 3
- B 2 and 3 only
- C 2 only
- D 1 only

31 Which of the following is NOT a feature of a just-in-time system?

- A No holding of buffer stock
- B Long production runs
- C Geographical proximity to suppliers
- D High quality raw materials inventory

Case Scenario Questions

Inventory management systems

1 Cat Co currently places one order per month for 10,000 components which are used in its manufacturing processes. The cost per component is \$7.50, the cost of ordering is \$200 per order and the cost of holding components in inventory is \$1.00 per component per year. Warehouse space is rented on a daily basis.

The supplier has offered a bulk purchase discount of 3.6% on orders of 30,000 or more components. If the bulk purchase discount is taken, the cost of holding components in inventory would increase to \$2.20 per component per year.

The supplier has also offered a 2% settlement discount for payment within 20 days. Cat Co currently takes 60 day's credit from the supplier and has an overdraft rate of 15% per year. Assume there are 360 days in one year.

(1) What is the total annual cost of ordering, purchasing and holding inventory if the bulk purchase discount is taken?

- A \$883,400
- B \$901,400
- C \$934,400
- D \$936,160

(2) Which of the following is most likely if Cat Co changes from renting warehouse space on a daily basis to signing a long-term lease?

- A Financial flexibility will increase
- B Financial gearing will decrease
- C Insurance costs will decrease
- D The space required will increase

(3) What is the annual value of the settlement discount (in percentage terms) and should the discount be accepted or rejected?

	Value of discount	Decision
A	18.4%	Reject
B	18.0%	Accept
C	18.4%	Accept
D	18.0%	Reject

(4) Which of the following are characteristics of an 'aggressive' policy relating to the level of investment in current assets?

- (i) Allowing extended credit to customers
 - (ii) Taking extended credit from suppliers
 - (iii) Holding no buffer inventory
- A (i) only
 - B (ii) and (iii) only
 - C (iii) only
 - D (i), (ii) and (iii)

Constructed Response Questions

1 Dusty Co has been looking at its inventory management, and specifically at component K.

Current position: Dusty Co purchases 1,500,000 units of component K each year and consumes the component at a constant rate. The purchase price of component K is \$14 per unit. The company places 12 orders each year. Inventory of component K in the financial statements of Dusty Co is equal to average inventory of component K. The holding cost of component K, excluding finance costs, is \$0.21 per unit per year. The ordering cost of component K is \$252 per order.

Economic order quantity: Dusty Co wishes to investigate whether basing ordering of component K on the economic order quantity will reduce costs.

Bulk order discount: the supplier of component K has offered Dusty Co a discount of 0.5% on the purchase price of component K, provided the company orders 250,000 units per order.

Other information: Dusty Co has no cash but has access to short-term finance via an overdraft facility at an interest rate of 3% per year. This overdraft currently stands at \$550,000.

Required:

- (a) (i) Calculate the annual holding and ordering costs of Dusty Co's current inventory management system. (1 mark)
- (ii) Calculate the financial effect of adopting the Economic Order Quantity as the basis for ordering inventory. (4 marks)
- (iii) Calculate the financial effect of accepting the bulk order discount. (4 marks)
- (iv) Recommend, with justification, which option should be selected by Dusty Co. (1 mark)
- (b) Discuss the key factors in determining working capital funding strategies. (10 marks)

(Total = 20 marks)

(ACCA FM September/December 2019)

2 Purdy Co uses 5 million litres of chemical X per year in its plastic manufacturing business. Chemical X costs Purdy Co \$2.50 per litre per year to store and each order requires a quality check that costs Purdy Co \$400 per order to perform.

Purdy Co currently purchases chemical X from a local supplier at a cost of \$3.00 per litre. It uses the economic order quantity (EOQ) model to determine order size and carries a safety inventory of five days' usage to protect against variations in lead time demand.

Purdy Co has recently been approached by a company offering to supply chemical X at a cost of \$2.25 per litre, if it will order in quantities of 1 million litres. The new supplier's credit terms are the same as the current supplier. Purdy Co estimates that its storage cost per litre and its quality check cost per order will be unchanged if it accepts this offer and that it will have sufficient storage space to cope with larger orders. However, it will need to increase its safety inventory to 15 days' usage if it accepts the new supplier's offer. Purdy Co operates for 365 days per year.

Required:

- (a) (i) Calculate the total annual costs relating to Chemical X for both the current and potential suppliers. (7 marks)
- (ii) Calculate the value of the average investment in Chemical X inventory for both the current and potential suppliers. (2 marks)
- (iii) Recommend, with justification, whether Purdy Co should change its supplier. (1 mark)
- (b) Discuss the conflict that exists between the objectives of liquidity and profitability in the management of working capital. (5 marks)
- (c) Explain the cash operating cycle and its relationship with the level of investment in working capital. (5 marks)

(Total = 20 marks)*(ACCA FM September/December 2022)*

3 Kandy Co is currently reviewing its working capital investment and two independent options are under consideration: a bulk purchase discount from one of its biggest suppliers, and offering an early settlement discount to its customers.

Bulk purchase discount: one of Kandy Co's biggest selling items is the gimble. Kandy Co pays \$30 for a gimble and annual sales demand is 18,000 units. At present, Kandy Co's order size is 1,000 gimbles and each order costs \$75 to place, excluding the purchase price. Annual holding costs are equivalent to 9% of the purchase price of a gimble. Due to the high level of demand for gimbles, Kandy Co always holds a buffer inventory sufficient to cover two days' demand.

Kandy Co has been offered a bulk purchase discount of 1.5% off the purchase price if the company increases its order size to 1,500 gimbles. Annual holding costs for the gimble would increase to 10% of the purchase price, and the cost of placing an order would increase by \$25 per order. If accepted, Kandy Co would still maintain buffer inventory covering two days' demand.

Early settlement discount: Kandy Co has total credit sales of \$45m per year, with all customers equal in sales value. Credit terms require payment within 30 days, but only 60% of customers pay on time, with the remaining 40% paying on average on 40 days. Kandy Co is considering offering an early settlement discount of 0.5% for payment within 10 days: if offered, 50% of customers would pay within 10 days, 20% would still pay to the normal 30-day terms, and 30% would continue paying on 40 days. Kandy Co finances short-term borrowing at 7% per year. Assume a 360-day year.

Required:

- (a) (i) Calculate the financial effect of accepting the bulk purchase discount. (6 marks)
- (ii) Calculate the financial effect of offering the early settlement discount. (4 marks)
- (iii) Comment on the financial acceptability of each option. (2 marks)
- (b) Discuss the conflict that exists between the objectives of liquidity and profitability in the management of working capital. (4 marks)
- (c) Explain the cash operating cycle and discuss its relationship with the level of investment in working capital. (4 marks)

(Total = 20 marks)*(ACCA FM September/December 2021)*

4 Nesud Co has credit sales of \$45m per year and on average settles accounts with trade payables after 60 days. One of its suppliers has offered the company an early settlement discount of 0.5% for payment within 30 days. Administration costs will be increased by \$500 per year if the early settlement discount is taken. Nesud Co buys components worth \$1.5m per year from this supplier.

From a different supplier, Nesud Co purchases \$2.4m per year of Component K at a price of \$5 per component. Consumption of Component K can be assumed to be at a constant rate throughout the year. The company orders components at the start of each month in order to meet demand and the cost of placing each order is \$248.44. The holding cost for Component K is \$1.06 per unit per year.

The finance director of Nesud Co is concerned that approximately 1% of credit sales turn into irrecoverable debts. In addition, she has been advised that customers of the company take an average of 65 days to settle their accounts, even though Nesud Co requires settlement within 40 days.

Nesud Co finances working capital from an overdraft costing 4% per year. Assume there are 360 days in a year.

Required:

- (a) Evaluate whether Nesud Co should accept the early settlement discount offered by its supplier. (4 marks)
- (b) Evaluate whether Nesud Co should adopt an economic order quantity approach to ordering Component K. (6 marks)
- (c) Critically discuss how Nesud Co could improve the management of its trade receivables. (10 marks)

(Total = 20 marks)

(ACCA F9 September 2016)

Answers

Objective Test Questions

1 C

WORKING

	\$
Raw material effect using purchases (100m x 50% x 5/365)	684,932
WIP effect using cost of goods sold (100m x 4/365)	1,095,890
Finished goods effect using cost of goods sold (100m x 5/365)	1,369,863
Reduction in inventory	3,150,685

Reduction in inventory held is approximately \$3.151m.

2 A and C

The economic order quantity (which in turn determines the optimal level of inventory) is determined by the holding cost per unit of inventory, the cost of placing an order for inventory, and annual demand. The current level of inventory and the re-order level do not themselves determine the optimal level, they follow from it, rather than driving it.

3 C

If lead time (the delay between placing and receiving an order for inventory) becomes more variable, the amount of safety stock (buffer) needed to reduce the risk of stockouts will increase. A decrease (not increase) in the cost of running out of inventory, an increase (not decrease) in holding cost, and a decrease (not increase) in the reliability of demand forecasting would each, if anything, point the other way.

4 B

WORKING

Reorder level = safety stock + (lead time x sales per week). $500 = \text{safety stock} + (3 \times 50)$, so safety stock = $500 - 150 = 350$ units.

5 B

According to the EOQ formula, an increase in the cost of holding inventory would lead to a smaller order quantity and hence a reduction in the average level of inventory. Increases in the cost of placing an order, in annual demand, or in lead time would not, of themselves, prompt management to reduce inventory levels.

6 C and E

The EOQ model assumes that holding costs depend on the average level of inventory held (as opposed to the maximum level), and that demand is constant throughout the year. It does not assume that the holding cost per unit, the cost of placing an order, or the purchase cost per unit vary with the quantity ordered, all of these are assumed to be fixed per unit/per order, which is precisely why quantity discounts and bulk-order cost changes need to be evaluated as a separate comparison outside the basic EOQ formula.

7 C

Annual demand is a key input into the EOQ formula, and the re-order level = lead time (in days) x demand per day, so both are influenced by demand. The EOQ does not determine the ROL (or vice versa): they are two separate, independently calculated quantities. Lead time affects the ROL directly, but does not enter the basic EOQ formula at all.

8 A

The EOQ is independent of any decision about whether safety (buffer) stock needs to be held; it is driven purely by annual demand, holding costs and order costs. Safety stock is a separate concept, relevant to calculating the re-order level rather than the EOQ itself.

9 C

Per the EOQ formula, $EOQ = \sqrt{2 \times Co \times D/Ch}$, a decrease in the holding cost (Ch) would increase the EOQ, since Ch appears in the denominator. A decrease in the cost per order or in quantity demanded would decrease (not increase) the EOQ, since both appear in the numerator. The level of buffer inventory does not feature in the EOQ formula at all.

10 D

Considering the EOQ formula, if the cost of ordering (Co) decreases, the EOQ (x) will also decrease. The annual ordering cost ($D \times Co/x$) will also decrease.

11 C**WORKING**

Holding cost per unit = $8\% \times \$40 = \3.20 . $EOQ = \sqrt{2 \times \$160 \times 9,000/\$3.20} = \sqrt{900,000} = 949$ units (to the nearest unit).

12 1,140 units**WORKING**

The buffer inventory held is not needed for the EOQ calculation (it would be included as part of the calculation for total inventory costs, but does not affect the EOQ itself). $Co = \$65$, $D = 300,000$, $Ch = \$2.50 \times 12 = \30 (the period of demand and of the holding cost must match, here both calculated as annual figures).

$EOQ = \sqrt{2 \times \$65 \times 300,000/\$30} = \sqrt{1,300,000} = 1,140$ units (to the nearest 10 units).

13 D**WORKING**

$EOQ = \sqrt{2 \times Co \times D/Ch} = \sqrt{2 \times \$250 \times 625,000/\$0.50} = \sqrt{625,000,000} = 25,000$ units.

14 A

WORKING

Minimum inventory level = reorder level – (average usage x average lead time).

Average usage per week = 625,000 units/50 weeks = 12,500 units.

Average lead time = 2 weeks.

Reorder level = 35,000 units.

Minimum inventory level = 35,000 – (12,500 x 2) = 10,000 units.

15 1,600 ball bearings

WORKING

Annual demand $D = 40 \times 250 = 10,000$ ball bearings.

Ordering cost $C_o = \$64$.

Holding cost per unit per year $C_h = 25\% \times \$2 = \0.50 .

EOQ = $\sqrt{(2 \times C_o \times D/C_h)} = \sqrt{(2 \times 64 \times 10,000/0.5)} = 1,600$ ball bearings.

16 \$22,219

WORKING

Total cost = annual purchase cost + annual ordering cost + annual holding cost.

Annual purchase cost = 10,000 units x \$2 = \$20,000.

Annual ordering cost = number of orders x cost per order = $(10,000/250) \times \$50 = \$2,000$.

Annual holding cost = average inventory level x cost to hold per unit per year = $[(250/2) + 50] \times \$1.25 = \218.75 .

Total cost = \$20,000 + \$2,000 + \$218.75 = \$22,218.75, i.e. \$22,219 to the nearest \$.

17 \$5,454

WORKING

Ordering cost = $12 \times \$267 = \$3,204$ per year (one order per month).

Monthly order = monthly demand = $300,000/12 = 25,000$ units. Buffer inventory = $25,000 \times 0.4 = 10,000$ units.

Average inventory excluding buffer inventory = $25,000/2 = 12,500$ units. Average inventory including buffer inventory = $12,500 + 10,000 = 22,500$ units.

Holding cost = $22,500 \times \$0.10 = \$2,250$ per year. Total cost = $\$3,204 + \$2,250 = \$5,454$ per year.

18 \$5,000

WORKING

$EOQ = \sqrt{(2 \times Co \times D/Ch)} = \sqrt{(2 \times 267 \times 300,000/0.10)} = 40,025$ units per order.

Number of orders per year = $300,000/40,025 = 7.5$ orders per year. Order cost = $7.5 \times \$267 = \$2,003$.

Average inventory excluding buffer inventory = $40,025/2 = 20,013$ units. Average inventory including buffer inventory = $20,013 + 10,000 = 30,013$ units (buffer inventory of 10,000 units as calculated for the current policy above).

Holding cost = $30,013 \times \$0.10 = \$3,001$ per year. Total cost = $\$2,003 + \$3,001 = \$5,004$ per year, i.e. \$5,000 to the nearest \$100.

19 C

WORKING

$EOQ = \sqrt{(2 \times Co \times D/Ch)} = \sqrt{(2 \times 267 \times 300,000/0.10)} = 40,025$ units. Number of orders per year = $300,000/40,025 = 7.5$ orders per year.

Order cost = $7.5 \times \$267 = \$2,003$. This version of Plot Co maintains no buffer inventory, so average inventory = $1/2 \times 40,025 = 20,012$ units.

Holding cost = $20,012 \times \$0.10 = \$2,001$ per year. Total cost = $\$2,003 + \$2,001 = \$4,004$ per year, i.e. \$4,000 to the nearest \$100.

20 B

WORKING

Saving in ordering cost = $250 \times (12 - 4) = \$2,000$ per year (orders/year fall from $240,000/20,000 = 12$ to $240,000/60,000 = 4$).

Increase in holding cost = $1 \times [(60,000/2) - (20,000/2)] = \$20,000$ per year.

Annual saving from the discount = $240,000 \times \$25 \times 0.5\% = \$30,000$ per year.

Benefit of accepting the discount = $2,000 - 20,000 + 30,000 = \$12,000$ per year.

(ACCA FM March/June 2022)

21 40 units

WORKING

The chief buyer's salary relates to all product lines, not just this one, so it is not incremental to this order decision and must be excluded from the ordering cost, $Co = \$10$ (the supplier's delivery cost per order only).

Holding cost Ch = annual storage cost + cost of capital tied up = $\$2 + (10\% \times \$480) = \$2 + \$48 = \$50$ per unit per year.

$EOQ = \sqrt{(2 \times \$10 \times 4,000/\$50)} = \sqrt{1,600} = 40$ units.

22 77 units

WORKING

With no buffer inventory held, the amount on hand when a new order is placed equals the consumption during the one-week lead time: $1/52 \times 4,000 = 76.9$, i.e. 77 units.

23 A**WORKING**

Usage per week = $400,000/50 = 8,000$ units.

Consumption during the 3-week lead time = $8,000 \times 3 = 24,000$ units.

Buffer (safety) inventory = reorder level – lead-time usage = $30,000 - 24,000 = 6,000$ units.

24 \$30,200**WORKING**

Ordering cost = $\$200 \times (600,000/100,000) = \$1,200$ per year.

Consumption during the one-week lead time = $600,000/50 = 12,000$ units. Buffer inventory = reorder level – lead-time usage = $20,000 - 12,000 = 8,000$ units.

Average inventory held during the year = $8,000 + (100,000/2) = 58,000$ units. Holding cost = $58,000 \times \$0.50 = \$29,000$ per year.

Total cost = ordering cost + holding cost = $\$1,200 + \$29,000 = \$30,200$ per year.

Question 25

Statement	Is an assumption	Is not an assumption
Constant demand	✓	
Zero lead time		✓

The EOQ formula assumes demand is constant throughout the year. It does not assume lead time is zero, lead time simply needs to be known and constant (it is used, together with demand, to calculate the re-order level, not the EOQ itself, and could equally be a positive number of days or weeks).

26 A

The main objective of just-in-time (JIT) inventory management is to reduce the level of inventory held, and hence holding costs. Because very little (if any) buffer inventory is held under JIT, the risk of stockouts actually increases rather than decreases, and JIT requires much closer integration with (not reduced dependence on) suppliers, since deliveries must be reliable and frequent.

27 A and C

JIT aims to reduce the capital tied up in inventory (since very little inventory is held) and to eliminate activities that do not add value, as part of a broader lean-production philosophy. JIT requires a flexible (not inflexible) production process able to respond to demand in small batches, and inventory ordering costs are typically higher (not lower) under JIT, since frequent small deliveries replace fewer, larger orders.

28 D

Obsolete inventory is minimised under JIT, since very little inventory is held at any time. Quality control becomes more important (not eliminated) under JIT, since faulty goods are very costly to a business with no buffer inventory to fall back on. Finance costs of holding inventory are minimised (not held constant) under JIT. Raw materials need to be delivered more frequently, in smaller loads, to avoid holding large amounts of inventory, so delivery frequency increases, it does not reduce.

29 \$907,400

WORKING

Current cost = purchase cost + order cost + holding cost.

Purchase cost = 120,000 units x \$7.50 = \$900,000 per year.

Order costs = number of orders x fixed order cost = (120,000/10,000) x \$200 = \$2,400 per year.

Holding cost = average inventory level x cost per unit per year = (10,000/2) x \$1 = \$5,000 per year.

Total current cost = \$900,000 + \$2,400 + \$5,000 = \$907,400 per year.

30 C

JIT requires close integration with suppliers, which increases (not decreases) the level of dependence on them. Because the aim of JIT is to eliminate the holding of inventory, the operating cycle shortens; this is a genuine benefit. However, with very little buffer inventory held, the risk of stockouts rises rather than falls.

31 B

A low inventory level under JIT requires short ('just-in-time') production runs, not long ones, long production runs are associated with traditional batch-based inventory management, which is what JIT is designed to move away from. No holding of buffer stock, geographical proximity to suppliers (to enable frequent, reliable, short-notice deliveries) and high quality raw materials (since there is no buffer to fall back on if defective materials are received) are all genuine features of a JIT system.

Case Scenario Questions

1 (1) B

WORKING

Number of orders per year = $120,000/30,000 = 4$ orders per year. Ordering cost = $4 \times \$200 = \800 per year.

Holding cost = $(1/2 \times 30,000) \times \$2.20 = \$33,000$ per year.

Annual cost of components = $120,000 \times \$7.50 \times (1 - 3.6\%) = 120,000 \times \$7.50 \times 0.964 = \$867,600$ per year.

Total cost = $\$867,600 + \$800 + \$33,000 = \$901,400$ per year.

(2) D

Currently, renting warehouse space daily allows Cat Co to match its space to its actual needs, renting less space when inventory is low and more when inventory is high. If a long-term lease were signed instead, the warehouse would have to be large enough to store the maximum level of inventory rather than the average level, so the space required would increase.

(3) C

WORKING

40-day value of the discount = $2/98 = 2.04\%$. Annualised value = $2.04\% \times 360/40 = 18.4\%$.

The value of the discount exceeds Cat Co's cost of short-term finance (15% per year), so the discount should be accepted.

(4) B

An 'aggressive' policy refers to holding low levels of current assets. Allowing extended credit to customers would increase working capital, so this is not part of an aggressive policy. Taking extended credit from suppliers and holding no buffer (safety) inventory are both consistent with an aggressive policy.

Constructed Response Questions

1 (a) (i) Current annual holding and ordering costs

WORKING

Order size = $1,500,000/12 = 125,000$ units. Average inventory = 62,500 units.

Holding cost = $62,500 \times \$0.21 = \$13,125$.

Ordering cost = $12 \times \$252 = \$3,024$.

Total = \$16,149 per year.

(ii) Financial effect of adopting the EOQ

WORKING

EOQ = $\sqrt{(2 \times \$252 \times 1,500,000/\$0.21)} = \sqrt{3,600,000,000} = 60,000$ units.

Orders per year = $1,500,000/60,000 = 25$.

Average inventory = 30,000 units.

Holding cost = $30,000 \times \$0.21 = \$6,300$.

Ordering cost = $25 \times \$252 = \$6,300$.

Total operating cost = \$12,600, a reduction of $\$16,149 - \$12,600 = \$3,549$ per year.

The reduction in average inventory ($62,500 - 30,000 = 32,500$ units) also releases finance: $32,500 \times \$14 = \$455,000$ less tied up in inventory, saving $3\% \times \$455,000 = \$13,650$ per year in overdraft interest.

Overall EOQ saving = $\$3,549 + \$13,650 = \$17,199$ per year.

(iii) Financial effect of accepting the bulk order discount

WORKING

Discounted price = $\$14 \times (1 - 0.5\%) = \13.93 per unit.

Orders per year = $1,500,000/250,000 = 6$.

Average inventory = 125,000 units.

Holding cost = $125,000 \times \$0.21 = \$26,250$.

Ordering cost = $6 \times \$252 = \$1,512$.

Total operating cost = \$27,762, an increase of $\$27,762 - \$16,149 = \$11,613$ per year.

The increase in average inventory value ($125,000 \times \$13.93 - 62,500 \times \$14 = \$1,741,250 - \$875,000 = \$866,250$) increases finance needed, costing $3\% \times \$866,250 = \$25,988$ per year in extra overdraft interest.

The value of the discount itself = $1,500,000 \times \$14 \times 0.5\% = \$105,000$ per year.

Overall net benefit of the bulk order discount = $\$105,000 - \$11,613 - \$25,988 = \$67,399$ per year.

(iv) Recommendation

The bulk order discount, with an overall net benefit of \$67,399 per year, is financially preferable to adopting the EOQ, which saves only \$17,199 per year. Dusty Co should therefore accept the bulk order discount and increase its order size to 250,000 units.

(b) Key factors in determining working capital funding strategies

Permanent and fluctuating current assets: one key factor is the distinction between permanent and fluctuating current assets. Permanent current assets represent the core level of current assets needed to support normal levels of business activity, for example, the level of trade receivables associated with the normal level of credit sales and existing terms of trade. Business activity will be subject to unexpected variations, however, such as some customers being late in settling their accounts, leading to unexpected variations in current assets. These can be termed fluctuating current assets.

Relative cost and risk of short-term and long-term finance: a second key factor is the relative cost of short-term and long-term finance. The normal yield curve suggests that long-term debt finance is more expensive than short-term debt finance, for example because of investor liquidity preference or default risk. Provided the terms of loan agreements are adhered to and interest is paid when due, however, long-term debt finance is a secure form of finance and hence low risk. While short-term debt finance is lower cost than long-term debt finance, it is higher risk, for example, an overdraft is technically repayable on demand, while a short-term loan is subject to the risk that it may be renewed on less favourable terms than those currently enjoyed.

Matching principle: a third key factor is the matching principle, which states that the maturity of assets should be reflected in the maturity of the finance used to support them. Short-term finance should be used for fluctuating current assets, while long-term finance should be used for permanent current assets and non-current assets.

Relative costs and benefits of different funding policies: a fourth key factor is the relative costs and benefits of different funding policies. A matching funding policy would use long-term finance for permanent current assets and non-current assets, and short-term finance for fluctuating current assets. A conservative funding policy would use long-term finance for permanent current assets, non-current assets and some of the fluctuating current assets, with short-term finance being used for the remaining fluctuating current assets. An aggressive funding policy would use long-term finance for the non-current assets and part of the permanent current assets, and short-term finance for fluctuating current assets and the balance of the permanent current assets.

A conservative funding policy, using relatively more long-term finance, would be lower in risk but lower in profitability. An aggressive funding policy, using relatively more short-term finance, would be higher in risk but higher in profitability. A matching funding policy would balance risk and profitability, avoiding the extremes of a conservative or an aggressive funding policy.

Other key factors: other key factors in working capital funding strategies include managerial attitudes to risk, previous funding decisions and organisation size. Managerial attitudes to risk can lead to a company preferring one working capital funding policy over another, for example, a risk-averse managerial team might prefer a conservative working capital funding policy. Previous funding decisions dictate the current short-term/long-term financing mix of a company. Organisational size can be an important factor in relation to, for example, access to different forms of finance in support of a favoured working capital funding policy.

(ACCA FM September/December 2019)

2 (a) (i) Total annual costs relating to Chemical X**WORKING**

EOQ (current supplier) = $\sqrt{(2 \times \$400 \times 5,000,000 / \$2.50)} = \sqrt{1,600,000,000} = 40,000$ litres. Orders per year = $5,000,000 / 40,000 = 125$.

Safety inventory (current) = $5,000,000 / 365 \times 5 = 68,493$ litres. Safety inventory (new supplier) = $5,000,000 / 365 \times 15 = 205,479$ litres.

	Current supplier	New supplier
Purchase cost per litre	\$3.00	\$2.25
Order size	40,000 litres (EOQ)	1,000,000 litres (imposed)
Orders per year	125	5
Holding cost (Ch x order size/2)	\$50,000	\$1,250,000
Ordering cost (Co x orders/year)	\$50,000	\$2,000
Purchase cost (price x annual demand)	\$15,000,000	\$11,250,000
Safety inventory holding cost	\$171,233	\$513,699
Total annual cost	\$15,271,233	\$13,015,699

Switching to the new supplier saves $\$15,271,233 - \$13,015,699 = \$2,255,534$ per year (around 15% of current procurement cost).

(ii) Average investment in Chemical X inventory**WORKING**

Current: average inventory = $40,000 / 2 + 68,493 = 88,493$ litres, valued at \$3.00 per litre = \$265,479.

New supplier: average inventory = $1,000,000 / 2 + 205,479 = 705,479$ litres, valued at \$2.25 per litre = \$1,587,328.

The investment in Chemical X inventory would increase by $\$1,587,328 - \$265,479 = \$1,321,849$.

(iii) Recommendation

Although switching supplier saves over \$2.25m per year in procurement costs, it requires an extra \$1.3m of investment tied up in inventory, a significant issue given that Purdy Co is already under pressure from its bank to reduce its overdraft. Purdy Co should therefore consider carefully whether it can fund this additional working capital investment before switching, rather than switching on cost grounds alone.

(b) Conflict between liquidity and profitability

Working capital management involves the management of a company's current assets and current liabilities, principally inventory, accounts receivable, cash and accounts payable. When managing working capital, a company faces twin objectives. One objective is to earn profits in order to provide satisfactory returns for its investors and contribute towards the maximisation of shareholder wealth. Another is to generate enough cash to allow the business to meet its financial commitments, such as wages, salaries, accounts payable, interest and tax.

These two objectives can often conflict. For example, to increase profitability, a firm may increase its inventory levels in order to be able to offer customers prompt delivery. In theory, this will be attractive to customers and sales and profits should increase. However, by increasing inventory levels the company ties up more cash in inventory, thus decreasing its ability to meet its short-term commitments.

On the other hand, a company which pressurises customers for payment by aggressive management of accounts receivable may find that its cash resources increase as customers pay more promptly, but at the same time its sales and profits may fall as customers turn to other suppliers who offer more generous credit terms.

Cash management involves a similar trade-off. Cash is often regarded as 'an idle asset' as it earns no return. Companies often invest their cash resources to earn some return and increase profitability. By doing so, they usually have to tie up the cash for a fixed amount of time (otherwise the interest earned would be very small), and hence damage their liquidity.

Finally, companies which take long periods of credit from suppliers may find their liquidity improves, but this can be to the detriment of profits as settlement discounts may be lost and supplier relationships damaged. When managing working capital, firms need to be aware of the conflict between the twin objectives. A sensible approach is to see the need for sufficient liquidity as a constraint on working capital management, and then to seek to maximise profits subject to this constraint.

(c) The cash operating cycle and its relationship with working capital investment

The cash operating cycle (sometimes known as the working capital cycle) measures the interval between the company paying out cash at the start of its production process (in manufacturing, this is normally paying suppliers for raw materials) and receiving cash from customers when products are sold and paid for. It is calculated by adding inventory days (for raw materials, work in progress and finished goods) to accounts receivable days and then deducting accounts payable days.

The length of a company's cash operating cycle has two major drivers. First, the type of industry a company operates in: house construction companies often have very long cash operating cycles, since their business involves buying land, gaining permission to build, building houses and infrastructure, selling, and then collecting cash from sales, a process that often takes more than a year. Large supermarket chains, on the other hand, often have very short cash operating cycles: their buying power allows them to enjoy long credit periods from suppliers, while inventory holding periods and accounts receivable are very short, and often they have negative cash operating cycles.

The efficiency of a company's working capital management will also affect the length of the cycle. A company which has poor control of inventory and accounts receivable, yet which pays its suppliers promptly, will find that it has a longer cash operating cycle than competitors who manage their working capital well.

The longer a company's cash operating cycle, the greater its investment in net working capital will be. If inventory is held for a large number of days, the amount of money invested in it will be high. Similarly, the longer it takes accounts receivable to be settled, the more money the company will have invested. Finally, a short accounts payable period will mean that the company takes little credit from suppliers, and hence fails to reduce its net working capital investment.

(ACCA FM September/December 2022)

3 (a) (i) Financial effect of the bulk purchase discount**WORKING**

Current: buffer inventory = $18,000 \times 2/360 = 100$ units. Average inventory = $1,000/2 + 100 = 600$ units.

Ordering cost = $(18,000/1,000) \times \$75 = \$1,350$.

Holding cost = $600 \times (9\% \times \$30) = 600 \times \$2.70 = \$1,620$.

Purchase cost = $18,000 \times \$30 = \$540,000$.

Total = \$542,970.

With discount: discounted price = $\$30 \times (1 - 1.5\%) = \29.55 . Average inventory = $1,500/2 + 100 = 850$ units.

Ordering cost = $(18,000/1,500) \times (\$75 + \$25) = 12 \times \$100 = \$1,200$.

Holding cost = $850 \times (10\% \times \$29.55) = 850 \times \$2.955 = \$2,512$.

Purchase cost = $18,000 \times \$29.55 = \$531,900$.

Total = \$535,612.

The bulk purchase discount saves $\$542,970 - \$535,612 = \$7,358$ per year.

(ii) Financial effect of the early settlement discount**WORKING**

Current receivables = $[(60\% \times \$45m \times 30) + (40\% \times \$45m \times 40)]/360 = (\$810m + \$720m)/360 = \$4,250,000$. Finance cost = $\$4,250,000 \times 7\% = \$297,500$.

Receivables after offering the discount = $[(50\% \times \$45m \times 10) + (20\% \times \$45m \times 30) + (30\% \times \$45m \times 40)]/360 = (\$225m + \$270m + \$540m)/360 = \$2,875,000$. Finance cost = $\$2,875,000 \times 7\% = \$201,250$.

Finance cost saving from lower receivables = $\$297,500 - \$201,250 = \$96,250$. Cost of the discount itself = $\$45m \times 0.5\% \times 50\%$ (only those taking it) = $\$112,500$.

The early settlement discount produces a net cost of $\$112,500 - \$96,250 = \$16,250$ per year.

(iii) Comment on financial acceptability

The bulk purchase discount is financially acceptable, saving Kandy Co \$7,358 per year, and should be accepted. The early settlement discount is not financially acceptable; it costs Kandy Co a net \$16,250 per year, since the discount given away exceeds the finance cost saved from collecting cash more quickly, and should be rejected.

(b) Conflict between liquidity and profitability

Working capital management involves the management of a company's current assets and current liabilities, principally inventory, accounts receivable, cash and accounts payable. When managing working capital, a company faces twin objectives. One objective is to earn profits in order to provide satisfactory returns for its investors and contribute towards the maximisation of shareholder wealth. Another is to generate enough cash to allow the business to meet its financial commitments, such as wages, salaries, accounts payable, interest and tax.

These two objectives can often conflict, and Kandy Co's own bulk purchase discount decision illustrates the tension well. Increasing the order size to 1,500 gimbles reduces the purchase price and lowers total procurement and ordering costs, improving profitability. However, it also increases average inventory held from 600 to 850 units, which ties up additional cash in inventory and increases the funding required for working capital, worsening liquidity even as profitability improves.

More generally, a firm may increase inventory levels in order to offer customers prompt delivery: in theory this is attractive to customers and increases sales and profits, but it ties up more cash and decreases the company's ability to meet its short-term commitments. Conversely, a company which pressurises customers for payment by aggressive management of accounts receivable may find its cash resources increase as customers pay more promptly, but its sales and profits may fall as customers turn to other suppliers offering more generous credit terms.

Cash management involves a similar trade-off: cash is often regarded as 'an idle asset' as it earns no return, so companies often invest cash resources to earn some return, but doing so usually requires tying up the cash for a fixed period, which can damage liquidity. When managing working capital, firms need to be aware of the conflict between the twin objectives of liquidity and profitability. A sensible approach, as illustrated by Kandy Co's decision, is to see the need for sufficient liquidity as a constraint on working capital management, and then to seek to maximise profits subject to that constraint.

(c) The cash operating cycle and its relationship with working capital investment

The cash operating cycle (sometimes known as the working capital cycle) measures the interval between a company paying out cash at the start of its production process (in manufacturing, normally paying suppliers for raw materials) and receiving cash from customers when products are sold and paid for. It is calculated by adding inventory days (for raw materials, work in progress and finished goods) to accounts receivable days and then deducting accounts payable days.

For Kandy Co, the gimble represents a significant component of inventory investment, and any change to the order size and holding policy for the gimble, such as the bulk purchase discount under consideration, directly affects the length of the cash operating cycle by changing the number of days' inventory held.

The length of a company's cash operating cycle has two major drivers. First, the type of industry a company operates in: some industries inherently require long production and selling processes, giving rise to long cash operating cycles, while others (such as large supermarket chains) enjoy short or even negative cash operating cycles because of strong buying power over suppliers combined with fast-moving inventory and prompt cash sales. Second, the efficiency of a company's working capital management: a company with poor control of inventory and accounts receivable, yet which pays its suppliers promptly, will have a longer cash operating cycle than a competitor that manages working capital well.

The longer a company's cash operating cycle, the greater its investment in net working capital will be. If inventory is held for a large number of days, as would be the case if Kandy Co increases its order size under the bulk purchase discount, the amount of money invested in it will be higher. Similarly, the longer it takes accounts receivable to be settled, the more money the company will have invested, while a short accounts payable period means the company takes little credit from suppliers and so fails to reduce its net working capital investment.

(ACCA FM September/December 2021)

4 (a) Evaluation of the early settlement discount offered by the supplier**WORKING**

Relevant trade payables before the discount = $\$1,500,000 \times 60/360 = \$250,000$.

Relevant trade payables after the discount = $\$1,500,000 \times 30/360 = \$125,000$.

Reduction in trade payables = $\$250,000 - \$125,000 = \$125,000$.

Increase in finance cost from paying suppliers sooner = $\$125,000 \times 4\% = \$5,000$.

Increase in administration costs = $\$500$.

Total additional cost of taking the discount = $\$5,000 + \$500 = \$5,500$.

Discount received from the supplier = $\$1,500,000 \times 0.5\% = \$7,500$.

Net benefit of accepting the discount = $\$7,500 - \$5,500 = \$2,000$ per year. Since there is a net benefit, Nesud Co should accept the early settlement discount offered by its supplier.

(b) Evaluation of adopting the EOQ for Component K**WORKING**

Annual demand = $\$2,400,000/\$5 = 480,000$ units per year. Currently ordered monthly: $480,000/12 = 40,000$ units per order.

Current ordering cost = $12 \times \$248.44 = \$2,981$ per year.

Average inventory = $40,000/2 = 20,000$ units.

Current holding cost = $20,000 \times \$1.06 = \$21,200$ per year.

Total current cost = $\$24,181$ per year.

EOQ = $\sqrt{(2 \times \$248.44 \times 480,000/\$1.06)} = 15,000$ units.

Orders per year = $480,000/15,000 = 32$.

Ordering cost = $32 \times \$248.44 = \$7,950$ per year.

Average inventory = $15,000/2 = 7,500$ units.

Holding cost = $7,500 \times \$1.06 = \$7,950$ per year.

Total EOQ cost = $\$15,900$ per year.

Adopting the EOQ reduces Nesud Co's total ordering and holding costs for Component K by $\$24,181 - \$15,900 = \$8,281$ per year. Nesud Co should therefore adopt the EOQ as the basis for ordering Component K, subject to the same practical caveats that apply to any EOQ-based analysis, in particular, that demand, holding costs and ordering costs will not necessarily remain perfectly constant in practice.

(c) Critical discussion of how Nesud Co could improve the management of its trade receivables

Credit analysis: before granting credit to a new customer, Nesud Co should assess that customer's creditworthiness. This can be done by taking up references from the customer's bank and from other companies that already trade with the customer, by obtaining a report from a credit reference agency, by analysing the customer's published financial statements and payment record, by using an in-house credit scoring system, and, for larger orders, by visiting the customer's premises.

Credit control: once credit has been granted, Nesud Co should maintain adequate control over trade receivables. This includes regularly preparing an aged receivables analysis to identify overdue accounts, monitoring individual customers' credit utilisation against the credit limits set for them, and reviewing credit limits periodically in light of customers' payment history and changing circumstances.

Collection of amounts owed: Nesud Co should have clear, consistently-applied procedures for collecting overdue amounts, such as issuing regular statements and reminder letters, following up promptly by telephone, and escalating persistently overdue accounts to legal action or to a debt collection agency where necessary. Nesud Co needs to be sensitive to the balance between the costs of these collection procedures (including the risk of damaging customer relationships) and the benefit of the amounts recovered.

Factoring of trade receivables: Nesud Co could also consider outsourcing its receivables management to a factor. A factor can provide administration of the sales ledger, credit control and debt collection services, protection against bad debts (with non-recourse factoring), and finance in the form of an advance of up to around 80% of the value of eligible receivables, which would improve Nesud Co's cash flow immediately rather than waiting for the full 65-day average collection period.

(ACCA F9 September 2016)

Chapter 9: Working Capital Management: Accounts Receivable and Payable

Questions

Objective Test Questions

The objectives of accounts receivable management

1 The management of XYZ Co has annual credit sales of \$20m and accounts receivable of \$4m. Working capital is financed by an overdraft at 12% interest per year. Assume 365 days in a year.

What is the annual finance cost saving if the management reduces the collection period by 60 days?

- A \$394,521
- B \$85,479
- C \$78,904
- D \$68,384

2 Amicable Wireless Co offers customers credit terms of 2% discount for payment within 10 days or full payment within 25 days. 60% of Amicable's customers take the 2% discount and pay on day 10. The remainder of Amicable's customers pay on day 30.

What is Amicable's receivables collection period?

- A 12 days
- B 16 days
- C 18 days
- D 20 days

3 The following information has been extracted from a company's financial records:

	Opening balance \$	Closing balance \$
Cash	3,900	3,000
Cash equivalents	3,800	4,400
Accounts receivable	14,600	12,900
Total current assets	22,300	20,300
Revenues	103,200	
Expenses	20,430	
Net income	82,770	

What is the company's average receivables collection period (to the nearest day)? Assume a 365-day year.

- A 59 days
- B 51 days
- C 49 days
- D 47 days

4 Foster Co is considering implementing a new cash collection system at a cost of \$80,000 per year. Annual sales are \$90m and the new system will reduce collection time by three days.

If Foster can invest surplus funds at 8% per year, what is the net gain/(loss) per year from implementing the new system? Assume a 360-day year.

- A \$60,000
- B (\$20,000)
- C (\$60,000)
- D (\$140,000)

Accounts receivable, establishing a credit policy

5 Which TWO of the following would be key aspects of a company's accounts receivable credit policy?

- A Assessing creditworthiness
- B Checking credit limits once a year
- C Invoicing promptly and collecting overdue debts
- D Delaying payments to obtain a 'free' source of finance

6 PKA Co is a European company that sells goods solely within Europe. Customers are allowed 30 days' credit, but the average accounts receivable period in the last financial year was 75 days, in line with the industry average. Bad debts as a percentage of sales, which are all on credit, increased in the last financial year from 5% to 8%. The accounts receivable department is currently short staffed.

What are the best ways for PKA Co to improve the management of accounts receivable?

- (1) Assess the creditworthiness of new customers**
- (2) Introduce early settlement discounts**
- (3) Take legal action against the slow payers and non-payers**

- A 1 and 2 only
- B 2 only
- C 1 and 3 only
- D 1, 2 and 3

Accounts receivable, calculations (early settlement discount evaluation)

7 Ling Co has annual credit sales of \$4,500,000 and on average customers take 60 days to pay, assuming a 360-day year. As a result, Ling has a trade receivables balance of \$750,000. Ling relies on an overdraft to finance this at an annual interest rate of 8%.

Ling is considering offering an early settlement discount to its customers of 0.5% for payment in 30 days. It expects that 25% of its customers (representing 35% of the annual credit sales figure) will pay on 30 days in order to obtain the discount.

If Ling introduces the proposed discount, what will be the NET saving?

\$

8 Wallace Co has annual credit sales of \$4,500,000 and on average customers take 60 days to pay, assuming a 360-day year. As a result, Wallace Co has a trade receivables balance of \$750,000. The company relies on an overdraft to finance this at an annual interest rate of 10%.

Wallace Co is considering offering an early settlement discount of 1% for payment in 30 days. It expects that 25% of its customers (representing 35% of the annual credit sales figure) will pay in 30 days in order to obtain the discount.

If Wallace Co introduces the proposed discount, what will be the NET impact?

- A \$1,875 saving
- B \$1,875 cost
- C \$2,625 saving
- D \$2,625 cost

9 A Co offers a quick payment discount of 2% if customers pay within 15 days. Customers currently take an average of 45 days to pay their credit balance.

What is the annual effective cost of the discount (to one decimal place of a percentage)?

- A 27.9%
- B 20.4%
- C 17.8%
- D 2.0%

10 Plot Co is considering offering a 2% early settlement discount to its customers. Currently sales are \$10 million and customers take 60 days to pay. Plot Co estimates half the customers will take up the discount and pay cash. Plot Co is currently financing working capital using an overdraft on which it pays a 10% charge. Assume 365 days in a year.

What will be the effect of implementing the policy?

- A Benefit of \$17,808
- B Cost of \$17,808
- C Benefit of \$82,192
- D Benefit of \$182,192

11 Hall Co is a retail organisation that has found its working capital position deteriorating as business customers take longer and longer to pay. Hall Co is considering offering customers a 1.5% discount for payment within one month, in respect of receivables who currently take two months or three months to pay.

What would be the annual cost of that discount in respect of the 2-month and 3-month receivables respectively?

	2-month receivables	3-month receivables
A	19.89%	9.49%
B	9.49%	4.64%
C	4.64%	9.49%
D	9.49%	19.89%

Accounts receivable, invoice discounting and factoring

12 Identify, by clicking on the relevant box in the table below, whether each of the following statements about debt factoring and invoice discounting is true or false.

Statement	True	False
Factoring is with recourse whereas discounting is without recourse		
Invoice discounting is usually performed on the entire receivables ledger		
Both are relatively cheap sources of finance		
Only factoring involves outsourcing the administration of the receivables ledger		

13 The CFO of a company is concerned about the company's accounts receivable turnover ratio. The company currently offers customers terms of 3% discount for settlement within 10 days or full payment within 30 days.

Which of the following strategies is most likely to improve the company's accounts receivable turnover ratio?

- A Using invoice discounting
- B Changing customer terms to a 1% discount for settlement within 10 days
- C Entering into a factoring agreement with a finance company
- D Changing customer terms to a 3% discount for settlement within 20 days

14 A factor has offered the following arrangement to Dino Co, which has annual sales of \$1,500,000 and an average receivables period of one month: the factor will advance 80% of the value of receivables at a 10% annual interest rate and charge a fee of 2% on revenue. Dino Co estimates that the company would save \$24,000 in administration expenses over the year.

What is the annual net cost of factoring?

- A \$12,000
- B \$14,800
- C \$16,000
- D \$20,000

15 What does invoice discounting normally involve?

- A Offering a cash discount for early settlement of invoices
- B Selling all invoices to a finance house who takes over administration of the receivables ledger
- C Receiving a cash advance from a finance house based on a percentage of the value of selected invoices
- D Writing off an invoice, partly or in total, as a bad debt

16 When accounts receivable are sold to a debt factoring company, which of the following is LEAST likely?

- A Cash will be received sooner than if receivables were not sold
- B The accounts receivable will be sold above their face value
- C The debt factoring company will make a profit on the service provided
- D Working capital will decrease

17 In order to improve the management of receivables, PKA Co is considering using a debt factor on a 'with-recourse' basis.

Which of the following are benefits of 'with-recourse' factoring for PKA Co?

- (1) A fall in bad debts**
- (2) A reduction in accounts receivable staffing costs**
- (3) An improvement in short-term liquidity**

- A 2 only
- B 1, 2 and 3
- C 2 and 3 only
- D 1 and 3 only

18 L Co is considering whether to factor its sales invoices. A factor has offered L Co a non-recourse package at a cost of 1.5% of sales and an admin fee of \$6,000 per year. Bad debts are currently 2% of sales per year and sales are \$1.5 million per year.

What is the cost of the package to L Co?

\$

19 Which of the following statements is/are correct?

- (1) Factoring with recourse provides insurance against bad debts.**
- (2) The expertise of a factor can increase the efficiency of trade receivables management for a company.**

- A 2 only
- B 1 only
- C Neither 1 nor 2
- D 1 and 2

Accounts payable, managing trade credit

20 A supplier offers a 3% discount for payment within 10 days or full payment within 45 days.

Assuming a 360-day year, what is the simple annualised cost of not taking the discount?

- A 37.1%
- B 36.0%
- C 31.8%
- D 24.7%

21 Scrimpy Co buys materials from Frugal Enterprises. Frugal offers discount terms of 2% discount for payment within 10 days or full payment within 30 days.

Assuming a 360-day year, what is the annual percentage cost associated with Scrimpy's failure to take advantage of the discount offered by Frugal?

- A 42.8%
- B 43.5%
- C 43.8%
- D 44.6%

22 A training company currently takes 30 days' credit from its freelance tutors. Annual fees to one tutor amount to \$86,000 and the company pays overdraft interest at 11%. The tutor has offered a 1.75% discount for settlement within 7 days. There are 365 days in the year.

What is the effect of accepting the discount on the company's profit (to the nearest \$)?

- A Benefit of \$1,505
- B Benefit of \$909
- C Cost of \$596
- D Cost of \$1,505

23 Generally, increasing the accounts payable payment period suggests advantage is being taken of available credit, but there are risks involved.

Which of the following is unlikely to be one of the risks involved in increasing the accounts payable payment period?

- A Customer bargaining power increasing
- B Losing supplier goodwill
- C Losing prompt payment discounts
- D Suppliers increasing the price to compensate

24 Colix Co, a major supplier of Component M to Amax Co, has offered a 1% early settlement discount for payment within 30 days. Amax Co currently takes 72 days to settle outstanding invoices with Colix Co, and its trade payables to Colix Co currently stand at \$3.2m. Amax Co has a cost of short-term finance of 6% through an overdraft and no surplus cash. Assume that there are 360 days in each year.

Using days as a basis, what is the financial consequence of accepting the early settlement discount?

- A Benefit of \$48,000 per year
- B Cost of \$32,000 per year
- C Cost of \$80,000 per year
- D Benefit of \$80,000 per year

(ACCA FM March/June 2022)

25 Plot Co sells Product Q, with sales occurring evenly throughout the year. The annual demand for Product Q is 456,000 units and Plot Co purchases this product at \$1 per unit on 60 days' credit. The supplier has offered an early settlement discount of 1% for settlement of invoices within 20 days. Plot Co holds no cash and finances working capital with short-term finance costing 5% per year. Assume that there are 365 days in each year.

What is the increase in Plot Co's annual finance costs if it accepts the early settlement discount for Product Q (to the nearest \$100)?

- A \$1,300
- B \$2,300
- C \$2,500
- D \$3,700

26 Plot Co (same data as above, Product Q, annual demand 456,000 units at \$1 per unit on 60 days' credit, supplier's early settlement discount of 1% for settlement within 20 days).

Which TWO of the following are potential advantages for Plot Co of accepting the early payment discount for Product Q?

- A A shortening in the operating cycle
- B An increase in profit margins
- C Improved business reputation
- D An increase in the current ratio

27 Which of the following is NOT a potential hidden cost of increasing credit taken from suppliers?

- A Damage to goodwill
- B Early settlement discounts lost
- C Business disruption
- D Increased risk of bad debts

Accounts receivable and payable, managing foreign trade

28 In relation to managing foreign accounts receivable, which of the following statements is correct?

- A Companies expecting to receive foreign currency from foreign accounts receivable will be concerned about the risk of the foreign currency appreciating against the domestic currency
- B The safest way to conduct business with foreign accounts receivable is by making a sale where goods are shipped and delivered before payment is due (open account)
- C There is no need to assess the creditworthiness of foreign customers if a company has export credit insurance for foreign accounts receivable
- D Discounting bills of exchange can reduce foreign accounts receivable default risk

29 Which of the following is LEAST likely to be used in the management of foreign accounts receivable?

- A Letters of credit
- B Bills of exchange
- C Invoice discounting
- D Commercial paper

Case Scenario Questions

Accounts receivable, invoice discounting and factoring

1 Widnor Co has credit sales of \$26,750,000 per year and 1% of these turn into bad debts. Customers are offered 50 days' credit but on average take an additional 10 days. Widnor Co has a cost of short-term finance of 5% per year.

The finance director is considering a proposal from a factoring company which would manage Widnor Co's sales ledger on a with-recourse basis. The factor believes it can reduce the average trade receivables collection period to 35 days, while cutting bad debts by 70%.

The factor would also advance Widnor Co 80% of the value of invoices raised at an interest rate of 7% per year. Assume that there are 360 days in each year.

(1) What is the level of Widnor Co's bad debts expense if it accepts the factor's offer?

\$

(2) What is Widnor Co's overall cost of financing receivables if it accepts the factor's offer (to the nearest \$000)?

- A \$1,766,000
- B \$1,498,000
- C \$172,000
- D \$123,000

(3) Which of the following methods could Widnor Co use to assess the creditworthiness of potential customers?

- (i) Trade references
 - (ii) Credit rating agencies
 - (iii) Financial statements analysis
 - (iv) Analysis of the yield curve
- A (i), (ii), (iii) and (iv)
 - B (i), (ii) and (iii) only
 - C (iii) and (iv) only
 - D (i) and (ii) only

Constructed Response Questions

1 Oscar Co designs and produces tracking devices. The company is managed by its four founders, who lack business administration skills.

The company has revenue of \$28m, and all sales are on 30 days' credit. Its major customers are large multinational car manufacturing companies and are often late in paying their invoices. Oscar Co is a rapidly growing company and revenue has doubled in the last four years. Oscar Co has focused in this time on product development and customer service, and managing trade receivables has been neglected.

Oscar Co's average trade receivables are currently \$5.37m, and bad debts are 2% of credit sales revenue. Partly as a result of poor credit control, the company has suffered a shortage of cash and has recently reached its overdraft limit. The four founders have spent large amounts of time chasing customers for payment. In an attempt to improve trade receivables management, Oscar Co has approached a factoring company.

The factoring company has offered two possible options.

Option 1: Administration by the factor of Oscar Co's invoicing, sales accounting and receivables collection, on a full recourse basis. The factor would charge a service fee of 0.5% of credit sales revenue per year. Oscar Co estimates that this would result in savings of \$30,000 per year in administration costs. Under this arrangement, the average trade receivables collection period would be 30 days.

Option 2: Administration by the factor of Oscar Co's invoicing, sales accounting and receivables collection on a non-recourse basis. The factor would charge a service fee of 1.5% of credit sales revenue per year. Administration cost savings and average trade receivables collection period would be as Option 1. Oscar Co would be required to accept an advance of 80% of credit sales when invoices are raised at an interest rate of 9% per year.

Oscar Co pays interest on its overdraft at a rate of 7% per year and the company operates for 365 days per year.

Required:

- (a) Calculate the costs and benefits of each of Option 1 and Option 2 and comment on your findings. (8 marks)
- (b) Discuss reasons (other than costs and benefits already calculated) why Oscar Co may benefit from the services offered by the factoring company. (6 marks)
- (c) Discuss THREE factors which determine the level of a company's investment in working capital. (6 marks)

(Total = 20 marks)

(ACCA F9, adapted, September/December 2018)

2 The finance director of Widnor Co has been looking to improve the company's working capital management. Widnor Co has revenue from credit sales of \$26,750,000 per year and, although its terms of trade require all credit customers to settle outstanding invoices within 40 days, on average customers have been taking longer. Approximately 1% of credit sales turn into bad debts which are not recovered.

Trade receivables currently stand at \$4,458,000 and Widnor Co has a cost of short-term finance of 5% per year.

The finance director is considering a proposal from a factoring company, Nokfe Co, which was invited to tender to manage the sales ledger of Widnor Co on a with-recourse basis. Nokfe Co believes that it can use its expertise to reduce average trade receivables days (i.e. collection period) to 35 days, while cutting bad debts by 70% and reducing administration costs by \$50,000 per year. A condition of the factoring

agreement is that the company would also advance Widnor Co 80% of the value of invoices raised at an interest rate of 7% per year. Nokfe Co would charge an annual fee of 0.75% of credit sales.

Required:

- (a) Advise whether the factor's offer is financially acceptable to Widnor Co. (7 marks)
- (b) Briefly discuss how the creditworthiness of potential customers can be assessed. (3 marks)
- (c) Discuss how risks arising from granting credit to foreign customers can be managed and reduced. (10 marks)

(Total = 20 marks)

(ACCA F9 June 2015, amended)

3 ZXC Co currently has income of \$30 million per year, of which 80% is from credit sales, and a net profit margin of 10%. Due to fierce competition, ZXC Co has lost market share and is looking for ways to win back former customers and to keep the loyalty of existing customers. The sales director has pointed out that a major competitor of ZXC Co currently offers an early settlement discount of 0.5% for settlement within 30 days, while ZXC Co itself does not offer an early settlement discount. He suggests that if ZXC Co could match this early settlement discount, annual income from credit sales would increase by 20%.

Credit customers of ZXC Co take an average of 51 days to settle invoices. Approximately 0.5% of the company's credit sales have historically become bad debts each year and written off as irrecoverable. The finance director has been advised that offering an early settlement discount of 0.5% for payment within 30 days would increase administration costs by \$35,000 per year, while 75% of credit customers would be likely to take the discount. The credit controller believes that bad debts would fall to 0.375% of credit sales if the early settlement discount were introduced.

ZXC Co has an average short-term cost of finance of 4% per year. Assume that there are 360 days in each year.

Required:

- (a) Evaluate whether ZXC Co should introduce the early settlement discount. (6 marks)
- (b) Discuss TWO ways in which a company could reduce the risk associated with foreign accounts receivable. (4 marks)

(Total = 10 marks)

(ACCA F9 September/December 2015)

4 KXP Co is an e-business which trades solely over the internet. In the last year the company had credit sales of \$15 million. All sales were on 30 days' credit to commercial customers.

Extracts from the company's most recent statement of financial position relating to working capital are as follows:

	\$'000
Trade receivables	2,466
Trade payables	2,220
Overdraft	3,000

In order to encourage customers to pay on time, KXP Co proposes introducing an early settlement discount of 1% for payment within 30 days, while increasing its normal credit period to 45 days. It is expected that, on average, 50% of customers will take the discount and pay within 30 days, 30% of customers will pay after 45 days, and 20% of customers will not change their current paying behaviour.

KXP Co currently orders 15,000 units per month of Product Z, demand for which is constant. There is only one supplier of Product Z and the cost of Product Z purchases over the last year was \$540,000. The supplier has offered a 2% discount for orders of Product Z of 30,000 units or more. Each order costs KXP Co \$150 to place and the holding cost is 24 cents per unit per year. KXP Co has an overdraft facility charging interest of 6% per year.

Required:

- (a) Calculate the net benefit or cost of the proposed changes in trade receivables policy and comment on your findings. (5 marks)
- (b) Calculate whether the bulk purchase discount offered by the supplier is financially acceptable and comment on the assumptions made by your calculation. (5 marks)
- (c) Identify and discuss the factors to be considered in determining the optimum level of cash to be held by a company. (5 marks)
- (d) Discuss the factors to be considered in formulating a trade receivables management policy. (5 marks)

(Total = 20 marks)

(ACCA December 2012, amended)

5 Constant Co has annual credit sales of \$45 million. Trade receivables currently stand at \$7.5 million, and bad debts are 2% of credit sales (i.e. \$900,000 per year). Constant Co is considering two alternative proposals to improve the management of its receivables. Constant Co's cost of short-term finance is 4% per year.

Proposal 1 - early settlement discount: Constant Co would offer a 1.25% discount to customers who pay within 30 days, instead of the current average of 60 days. It is estimated that 70% of customers (by value) would take up this discount. Bad debts would fall by \$225,000 per year.

Proposal 2 - factoring: a factor has offered to manage Constant Co's receivables for an annual fee of 1.5% of credit sales, and would advance 80% of the value of receivables at an interest rate of 5% per year (Constant Co's normal short-term finance cost is 4%, so this represents an additional cost on the advance). The factor's services would reduce the trade receivables collection period to 40 days and would reduce bad debts to 0.5% of credit sales.

Required:

- (a) (i) Evaluate the financial effect of the early settlement discount proposal. (4 marks)
- (ii) Evaluate the financial effect of the factoring proposal. (5 marks)
- (iii) Advise, on financial grounds, which of the two proposals Constant Co should choose. (1 mark)
- (b) Discuss THREE benefits to Constant Co of centralising its treasury management function. (6 marks)
- (c) Discuss TWO cash management models that Constant Co could use to manage its cash balances. (4 marks)

(Total = 20 marks)

Answers

Objective Test Questions

1 A

WORKING

Reduction in receivables = $60/365 \times \$20\text{m} = \$3,287,671$. Finance cost saving = $\$3,287,671 \times 12\% = \$394,521$.

Tutorial note: the question says the collection period is reduced BY 60 days, not TO 60 days; it falls from 73 days ($\$4\text{m}/\$20\text{m} \times 365$) to 13 days ($73 - 60$).

2 C

WORKING

$(60\% \times 10 \text{ days}) + (40\% \times 30 \text{ days}) = 6 + 12 = 18 \text{ days}$.

3 C

WORKING

The average collection period is the number of days revenue in trade receivables. Average receivables = $(14,600 + 12,900)/2 = \$13,750$. Average receivables collection period = $\$13,750/\$103,200 \times 365 = 48.6$ days, i.e. 49 days to the nearest day.

Tutorial note: a receivables collection period should be based on credit sales, unless otherwise stated, assume that all sales are on credit.

4 B

WORKING

	\$
Cost of system	80,000
Investment income ($3/360 \times \$90\text{m} \times 8\%$)	(60,000)
Net loss per year	20,000

5 A and C

Assessing the creditworthiness of new customers, and invoicing promptly and collecting overdue debts, are both key aspects of a credit policy for accounts receivable. Checking credit limits only once a year is too infrequent and may not be appropriate, for example, if a customer is known to be struggling, limits should be reviewed sooner. Delaying payments to obtain a 'free' source of finance is an aspect of a company's accounts payable policy, not its accounts receivable policy.

6 A

The key to reducing the percentage of bad debts is to assess the creditworthiness of customers. Since the industry average accounts receivable period is 75 days, PKA needs to be careful not to lose business as a result of over-stringent credit control action (such as legal action). A good approach would instead be to encourage early payment, for example through early settlement discounts.

7 \$2,625**WORKING**

Reduction in receivables = $\$4.5\text{m} \times 30/360 \times 35\% = \$131,250$.

Interest saved = $\$131,250 \times 8\% = \$10,500$.

Cost of discount = $\$4.5\text{m} \times 35\% \times 0.5\% = \$7,875$.

Net saving = $\$10,500 - \$7,875 = \$2,625$.

8 D**WORKING**

Reduction in receivables = $\$4,500,000 \times 30/360 \times 35\% = \$131,250$.

Interest saved at 10% = $\$131,250 \times 10\% = \$13,125$.

Cost of discount = $\$4,500,000 \times 35\% \times 1\% = \$15,750$.

Net cost = $\$13,125 - \$15,750 = \$2,625$ cost. Although this is the same underlying dataset as the Ling Co question above, the higher discount rate (1% vs 0.5%) and higher cost of finance (10% vs 8%) here produce the opposite result, a net cost rather than a net saving.

9 A**WORKING**

By taking the discount, a customer pays on day 15 instead of day 45, i.e. 30 days earlier, in exchange for giving up 2% of the invoice value.

Effective annual cost = $(1 + 2/98)^{365/30} - 1 = 27.9\%$.

10 B**WORKING**

Current receivables = $\$10\text{m} \times 60/365 = \$1,643,836$. Interest on current receivables at 10% = $\$164,384$.

If half the customers pay cash (taking the discount), revised receivables = $50\% \times \$10\text{m} \times 60/365 = \$821,918$.

Reduction in receivables = $\$1,643,836 - \$821,918 = \$821,918$.

Interest saved = $\$821,918 \times 10\% = \$82,192$.

Cost of discount = $50\% \times \$10\text{m} \times 2\% = \$100,000$. Net cost = $\$100,000 - \$82,192 = \$17,808$ cost.

11 A**WORKING**

Taking the discount means paying on day 30 instead of day 30/60/90, here, customers who currently pay after two months save one month, and customers who currently pay after three months save two months, by paying within one month instead in exchange for a 1.5% discount.

	\$
Amount paid if discount taken (per \$100 of invoice)	98.50
Discount given up by NOT taking it	1.50

2-month receivables (1 month saved): annual rate = $(1 + 1.5/98.5)^{12/1} - 1 = 19.89\%$.

3-month receivables (2 months saved): annual rate = $(1 + 1.5/98.5)^{12/2} - 1 = 9.49\%$.

Tutorial note: these figures are derived from first principles below, and confirm option A.

12 WORKING

Statement	Answer
Factoring is with recourse whereas discounting is without recourse	FALSE
Invoice discounting is usually performed on the entire receivables ledger	FALSE
Both are relatively cheap sources of finance	FALSE
Only factoring involves outsourcing the administration of the receivables ledger	TRUE

Debt factoring is usually performed on the entire receivables ledger and the factor takes over responsibility for debt collection and credit control; invoice discounting is normally used only for selected high-value invoices, with administration retained by the client.

13 C

The accounts receivable turnover ratio is expressed as sales/accounts receivable. A reduction in accounts receivable would improve (increase) the turnover ratio. Factoring would probably reduce the receivables collection period, and hence the level of receivables, thereby improving the company's accounts receivable turnover ratio.

14 C

WORKING

Average receivables = $\$1,500,000/12 = \$125,000$.

Annual finance charge = $\$125,000 \times 80\% \times 10\% = \$10,000$.

Service fee = $\$1,500,000 \times 2\% = \$30,000$.

Saved admin cost = $\$24,000$.

Net cost = $\$10,000 + \$30,000 - \$24,000 = \$16,000$.

15 C

Invoice discounting involves presenting selected sales invoices to a finance house and receiving a short-term loan based on an agreed percentage (usually up to 80%) of their nominal value. Responsibility for administering the receivables ledger is retained by the company, unlike debt factoring where administration is passed to the factor.

16 B

The factor would be prepared to provide a cash advance up to around 80% of the value of receivables, not above their face value.

17 C

A factor should be able to accelerate receipts so that they are in line with PKA's terms of trade, which will reduce accounts receivable staffing costs and improve liquidity. With-recourse factoring does not remove

the risk of bad debts, so statement (1) is incorrect. Customer relationships are more likely to deteriorate than improve when a factor takes over chasing debts.

18 \$28,500

WORKING

Cost of the package = (sales $\times$ 1.5%) + \$6,000 = (\$1.5m $\times$ 1.5%) + \$6,000 = \$22,500 + \$6,000 = \$28,500. Under this non-recourse package, the factor bears the risk of bad debts.

19 A

Statement (1) is incorrect: it is factoring with NO recourse (non-recourse), not with recourse, that provides insurance against bad debts. Statement (2) is correct: a factor's specialist expertise can improve the efficiency of trade receivables management.

20 C

WORKING

If the discount is refused, it costs \$3 to borrow \$97 from the supplier for an extra 35 days (45 - 10). The 35-day cost of refusing the discount is $3/97 = 3.09\%$. The simple annual rate is therefore 31.8% ($3/97 \times 360/35$).

Tutorial note: the simple annual rate was asked for, not the compound annual rate. Had the compound annual rate been asked for, the correct formula would have been $(1 + 3/97)^{360/35} - 1 = 36.8\%$.

21 C

WORKING

The discount period saved by refusing the discount is $30 - 10 = 20$ days. Compound annual cost = $(1 + 2/98)^{360/20} - 1 = 43.9\%$, i.e. 43.8% to the nearest option.

Tutorial note: this figure is derived from first principles below, and is closest to option C.

22 B

WORKING

Current account payable = \$86,000 $\times$ 30/365 = \$7,068. New account payable if discount taken = \$86,000 $\times$ 7/365 = \$1,649.

	\$
Increased interest expense $((\$1,649 - \$7,068) \times 11\%)$	(596)
Discount received $(\$86,000 \times 1.75\%)$	1,505
Benefit	909

23 A

Customer bargaining power increasing is associated with an increased receivables collection period, not with an increased payables payment period, so this is not a risk of increasing the accounts payable payment period. Losing supplier goodwill, losing prompt payment discounts, and suppliers increasing prices to compensate are all genuine risks of taking longer to pay.

24 A**WORKING**

Annual purchases of Component M = $\$3.2\text{m} \times 360/72 = \$16,000,000$ per year.

Revised trade payables = $\$16\text{m} \times 30/360 = \$1,333,333$.

Increase in financing cost = $6\% \times (\$3,200,000 - \$1,333,333) = \$112,000$.

Discount saving = $\$16\text{m} \times 1\% = \$160,000$. Benefit = $\$160,000 - \$112,000 = \$48,000$ per year.

(ACCA FM March/June 2022)

25 C**WORKING**

Product Q trade payables at end of year (before discount) = $456,000 \times \$1 \times 60/365 = \$74,959$. Product Q trade payables after discount = $456,000 \times \$1 \times 0.99 \times 20/365 = \$24,736$.

Decrease in Product Q trade payables = $\$74,959 - \$24,736 = \$50,223$. Increase in financing cost = $\$50,223 \times 5\% = \$2,511$, i.e. \$2,500 to the nearest \$100.

26 B and C

Accepting the discount improves profit margins (the discount received reduces the cost of purchases) and can improve business reputation with the supplier. It does NOT shorten the operating cycle, reducing the payables payment period actually lengthens the operating cycle. Since Plot Co has no cash and funds the payment from short-term financing, one current liability (trade payables) is simply replaced by another, so there is no impact on the current ratio.

27 D

Increased risk of bad debts relates to accounts receivable, not accounts payable, so it is not a hidden cost of taking extended credit from suppliers. Damage to supplier goodwill, losing prompt payment (early settlement) discounts, and disruption to the business relationship (e.g. reduced priority in times of shortage) are all genuine hidden costs of increasing credit taken from suppliers.

28 D

It is correct that discounting bills of exchange can reduce foreign accounts receivable (FAR) default risk, since it reduces the level of investment in FAR. Companies expecting to receive foreign currency from FAR are concerned about the risk of the foreign currency depreciating, not appreciating, against the domestic currency, so option A is incorrect. Open account terms (goods shipped and delivered before payment is due) are in fact one of the riskiest ways to conduct foreign trade, not the safest. Export credit insurance reduces the impact of non-payment, but does not remove the need to assess creditworthiness in the first place.

29 D

Commercial paper is a source of short-term finance for a company's own funding needs and is not directly applicable to the management of foreign accounts receivable. Letters of credit, bills of exchange and invoice discounting are all genuine tools used to manage the risk and investment associated with foreign accounts receivable.

Case Scenario Questions**1 (1) \$80,250****WORKING**

As the factoring is 'with recourse', Widnor retains the bad debts risk, although the level of bad debts is expected to fall by 70%. New level of bad debts = $0.3\% \times \$26,750,000 = \$80,250$.

(2) C**WORKING**

Revised level of receivables = $\$26,750,000 \times 35/360 = \$2,600,694$.

Revised cost of financing receivables = $(\$2,600,694 \times 80\% \times 7\%) + (\$2,600,694 \times 20\% \times 5\%) = \$145,639 + \$26,007 = \$171,646$, i.e. \$172,000 to the nearest \$000.

(3) B

Trade references, credit rating agencies and financial statements analysis can all be used to assess creditworthiness. The yield curve relates to the relationship between short and long-term interest rates and is not related to analysing the credit risk of potential customers.

Constructed Response Questions**1 (a) Costs and benefits of Options 1 and 2****WORKING**

Current trade receivables = \$5,370,000.

Revised trade receivables (both options) = $\$28,000,000 \times 30/365 = \$2,301,370$.

Reduction in receivables = $\$5,370,000 - \$2,301,370 = \$3,068,630$.

Option 1	\$
Reduction in financing cost ($3,068,630 \times 7\%$)	214,804
Reduction in admin costs	30,000
Benefits	244,804
Factor's fee ($28,000,000 \times 0.5\%$)	(140,000)
Net benefit	104,804

Option 2	\$
Reduction in financing cost ($3,068,630 \times 7\%$)	214,804
Reduction in admin costs	30,000
Bad debts saved ($28,000,000 \times 2\%$)	560,000
Benefits	804,804
Increase in finance cost ($2,301,370 \times 80\% \times 2\%$)	(36,822)
Factor's fee ($28,000,000 \times 1.5\%$)	(420,000)
Net benefit	347,982

Both options are financially acceptable to Oscar Co, with Option 2 offering the greater benefit (\$347,982 vs \$104,804), which should be accepted.

(b) Reasons why Oscar Co may benefit from the factoring company's services

Economies of specialisation: factors specialise in trade receivables management, offering 'economies of specialisation'. They are experts at getting customers to pay promptly and may achieve payment periods and bad debt levels which clients could not achieve themselves, the factor may be able to persuade the large multinational companies which Oscar Co supplies to pay on time.

Scale economies: because of the scale of their operations, factors are often able to manage receivables more cheaply than clients such as Oscar Co could on their own. Factor fees, even after allowing for the factor's profit margin, can be less than the client's own receivables administration cost.

Free up management time: factoring can free up management time to focus on more important tasks, a major benefit for Oscar Co, where the four founders are currently spending large amounts of time chasing customers for payment.

Bad debts insurance: insurance against bad debts (under non-recourse factoring) shields clients from non-payment. This is particularly attractive to smaller companies who may not be able to absorb the financial shock of a large bad debt, as could well be the case for Oscar Co, as a small company supplying much larger car manufacturers.

Accelerate cash inflow: factor finance can be useful to companies that have exhausted other sources of finance, useful to Oscar Co given it has recently reached its overdraft limit.

Finance through growth: although factor finance is generally more expensive than a bank overdraft, the funding level is linked to the company's volume of sales, helping to finance expansion and protecting against overtrading, a major potential advantage for a rapidly growing company like Oscar Co.

(c) Factors determining the level of a company's investment in working capital

A company's working capital investment is equal to the sum of its inventories and its accounts receivable, less its accounts payable. Three key factors determine its level.

The nature of the industry and the length of the working capital cycle: some businesses (e.g. housebuilding) have long production processes which inevitably lead to long working capital cycles and large investments in working capital. Others (e.g. supermarkets) buy on long credit terms, turn over inventory rapidly and sell for cash, often receiving payment from customers before they need to pay suppliers, giving little or even negative investment in working capital.

Working capital investment policy: a conservative approach offers long credit periods to customers, carries high inventory levels, and pays suppliers promptly, increasing working capital investment; an aggressive approach does the opposite, minimising working capital investment but increasing risk (e.g. stock-outs, lost sales, damaged supplier relationships).

The length of the cash operating cycle: the longer the cycle (inventory days plus receivables days less payables days), the greater the investment in working capital required, since cash is tied up for longer between paying for inputs and receiving cash from customers.

(ACCA F9, adapted, September/December 2018)

2 (a) Evaluation of the factor's offer

The factor's offer will be financially acceptable to Widnor Co if it results in a net benefit rather than a net cost.

	\$
Current trade receivables	4,458,000
Revised trade receivables (26,750,000 x 35/360)	2,600,694
Reduction in trade receivables	1,857,306

	\$
Reduction in financing cost ($1,857,306 \times 5\%$)	92,865
Saving in bad debts ($26,750,000 \times 1\% \times 70\%$)	187,250
Reduction in administration costs	50,000
Benefits	330,115
Increase in financing cost ($2,600,694 \times 80\% \times (7\% - 5\%)$)	(41,611)
Factor's annual fee ($26,750,000 \times 0.75\%$)	(200,625)
Net benefit	87,879

The factor's offer is therefore financially acceptable to Widnor Co, as it results in a net benefit of \$87,879 per year.

(b) Assessing the creditworthiness of potential customers

The creditworthiness of potential customers can be assessed from a range of sources of information. References are useful in this respect, and potential customers should supply a bank reference and a trade or other reference when seeking credit on purchases. Another source of information is the credit rating of the potential customer, which can be checked by a credit rating agency or credit reference agency. For larger potential customers, a file can be opened where additional information can be located, evaluated and stored, such as the annual report and accounts of the potential customer, press releases and so on.

(c) Managing risks arising from granting credit to foreign customers

Foreign debts raise special problems. When goods are sold abroad, the customer might ask for credit. Exports take time to arrange, and there might be complex paperwork. Transporting the goods can be slow, if they are sent by sea. These delays in foreign trade mean that exporters often build up large investments in inventories and accounts receivable, which have to be financed somehow. The risk of bad debts can also be greater with foreign trade than with domestic trade, if a foreign customer refuses to pay, the exporter must pursue the debt in the customer's own country, subject to the laws of that country.

A company can reduce its investment in foreign accounts receivable by insisting on earlier payment, or by arranging for a bank to give cash for a foreign debt sooner than the exporter would otherwise receive payment. Where the exporter asks its bank to handle collection of a bill of exchange or cheque on its behalf, the bank may be prepared to make an advance against the collection, typically 80% to 90% of its value. Negotiation of bills or cheques is similar, but used where the instrument is payable outside the exporter's own country. Discounting bills of exchange is where a bank buys the bill before it is due and credits the value of the bill, after a discount charge, to the company's account.

Export factoring can be used where the exporter pays for a factor's specialist expertise in order to reduce bad debts and the level of investment in foreign accounts receivable. Documentary credits (letters of credit) provide a secure, low-risk method of obtaining payment: the buyer requests a bank in its own country to issue a letter of credit in favour of the exporter, and the issuing bank, by doing so, guarantees payment to the exporter provided the terms and conditions of the credit are met. Countertrade, exchanging goods for other goods rather than for cash, is another means of financing trade.

Export credit insurance protects against the risk of non-payment by foreign customers for reasons outside the customer's control; if a credit customer defaults, pursuing the case through the courts can be lengthy, so this insurance shifts that risk to the insurer, though premiums are often high and the benefit may not always justify the cost.

(ACCA F9 June 2015, amended)

3 (a) Evaluation of the early settlement discount**WORKING**

Current credit sales income = \$30,000,000 x 80% = \$24,000,000.

Credit sales income after introducing the discount = \$24,000,000 x 120% = \$28,800,000.

Increase in income = \$4,800,000, so increase in net profit = \$4,800,000 x 10% = \$480,000.

Current level of bad debts = \$24,000,000 x 0.5% = \$120,000 per year. Revised level of bad debts = \$28,800,000 x 0.375% = \$108,000 per year, a saving of \$12,000 per year.

	\$
Trade receivables taking discount (28,800,000 x 75% x 30/360)	1,800,000
Trade receivables not taking discount (28,800,000 x 25% x 51/360)	1,020,000
Revised level of trade receivables	2,820,000
Current trade receivables (24,000,000 x 51/360)	3,400,000
Reduction in trade receivables	580,000

	\$	\$
Reduction in financing costs (580,000 x 4%)	23,200	
Increase in net profit	480,000	
Reduction in bad debts	12,000	
Benefits		515,200
Increase in administration costs	35,000	
Cost of discount (28,800,000 x 0.5% x 75%)	108,000	
Costs		(143,000)
Net benefit of the proposed discount		372,200

The early settlement discount produces a net benefit of \$372,200 per year, so ZXC Co should introduce it.

(b) Reducing the risk associated with foreign accounts receivable

A company could reduce the risk associated with foreign accounts receivable, such as export credit risk, by reducing the level of investment in them, for example, by using bills of exchange. If payment by the foreign customer is linked to bills of exchange, these can be discounted (sold to the bank at a discount to their face value, with the company getting cash immediately, decreasing the outstanding level of trade receivables) or negotiated (the bank makes an advance of cash to the company, with the debt settled when the bills are paid).

The risk of non-payment can also be reduced by raising an international letter of credit (documentary credit) linked to the contract for the sale of goods, which could be confirmed (guaranteed) by a bank in the foreign customer's country. The exporting company could also arrange export credit insurance against the risk of non-payment for reasons outside the customer's control, and reduce the risk of bad debts by performing the same creditworthiness assessment processes on foreign credit customers as domestic ones, such as seeking credit references and bank references.

(ACCA F9 September/December 2015)

4 (a) Cost/benefit of changing the trade receivables policy**WORKING**

Receivables paying within 30 days = $50\% \times \$15\text{m} \times 30/365 = \$616,438$.

Receivables paying after 45 days = $30\% \times \$15\text{m} \times 45/365 = \$554,795$.

Total receivables of customers changing their payment pattern = $\$616,438 + \$554,795 = \$1,171,233$.

Original value of these receivables (the 80% of customers changing behaviour) = $80\% \times \$2,466,000 = \$1,972,800$.

	\$
Original value of receivables changing behaviour	1,972,800
Revised value of these receivables	1,171,233
Reduction in trade receivables	801,567

	\$
Reduction in financing cost ($801,567 \times 6\%$)	(48,094)
Cost of early settlement discount ($50\% \times 15,000,000 \times 1\%$)	75,000
Net cost of changing trade receivables policy	26,906

Tutorial note: alternatively, the reduction in receivables can be found using collection periods, current collection period = $\$2,466\text{k}/\$15,000\text{k} \times 365 = 60$ days.

Revised weighted-average collection period = $(50\% \times 30) + (30\% \times 45) + (20\% \times 60) = 15 + 13.5 + 12 = 40.5$ days.

Decrease = 19.5 days.

Reduction in receivables = $\$15\text{m} \times 19.5/365 = \$801,370$ (small difference due to rounding).

The net cost of \$26,906 means the proposed change in trade receivables policy is not, on these figures alone, financially worthwhile, so KXP Co should not adopt it. However, the analysis excludes bad debts and assumes constant sales throughout the year, both of which the company may need to take into account. Given that receivables are on average failing to meet the current 30-day credit period, KXP Co may still wish to consider how its trade receivables policy could be changed to encourage earlier payment.

(b) Bulk purchase discount**WORKING**

Total annual cost of the inventory policy = cost of materials + ordering cost + holding cost. KXP Co requires 180,000 units of Product Z per year ($12 \times 15,000$).

Current policy	\$
Cost of materials	540,000
Ordering cost (12 orders $\times$ \$150)	1,800
Holding cost ($\$0.24 \times 15,000/2$)	1,800
Total annual cost	543,600

Proposed policy (orders of 30,000 units)	\$
Cost of materials ($540,000 \times 98\%$)	529,200
Ordering cost (6 orders $\times$ \$150)	900
Holding cost ($\$0.24 \times 30,000/2$)	3,600
Total annual cost	533,700

Net benefit of taking the bulk purchase discount = \$543,600 - \$533,700 = \$9,900 per year.

The bulk discount should therefore be accepted. However, KXP Co may wish to evaluate the appropriateness of a number of key assumptions before doing so: that demand for Product Z is constant throughout the year and does not change from year to year, and that ordering costs and holding costs are both constant throughout the year.

(c) Factors determining the optimum level of cash to hold

The level of cash required for the company's operations: this includes holding enough cash to pay for transactions expected to occur during the period (including paying suppliers and finance costs), which can be achieved by drawing up a cash budget, and to cover unexpected expenditure and uncertainty, a precautionary 'buffer' for unexpected events, estimated from previous experience, in addition to the amounts forecast in the cash budget.

The availability of finance: not all sources of finance are available to every company. A small or medium-sized company, for example, may not be able to obtain or extend bank loans as easily, and an unlisted company will find it difficult and expensive to raise funds through issuing securities. Where raising new finance is difficult and/or expensive, a company will need to hold more cash.

The availability and attractiveness of other uses for the cash: rather than holding cash for no return, a company usually has the option of placing it on deposit with a bank, investing it in short- or long-term debt instruments, or investing in equity shares of listed companies. How far the company pursues these alternatives depends on the amount of investment required, the expected level of return, the term to maturity and the ease of realising the investment. A company may also wish to hold cash so that it can take advantage of an unexpected speculative opportunity when one arises.

(d) Factors to consider in formulating a trade receivables management policy

The total credit: each company must determine the level of total credit it is willing to offer, balancing maximising revenue from customers against minimising the finance costs of funding the credit period and minimising bad debts. A longer credit period may attract more sales but increase finance costs; a shorter period reduces the need for additional finance but may lose sales opportunities. The cost of the additional finance, whether bank overdraft interest, loans or equity, must be considered.

Credit control: companies need a policy for assessing the creditworthiness of customers, verifying that new customers are creditworthy before concluding a sale to reduce the risk of default. This may involve requiring references for new customers, checking credit ratings through a credit rating agency, and offering a lower level of credit to new customers; a credit rating system may be devised to set the appropriate level of credit based on customer characteristics.

Collection: a credit policy can only be maintained if it is policed effectively and amounts owing are collected. The company needs to monitor customers' payment records to ensure credit limits are maintained, with a regular aged receivables analysis and any breaches brought to the attention of the credit controller. How tightly the policy is policed depends on factors such as the number of customers requiring more credit and the extent of the company's exposure to accounts receivable, and the costs of collection, internal or external, must not exceed the amount collected.

Changes to the credit policy: the credit policy needs to be reviewed regularly and revised as economic conditions and customer payment patterns change. The company may wish to assess whether it is beneficial to offer an early payment discount to encourage earlier payment, or to extend the credit period to encourage custom, the associated costs and impact on working capital must be considered, and a change should only go ahead once its financial benefit outweighs its additional costs.

(ACCA December 2012, amended)

5 (a) (i) Early settlement discount**WORKING**

Cost of the discount = $70\% \times \$45,000,000 \times 1.25\% = \$393,750$.

Benefit 1 - reduction in bad debts = \$225,000 (given).

Benefit 2 - reduction in financing cost from faster collection: current receivables days = $(\$7,500,000 / \$45,000,000) \times 360 = 60$ days.

New (blended) receivables days = $(30 \text{ days} \times 70\% \text{ taking the discount}) + (60 \text{ days} \times 30\% \text{ not taking it}) = 21 + 18 = 39$ days.

New receivables = $(39/360) \times \$45,000,000 = \$4,875,000$.

Reduction in receivables = $\$7,500,000 - \$4,875,000 = \$2,625,000$.

Financing cost saving = $4\% \times \$2,625,000 = \$105,000$.

	\$
Reduction in bad debts	225,000
Reduction in financing cost	105,000
Total benefits	330,000
Cost of the discount	(393,750)
Net effect	(63,750)

The net effect is a cost of \$63,750 per year, so the early settlement discount proposal is not worthwhile on financial grounds alone.

(ii) Factoring**WORKING**

Benefit 1 - reduction in bad debts = $(2.0\% - 0.5\%) \times \$45,000,000 = \$675,000$.

Benefit 2 - reduction in financing cost from faster collection: new receivables = $\$45,000,000 \times (40/360) = \$5,000,000$.

Reduction in receivables = $\$7,500,000 - \$5,000,000 = \$2,500,000$.

Financing cost saving = $4\% \times \$2,500,000 = \$100,000$.

Cost 1 - factor's fee = $1.5\% \times \$45,000,000 = \$675,000$.

Cost 2 - additional interest cost on the advance: the factor charges 5% versus Constant Co's normal 4% cost of short-term finance, a 1% additional cost, on the 80% of the new (reduced) receivables balance that is advanced = $1\% \times 80\% \times \$5,000,000 = \$40,000$.

	\$
Reduction in bad debts	675,000
Reduction in financing cost	100,000
Total benefits	775,000
Factor's fee	(675,000)
Additional interest cost on advance	(40,000)
Total costs	(715,000)
Net effect	60,000

The net effect is a benefit of \$60,000 per year, so the factoring proposal is financially worthwhile.

(iii) Advice

The factoring proposal is financially preferable, since it produces a net annual benefit of \$60,000, compared with a net annual cost of \$63,750 under the early settlement discount proposal (on an own-figure-rule basis from parts (i) and (ii)).

(b) Benefits of centralising treasury management

Economies of scale/specialist expertise: a central treasury team of specialists can be maintained more cost-effectively than treasury staff being duplicated in every division/subsidiary, improving the quality of financial risk management (e.g. foreign exchange, interest rate hedging) while reducing overall staffing costs.

Cash pooling/netting: centralising cash management allows surplus cash in one part of the business to fund a shortfall elsewhere (cash pooling), and allows intra-group payments to be netted off, reducing the group's overall external borrowing requirement and the associated interest cost, and reducing the number and cost of cross-border transactions.

Stronger negotiating power and consistent policy: a central treasury function can negotiate more favourable terms with banks and counterparties due to the larger, consolidated volume of business it can offer, and can apply consistent risk-management policies (e.g. hedging strategy) across the whole group rather than inconsistent, ad hoc decisions being made locally.

(c) Cash management models

Baumol model: treats cash management like an inventory/EOQ problem, balancing the transaction cost of converting securities into cash against the opportunity (holding) cost of holding cash, to determine the optimal amount of cash to raise each time by selling securities; it assumes cash outflows are constant and predictable over time, which limits its realism for businesses (like Constant Co) with variable day-to-day cash flows.

Miller-Orr model: sets an upper limit, a lower limit and a return point for the cash balance.

When cash reaches the upper limit, securities are purchased to bring the balance down to the return point, and when cash falls to the lower limit, securities are sold to bring the balance back up to the return point.

Unlike the Baumol model, it is designed for cash flows that fluctuate unpredictably day to day, making it more suitable in practice for many businesses, though it still relies on estimates of the variance of cash flows and the cost of transactions/interest rates that may be hard to forecast accurately.

Chapter 10: Working Capital Management: Cash and Funding Strategies

Questions

Objective Test Questions

Reasons for holding cash

1 Although cash needs to be invested to earn returns, businesses need to keep a certain amount readily available.

Which TWO of the following are reasons for holding cash?

- A Movement motive
- B Transactions motive
- C Precautionary motive
- D Asset motive

Cash budgets and cash flow forecasts

2 Evans Co has average daily cash outflows of \$3m. A new cash management system can add two days to the cash payments schedule. Evans can earn 10% per year on surplus funds.

What is the maximum amount that Evans should be willing to pay each year for this cash management system?

- A \$3,000,000
- B \$1,500,000
- C \$600,000
- D \$150,000

3 The following items were extracted from a company's budget for next month:

	\$
Purchases on credit	360,000
Expected decrease in inventory over the month	12,000
Expected increase in trade payables over the month	15,000

What is the budgeted payment to trade payables for the month?

- A \$333,000
- B \$345,000
- C \$357,000
- D \$375,000

4 A company commenced business on 1 April. Sales in April were \$20,000, but this is expected to increase by 2% each month. Credit sales amount to 60% of total sales.

The credit period allowed is one month. Bad debts are expected to be 3% of credit sales, but other customers pay on time.

How much cash is expected to be received in May?

- A \$19,560
- B \$19,640
- C \$19,800
- D \$20,160

5 JP Co has budgeted that sales will be \$300,100 in January 20X2, \$501,500 in February, \$150,000 in March and \$320,500 in April. Half of sales will be credit sales. 80% of receivables are expected to pay in the month after sale, 15% in the second month after sale, while the remaining 5% are expected to be bad debts. Receivables who pay in the month after sale can claim a 4% early settlement discount.

What level of sales receipts should be shown in the cash budget for March 20X2 (to the nearest \$)?

\$

(ACCA FM September/December 2021)

6 A cash budget was drawn up as follows:

	October \$	November \$	December \$
Receipts:			
Credit sales	20,000	11,000	14,500
Cash sales	10,000	4,500	6,000
Payments:			
Suppliers	13,000	4,200	7,800
Wages	4,600	2,300	3,000
Overheads	3,000	1,750	1,900
Opening cash	500		(balance b/f)

What is the closing cash balance for December budgeted to be?

- A \$20,550 overdraft
- B \$21,050 overdraft
- C \$22,450 deposit
- D \$24,950 deposit

7 A company is preparing its cash flow forecast for the next financial period.

Which THREE of the following items should be included in the calculations?

- A A corporation tax payment
- B A dividend receipt from a short-term investment
- C The loss made on the disposal of an item of machinery
- D A bad debt written off
- E An increase in a provision
- F The receipt of funding for the purchase of a new vehicle

8 Gaimett Co has some surplus cash. The managers have decided that they will hold on to \$600,000 of the available cash for 3 months, as they know that a good investment opportunity will become available at that time.

Which of the following should NOT be included in a cash flow forecast?

- A Payment of a one-off dividend
- B Receipt from an insurance company to cover the value of stolen inventory
- C Tax-allowable depreciation allowed in relation to a new asset purchase
- D Buy-back of shares

Cash management models (e.g. Baumol model, Miller-Orr model)

9 Identify, by clicking on the relevant box in the table below, whether the following statements about the Miller-Orr cash management model are true or false.

Statement	True	False
The greater the variability in cash flows, the greater is the spread between the upper and lower cash balance limits		
When short-term investments are liquidated, the company's cash balance should return to the lower limit		
If interest rates rise, the spread between the upper and lower cash balance limits rises		

10 A company uses the Baumol cash management model. Cash disbursements are constant at \$20,000 each month. Short-term investments yield 5% a year, while cash held in the company's bank account earns zero interest. The transaction cost for each sale of short-term investments is \$30.

Working to the nearest \$100, what is the optimal amount to be transferred in each transaction?

- A \$500
- B \$1,700
- C \$4,900
- D \$17,000

11 Which of the following is an assumption of the Baumol model but not of the Miller-Orr model?

- A Constant usage of cash
- B Holding cash incurs an opportunity cost
- C Fixed commission for switching between cash and cash equivalents
- D Variability of cash flows

12 A company needs \$150,000 each year for regular payments. Converting the company's short-term investments into cash to meet these regular payments incurs a fixed cost of \$400 per transaction. These short-term investments pay interest of 5% per year, while the company earns interest of only 1% per year on cash deposits.

According to the Baumol Model, what is the optimum amount of short-term investments to convert into cash in each transaction?

- A \$38,730
- B \$48,990
- C \$54,772
- D \$63,246

(ACCA FM June 2015)

13 Nelson Co has set up a new subsidiary which is expected to require \$3m in cash over the next year. Nelson will provide this cash by selling short-term investments, which yield 5% per year, and making transfers to the subsidiary's bank account, which pays interest at 1% per year. There is a fixed commission for selling investments of \$50 per transaction.

What is the economic transfer using the Baumol model?

- A \$77,460
- B \$86,603
- C \$7,746
- D \$8,660

14 The daily standard deviation of a company's cash flows is \$5,000. The company receives a yield of 5% per year on its short-term investments and pays a fixed commission of \$30 when buying or selling investments. Management have set a lower limit for cash of \$5,000.

What is the company's upper limit for cash, using the Miller-Orr model?

- A \$48,050
- B \$53,050
- C \$5,085
- D \$11,722

15 The treasury department in TB Co has calculated, using the Miller-Orr model, that the lowest cash balance they should have is \$1 million, and the highest is \$10 million. If the cash balance goes above \$10 million they transfer the cash into money market securities.

Are the following true or false?

Statement	True	False
When the balance reaches \$10 million they would buy \$6 million of securities		
When the cash balance falls to \$1 million they will sell \$3 million of securities		
If the variance of daily cash flows increases, the spread between upper and lower limit will be increased		

(ACCA FM September/December 2021)

Short-term investment and borrowing solutions

16 The CFO of Lang Co wants to earn a higher return on the company's cash holdings.

Which of the following comparable maturity investments will earn Lang the highest expected return?

- A Certificates of deposit
- B Treasury bills
- C Commercial paper
- D Bank deposits

17 Which of the following securities has the LEAST amount of default risk?

- A Corporate loan notes
- B Treasury bills
- C Commercial paper
- D Bills of exchange

18 Which of the following would LEAST likely be a function of a treasury department?

- A Managing relationships with banks
- B Liquidity management including investment of surplus funds
- C Currency management
- D Deciding appropriate asset depreciation rates

19 Widnor Co is considering a proposal from a factoring company that would improve the company's cash position, and Widnor's finance director is evaluating alternative short-term investments.

Which of the following statements are true or false?

Statement	True	False
A certificate of deposit is an example of a money market instrument		
Commercial paper is high-risk debt issued by a company		
Treasury bills are bought and sold on a discount basis		

20 Thrifty Co's cash budget highlights a short-term surplus in the near future.

Which of the following actions would be appropriate to make use of the surplus?

- A Pay suppliers earlier to take advantage of any prompt payment discounts
- B Buy back the company's shares
- C Increase payables by delaying payment to suppliers
- D Invest in a long-term deposit bank account

21 Gaimett Co's managers have decided that they will hold on to \$600,000 of available cash for 3 months, as they know that a good investment opportunity will become available at that time.

Which TWO of the following investments would be suitable to make use of this cash over this period?

- A Certificates of deposit
- B Listed company shares
- C Unlisted company shares
- D Instant access bank deposit account
- E Listed company debt

Strategies for funding working capital

22 Cain Co and Abel Co are two companies in the same industry. Each company finances its working capital using a mixture of long-term debt and short-term finance. Extracts from their working capital data are as follows:

	Cain Co	Abel Co
Maximum working capital requirement (\$m)	20	24
Minimum working capital requirement (\$m)	15	18
Long-term debt used to finance working capital (\$m)	17	18

Which of the following correctly describes the working capital funding strategy used by each company?

- A Cain Co: aggressive; Abel Co: matching
- B Cain Co: conservative; Abel Co: aggressive
- C Cain Co: conservative; Abel Co: matching
- D Cain Co: matching; Abel Co: aggressive

Case Scenario Questions

Topics 2-3 Cash budgets, cash flow forecasts and cash management models

1 King Co is a decentralised organisation whose different divisions have approached the matter of cash control in different ways, partly because of their different circumstances. Assume 365 days in a year.

The Western division is facing a two-year major capital investment programme for which significant funds need to be raised to meet the steady demand for cash. The division intends to use the Baumol model to decide when to raise funds, which will be done by selling off investments currently earning 5% per year. The transaction cost of these sales will be \$500 per transaction and the total amount needed over the two years is \$2,000,000.

The Alpine division has no significant investment plans but finds itself regularly having to either sell investments to make funds available or invest surplus cash. This is because of the considerable variation in daily cash inflows, which has been quantified as having a standard deviation of \$7,000. As a result, the division uses the Miller-Orr model to determine when to invest and when to make sales, using the same transaction cost and investment rate as the Western division.

The Southern division has just been created to manufacture a product used by other divisions but mainly sold to outside customers. The coming month, Month 1, will be spent converting and equipping existing corporate premises, at a total cash cost of \$800,000. Production will start in Month 2 and sales in Month 3. Planned sales volumes are as follows.

Month	3	4	5	6	7
Sales volume (000s)	10	11	13	16	20

The product will sell for \$24 per unit. 20% of sales will be for cash. Of the credit sales, 25% of credit customers (including other divisions) pay in the month following the sale, 50% will pay one month later and 25% after another month.

(1) How much finance should the Western division raise in a single tranche according to the Baumol model?

- A \$14,142
- B \$20,000
- C \$141,421
- D \$200,000

(2) What does the Miller-Orr model suggest is the spread between the upper and lower limits of cash levels that the Alpine division should maintain?

- A \$1,123
- B \$8,028
- C \$21,489
- D \$153,569

(3) What will be the Southern division's cash receipts from sales in Month 5?

\$

(4) It is important to distinguish between a cash budget and a cash forecast. What is the validity of the following statements?

Statement 1: A cash forecast is an estimate of cash receipts and payments for a future period under existing conditions before taking account of possible actions to modify cash flows, raise new capital, or invest surplus funds.

Statement 2: A cash budget is a commitment to a plan for cash receipts and payments for a future period after taking any action necessary to bring the preliminary cash forecast into conformity with the overall plan of the business.

- A Both statements are true
- B Statement 1 is true, Statement 2 is false
- C Statement 1 is false, Statement 2 is true
- D Both statements are false

(5) Which of the following is NOT an advantage to King Co of having a centralised treasury function?

- A Better foreign currency risk management
- B Borrowing can be made at more advantageous rates
- C Increases individual division's responsibility for cash
- D The treasury function can be run as a profit centre to raise additional profits

Constructed Response Questions

1 Flit Co is preparing a cash flow forecast for the three-month period from January to the end of March. The following sales volumes have been forecast:

	December	January	February	March	April
Sales (units)	1,200	1,250	1,300	1,400	1,500

Notes:

- 1 The selling price per unit is \$800 and a selling price increase of 5% will occur in February. Sales are all on one month's credit.
- 2 Production of goods for sale takes place one month before sales.
- 3 Each unit produced requires two units of raw materials, costing \$200 per unit. No raw materials inventory is held. Raw material purchases are on one month's credit.
- 4 Variable overheads and wages equal to \$100 per unit are incurred during production, and paid in the month of production.
- 5 The opening cash balance at 1 January is expected to be \$40,000.
- 6 A long-term loan of \$300,000 will be received at the beginning of March.
- 7 A machine costing \$400,000 will be purchased for cash in March.
- 8 Flit Co operates for 365 days per year.

Required:

- (a) Calculate the cash balance at the end of each month in the three-month period. (5 marks)
- (b) Calculate the forecast current ratio at the end of the three-month period. (2 marks)
- (c) Assuming that Flit Co expects to have a short-term cash surplus during the three-month period, discuss whether this should be invested in shares listed on a large stock market. (3 marks)
- (d) Explain how the Baumol model can be employed to reduce the costs of cash management. (5 marks)
- (e) Renpec Co, a subsidiary of Flit Co, has set a minimum cash account balance of \$7,500. The average cost to the company of making deposits or selling investments is \$18 per transaction and the standard deviation of its cash flows was \$1,000 per day during the last year. The average interest rate on investments is 5.11% per year.

Determine the spread, the upper limit and the return point for the cash account of Renpec Co using the Miller-Orr model, and explain the relevance of these values for the cash management of the company. (5 marks)

(Total = 20 marks)

(ACCA F9 December 2014, amended)

2 Guillermo Co is a medium-sized company that manufactures a single product. The product should take two days to manufacture.

Figures from its most recent financial statements are as follows:

	\$000
Revenue	6,000
Inventory, raw material	120

	\$000
Inventory, work in progress	20
Inventory, finished goods	80
Trade receivables	850
Trade payables	162

Credit sales account for 75% of the total revenue and Guillermo Co's gross profit margin is 60%. Guillermo Co has a large number of credit customers but there is no separate credit control function, and chasing of overdue debt is left to the relevant customer account manager. The finance manager considers this to be satisfactory to enable the finance department to concentrate on more important aspects of the business. Standard credit terms for credit customers is 30 days and no credit accounts are currently suspended. All customer account managers are paid a bonus based on the level of sales achieved.

Purchases

Raw material purchases in the year totalled \$2,160,000 and 90% of this was purchased on credit. Raw material is the only supply Guillermo Co purchases on credit. Guillermo Co has always tried to maintain good supplier relations and the average credit term from suppliers is 35 days.

Cash management

The management of Guillermo Co has decided it wishes the minimum cash balance to be \$40,000, but the cash balance is volatile with a daily standard deviation of \$15,000.

If extra cash is needed then Guillermo Co will sell tradeable securities from an investment portfolio it owns. The transaction cost for a purchase is the same as for a sale, at \$55 per transaction.

Interest rates are currently 9% per year. Assume a 360-day year for ALL calculations.

Required:

- (a) Calculate Guillermo Co's cash operating cycle. (5 marks)
- (b) Discuss THREE ways in which Guillermo Co could reduce its cash operating cycle. (6 marks)
- (c) Using the Miller-Orr model, calculate the target cash balance for Guillermo Co. (3 marks)
- (d) Discuss the motives for a business to hold cash. (6 marks)

(Total = 20 marks)

(ACCA FM September/December 2024)

Answers

Objective Test Questions

1 B and C

Transactions motive is to have enough cash to pay for day-to-day transactions. Precautionary motive is to hold cash to pay for unexpected items, e.g. a machine breakdown. 'Movement motive' and 'asset motive' are not recognised motives for holding cash.

2 C

WORKING

	\$000
Average daily cash outflows	3,000
Days added to payments schedule	x 2
Excess funds	6,000
Yield on excess funds	x 10%
Maximum acceptable cost for cash management system	600

3 B

WORKING

Budgeted payment to trade payables = purchases on credit less the expected increase in trade payables over the month (since a rise in payables means less cash is paid out than the value of purchases made) = \$360,000 - \$15,000 = \$345,000. The expected decrease in inventory is not relevant to the trade payables payment calculation.

4 C

WORKING

May sales = \$20,000 x 1.02 = \$20,400. May cash sales (received immediately) = \$20,400 x 40% = \$8,160.

April credit sales, collected in May = \$20,000 x 60% = \$12,000, less 3% bad debts = \$12,000 x 97% = \$11,640.

Total cash received in May = \$8,160 + \$11,640 = \$19,800.

5 \$290,084

WORKING

Receipts for March	\$
50% x March sales for cash (50% x \$150,000)	75,000
80% x February credit sales less 4% discount (50% x 80% x \$501,500 x 96%)	192,576
15% x January credit sales (50% x 15% x \$300,100)	22,508
Total	290,084

(ACCA FM September/December 2021)

6 D**WORKING**

	October \$	November \$	December \$
Total receipts	30,000	15,500	20,500
Total payments	(20,600)	(8,250)	(12,700)
Net cash flow	9,400	7,250	7,800
Balance b/f	500	9,900	17,150
Balance c/f	9,900	17,150	24,950

Total receipts = credit sales (collected in the same month, per the question's own presentation) plus cash sales; total payments = suppliers plus wages plus overheads. The December closing balance of \$24,950 is a deposit (positive balance), giving option D.

7 A, B and F

A corporation tax payment, a dividend receipt from a short-term investment, and the receipt of funding for the purchase of a new vehicle are all genuine cash flows and should be included in a cash flow forecast. The loss on disposal of machinery, a bad debt written off, and an increase in a provision are all non-cash accounting items and should be excluded.

8 C

Tax-allowable depreciation is not a cash flow and should not be included in a cash flow forecast. The payment of a dividend, an insurance receipt, and a share buy-back are all genuine cash flows and should be included.

9 WORKING

Statement	Answer
The greater the variability in cash flows, the greater is the spread between the upper and lower cash balance limits	TRUE
When short-term investments are liquidated, the company's cash balance should return to the lower limit	FALSE
If interest rates rise, the spread between the upper and lower cash balance limits rises	FALSE

Daily variance of cash flows and the interest rate on investments are key inputs into the formula for the 'spread'. Since interest rate appears in the denominator of the spread formula, a rise in interest rates DECREASES the spread, not increases it. When short-term investments are liquidated (sold for cash), the cash balance should rise to the 'return point', not merely to the lower limit.

10 D**WORKING**

Baumol model: $Q = \text{square root of } (2 \times CO \times D / CH)$, where $D = \text{annual demand for cash} = \$20,000 \times 12 = \$240,000$, $CO = \$30$, $CH = 5\% = 0.05$.

$Q = \text{square root of } (2 \times 30 \times 240,000 / 0.05) = \text{square root of } 288,000,000 = \$16,971$, i.e. \$17,000 to the nearest \$100.

11 A

Like the EOQ model, the Baumol model assumes constant usage/demand of cash. The Miller-Orr model, by contrast, can deal with variability in cash flows, so constant usage is not one of its assumptions. Both

models recognise that holding cash incurs an opportunity cost and both incorporate a fixed transaction cost for switching between cash and investments.

12 C**WORKING**

$D = \$150,000$, $CO = \$400$, $CH = 5\% - 1\% = 4\% = 0.04$ (the opportunity cost of holding cash is the excess of the return foregone on short-term investments over the interest earned on cash deposits).

$Q = \text{square root of } (2 \times 400 \times 150,000 / 0.04) = \text{square root of } 3,000,000,000 = \$54,772$.

(ACCA FM June 2015)

13 B**WORKING**

Transaction cost (CO) = \$50.

Annual demand for cash (D) = \$3,000,000.

Opportunity cost of holding cash (CH) = $5\% - 1\% = 4\% = 0.04$.

Economic transfer = square root of $(2 \times 50 \times 3,000,000 / 0.04) = \text{square root of } 7,500,000,000 = \$86,603$.

14 B**WORKING**

Variance = $\$5,000^2 = \$25,000,000$ per day. Daily interest rate = $5\% / 365 = 0.0137\%$ per day.

Spread = $3 \times [(3/4 \times \text{transaction cost} \times \text{variance}) / \text{daily interest rate}]^{1/3} = 3 \times [(0.75 \times 30 \times 25,000,000) / 0.000137]^{1/3} = \$48,050$.

Upper limit = lower limit + spread = $\$5,000 + \$48,050 = \$53,050$.

15 WORKING

Statement	Answer
When the balance reaches \$10 million they would buy \$6 million of securities	TRUE
When the cash balance falls to \$1 million they will sell \$3 million of securities	TRUE
If the variance of daily cash flows increases, the spread between upper and lower limit will be increased	TRUE

TB Co's spread = $\$10\text{m} - \$1\text{m} = \$9\text{m}$. The return point = lower limit + $(1/3 \times \text{spread}) = \$1\text{m} + (1/3 \times \$9\text{m}) = \$4\text{m}$.

When the upper limit is reached, sufficient securities are purchased to reduce the cash balance back to the return point: $\$10\text{m} - \$4\text{m} = \$6\text{m}$. When the lower limit is reached, sufficient securities are sold to increase the cash balance back to the return point: $\$4\text{m} - \$1\text{m} = \$3\text{m}$.

An increase in variance increases the spread, since variance appears in the numerator of the spread formula.

(ACCA FM September/December 2021)

16 C

The riskier an investment is, the higher the expected return. Of the four choices given, commercial paper carries the greatest risk to an investor and therefore offers the highest expected return. Commercial paper is an unsecured, short-term debt instrument issued by a corporation.

17 B

Default risk is the risk that the security will not be repaid because the issuing entity is insolvent or illiquid. Treasury bills are short-term debts issued by the government, which has virtually no risk of being insolvent or illiquid, so they carry the least default risk of the four options.

18 D

Deciding appropriate asset depreciation rates would usually be a function of the finance/accounting department, not the treasury department. Managing bank relationships, liquidity management (including investment of surplus funds) and currency management are all genuine treasury functions.

19 WORKING

Statement	Answer
A certificate of deposit is an example of a money market instrument	TRUE
Commercial paper is high-risk debt issued by a company	FALSE
Treasury bills are bought and sold on a discount basis	TRUE

Commercial paper is short-term, low-risk debt issued by companies with high credit ratings, not high-risk debt.

20 A

Paying suppliers early to take advantage of prompt payment discounts is a genuine, appropriate short-term use of a temporary cash surplus. Buying back shares and investing in a long-term deposit account are both solutions more suited to a long-term cash surplus, if the cash is needed again in the short term, it would not be readily available if either of these options had been used. Increasing payables by delaying payment to suppliers would help with a short-term cash shortage, not a surplus.

21 A and D

The cash is only available for investment over a 3-month period and will be needed at that point, so long-term investments are unsuitable, and high-risk investments where some or all of the cash may be lost should be avoided. Certificates of deposit and an instant access bank deposit account would both be suitable under these circumstances. Listed company shares and listed company debt would both be too risky, as their market values could fall, meaning the entire cash sum may not be returned; unlisted company shares may also be difficult to sell in time, as well as being high risk.

22 C

A company can fund its working capital using a mixture of long-term and short-term finance, and the balance between the two determines whether the strategy is aggressive, matching or conservative. Under a conservative strategy, long-term finance funds all of the permanent current assets plus part of the fluctuating current assets. Under a matching strategy, long-term finance funds the permanent current assets only, with short-term finance funding the fluctuating current assets. Under an aggressive strategy, short-term finance funds all of the fluctuating current assets plus part of the permanent current assets, so long-term finance does not even cover the minimum (permanent) working capital requirement.

Cain Co's minimum (permanent) working capital requirement is \$15m, but it uses \$17m of long-term debt to finance working capital. As long-term finance exceeds the permanent requirement, it also covers part of the fluctuating element, so Cain Co follows a conservative strategy.

Abel Co's minimum (permanent) working capital requirement is \$18m, and it uses exactly \$18m of long-term debt to finance working capital. As long-term finance exactly equals the permanent requirement, with short-term finance funding only the fluctuating element, Abel Co follows a matching strategy.

Case Scenario Questions

1 (1) C

WORKING

Since CH (the annual investment/interest rate) is expressed per year, D must also be an annual figure. Total demand over the two years is \$2,000,000, so annual demand $D = \$1,000,000$.

$Q = \text{square root of } (2 \times CO \times D / CH) = \text{square root of } (2 \times 500 \times 1,000,000 / 0.05) = \$141,421$.

(Using the total two-year demand of \$2,000,000 directly as D, instead of the annual figure, gives the distractor answer of \$200,000, a mismatch of time periods between D and CH.)

(2) D

WORKING

Daily variance = $\$7,000^2 = \$49,000,000$. Daily interest rate = $5\%/365 = 0.0137\%$ per day.

Spread = $3 \times [(0.75 \times \$500 \times \$49,000,000) / (0.05/365)]^{1/3} = \$153,569$.

(3) \$211,200

WORKING

Month	3	4	5	6	7
Sales volume (000s)	10	11	13	16	20
Sales revenue \$000 (x \$24)	240	264	312	384	480
Cash sales (20%)	48.0	52.8	62.4	76.8	96.0

Of credit sales (80% of revenue): 25% pay one month after sale, 50% pay two months after sale, 25% pay three months after sale.

Receipts in Month 5	\$000
Cash sales (Month 5)	62.4
1-month receivables: 25% of 80% of Month 4 sales ($25\% \times 264 \times 80\%$)	52.8
2-month receivables: 50% of 80% of Month 3 sales ($50\% \times 240 \times 80\%$)	96.0
Total	211.2

Total Month 5 cash receipts = \$211,200.

(4) A

Both statements are correct definitions, a cash forecast is the preliminary, unmodified estimate of cash flows, while a cash budget is the plan adopted after taking any corrective action needed to bring the forecast into line with the overall business plan.

(5) C

Centralising the treasury function REDUCES individual divisions' responsibility for cash, since responsibility is instead held centrally, so this is not an advantage of centralisation. A centralised treasury can achieve better foreign currency risk management (by netting opposing exposures across divisions), can borrow at more advantageous rates (through economies of scale), and can be run as a profit centre.

Constructed Response Questions**1 (a) Cash balance at the end of each month****WORKING**

	December	January	February	March	April
Sales (units)	1,200	1,250	1,300	1,400	1,500
Selling price (\$/unit)	800	800	840	840	
Sales (\$000)		960	1,000	1,092	

Sales are received one month after sale, so January receipts reflect December's units sold, February receipts reflect January's units sold, and March receipts reflect February's units sold, at the selling price applicable in the month of sale (including the 5% price rise from February).

	January	February	March
Prior month's sales (units)	1,200	1,250	1,300
Selling price (\$/unit)	800	800	840
Receivables receipts (\$)	960,000	1,000,000	1,092,000

Production takes place one month before sale, so December production is for January sales (1,250 units), January production is for February sales (1,300 units), and February production is for March sales (1,400 units). Each unit requires 2 units of raw material at \$200/unit, purchased on one month's credit and paid the following month; variable overheads/wages of \$100/unit are paid in the month of production.

	January \$000	February \$000	March \$000
Receivables	960	1,000	1,092
Loan			300
Total receipts	960	1,000	1,392
Raw materials (paid one month after production)	500	520	560
Variable costs (paid in month of production)	130	140	150
Machine			400
Total payments	630	660	1,110
Net cash flow	330	340	282
Opening balance	40	370	710
Closing balance	370	710	992

Raw materials are bought on one month's credit, so raw materials paid each month reflect the PREVIOUS month's production, at 2 units of raw material per unit produced costing \$200/unit; variable costs paid each month are the \$100/unit variable overheads and wages on THAT month's own production.

	January	February	March
Production one month earlier (for raw materials)	1,250	1,300	1,400
Raw materials paid (\$)	500,000	520,000	560,000

	January	February	March
Production this month (for variable costs)	1,300	1,400	1,500
Variable costs paid (\$)	130,000	140,000	150,000

Closing cash balances: January \$370,000, February \$710,000, March \$992,000.

(b) Forecast current ratio

Current assets: inventory at the end of the three-month period will be the finished goods for April sales of 1,500 units, valued at cost of production (\$400 materials + \$100 variable overheads = \$500 per unit) = $1,500 \times \$500 = \$750,000$.

Trade receivables will be March sales of $1,400 \times \$840 = \$1,176,000$.

Cash balance = \$992,000.

Current liabilities: trade payables will be the cash owed for March raw materials, i.e. 1,500 units (April production) $\times 2 \times \$200 = \$600,000$.

Current ratio = $(\$750,000 + \$1,176,000 + \$992,000) / \$600,000 = 4.9$ times.

(c) Should the short-term surplus be invested in listed shares?

When investing a cash flow surplus, a company should consider liquidity (how quickly and easily an asset can be converted into cash, shares in a listed company on a large stock market should be reasonably liquid), profitability (the company should seek a good return for the risk incurred, but a good return on shares usually requires a long-term holding period, whereas Flit Co's cash flow surplus is only temporary), and safety (the risk of the asset falling in value, share values can go down as well as up, which could lead to capital losses and cause problems meeting future cash outflows if the cash is needed in the short term, as is the case here).

Since the surplus is only short-term, investing in shares is inappropriate; placing the funds in a bank deposit account would be more appropriate.

(d) The Baumol model and cash management

A number of different cash management models indicate the optimum amount of cash a company should hold. The Baumol model is based on the idea that deciding on optimum cash balances is like deciding on optimum inventory levels, and it suggests the optimum amount to be transferred regularly from investments into the current account, applying the economic order quantity formula to cash.

Two types of cost are involved in obtaining cash: the fixed cost of each transfer (e.g. the transaction cost of selling investments or the cost of negotiating an overdraft) and the variable, opportunity cost of holding money as cash rather than as an interest-earning investment. The Baumol model balances these two costs to find the optimum transfer amount.

The Baumol approach has drawbacks for companies such as Flit Co:

In reality it is unlikely to be possible to predict cash requirements over future periods with much certainty.

No buffer of cash is allowed for, and there may be costs associated with running out of cash.

There may be other costs of holding cash that increase with the average amount held.

The model assumes constant transaction costs and interest rates.

(e) Renpec Co, the Miller-Orr model**WORKING**

Daily interest rate = $5.11\%/365 = 0.014\%$ per day.

Variance of cash flows = $(\text{standard deviation})^2 = \$1,000^2 = \$1,000,000$ per day.

Transaction cost = \$18 per transaction.

Spread = $3 \times [(0.75 \times 18 \times 1,000,000) / 0.00014]^{1/3} = \$13,757$.

	\$
Lower limit	7,500
Upper limit $(7,500 + 13,757)$	21,257
Return point $(7,500 + 13,757/3)$	12,086

Relevance of the values: the Miller-Orr model takes account of uncertainty in cash flows. Renpec Co's cash balance is allowed to vary freely between the lower and upper limits calculated by the model. If the cash balance reaches the upper limit, the firm buys sufficient securities to return the cash balance to the return point; when the cash balance reaches the lower limit, the firm sells securities to bring the balance back up to the return point. This helps Renpec Co decrease the risk of running out of cash while avoiding holding unnecessarily high, low-earning cash balances.

(ACCA F9 December 2014, amended)

2 (a) Cash operating cycle**WORKING**

Cost of sales = $\$6,000,000 \times (1 - 0.60) = \$2,400,000$.

Credit sales = $\$6,000,000 \times 75\% = \$4,500,000$.

Credit purchases = $\$2,160,000 \times 90\% = \$1,944,000$ (raw material is the only credit purchase, but the raw material inventory holding period itself is calculated against total raw material purchases of \$2,160,000, since it measures how long total raw material inventory is held, regardless of whether it was bought for cash or on credit).

	Days
Raw materials: $\$120,000/\$2,160,000 \times 360$	20
Work in progress: $\$20,000/\$2,400,000 \times 360$	3
Finished goods: $\$80,000/\$2,400,000 \times 360$	12
Receivables collection: $\$850,000/\$4,500,000 \times 360$	68
Payables payment: $\$162,000/\$1,944,000 \times 360$	(30)
Cash operating cycle	73

Cash operating cycle = $20 + 3 + 12 + 68 - 30 = 73$ days.

(b) Three ways to reduce the cash operating cycle

Improve credit control/receivables collection: at 68 days, the average receivables collection period is over twice the standard terms of 30 days. This suggests very poor credit control, particularly in relation to chasing up overdue debt. It appears the finance department is not taking ownership of this and is instead leaving it to account managers who are probably more interested in increasing sales than chasing old ones. Credit control should be brought within the finance department's responsibilities, and overdue accounts should be suspended until up to date.

Improve process efficiency: on average the product is taking three days to manufacture according to the work-in-progress calculation, which suggests the production process is running at only 67% efficiency (the product should take two days to manufacture). The production process should be examined to identify any bottleneck or other efficiency improvements that could be made. This would have knock-on benefits such as being able to reduce the level of finished goods held, as demand could be more readily met from production.

Raw material inventory holding: raw material holding of 20 days, given the three-day production period, would suggest that Guillermo Co's holding of raw material may be considered high. If so, raw material holding should be reduced, perhaps even considering whether just-in-time purchasing of raw material should be adopted.

Trade payables payment: the average credit term with trade suppliers is 35 days, yet Guillermo Co is paying its suppliers after 30 days on average. Paying earlier than the agreed term is detrimental to the cash cycle, and even though Guillermo Co likes to maintain good supplier relationships, this could still be achieved by paying on 35 days, improving the cash cycle by five days.

(Only three ways were required.)

(c) Miller-Orr model, target cash balance**WORKING**

Daily variance = $\$15,000^2 = \$225,000,000$.

Daily interest rate = $9\%/360 = 0.025\%$ per day (0.00025 as a decimal).

Transaction cost = \$55 per transaction.

Spread = $3 \times [(0.75 \times \$55 \times \$225,000,000) / 0.00025]^{1/3} = \$100,079$.

Target cash balance (return point) = $\$40,000 + (1/3 \times \$100,079) = \$40,000 + \$33,360 = \$73,360$.

(d) Motives for holding cash

Holding cash in a bank account does not usually generate high returns, and too high a cash balance is therefore undesirable, but there are three key motives for holding cash.

Transactions motive: cash should be kept in order to meet day-to-day expenses. Suppliers, for example, will require payment in cash in accordance with the agreed terms, and failure to do so could result in trade accounts being withdrawn, causing business disruption. Inability to pay employees' wages and salaries would also result in severe disruption due to industrial action such as a walkout of employees.

Precautionary motive: unexpected costs could occur which require a cash outlay, for example, an unexpected fine or claim against the business. Inability to settle such items would negatively impact the business in many ways, including reputational damage or even forced closure in extreme circumstances.

Investment/speculative motive: unexpected opportunities can arise which, while requiring cash outlay, could bring a financial return. Holding an amount of cash for such events could enable a business to take advantage of such opportunities, such as acquiring an undervalued company.

(ACCA FM September/December 2024, amended)



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